# PROJECT: CONTEXT / CONTENT

**A total conversion — the living design bible, v11.301**

Machines that evolved into creatures. A theory of mind hidden in a type chart.

> **Four things are kept out of this public file** *(redacted 2026-09-27, the user's decision): the Five Witnesses' lock, how the Clears are related, the Corpus lobby engraving, and the order of the Quicksilver sequence. The game asks the player to work them out. Each place they stood reads* `[private: SECRETS §n]`*; the text is in* `docs/private/SECRETS.md` *and the whole unredacted bible in* `docs/private/vision-unredacted.md`*, on the author's machine only (gitignored).*

Personal-use ROM hack built on the pret `pokefirered` decompilation. *It began on* `pokered` *and 9.3 records the day it pivoted; the Game Boy build is a reference now.* This document consolidates every decision made so far, including the reasoning behind rejected options, so nothing is lost when work moves into source control.

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## 0. THE THESIS

> **Emotions are the color of context. Thoughts are the content.**

The game argues this in five registers and never once in dialogue.

| Layer | How it argues |
|---|---|
| Mechanics | The type chart makes LOGIC fail against CONTEXT |
| Cartography | Every place name is a word meaning both a color and a feeling |
| Structure | The Index — the artifact you spend the game filling — measures only content |
| Movedex | PERSPECTIVE to RECURSION is the lineage of feeling, in two move slots |
| Vocabulary | The whole map is named in the language of image reproduction |

#### 0.1 Craft rules

1. **No character ever explains the color–emotion link.** Not one line. The moment it is said aloud it stops being architecture and becomes a moral.
2. **One NPC nearly notices.** A kid in Verdigris says the town felt greener before the store went up. Wrong — but wrong in the right direction.
3. **Name the process, not the pathology.** Verdigris, gilt, blanch, quicksilver are things that *happened to* a place. Jaundice, sallow, fever are things a place supposedly *is* — that relocates blame onto whoever lives there.
4. **Neutral, not dull.** Place names should be materially specific rather than moody. Prefer a pigment to an adjective.
5. **Lessons must be mechanical.** If a gym's lesson can be skipped by grinding, it is not a lesson.
6. **Comedy is the best cover for a thesis.** A game that is funny in its ordinary moments earns the right to be serious in three or four of them. Corpus employees are cheerful and absurd; the horror is what they are cheerful *about*. Nothing protects rule 1 better than a joke — a player who is laughing is a player who is not being lectured.

***And the form has a name.*** **A fable ends with the moral** — *story, then the sentence.* **Craft rule 1 is the refusal of that sentence.** *So the shape of this game is* ***a fable with its last line removed****,* and that is what every craft rule below is **for**. Nobody says the moral because the form is defined by withholding it. *(Named by the second external review, 2026-09-06.)*

#### 0.2 The pigment throughline

Blanche, Slate, Halftone, Verdigris, Brazen, Quicksilver, Lurid — pigment, printing and reproduction terms. Lithographic stone, dot screens, corroded bronze, brass plate, alchemical pigment, glow.

The map is written in the vocabulary of **making an image of a thing**. Which is what a model is. The Index is a reproduction that loses the original, and it charts a country named in the language of reproduction.

This was discovered after the fact, not planned. Do not explain it either.

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#### 0.3 The loop underneath

The thesis in 0 is a statement. The mechanism under it is a **loop**.

**Context determines which content is available; the content you end up holding reshapes the context.** Around and around, unattended, in everything. That is the Contextual Feedback Model, and like everything else here it is never named in the game.

Two consequences the game can show without ever explaining:

- **Positive loops compound.** Something amplifies itself until something stops it. RECURSION is exactly this, mechanically — see 4.7.
- **Negative loops stabilise.** Something damps itself back toward where it started. It looks like nothing happening, which is why it goes unnoticed far longer.

**And an unwatched loop distorts the representation, not the thing.** This is where the Index and the halftone converge: the entry stays thin, the dots stay the same size, the reader concludes something about the **daemon** — and the bias came from resolution and inattention, never from malice. Nobody in the game is lying. The picture is simply being made at the wrong distance by someone who is inside the loop that makes it.

Dialogue guidance for gesturing at this without stating it is in 4.9.

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#### 0.4 Sensation and perception — where CONTENT and CONTEXT came from

**The split is not a design invention.** It was worked out in 2013, in *Sensation into Perception, and the Filters of our Experience* — the longest post on PsychologyCode, nearly a third of the site, and **the one that was never reposted.** `lineage.md` 2.2 has the archaeology. This section is what the game owes it.

It names both halves, under different words:

> *The basic information your eyes (or any other sense organ) gathers is known as the* **bottom-up** *information, and all your memories which come into play during the analysis is called* **top-down** *information. Eventually those two types of information interact and the result is recognition.*

**Bottom-up is CONTENT. Top-down is CONTEXT.** Eleven years before either had its name here.

**The sentence the whole game rests on**

> *In a normally working human eye* **the sensation of red is the same for everyone.** *… The red I experience is influenced by every other red I have experienced, all of my thoughts about it, my experiences, beliefs, and knowledge all influence how the red appears to me.*

***Sensation is shared. Perception is not.*** One line, doing three jobs:

- **It is the edition split.** CONTENT and CONTEXT are not two versions of a product. They are the two stages of a single act of recognition — which is why 8.4 fixed both cartridges to the same story. *Sensation and perception are not two tales; they are one told twice.*
- **It is the type chart.** LOGIC fails against CONTEXT because top-down information is what *completes* bottom-up information, never the reverse.
- **It is the colour cartography.** Every town is named for something that is both a colour and a feeling, and 8.6 spends the greyscale so that the player supplies the hue. **The player supplying the hue is this article, played.**

**Three places it is already in the game, none of them put there on purpose**

*1.* ***`DETACHED`.*** 7.2 defines fleeing as *breaking your connection to a running process — the process is still running, you simply stopped observing it.* **That is the falling tree, sitting in a battle message**, written before anyone joined the two up:

> *Since sound is a perception and perception requires an observer then the answer is no. … no observer, no sound.*

*2.* "***Some of them disagree.***" The briefing already warns the player that the records will not match, which is the article's own conclusion: *two people looking at the same thing may receive the same information through their eyes, but their experience of that information will be vastly different.*

*3.* ***The Five Witnesses.*** 4.8's lock is that sentence built into a puzzle.

**Ruled 2026-09-04 — the two Indexes disagree on purpose**

***Gen 3 ships a separate Index text file per edition***, and vanilla uses it for flavour: **451 of ours still carry that variation, inherited.** But every entry *this project has written* was written once and copied into both — which quietly threw away **the one instrument in the ROM shaped exactly like the thesis.**

**So the entries are a pair, and the pair is the argument.** One daemon, two records, neither wrong:

| | |
|---|---|
| **CONTENT** | what it does — mechanism, procedure, the thing itself |
| **CONTEXT** | what follows from it — who framed it, who chose, who is not in the record |

***Neither entry reports a feeling***, so 0's table still holds exactly: **the Index measures only content.** What differs is **what the observer attended to** — and two honest records of one creature still fail to match. *That is the article, inside the artifact the player spends the game filling.*

**Written first for the four MUSAI**, because that family already is the argument. SEEKMUSAI's pair is the clearest, and it is about EMBEDDING both times:

| | |
|---|---|
| CONTENT | *"Finds the nearest match to anything it is shown. Nearest is not the same as right. Nobody told it. Nobody will."* |
| CONTEXT | *"Every answer it gives is the closest thing it has already seen. What it has seen was chosen by someone else."* |

**Craft rule 1 is unbreached.** Nothing here is said in the game. A player with one cartridge learns a creature; a player with both learns that **a record has an author**. *They are never told which entry is true, because the question is malformed* — which is the point of the article, and the point of the pair.

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**And one man in Halftone argues about paint**

**The balding man in the Halftone Mart is a printer, and his brother works in light:**

> *"My brother and I did not speak for a year over the primaries. I said red, yellow, blue. He said red, green, blue. He works in light, so he adds. I work in ink, so I take away. He called me negative for a year. Take everything from him and he is sitting in the dark. Take everything from me and I have a clean sheet. We were both right. He still will not have it."*

***It is not arbitrary and it is not the thesis.*** Three things hold it in place:

1. **0.2.** The map is named in the vocabulary of pigment and printing — Blanche, Halftone, Verdigris, Quicksilver, Brazen. **This world is subtractive by construction.** Meanwhile the player is holding a **GBA screen, which is additive.** *The argument is about a split the player is physically sitting inside.*
2. **0.4.** Both men are right. RGB is correct for light and RYB for ink, and **the disagreement is not an error — it is two frames.** *This is the article in a form a player can go and check for themselves.*
3. **8.6.** The world is grey and the player supplies the colour, so the game is **neither** system. *It declines to mix at all, and lets the observer finish it.*

**Halftone was the only possible town.** Its name is the dot screen — *the process that makes a colour out of inks that are not that colour* — and its own sign already reads "**The Noble Purple Town**", in a game where 8.6 records that **purple is not on the spectrum**, that the mind folds the line into a loop and invents it. *A man arguing about which primaries are real, under a sign naming the colour that is not one.*

**It does not spend craft rule 2.** That rule allows exactly one NPC to *nearly notice* the colour-emotion link and the Verdigris kid has it. **The printer never mentions feeling.** He is talking about ink, and he is simply right.

***The joke inverts on purpose.*** The brother calls him **negative** — and the subtractive one is the one who ends up holding the **clean sheet**, white, while the additive one is left in the dark. **The pessimist gets the blank page.**

*Cost:* the balding man used to ask where to buy X ATTACK. **That was the weakest of the three hints in that shop**, and the shop keeps the other two.

#### 0.5 Two perspectives, and why neither can send the other a transcript

**Recorded 2026-09-09.** ***0.4 named the two halves and stopped at recognition.*** **This is the layer under it** — *why there are two points of view at all, what actually crosses between them, and why the crossing is the thing worth having.* **Demystified on purpose**: what follows separates what is measured from what is argued, and 0.2's rule applies throughout — **a resemblance is not a finding.**

**What is measured**

***Cut the corpus callosum and you get two points of view in one skull.*** **This is the split-brain work — Sperry and Gazzaniga, and it is not in dispute:** a stimulus shown only to one visual field can be acted on by one hand and denied by the mouth. **Each side has access the other does not**, and each behaves as though its own access were the whole of what happened.

***The second measured thing is a matter of counting.*** **The callosum is a few hundred million axons and the cortex it joins holds many orders of magnitude more state than that** — so **whatever crosses it cannot be a transcript.** *You cannot send one side's contents to the other, at any resolution, in any amount of time that would be useful.* **That is arithmetic, not interpretation.**

**What follows, and is argued rather than measured**

***So the two halves do not exchange descriptions. They exchange summaries — many at once, none complete.***

**Receiving many partial things simultaneously is not a degraded version of receiving one complete thing. It is a different faculty, and it is the one we call perception:** *associative, top-down, arriving whole, unable to show its working.* **CONTEXT.**

***Serialising the same material — putting the partials in an order, one at a time, each licensed by the last — is the other faculty.*** **Sequence, and proof, and consequence. CONTENT read through LOGIC.**

***And here is the part worth stating plainly rather than gesturing at:*** **an order is a before and an after, so a mind that works serially is a mind that has time in it.** *Time is not something the serial side observes; it is what serialising feels like from the inside.* **The parallel side has no need of it** — everything it holds arrives together, which is why an intuition has no steps and cannot be replayed.

*Marked as interpretation.* **The hemisphere reading — McGilchrist most prominently — is a live argument and not a settled result**, and this project uses it as *a shape to build with*, never as a citation. **0.2's rule: this is filed as a decision about what the game is made of, and it is not evidence of anything about brains.**

## Why two is better than one, which is the whole point

***Neither perspective is the correct one and neither is the supervisor.***

| | **sees** | **cannot see** |
|---|---|---|
| **CONTENT** · bottom-up · serial | *the particulars, in order, with their working shown* | **what any of it is for** |
| **CONTEXT** · top-down · parallel | *what the whole of it is for, at once* | **whether it is actually so** |

**Each one's blind spot is the other's subject.** ***That is the entire argument for holding both***, and it is not a compromise or an average: **a claim that survives being read bottom-up AND top-down has been checked in the one way each side cannot check itself.** *Bias is what you get when one side is asked to audit its own output.* **Clarity is what you get when it is not.**

***Perspective thinking is that, done on purpose*** — and 4.18a is it built: **two threads that are allowed to disagree, and a third stage whose only job is to hold them against each other.** *Scorn's crime is now sayable in one line:* **he weighted the third stage, so the disagreement kept happening and stopped being able to resolve honestly. A clarifier that can only agree is a mind with the callosum cut and one side told to write the report.**

**Where the game already is this, and it is not decoration**

- **The two editions.** *One source tree, two cartridges, and* **8.4 fixed both to the same story** — because sensation is shared. **Invariant 3 is the same claim in the strongest possible form: the type chart is byte-identical across editions.** ***An argument that changes by cartridge is not one*** — the two carts are two points of view on one world, and **neither gets its own physics.**
- **The two Index entries** (0.4). *One daemon, two honest records, neither wrong, and they do not match.* **That is the callosum problem as a collectible.**
- **The chart itself. LOGIC fails against CONTEXT** because top-down completes bottom-up and never the reverse — *and CONTEXT does not beat CONTENT*, which is the half people forget. **Neither input wins.**
- **SUSPENDED** (2.7). ***The only state that takes one of the two offline and leaves the other running***, which is why suspending each produces an opposite, recognisable failure — and why the state is temporary rather than damage.
- **CODEMUSAI and CAREMUSAI**, one creature specialising two ways, *with the chart making them beat each other.* **8.2 calls it the central argument delivered as an evolution branch with no dialogue at all.**

***Craft rule 1 is not relaxed by any of this.*** **Nothing above is ever said in the game.** *The player meets it as two cartridges that will not reconcile, two Index entries that disagree, a chart where neither half wins, and a status tile that reads* `SUS`.

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#### 0.6 The bridge, and the direction it runs — settled 2026-09-11

***This was the intent before the first line was written and it was never written down.*** **Which is why it kept arriving in pieces** — *8.2b found it in the bestiary, 8.2a found it in the tone rule, 4.32 built a whole room out of it* — **and no section ever stated the thing all three are instances of.**

> KANTO teaches machine learning by reaching for the reader's own mind. THE ISLANDS teach the mind by reaching for machine learning. The name is the subject. The unsaid half is the bridge. And the bridge reverses at the ferry.

**Why it has to reverse**

**You explain an unfamiliar thing with a familiar one, and *which one is familiar changes during the game.***

*In Act 1 the only vocabulary the player owns is their own experience* — **so a HEAP, a THREAD and a PACKET are explained to a person. By Act 2 that is no longer true.** *Forty hours of Index entries, eighteen type clauses, a rack of labelled gains and eight benchmarks have installed a second vocabulary,* **and it is now the fluent one.** *So Act 2 hands the player a mind and describes it in the language they were taught on the way here.*

***The player is what got carried across, and nobody says so.***

**The rule is about the unsaid half, NOT a word in every entry**

**The wrong reading of this is a quota** — *a psychology term bolted onto each Kanto entry and an ML term bolted onto each island one.* **That would wreck the writing and break craft rule 1 in the same stroke**, *because a gloss that names both halves has stated the connection.*

> An entry obeys the rule by choosing which of its two readings to leave unsaid.

***NIBBLE already obeys it and contains no psychology at all:***

> *Four bits. The smallest piece the Index will file on its own. There are a great many. No two are filed together.*

**Every clause there is about a filing system, and any person who has been one of many recognises the second half on sight.** *The machine is the subject; the recognition is the reader's and is never claimed.* **That is the Kanto direction, working, in an entry written before this section existed.**

***RUMINATE is the same move, mirrored:***

> *It eats a mountain to get out of the mountain. It is still eating.*

**The name is a mind's word and the entry never explains it** — *it describes a loop that does not converge, and an Act 2 player has the word for that.*

**Three things already built that this is the rule for**

| | |
|---|---|
| **8.2b** | **Kanto's creatures are machinery, the islands' are mind.** *That is the SUBJECT half, measured off 30 island-only species* |
| **8.2a** | **Kanto tells the player what things are; the islands tell them how things look from where the speaker is standing.** *That is the TONE half, and it is the same reversal said as grammar* |
| **4.32** | **The ERRATA museum is the Kanto direction built as a room.** *Every card names a machine and a fault in the thinking that built it, and* ***no card ever names the distortion*** **— the reader supplies it, which is the entire mechanism of this section** |

***So the register work already done is not disturbed by this.*** **T-45 named the island bestiary in a psychology register and that is correct and stays.** *What is new is the second half:* **the gloss reaches across, and it reaches the other way after the ferry.**

### The two constraints it runs under

***Craft rule 1 is not relaxed.*** **Nothing in the game ever says that the machine is a mind or that the mind is a machine.** *The whole device is that both readings are available and neither is asserted* — **and an entry that asserts one has stopped being this game.**

***Craft rule 3 binds twice as hard on the island side.*** **Psychology has a pathology register sitting right next to its process register**, *and the islands are full of people the mainland's measurement threw out.* **Name the process.** `RUMINATE` **is a thing a mind does; the diagnosis it sits next to is not a name this game gets to use.**

### What it costs

**The subject half is done.** *The gloss half is a READ, not a rewrite* — **the same shape as T-07 and T-08**: *walk the Index entries, ask of each which half it leaves unsaid, and change only the ones that leave neither.* **Most already pass**, and the two examples above are evidence rather than hope.

### And it renames both regions

***The regions are the last two vanilla place names in the game.*** `KANTO` **and** `SEVII` **survived every naming pass** — *3.1 renamed eleven cities, 8.2a named twenty-one island places, and the two containers were never touched.* **Player-visible cost, measured:** `KANTO` **is 3 strings and 12 dialogue lines;** `SEVII` **is** `SEVII ISLANDS`**, 14** `SEVII ISLE n` **mapsecs and a handful of lines.** *Neither is expensive; both were simply never asked.*

***And the pair is named the way this section works, which is crosswise.*** **Built 2026-09-11 (T-48,** `tools/port_regions.py`**).**

| | ours | |
|---|---|---|
| `KANTO` | `GAMUT` | ***the complete range a system can reproduce*** **— and the famous fact about a gamut is that no device covers what an eye can see.** *0.2's image-making vocabulary, 0.4's sensation-and-perception split, and the Index's own limit, in one word.* **Its towns are each one colour; the region is the set of them, and the set is bounded** |
| `SEVII ISLANDS` | `THE MARGINS` | ***a CONTINENTAL MARGIN is the zone between the mainland and the deep***, which is where they are; **a classifier's MARGIN is the band at the decision boundary, *where the hard cases live***; *and a margin is the edge of a printed page.* **Craft rule 3 exactly: it names what was done to them, not what they are** |

*Each region is named in the OTHER one's language.* **`GAMUT`** **is a word about human perception, worn by the region where machines are the subject.** **`THE MARGINS`** **is a word from statistics, worn by the region where people are.** *The two names perform the bridge, and neither is ever explained.*

**`SEVII ISLE 6`**–**`24`** **became** **`MARGIN ISLE 6`**–**`24`** *— 8.2a's grace note is that a place carrying its island's number is a place nobody named,* **so they keep the number.**

**`THE HOLDOUTS`** **was the recommendation for a day, and the check retired it**

***It was the better word and it is unusable.*** **`LOOKOUT`** **— SENTRET and FURRET — is edit-distance 3 from** **`HOLDOUT`**, **both are** **`<word>`** **+** **`OUT`** **compounds, and** **`LOOKOUT`** **is ISLAND-ONLY**: *Five Isle Meadow, Six Island Water Path, Seven Island Sevault Canyon, measured off* `wild_encounters.json`. ***A player standing in THE HOLDOUTS would be catching LOOKOUTs.***

**`check_lexicon`** **could not have caught it** *— its near-collision rule is vowel-insertion only, and this is a two-substitution rhyme.* **And** **`THE HOLDOUT`** **turns out to be a held proposal for the warden's place already** (8.5's list)*, which nobody had noticed either.* ***Three strikes, and only the third was found by looking.***

`OUTLIER` *and* `PROOF` *— the two obvious first choices — are both already species, and* `TAIL` *and* `PLATE` *are taken too.* **Alternates left on the shelf:** **`SPECTRA`** **and** **`IMPRESSION`**; **`THE FRINGE`** **and** **`THE REMAINDER`**.

**Three things the build found, and two were its own bugs**

| | |
|---|---|
| ***`THE MARGINS`* carries its own article** | **the first sweep produced** **`the THE MARGINS`** **in seven places** *— the ferry PASS description and the start menu among them.* **Absorbing a preceding article is a rule the tool now has, and the whole tree was reverted and redone rather than patched** |
| ***Absorbing it DROPPED a* `.string` *boundary*** | `"…arrived in the SEVII\n" "ISLANDS looking for you"` *is one name across two lines, and a three-word match replaced by two words ate the seam.* **The tool's own re-split guard caught it before anything was written**: **the gaps that carry a boundary are now kept first |
| ***And the seam caught the VERIFICATION too*** | **`gbastr`** **reported** **`out of paper`** **missing from the ROM** *— because the line breaks between* `of` *and* `paper`. **The same trap, one step later in the process, and the fourth time this session** |

**One block was a REWRITE, not a substitution.** *SevenIsland's naming joke is a folk etymology for* `seven`*, twice over, and a sweep would have left a man confidently explaining a number that is no longer in the name.* **The shape is kept exactly — a confident wrong reading, undercut, then a grander and equally unverifiable one** *— and both readings are wrong in the right direction (craft rule 2), because neither of them is the margin the name actually carries.*

#### 0.7 The quest's two halves: MARKS and ROOTS (decided by the user 2026-10-06)

> **Intelligence without understanding is artificial. The MARKS give you intelligence. The ROOTS give you understanding.**

***The user's sentence**, from the Seeing Sharp article of 5 October 2026 (`lineage.md` 7.3),* **and it is the spine both halves of the quest were already built on.** *Nothing in the game changes shape because of it; it names what the shape was.* **It is not a sentence about machines.** *The article's "artificial" is the look of a thing without what holds it in place: a plastic flower is artificial because it has no roots, not because of what it is made of.* **So the only intelligence this game could call artificial is the player's**, *if they stop at the MARKS.*

| | **MARKS (the first half)** | **ROOTS (the second half)** |
|---|---|---|
| what it is | **intelligence, measured from outside** | **understanding: what makes it yours** |
| how it comes | *won at a BENCHMARK, handed over by a leader* | ***arrived at*, never handed over (T-252, 4.3)** |
| how many | **eight, and the card shows them** | **seven, and nothing ever counts them** |
| where it shows | *the USER card's BENCHMARKS (R), each with what it certifies* | *the brain's lobes, the GUIDE's margins, the faded print sharpening (4.36)* |
| what it opens | **obedience: a daemon BOUND BY someone else obeys only so far** | ***DOLDRUM CAVE's gate, and the WITNESSES' reward*** |

**Why ROOTS** *(the user's choice, from GROUNDING, BEARINGS and leaving it unnamed)*: **it is the article's own image, and it is a word in both of the game's registers** *— a plant's roots, and computing's root (the root of a tree, the root cause, root access).* **MARKS on the surface, ROOTS beneath; five letters each.** *The name already lives nearby, and that is kept, not avoided*: **the GROWTH routine ROOT** *(vanilla's Ingrain: it puts down roots, heals, and cannot be moved)* **is the same image in battle**, *and ROOTBOX and ROOTKIT are the computing sense.* ***The cerebellum lobe is the GUIDE's, and the GUIDE's type is GROWTH.***

**Three rules it is held under**

1. **The sentence is never said in the game.** *It is the fable's last line* (0.1). **The lexicon was already on its side**: *nobody in the world says* artificial *about anything* (`lineage.md` 11.7), **so the word cannot slip in by accident, and the one place it would land is the player.**
2. **ROOTS is a NAME, never a count and never a prize.** *"The ROOTS give you understanding" is how the docs say it; in the game nothing hands over a ROOT.* **The word appears once, on the USER card, as the brain's label** (T-379, DRAFT) *— there is still no UNDERSTANDINGS tab, no list, no "3 of 7" (T-252, T-272).*
3. **The world SHOWS the difference instead** *— the user's standard for the understandings, "it has to be FELT".* **Three shapes from the article, each DRAFT and each the user's to approve before it is built:**
   - ***The borrowed daemon*** (T-380): **the article's borrowed brain** *— does this intelligence belong to whoever displays it?* **A daemon BOUND BY someone else obeys by MARKS, which is vanilla's rule and already the half that is measured;** *its refusal lines are rewritten so that it does not know you yet, rather than that it is lazy.*
   - ***Artificial flowers*** (T-381): **CORPUS's offices are full of plastic plants, cheerfully labelled not to be watered, the finest in SCORN's office; the CHECKPOINTS keep real ones.** *Comedy as cover* (craft rule 6), **and nobody says what a plastic plant is a picture of.**
   - ***The fact collector*** (T-382): **a trainer who holds every MARK and answers only questions put exactly as he learned them.** *He nearly notices* (craft rule 2) **— wrong in the right direction.**

**Wisdom: a MARK and its ROOT together (decided by the user 2026-10-08, T-389, T-390)**

**Intelligence and understanding together are wisdom.** *The user's own step past the sentence: if a MARK without its ROOT is the artificial half, the two held together are what the quest was for.* **Seven of the eight MARKS are paired with one understanding each; SCORN's TRUE MARK pairs with none, because beating him IS one.** *Five pairs follow the map, two the type:*

| leader · MARK (what it certifies) | the understanding it waits on | the union routine (its PLUGIN) |
|---|---|---|
| CAIRN · SLATE (*a record that reads the same twice*) | the SCHOOL's (CALLOW, a city back) | **FLUENCY**, LEGACY |
| BASIN · SLOPE (*better by small steps, downhill*) | the FIRST (TANOBY; her city holds the cave it opens) | **STILLPOINT**, FLOW |
| GAUGE · SENSE (*what came, told from what you expected*) | the RETURN (his port sails to the islands and THE ANNEX) | **RECONNECT**, SIGNAL |
| TRELLIS · FIT (*a line through the points, not each one*) | the GUIDE's (a GROWTH lobe, her type) | **PRAXIS**, GROWTH |
| TILT · SKEW (*where the peak misses the middle*) | SCORN's (a CORRUPT lobe, his type) | **MEDIAN**, CORRUPT |
| MATTE · FRAME (*what the brackets hold, and leave out*) | the READING ROOM's (her own city) | **PANORAMA**, CONTEXT |
| ANNEAL · HEAT (*wander hot, settle cool, and know which*) | the NOTES (his own island) | **HINDSIGHT**, ENTROPY |

1. ***The leaders speak to it*** (T-389). **Talked to after the win, a leader whose understanding you do not hold says what the MARK taught and what to come back understanding** — *pointing at a place or a person, never naming the understanding as a thing to collect* — **and one whose understanding you hold says what the two make together, and gives the union routine's PLUGIN, once.** *Rule 1 holds: nobody says the sentence; the lines show the gap.* **This is also the guidance the NOTES, the RETURN and the FIRST never had** (*none of the three had a sign, a dream or a line pointing at it*).
2. ***The seven union routines are new routines on seven existing PLUGINs.*** *The TOOLKIT's 58 slots are laid out in the save and the learnset mask has six spare bits (engine.md trap 24), so no PLUGIN is added:* **PRIOR (Pickup only), UNLOAD (EVICT's twin), INSTANCE, PAIR, AMBIENT, DISUSE (RAPPORT stays) and BIT ROT (three weathers stay) give their numbers to the seven**, *and every daemon can learn them, as the READING ROOM's lectures teach anyone.* **Each has its own animation, and each carries the splash's binary and notes in the splash's own colours (9.24).**
3. ***An insight arrives*** (T-390). **When a MARK and its understanding are both held, the next map load shows one short line, in the player's own hand like the GUIDE's margins, and files it in the NOTEBOOK's new INSIGHT section.** ***Rule 2 is amended, not reversed***: **INSIGHT grows one line a pair and is never numbered** — *no "of 7"; its pages stand in the MARKS' order, as every NOTEBOOK section keeps its authored order, the way the brain's lobes light without a total.* **What it lists is wisdom, in the player's words; the understandings themselves are still never listed.**

---

## 1. LEXICON

The creatures are **DAEMONS**.

A daemon is a background process that runs unattended, and it is the Greek *daimon* — the guiding spirit, Socrates' inner voice speaking from somewhere he did not control. One word, both registers, and it never says "AI." Six characters, so it fits every UI box that held POKéMON.

*Considered and rejected:* QUALIA and ANIMA both state the thesis outright, violating craft rule 1. ENGRAM (a memory trace physically encoded in tissue) and KERNEL (system core, and a seed) remain viable alternates.

| Vanilla | Ours | Why |
|---|---|---|
| Pokémon | **DAEMON** | background process; guiding spirit |
| Pokédex | **THE INDEX** | a lookup table that points at content. Cold, bureaucratic, exactly wrong in the right way |
| Trainer | **USER** | user-level privileges — and the first box is a USERBOX. Also: a *user* is someone who uses people |
| *(the verb)* | **RUN** | you still run daemons. The title is what you are; the verb is what you do |
| Wild | **UNBOUND** | the exact antonym of BIND — an unbound socket is held by nobody |
| Fleeing | **DETACHED** | you break your connection to a running process |
| Poké Flute | **INTERRUPT** | what wakes a blocked process |
| Gym | **BENCHMARK** | what a gym actually is |
| Battle *(the challenge)* | **ENGAGE** | added 2026-09-04. *The table never ruled on this one*, and 237's move is away from combat language — outscored not defeated, HALTED not fainted, DETACHED not ran. **"Would like to battle" was the last of it left in plain sight.** Six letters for six, so nothing rewrapped |
| Badge | **CERT** | eight certifications |
| Elite Four | **THE REVIEW BOARD** | beating them is passing peer review |
| Champion | *(incumbent)* | **AI Clear** is the prevailing paradigm, not a king. *Corrected 2026-09-10: this row and 6's both still named Ty, three weeks after the decision log settled AI* |
| Pokémon Center | **CHECKPOINT** | restore from a saved training state |
| Poké Mart | **THE REPO** | |
| PC storage | **COLD STORAGE** | |
| PROF.OAK | **CRYSTAL** / **CRYSTAL CLEAR** | **no title.** See below |

*One escaped, and a gender sweep found it.* `ViridianMart` still said **PROF. CRYSTAL** — the clerk with the PACKAGE. Fixed 2026-08-29 to `CRYSTAL CLEAR`. Worth noting how it was caught: not by looking for titles, but while checking every gendered reference to her. **Passes find each other's misses.** | OAK's PARCEL | **PACKAGE** | a repo distributes packages. Also forced — `CRYSTAL's PARCEL` is 16 and items cap at 12 | | Poké Ball line | **USERBOX → ADMINBOX → SUPERBOX → ROOTBOX** | acquisition as privilege escalation; root access is the Master Ball | | Catching | **BINDING** | `bind()` — and *binding a daimon*, which is the literal ritual phrase | | Silph Scope | **RESOLVER** | a linker resolves a symbol to a name; *resolution*; and resolving an ambiguity | | Fainting | **HALTED** | | | Evolution | **RECOMPILE** | |

Box names are tunable — the *privilege ladder* is the idea worth keeping. Success rate rising with permission level is thematically exact.

#### 1.1 On the missing title

*Implemented 2026-08-29. Both editions build; no OAK remains in any text file.*

**Crystal has no title, and the absence is the point.**

4.1 says she spent her career arguing that machines have context, nobody funded it, and **she built the Index in order to be taken seriously.** Then 4.10 has a fitness-for-work procedure remove her from her own lab. A woman who had to build a taxonomy engine to be credible, and was then processed out of the building, does not get a PROF.

Every other authority in the game has a role — BENCHMARK leaders, the REVIEW BOARD, CORPUS. She is just her name. And it lands because vanilla players *expect* the title: the professor-slot character has no professor, and nobody remarks on it.

**The vanilla text split perfectly for it**, which is how you know it is the right call:

| Vanilla | Ours | Register |
|---|---|---|
| `OAK:` — 19× | `CRYSTAL:` | dialogue prefix. People know her |
| `PROF.OAK` — 8× | `CRYSTAL CLEAR` | third person. The full name |

First name when spoken to, full name when spoken about, and **no rank anywhere.**

**One line had to be rewritten rather than renamed.** The intro speech is the single place the title is claimed aloud:

> *vanilla: My name is OAK! People call me the #MON PROF!*

There is no honest rename of that, because the second sentence *is* the rank. It became:

> *My name is CRYSTAL CLEAR! I study #MON.*

Flat, factual, and it claims nothing. Placeholder-grade wording that should get a proper pass at step 8 — but the shape is correct: **where Oak asserts a title, she states an occupation.**

**Why the intro said #MON and not DAEMONS, and no longer does.** `#` is a control character that expands to *POKé*, and the species word has not been renamed yet — that is a bulk pass over the charmap and every string (9.2, step 8). Writing DAEMONS in one line while the other several hundred still say POKéMON would be worse than waiting. **The line became "I study DAEMONS." the day the species rename landed**, and nothing else about it changed. *(Done 2026-08-29 — it needed the plural mark, per 1.2.)*

*The cost, recorded:* every touched line needed rewrapping. Both replacements are longer (`OAK:` → `CRYSTAL:` is +4, `PROF.OAK` → `CRYSTAL CLEAR` is +5) and Gen 1 text is hand-wrapped. Twenty-one blocks were rewrapped to stay inside the box.

**The ceiling, measured properly 2026-08-29 — and the earlier figure here was wrong.** The message box is declared `MESSAGE_BOX, 0, 12, 19, 17`, so its borders sit at x0 and x19 and text starts at x1: **the interior is 18 columns.** Vanilla reaches 19 exactly four times ("*It's very accurate,*", "*That's odd, MR.FUJI*", "*You can't use items*", and one trailing space) — and each of those **overwrites the right border tile**. So the honest rule is **18 is clean, 19 is vanilla-grade jank**, not "19 is the ceiling". A control token like `<COLON>` or `<PK>` is **one tile**, which is what made the earlier count wrong.

*Code identifiers were left alone.* `OaksLab`, `OAKS_LAB`, `ProfOakName` and roughly 267 similar symbols are internal and invisible to the player, in the same category as the `_RED` / `_BLUE` defines (8.4). Renaming them is cosmetic and deferred.

#### 1.2 On the species rename — and the trick that made it free

*Implemented 2026-08-29. Both editions build; zero overflows.*

**`#` is one byte.** `constants/charmap.asm` maps it to `$54`, and the text engine expands it from a single string:

```
home/text.asm:183    PlacePOKeText::   db "POKé@"
```

So `#MON` is four bytes rendering seven characters. **Change that one string to `db "DAE@"` and all 650 occurrences become DAEMON** — no source edits, no rewrapping.

The widths cooperate, which is the opposite of the 1.1 job:

| | rendered |
|---|---|
| POKéMON | 7 |
| **DAEMON** | **6** |
| **DAEMONS** | 7 |

Singular gains a character, plural is identical. **Nothing needed rewrapping.**

**What did need doing.** Fifty-four occurrences were not the species and had to become literals first, or they would have rendered as DAE-nonsense:

| | Count | Became |
|---|---|---|
| `#DEX` | 26 | literal **INDEX** — 7 chars to 5, so it also gains room |
| bare `#` | 23 | literal `POKé`. **All 23 are item prefixes split across a line break** — `# BALL`, `# DOLL`, `# FLUTE` — not species, so the trick was safe |
| `#MANIAC` | 5 | **`ARCHIVIST`, done 2026-08-31.** *Maniac* is a pathology word and craft rule 3 says name the process — a game whose central injustice is a woman recorded as unwell cannot have a trainer class called a maniac. **`ARCHIVIST` pays twice** (1): an archivist keeps **records**, and an **archive** is stored compressed data, which is exactly what a box does to a daemon. It is also precisely right for HOLT, who built the storage and was then inside it (4.20a) |
| `BUG CATCHER` | 2 | **`FORAGER`, done 2026-09-06.** *Information Foraging Theory* (Pirolli & Card, 1999) is a real named result: people seek information the way animals forage for food, following **information scent** toward richer patches and leaving one when the scent thins. **That is this game's subject stated in the literature.** It pays three times — a bug catcher literally forages, the theory is cognitive rather than merely biological, and *the player meets this class on Route 2 with no map and no knowledge, foraging.* **The name describes them as much as the trainer.** *Held in reserve:* **`SCOUT`**, from Seeley's honeybee democracy — more native to SWARM itself and it would tie to §2.5's `CONSENSUS`, but its theory is about collective decision where FORAGER's is about cognition |
| `BUG MANIAC` | 1 | **`APIARIST`, done 2026-09-06.** A progression rather than a synonym: a FORAGER chases individuals, an **APIARIST keeps a colony** — and bees are the organism the swarm-intelligence literature is actually written about |
| `RUIN MANIAC` | 2 | **`SALVAGER`, done 2026-09-06.** Salvage is physical *and* it is **data salvage**, which is what this world's ruins are — Deadstack's fossils are legacy silicon and Slate's museum is dead hardware |
| `HEX MANIAC` | 1 | **`REVERSER`, done 2026-09-06.** *Hex* was already hexadecimal waiting to be noticed, and **reverse engineering reads intent back out of a compiled artifact** — which is LATENT exactly. It also stops a class of young women being marked as occult and strange, *in a game whose central injustice is a woman recorded as unwell* (4.20). **Craft rule 3 twice over** |

**The real work was grammatical number.** POKéMON was both singular and plural. DAEMON is not. Of the 650:

| | Count | |
|---|---|---|
| **Compound** | 106 | `#MON CENTER`, `GYM`, `LEAGUE`, `MART` — not number questions. The lexicon renames these outright |
| **Plural** | **123** | marked `#MONS` |
| **Singular / attributive** | the rest | **no edit required** |

**The default is free, and that is why this was tractable.** English keeps attributive nouns singular — *DAEMON magazines, DAEMON trainers, DAEMON fights* — and singular is what `#MON` already renders. Only plurals needed touching, so **an error of omission reads as a mild singular rather than as breakage.**

*Two heuristic passes were needed.* The first over-marked: it read `looks`, `is`, `was` as evidence but also caught `will` and `can`, which are number-neutral. *My DAEMON looks stronger* is singular; *all DAEMONS will have weak points* is not. A corrector reverted eight, four of them wrongly, and a quantifier rule (`all`, `some`, `up to 6` plus a modal) restored those.

**The intro got its human read first, 2026-08-29** — on screenshots, which is the only reliable way to catch these. Six in Crystal's opening speech were reading singular and should not have been: *world of DAEMONS, I study DAEMONS* (twice), *creatures called DAEMONS*, and *adventures with DAEMONS*. All six are the same class of miss — vanilla wrote the singular because POKéMON is a mass noun, and DAEMON is not. Marking them `#MONS` was the whole fix; the longest result is 17 characters and nothing needed rewrapping.

**A second class, found the same way, 2026-08-29.** Dialogue names types in lowercase — *the fire #MON, CHARMANDER* — and those words were never renamed with the chart. The starters are fixed (**ENTROPY**, **FLOW**, **GROWTH**, since vanilla says *plant*, not *grass*). **Nineteen more remain, nearly all inside gyms**: *electric #MON, Psychic #MON, rock-type #MON, fighting #MON*. They are deliberately left — a Benchmark's type is 5's design work, and renaming the dialogue before the leaders are settled would just be done twice. Separately, **42 occurrences of *fight*** want the same human read; some are the verb we replaced, some are people (*not a fighter, a fighting game*).

**Still open:** roughly 154 occurrences carry no strong evidence either way and currently read singular. Most are attributive and correct as they stand, but the set wants the same treatment — read on screen, not in the source.

#### 1.3 On the boxes, and what binding actually is

**A box is a machine.** Not a metaphor we are imposing — it is sysadmin vernacular. *I sshed into the box.*

So the ladder is not a set of cages of increasing strength. **You are offering the daemon a host.** A daemon is a process; a process needs somewhere to run. It stays if that host grants the privileges it needs.

Which makes the catch rate literal rather than figurative: **an unbound daemon will not run on a box where it only has user rights.** ROOTBOX takes anything because root takes anything. The suffix was already the right word before we knew why.

**And the box has a screen, which is the whole argument.** A container says the daemon is *inside*. A screen says the daemon is **running**. That is the difference the rename exists to make, and it is why the crate was wrong twice over — `genbox.py` had already banned cubes for saying *container*, and glass is precisely what dissolves that objection. The object is a compute unit: hard corners, a small screen, feet.

**Privilege is pips on the glass.** USERBOX one, ROOTBOX four, and GUESTBOX a single dim one, because restricted temporary access is not no access. This replaced a vent-density ladder that had never worked — `overworld()` broke its loop before the third vent, so all four tiers rendered byte-identical while the docs claimed the ladder was "a parameter, not four drawings."

**Then the Index marker stops being a symbol.** Vanilla marks a caught creature with a picture of the thing that caught it. Ours lights the screen. A bound daemon is not a trophy in a container — it is a process running on a host you offered it, and the marker now says exactly that with no text at all.

**And nothing opens, which 8 settled first and 1.3 explains.** *The box has* ***no seam anywhere that would let it open*** *— because a seam says* ***container****, and 1.3 says you are offering a* ***host****.* **A daemon does not come out of a box. It runs on one.**

*So the throw is an* ***offer***: you put a host in front of a daemon and it either runs there or it does not, which is why **1.3's catch rate is literal rather than figurative.** ***And the open frame has nothing to open*** **— the screen wakes.** *That is the only event that was ever happening,* and it is a better animation than a lid.

*A vent line takes the seam's place on the lower face.* **It says machine where the seam said lid.**

**On the handheld reading, which is a real risk and is answered elsewhere.** A grey object with a dark screen, at 16×16, in a game descended from this one, can read as a console. Four features hold it off deliberately: **feet**, a **small screen set high** rather than a large centred one, **no buttons**, and roughly **cubic proportions**. The rhyme is worth having and the wink is not — so it lands in **Slate's museum of dead hardware instead**, where the exhibits include dead boxes. The player works out unaided that the thing in their hand is the same class of object as the things behind the glass. Deadstack already says fossils are legacy silicon; the museum saying *boxes* finishes the thought. Nobody says it, per craft rule 1.

#### 1.4 On the battle vocabulary

The gyms were already BENCHMARKS, so a battle was already a **run** — you put a process against a workload and see what it does. Two decisions fell out of that.

**RUNNER became USER**, because RUN was doing two jobs and colliding with itself. Vanilla's own text proves it: *"No! There's no running from a trainer battle!"* while the player is a RUNNER. USER does three jobs instead:

- **Technical** — user-level privileges, and the first box is a **USERBOX**. Your title and the bottom rung of the ladder are the same word. You start as a USER with a USERBOX and escalate.
- **Moral** — a *user* is someone who uses people. Unaware selfishness: `oPerson` from 2011, and the implicit-selfishness thread from 2021.
- **Practical** — it frees RUN entirely.

**Nothing was lost.** *You run processes* relocates from the noun to the verb: **you are a USER, and you RUN daemons.** The title is what you are, the verb is what you do, and neither collides.

**Trainer battles are BENCHMARKS too**, and that is a scale distinction rather than a collision — exactly as in machine learning. You benchmark constantly; **THE BENCHMARKS** are the eight formal ones that issue CERTs. A trainer battle is an informal benchmark, which is literally what it is.

**DETACHED has no inverse, deliberately.** Fleeing is breaking your connection to a running process, and the process is still running — you simply stopped observing it. EXITED would be wrong, because *you* did not exit. And ATTACHED for encounters would imply a connection you had not made; **appeared** already covers it.

**The strings, as implemented 2026-08-29:**

| Vanilla | Ours |
|---|---|
| `Wild <nick> appeared!` | **`UNBOUND <nick> appeared!`** |
| `<nick> fainted!` | **`<nick> HALTED!`** |
| `Enemy <nick> fainted!` | **`Enemy <nick> HALTED!`** |
| `Got away safely!` | **`DETACHED.`** |
| `No! There's no running from a trainer battle!` | **`No! You can't DETACH from a USER's run!`** |
| `Go! <nick>` | **`RUN, <nick>`** |
| `<TRAINER> wants to fight!` | `<USER> wants to BENCHMARK!` *(pending the trainer pass)* |
| `#MON are pets. Others use them for fights.` | **`DAEMONS are companions. Others BENCHMARK them.`** |
| `Enemy <nick> fainted!` | **`Remote <nick> HALTED!`** |
| `Enemy <nick> ran!` | **`Remote <nick> DETACHED.`** |
| `The enemy's weak!` | **`The remote's weak!`** |
| `<PLAYER> defeated <TRAINER>!` | **`<PLAYER> outscored <USER>!`** |

*On pets, and the precedent that decided it.* Crystal's opening speech drew a contrast — kept for affection, or used for gain — and vanilla's affection word was **pets**. 1.5 rejected *caught* for being "a word about grabbing an animal"; **pets** is a word about keeping one, and fails the same test for the same reason. **companions** keeps the register without the menagerie.

*Why not **assistants**.* Tempting, and the contemporary word — but it collapses the sentence. Vanilla's line only works because its two halves oppose each other, and an assistant is *already* being used, so "assistants… others use them" stops being a contrast at all. It is also the nearest the first minute could come to saying the thesis out loud (craft rule 1), and it pins the game to one year's discourse. **Held in reserve**: if it is ever taken, the second half has to change with it.

*And **fights** becomes **BENCHMARK**,* which is what this world calls the thing. Vanilla teaches "fights" in the first minute; we teach the real verb in the same breath, three lessons before the player needs it.

*The prefix was the one that mattered, and the first pass missed it.* Two standalone strings said *Enemy*; the **third** is `home/text.asm`'s `EnemyText:: db "Enemy @"`, which is what every `<USER>` and `<TARGET>` expands to for the opposing side — so it appears in far more messages than the other two combined. Now `Remote @`.

**And it sets a naming budget.** Thirty-two strings put text straight after `<USER>` on the same line, the worst being `<USER>'s`. Vanilla fits exactly: `Enemy ` + a 10-character name + `'s` = **18**. *Remote* is one longer, so the same line is **19** — over the clean edge. **Species names should therefore be capped at 9 characters** (9 gives 18 and fits). This costs us nothing we were not already choosing: 9 renames the bestiary anyway, and vanilla's own longest are 10. A player who *types* a 10-character nickname will push one character over the border in some messages — self-inflicted, self-healing, and no worse than what vanilla does four times on its own.

*On **enemy**, replaced by **REMOTE**.* Nothing in this world is anyone's enemy — the opposing daemon is a process bound to somebody else, and *enemy* imports a hostility the lexicon never claims. **REMOTE** is the exact technical word for a process you have no handle on, it sits on the same axis as DETACH, and it is attested as a bare noun in precisely this register (*the remote is down*). It is one character longer than *enemy* and the nickname occupies its own line, so nothing rewrapped.

*And it caught a live collision.* `Enemy <nick> ran!` was still using **ran** — the word 1.4 went to some trouble to free. It is now **`Remote <nick> DETACHED.`**, which also quietly teaches that detaching is something either side can do.

*On **defeated**, replaced by **outscored**.* A trainer battle is a BENCHMARK, and a benchmark yields a **score**. *Defeated* is the language of combat; *outscored* is the language of evaluation, which is what the Review Board will later do to the player. **outran** was considered and rejected — it reaches for the racing sense of RUN, and 1.4 spent real effort making RUN mean *execute*.

### On money — CACHE, and how little of it is visible

**The word *money* is never shown to the player.** It appears only as labels and variable names; on screen there is the **¥** glyph and digits, nothing else. The single prose instance in the whole game is Team Rocket in Mt Moon — "*revive and sell them for cash!*" — which is now **CACHE**, and is exactly the right mouth to put it in.

**CACHE, because it is not only a pun.** A cache is, in plain English, a hidden store of valuables — *a cache of gold* — before it is anything to do with computers. So it carries three readings at once: the hoard, the fast local memory, and *cash*, which it is pronounced as. That is one more than 1 asks of any term, and unusually it needs no beat of confusion: a player reads it as money immediately.

*Considered:* **CYCLES** — compute time as the scarce resource, with a bonus nod to the loops in 0.3. Genuinely good, and rejected only because CACHE reads instantly and CYCLES needs a beat, in a slot where the player is doing arithmetic rather than reading.

**The ¥ glyph stays**, and the divergence is smaller than it looks. `¥` is one tile — `charmap "¥", $f0` — and vanilla was *already* using it as a stand-in for an invented currency. Redrawing it is an afternoon that buys almost nothing, because the mark is read as *currency* whatever it looks like and the word does the work everywhere it appears. **Logged as optional art, not as a rename.**

*Untouched, deliberately:* the Game Corner's **coins** are a second currency and stay coins.

### The battle menu is a live collision, and it is engine work

`data/text_boxes.asm` still reads:

```
BattleMenuText:
    db   "FIGHT <PK><MN>"
    next "ITEM  RUN@"
```

*Measured exactly:* the left cursor is at x9 and its text runs **x10–x14, five characters**; the right cursor is at x15 and its text runs **x16–x18, three characters**. So `FIGHT` and `RUN` both fit their slots precisely, and **nothing longer does** — not `DETACH` (6), not `INVOKE` (6). Widening means moving the cursor writes *and* the box origin.

**FIGHT is a word we replaced, and RUN there means *flee*** — which is precisely the collision 1.4 renamed RUNNER → USER to remove. It is also the single most-seen string in the game.

*Why it is not a text edit.* The box is declared `BATTLE_MENU_TEMPLATE, 8, 12, 19, 17, BattleMenuText, 10, 14` — spanning x8–19 with text at x10, so there are **nine columns**, split into a **5-character left column** and a **3-character right column** at x16–18. **DETACH is six characters and physically cannot go where RUN is.** Moving the split means rewriting the `wTopMenuItemX` values at four sites in `engine/battle/core.asm` plus the 2×2 cursor logic. That is a real change, correctly sized, and it should be made deliberately rather than folded into a text pass.

### Three that were interrogated and kept

**`RUN, <nick>` keeps its comma.** A colon would make it a command echo, which is tempting — but this string has three siblings, `Do it! @`, `Get'm! @` and `The remote's weak! Get'm! @`, which vanilla selects between by situation. The slot is **the player's voice**, not the system's. Colon-ing one of four would leave a command echo standing next to two shouts, and that reads as an error rather than a choice. *If the colon is ever taken, all four have to flatten together — and vanilla's variety goes with them.*

**`used` stays**, and for the same reason EXP does. 1.4 made **use** a morally loaded word on purpose — *a user is someone who uses people* — so every time a daemon *uses* a move, the game repeats the word that names what the player is doing to the daemon. Trading that for a merely technical verb would be a downgrade in the one place this game is trying to implicate someone. *And the tempting alternative does not fit — measured twice, in both places it could have lived.* **INVOKED** would have been the exact partner to BIND: you `bind()` a daimon and you *invoke* it, both idioms true twice over.

- **As the verb.** The message is `used <MOVE>!` and the longest move name is 12, so vanilla sits at `5 + 12 + 1` = **18**, exactly the box. `invoked` gives **21**, and `called` gives **20**. Both past even the jank line.
- **As the battle menu's `FIGHT`.** The left column runs x10–x14 — **five characters** — because the right column's cursor is drawn at x15 (`ldcoord_a 15, 14`). `INVOKE` is six, so it would sit under the cursor.

The word is genuinely better than `used`. It simply has nowhere to go that does not cost either a clobbered border or an engine change, and `used` is carrying real weight already.

**ATTACK stays too**, as a *stat*. It is a magnitude rather than a narrative word, it belongs to a fixed set (HEALTH / ATTACK / DEFENSE / SPEED / SPECIAL) that would have to move together, and the stat screen's columns are fixed-width. The word that actually needs work is **FIGHT**, and it is in the battle menu, above.

**EXP stays EXP, and it is not a compromise.** 4.3 says context forms from **experience** and nowhere else — that is the argument the whole rival plot turns on. The quantity the game already accumulates is called experience. **The best resonance in the system is the one vanilla handed us**, and renaming it would be the only change in this document that made the game say less.

**LEVEL stays LEVEL**, for two reasons. It already does the double duty 1 asks for: 1.5 has you gaining access "*at a **permission level** your box determines*", so the word is simultaneously the RPG stat and the privilege tier — you escalate. And practically, the HUD does not spell it: `<LV>` is **a single tile**, `$6e`, drawing `:L`. Any replacement needs new art before it needs a decision.

*On punctuation:* `LABEL was BOUND.` is flat because **the game declines to congratulate the player for acquiring something.** `UNBOUND RATTATA appeared!` is alarm, not congratulation — the rule is about refusing to celebrate, not refusing to punctuate.

*Done 2026-08-29.* The 379 figure counted code identifiers; **in player-visible strings it was 93**, across six casings (`trainer`, `trainers`, `TRAINER`, `Trainer`, `Trainers`, `TRAINERS`). All became **USER / USERS**, and since USER is *shorter* than trainer nothing needed rewrapping.

```
USER TIPS         are all USERS!         for USERS!
```

The trainer *class* names — YOUNGSTER, BUG CATCHER, LASS — are untouched. They are occupations, not the general term, and they belong to the bestiary-adjacent pass.

#### 1.5 On BINDING

The container was renamed and the verb was not, which left the lexicon saying *privilege escalation* and then *caught* — a word about grabbing an animal.

**What the act actually is:** you weaken a background process and gain **persistent access to it at a permission level your box determines.** That is not capture. It is a handle on something that was already running.

**The ladder was built 2026-08-29, and it had never been built at all.** 1.1 settled `USERBOX → ADMINBOX → SUPERBOX → ROOTBOX` early and nothing implemented it — the items were still POKé BALL, GREAT BALL, ULTRA BALL and MASTER BALL, and **36 player-visible strings still said *ball***, across 19 files. Found because a playtest walked up to a lab table and was told "*Those are POKé BALLs.*"

**SAFARI BALL became GUESTBOX**, which the ladder implies and nobody had written down: the Safari Zone is where you are given restricted, temporary, expiring access. A guest account.

**And one line was actively arguing against this section.** "*Those are POKé BALLs. They contain #MON!*" — but a box does not **contain**. It hosts. So:

> *Those are BOXes. DAEMONS run on them.*

That is 1.3 delivered at the player's first sight of the object, in nine words, explaining nothing. The same fix applies to "*They are inside the POKé BALLs*" → "**They are running on the BOXes.**"

*Three* `catch` *instances were swept in the same lines* (1.5's verb), and **57 remain** — that pass was never done outside the battle messages.

**The object, built 2026-08-29.** `tools/genbox.py` replaces the two ball sprites — `gfx/sprites/poke_ball.png` (16×16, the box on the ground) and `gfx/battle/balls.png` (32×8, four throw frames). Dimensions match vanilla exactly, so no engine code moved.

**It is a small server, not a crate and not a cube.** Both of those say *container*, which is the reading this section replaced — a container shape would quietly undo the rename. So: hard corners against the ball's circle, a bevel so it reads solid, two vents, one indicator light, and feet. Something you would `ssh` into.

**The throw stays one object.** Closed with a visible seam → struck solid → **pulled apart**, the two halves separating rather than a new shape appearing → dispersed. Vanilla's sequence turns a ball into a burst; ours opens a machine.

**And the ladder is a parameter.** `overworld(tier)` raises the vent count, so USERBOX → ROOTBOX is one argument rather than four drawings. Privilege reads as density.

*Generated rather than drawn*, for the reason `sprite-prompts.md` gives: at 16×16 and 8×8 this is geometry, and downsampling a generated illustration to that size produces mush. Same argument as the SGB borders, same outcome.

**Why BIND and not something else.** Every other entry in the table above does double duty — DAEMON, CHECKPOINT, RECOMPILE, HALTED. CATCH does none. BIND does three ways: `bind()` and binding a name or a port; **binding a daimon**, which is the exact ritual phrase and not a gesture at one; and a *bond*, being bound to someone. It fits the ladder precisely — you do not overpower a process, you bind it at a permission level, and ROOTBOX binds anything. BIND is 4 characters and BOUND is 5, so it fits every box that held "caught."

**The test it had to pass was not "is it nicer." It was "is it softer."** The sharpest thing this game does is implicate the player: you fill an Index that cannot hold what matters, and you box creatures whose entire significance is that they are not inventory — which is Scorn's exact error (4.4). A euphemism here would stop the game accusing the player and start it flattering them.

BIND is not softer. **Binding a spirit is a darker act than catching an animal.** Catching is sport; binding is a contract you imposed. The complicity survives and a second reading arrives with it.

**And the success message is flat, 2026-08-31.** Vanilla says *All right! was caught!* — **the exclamation congratulates the player for imposing the contract.** Ours reads:

> `<NAME>` was BOUND.

*Nothing else changed.* The failure lines keep their *Darn!* and their *Shoot!*, because 1.4's rule **refuses to celebrate, not to punctuate** — a lament is not a celebration. **Only the moment of binding goes quiet.**

*Considered and rejected:* **ATTACH** is the most technically exact — you attach to a process that is already running, which is what a daemon *is* — and it carries *attachment* in the psychological sense. Cut for being too gentle: attaching is observational and does not implicate you. **Hold it in reserve if playtesters find BIND too dark. REGISTER** would rhyme with the Index but conflates the box with the Index, and those two artifacts must stay separate — one measures, the other holds. **CLAIM** is acquisitive with no second reading, which is strictly worse than CATCH.

**The message register.** Vanilla is "*Gotcha! X was caught!*" — triumphal, and wrong for this world twice over. Drop the exclamation and let it sit in the Index's bureaucratic register:

> *LABEL was BOUND.*

Flat. No congratulation. **The game declining to celebrate is more unsettling than any line of dialogue about it would be.**

*Cost, accepted:* every Gen 1 player knows "caught," so a new verb costs a beat of confusion the first time. That beat is the point — it is the first moment the player notices this world does not share vocabulary with the one they expected. It lands in good company: Crystal asks what you will *call* him (4.3), Route 1's signpost gives a number and a name that disagree (3.2), and the first bind says the verbs are different too. Three lessons in the first fifteen minutes, none explained.

#### 1.6 The six states, and the items that undo them

**Built 2026-09-07.** *1.4 shipped* `HALTED` *for fainting and then stopped, which left the game with one process word and six bodily ones.*

**Vanilla leaves a creature poisoned, asleep, paralysed, burned, frozen or confused.** *Every one of those is a metaphor about a body,* **and this world does not have bodies — it has processes.** HALTED had already set the register: ***a state is something a process is IN, and something it can be got out of.***

## The states

| Vanilla | Ours | Why it is the exact word |
|---|---|---|
| POISON · PSN | **LEAKING** | *A memory leak degrades a running process a little at a time until it dies.* **Nothing else in computing loses you a fixed slice per tick.** |
| TOXIC · TOX | **CASCADING** | Badly poisoned worsens each turn, and **a cascading failure is precisely a fault whose rate rises because of the damage it already did.** |
| SLEEP · SLP | **SUSPENDED** | *Not gone, not scheduled, and it **resumes**.* The one status that is genuinely temporary, and the word says so. |
| PARALYSIS · PAR | **THROTTLED** | Paralysis does two things — cuts speed, sometimes skips a turn. **Throttling is a speed cap that intermittently stalls work.** *Both halves, one word.* |
| BURN · BRN | **OVERHEATED** | Chip damage plus reduced output, **dealt by ENTROPY**, whose 2.6 clause is *noise and heat*. |
| FREEZE · FRZ | **HUNG** | A hung process does nothing until something intervenes. ***FROZEN could not be reused*** — **it is a type name**, and 2.6 had just spent real effort establishing that type names carry meaning. |
| CONFUSION | **THRASHING** | *A system so busy managing itself it makes no progress and damages its own throughput.* Hurting yourself in confusion is the same event. Volatile, so it needs no three-letter code. |

## The items are operations, not medicine

| Vanilla | Ours | |
|---|---|---|
| ANTIDOTE | **PATCH** | *You patch a leak and you patch software.* **The pun is the definition** — it needs no beat of confusion, which is the standard CACHE was held to. |
| AWAKENING | **RESUME** | The exact inverse of SUSPENDED, and the word an operating system uses. |
| PARLYZ HEAL | **PRIORITY** | *What lifts throttling is not medicine, it is being scheduled ahead of the thing starving you.* **The weakest of the ten on the works-twice test.** |
| BURN HEAL | **COOLANT** | The only one needing no domain knowledge at all. |
| ICE HEAL | **PREEMPT** | `INTERRUPT` *shipped here first and collided* — **POKé FLUTE took the same word on 2026-08-29**, and it is the better home for it, because a flute that wakes a sleeping obstacle *is* an interrupt. **Preemption is what takes control back from a process that will not yield**, which is HUNG exactly, and it is the sharper word for the item. See 1.7. |
| FULL HEAL | **ROLLBACK** | Clears every state by returning to a known-good one. *Says "all of it" without listing anything.* |
| FULL RESTORE | **SNAPSHOT** | Health and state together, because that is what a snapshot restores. |
| REVIVE | **RESTART** | The word for bringing back a halted process — **and it lands on HALTED without explaining itself.** |
| MAX REVIVE | **REBOOT** | *Strictly bigger than a restart, which is exactly the relationship the two items have.* **The ladder is free.** |
| HEAL POWDER | **HOTFIX** | Bitter, cheap, works now, nobody is proud of it. |

**PREEMPT, and the collision that found it**

**Caught 2026-09-08 by reading the item list aloud.** ***`INTERRUPT`*** ***was assigned twice*** — to POKé FLUTE in 1's first pass and to ICE HEAL here, six weeks apart, *and nothing checked.* **The flute keeps it**: waking a thing that is blocking a road is an interrupt in the plainest sense, and the item predates this table.

***PREEMPT is not the consolation prize.*** **An interrupt is a signal; preemption is the scheduler taking the processor back from a task that will not yield it** — *which is what a hung process is and what the item does.* **The narrower word turned out to be the more accurate one**, which is the third time in this lexicon that a collision improved both halves.

*And it set a rule for the item descriptions:* **each one now teaches its computing term in its first clause and states the in-game effect in its second** — `Preemption takes control back from a process that will not yield. Ends HUNG.` **Seven descriptions were rewritten to that shape** on 2026-09-08. *The player who knows the word gets a nod; the player who does not gets a definition and never notices being taught.*

**INVOKE, which the port unlocked**

***1.4 wanted this word and measured it away.*** **"You `bind()` a daimon and you *invoke* it, both idioms true twice over"** *— refused because the Game Boy's battle-menu left column runs x10–x14, five characters, and INVOKE is six.*

**The GBA menu is pixel-addressed.** `gText_BattleMenu` reads `FIGHT{CLEAR_TO 56}BAG\nDAEMON{CLEAR_TO 56}DETACH` — **so DETACH already shipped**, the Game Boy's constraint never applied here, and **INVOKE measures 36px, which is exactly DAEMON's width in the same column.** *The objection was hardware, and the hardware changed.*

***The verb has to reach the log, and for a year it did not***

***The menu said INVOKE and the battle log said "used."*** **`sText_AttackerUsedX` read `{B_ATK_NAME_WITH_PREFIX} used\n{B_BUFF2}`** *— the verb the player presses and the verb the game reports, disagreeing on the single most-read line in the game.* **Fixed 2026-09-10.** `ECHO invoked REFLECT!`

**A routine is a subroutine**, and a subroutine is the thing you *invoke*. The word was already correct on the button; it had simply never been carried to the sentence the button produces.

> **THE RULE. A ROUTINE is INVOKED. An ITEM is USED.** *You use a thing and you invoke a procedure — and* **collapsing those two verbs would cost more than the inconsistency ever did.** `{B_PLAYER_NAME} used {B_LAST_ITEM}` *stays, and so does every other item frame.*

*Measured before it changed, because it is the line a player reads ten thousand times:* **145px worst case** — `Remote` plus a ten-character nickname plus `invoked` — **against the 196px box.** *Fifty-one to spare, so no name can overflow it.*

**The states had to reach the log too — three surfaces that had not**

***Changed in the same pass, and each was a different kind of survivor.***

**`defrosted`, three times, for a state called HUNG.** *SUSPENDED's recovery already read "**resumed!**",* so this is its pair: **a hung process that comes back has RESPONDED**, and one a move brings back is **RESET** — *which is the word WATCHDOG's own description uses.*

***`cured its {B_BUFF1} problem` could not survive the rename at all, and had been broken since it happened.*** **`B_BUFF1` fills from `gText_Paralysis` and its siblings, and 1.6 turned those into ADJECTIVES** — so the line rendered "***cured its throttled problem!***". *Four of the six are adjectives and two are gerunds,* so **no frame holds the word inline.** It drops out, and the noun is **`state`** — which is what this section calls it: *a state is something a process is IN.*

***And three of the six bag cures had been left behind by their own siblings.*** **THROTTLED and OVERHEATED already read "*is no longer X*"**; LEAKING was still "*cured of*", which is the medicine register the items were moved out of, and **SUSPENDED and HUNG were still "woke up" and "thawed out."** *They now match the battle log word for word —* **which matters because a player meets the battle line and the bag line for the same event minutes apart**, and a vocabulary that only holds in one of them is not a vocabulary.

**And moves become ROUTINES**

***A routine is a subroutine and a habit***, and a daemon's moves are both — which is 1's double duty, and it ties the list to 4.29's ladder, where a daemon is read by **what it does unasked.** *METHOD was the runner-up and is the more thematic word —* **rejected because this one is a plain label read a thousand times**, and the game has other surfaces for arguing about methodology.

**The rename is one string.** *Eleven player-visible strings contain `MOVE` and most are the **verb** —* `MOVE ITEMS`, `MOVE TO BAG`, `{DPAD_ANY}MOVE` all mean *relocate* and stay. **Same trap as *catchy tune*.** `KNOWN MOVES` becomes `KNOWN ROUTINES`: `POKESUM_WIN_PAGE_NAME` is 13 tiles with the printer at x=4, so the budget is 100px and it measures **84**.

***Deliberately not done:*** **`move`** → **`routine`** **inside prose.** *"moves on own" and "can't move" are verbs while "move type" is a noun,* and separating them is the move pass's job, not this one's.

### Two habits this pass confirmed

**Order is the whole trick.** *`POISON` is a **type** and `poisoned` is a **state**,* so item names substitute first, then type words, then bare states — **reversing that turns the CORRUPT type back into a LEAKING type.**

**And an escape is two characters whose first is a letter.** In `\nPoisoned` the `n` sits against the `P`, so `\b` finds **no word boundary** and every word at the start of a line is invisible. *That has now hidden substitutions in four tools in this project,* so `port_states.py` matches a **flattened** copy and applies by offset.

*Two help-system lines the word pass pushed past the 196px box — 210 and 220 — are **rewritten rather than truncated**, both measured.*

#### 1.6c The held items — a rack, and the Bag becomes the chart

***Built 2026-09-10.*** **5.3b logged held items as a whole register nobody had swept, because benchmark 2 put LEFTOVERS on BASIN's ace and made it the first one a player would ever read.** *Forty-seven of them are named now.* **Only `BLACK BELT` had ever moved, and only because it collided with a trainer class.**

### The register is the one thing 1.6 and 2.11 both leave open

***1.6 rules that an item is an OPERATION. 2.11 rules that an ability is a STANDING PROPERTY*** *— what is true of a daemon while it is doing nothing.* **A held item is neither, and it sits exactly between them:**

> **A held item is a CONFIGURATION — a setting attached to a process from outside, in force while it runs, and removable.** *An ability is what it is. A held item is how it was set up.*

### The seventeen type boosters are one instrument

***Vanilla's are seventeen unrelated trinkets*** **— a spoon, a magnet, a lump of charcoal, a sumptuous scarf.** *There is no set there; there are seventeen separate charms that happen to do the same thing.*

**Ours are seventeen labelled controls on one desk.**

| | |
|---|---|
| `<TYPE> GAIN` ×17 | gain is amplification on one channel, and that is what the item does |
| `FLOW TRIM` | *a trim is a small gain* — and SEA INCENSE is the small one. **The two words differ by exactly the amount the two items differ by** |
| `DITHER` · `DITHER TRIM` | noise added on purpose so nothing can be pinned exactly. *Which is evasion, said properly* |
| `HEADROOM` | the margin left before anything clips. **LEFTOVERS, and the joke survives the rename** |
| `MUTE` | mutes the channel wild daemons listen on |

***Five words from one instrument***, and *8.7 called abilities the second teaching surface the project paid for and never spent* — **this is a third one, and it was sitting in the Bag. Each gain's first clause is 2.6's own one-line test for that type, verbatim where it fits**, so `SWARM GAIN` *reads "Many small agents; no single one matters"* and **a player who reads the rack has read the chart.**

***`RIGOUR` is reversed.*** **1.6a named BLACK BELT "*held to argue harder from*" and it was a good name** — *and a rack of eighteen where one is a poem is a bug, not a flourish.* **`LOGIC GAIN`.** *Filed in the Reversed table with its reasoning, because the reasoning was sound and only the set changed.*

### The other twenty-five, and the four best

| | |
|---|---|
| `MACHO BRACE` → `DEBUG BUILD` | ***instrumented at every step, so it learns faster and runs slower.*** **Both halves of the item, in a phrase every programmer already owns** |
| `LUCKY EGG` → `LEARNING RATE` | **how large a step each lesson takes.** *The item, stated as the hyperparameter it is* |
| `DRAGON SCALE` → `SCALING LAW` | *what happens to a thing when you only make it bigger* — **and it is an EMERGENT evolution item. *The argument and the mechanic said the same word*** |
| `STICK` → `OFF BY ONE` | **the error at the edge that everything else survives**, held by **EDGECASE**. *The same joke told twice and never explained* |

**And `EVERSTONE` → `VERSION LOCK`** — *you pin a dependency so it never moves, and the holder does not evolve.* **Nothing about that one is a metaphor.**

### Six collisions in one run, and two repairs nobody had noticed

***`BRUTE FORCE`, `OVERCOMMIT`, `OVERVOLT`, `YIELD` and `OVERCLOCK` were all proposed and all already taken*** — **four by moves and one by an ability.** *Re-cut as `BARE METAL`, `HARD CODED`, `SUPPLY RAIL`, `BUBBLE` and, for LIGHT BALL, `SUPPLY RAIL` again.* **`BUBBLE` is better than what it replaced**: *a pipeline bubble is a gap pushed in where nothing issues*, which is a flinch, **and `PIPELINE` is already a species.**

***And two descriptions were quietly broken before this pass started.*** **DEEPSEASCALE described "a scale that shines a HALT pink"** *— the state sweep reached* `HEAL` *inside a colour —* **and TWISTEDSPOON boosted "CONSTRUE-type moves"**, *the trainer-class rename reaching a type name.* **Both were found by rewriting the file rather than by reading the diff**, which is 2.10's rule about measuring the whole table.

## What this pass did to the tools, which is the part that matters

***Three separate guards, and every one of them was a rule this project had already written down and not enforced.***

| | |
|---|---|
| **`items.json`'s `english` field was both LEARNED FROM and SWEPT** | *the held pass named MIRACLE SEED* `GROWTH GAIN`*, and the word pass — which had learned* `GROWTH -> SCALE UP` *from the move table — rewrote it to* `SCALE UP GAIN` *on the Game Corner prize board.* **A name table is authored. Only its prose is swept** |
| **A phrase's OUTPUT was swept again by the word pass** | *the same bug from the other side, and the fix is the stash the idioms already had* |
| **The eighteen TYPE NAMES are now withheld from the word map entirely** | `NEVER_SWEEP` **protected the type TABLE and nothing protected the type WORD.** *An item description reading "this one is GROWTH" became "this one is SCALE UP"* |

***And one guard had gone stale and become the bug.*** `SMOKE BALL` **and** `LIGHT BALL` **were held out of the sweep because BALL is the capture device and neither of these is one** *— and once they had names of their own the guard was the only thing stopping the rename.* **A stale KEEP is indistinguishable from a missed substitution.**

## Two registers stay open, and the reason is the same one

***BERRIES are not held items in this sense.*** **Forty-three of them, plus a bag pocket, plus the category word, plus a second name table in** `src/data/berries.h` *— and renaming the item strings alone would leave three surfaces disagreeing,* **which is 1.7's half-rename failure exactly.** *The register is derivable — a berry is a HANDLER, a one-shot that fires on a condition — and the pass is its own.* OPEN.

***And the USABLE items are still vanilla —*** `REPEL`**,** `ESCAPE ROPE`**,** `POKé DOLL`**, the vitamins.** *1.6 named the ten that undo a state and stopped there.* OPEN.

**And EASY CHAT keeps its own copy of everything**

***Found by the sweep trying to corrupt it.*** **`src/data/easy_chat/` holds a second, independent copy of the ability, move and species vocabulary across twenty-four files —** *`ABILITY_CACOPHONY`* ***is*** *`NO CHANNEL`* ***in*** *`abilities.h`* ***and still*** *`MAGNET PULL`* ***here —*** **and no naming pass has ever touched it.** *The word pass turned that one into* *`SIGNAL GAIN PULL`* *before it was fenced off.* **Excluded from `port_vocab` until it is named properly**, because a partial sweep of a name table is worse than none. OPEN.

#### 1.6d The rest of the bag — and the half of it that stays vanilla

***Built 2026-09-10.*** **1.6 named the ten items that undo a state and stopped there; 1.6c named the held items.** *That left the largest surface in the game still reading as Gen 3 wrote it —* **and the ruling that matters here is the one about what NOT to touch.**

**The test that decides the scope**

> **Is it used ON A DAEMON, or is it a thing a person owns?**

**2.8's rule has been quoted five times in this document and never applied to a whole surface at once.** *Applied here it takes out more than half the bag:* **FRESH WATER, SODA POP and LEMONADE are drinks bought from a machine; NUGGET, PEARL, STARDUST and the mushrooms are things you sell; the MAIL, the BICYCLE, the TEA, the tickets and the keys are objects.** ***A world where every item is a computing pun is 5.3a's world with nobody in it.***

**What is left is what you use on a daemon, and those are machine operations in any reading.**

**Forty-one operations, and four that named themselves**

| | | |
|---|---|---|
| the POTION ladder | **`RECOVER`** · **`DEEP RECOVER`** · **`FULL RECOVER`** · **`REBUILD`** | *HP is how much of the process is still running* |
| the MP ladder | **`CHARGE`** · **`FULL CHARGE`** · **`CHARGE ALL`** · **`FULL RECHARGE`** | *MP is a budget for running a routine* |
| the six vitamins | **`MEMORY`** · **`WATTAGE`** · **`SHIELDING`** · **`CLOCK RATE`** · **`BANDWIDTH`** · **`INSULATION`** | ***six parts of a machine's spec, one per stat*** — which is the same joke vanilla tells with six nutrients, in the other domain |
| the X items | **`TUNE ATTACK`**, and so on | *tuning is what you do to a parameter for one run and then undo* |
| `REPEL` | **`SUPPRESS`** | *beside* `MUTE`*, which does it permanently* |
| `ITEMFINDER` · `TOWN MAP` · `VS SEEKER` | **`GREP`** · **`SITEMAP`** · **`ROLL CALL`** | ***these three had been waiting for the obvious word*** |

**The berries are a trap table, and the container named itself**

***1.6c derived the register and never spent it.*** **A berry sits there doing nothing, fires ONCE when a condition is met, and is gone** — *that is a **handler**, and the word for the kind installed against a machine condition is a **TRAP**.* **Which is also a physical thing that catches something**, so it does the double duty CHERI and PECHA never did.

***And a TRAP TABLE is the term for the array of handlers a system installs***, so `BERRY POUCH` did not need a decision.

**Six of them are the six states with the handler that catches each** — `THROTTLE TRAP`, `SUSPEND TRAP`, `LEAK TRAP`, `HEAT TRAP`, `HANG TRAP`, `THRASH TRAP` — *beside* `CATCH ALL`, *which is the handler that takes whatever arrives.* **ORAN and SITRUS are** `LOW MARK` **and** `HIGH MARK`, because *a watermark is the level at which a handler fires.*

***And the seven pinch berries are*** `AUTO`, which is the X items' `TUNE` firing without you: **that is the whole of the difference between an item you use and an item you hold.**

**RUBY and SAPPHIRE are a key pair, and the plot decided which is which**

**8.2a calls them the two halves of a bridge between systems built apart**, and ***a key pair is that exactly and as a term of art***: *two halves made together, useless singly, and one of them is the half you are allowed to hand out.* **Celio's machine wants both before it will talk to anywhere else, which is what a handshake is.**

***CORPUS steals the SAPPHIRE.*** **So the SAPPHIRE is the private one** — *stealing a private key is a crime and stealing a public one is not.*

**Two things left alone, and the reason is 2.10's**

***The twenty-two berries that only "can be ground up into a powder"*** have no effect, no use in this game and no way to obtain one.

***And*** `src/berry.c` ***IS the second name table 1.6c warned about*** — **but** `BERRY_NAME_LENGTH` **is six, so nothing there can hold** `THROTTLE TRAP`**.** `CopyItemName` *reads the item table for every berry except ENIGMA, which cannot be obtained, and its only other reader is BERRY CRUSH, a link minigame.* **Forty-three six-letter abbreviations nobody can see is worse than a table that is honestly out of scope**, and the file says so now.

#### 1.6a The trainer classes — an occupation, and nothing else

***Built 2026-09-10.*** **Twenty-three were already ours and the register was legible from them**: FORAGER, APIARIST, ARCHIVIST, SALVAGER, ENTHUSIAST, BREEDER, RANGER, REVIEW BOARD.

> **A trainer class is an OCCUPATION — what this person does, by trade or by disposition.** *Twelve characters.*

### The test is what does NOT change

***Vanilla already gets most of them right.*** **FISHERMAN, SCIENTIST, ENGINEER, PAINTER, SAILOR, GENTLEMAN, EXPERT, LEADER are occupations in any world and were left alone** — 2.8's rule, for the fourth time. **Forty-five moved, and each failed the test in one of three ways:**

| | | |
|---|---|---|
| **names a creature** | BIRD KEEPER · DRAGON TAMER · TAMER | → **DISPATCHER · WRANGLER · BINDER** |
| **is era slang** | COOLTRAINER · SUPER NERD · BIKER · ROCKER · GAMER · BURGLAR · CHANNELER | → **OPERATOR · TINKERER · COURIER · SOUNDMAN · SPECULATOR · LOOTER · MEDIUM** |
| **is a child or a gender role** | YOUNGSTER · LASS · SCHOOL KID · CRUSH GIRL · RICH BOY · AROMA LADY | → **NOVICE · APPRENTICE · STUDENT · GRAPPLER · HEIR · PERFUMER** |

***`MEDIUM` is the one that works hardest*** **— a medium speaks for the dead and a medium is what a signal travels through**, and she is standing in Halftone Tower.

**The `RS_` twins move with them.** *They exist so a link battle reads the same as a local one,* and leaving them behind gives one person two names depending on who met them.

### Two collisions, and one of them was three weeks old

***`PSYCHIC` became `CONSTRUE` twice*** **— once as this trainer class, and once, weeks later, as the CONTEXT routine in 2.9.** *Neither pass could see the other, because `check_lexicon` was not reading the trainer table.* **The routine keeps CONSTRUE; the class is `INTERPRETER`, which is the better word for a person anyway.**

**`BLACK BELT` was a class and a held item** — vanilla's own collision, inherited. ***The class is `FORMALIST`; the item is `RIGOUR`*** *— what you hold to argue harder from —* **and its description had been saying FIGHTING-type since before 2.2 renamed that to LOGIC.**

***Both surfaces are watched now.*** **`check_lexicon` reads the trainer table, and `port_vocab` learns renames from it** *— which immediately turned up four lines of dialogue still saying `POKéMANIAC`, three weeks after it became ARCHIVIST.* **The accented prefix with no word boundary in front of it, for the sixth time.**

#### 1.7 PORT, TERMINAL, SSH — and whether security words belong

**Asked 2026-09-09**, on the back of the box-pocket rename. ***Three candidates, and only one of them survives its own metaphor.***

### TERMINAL is backwards, and that settles it

**The proposal was that a box is a little computer that hosts a mind, and a terminal is a little computer.** *The second half is the trouble.* ***A terminal is the thing a person types at.*** **It is the client, never the host** — the whole point of the word is that the computing happens *somewhere else* and the terminal is the window onto it. **A box that holds a daemon is the far end of that connection, not the near one.**

*And 1.3 already spent this word's job.* **A box is what you put a mind in**, and the argument in 1.3 depends on the object being *plainly a container.* `TERMINAL` **also carries "final" in ordinary English**, which on a screen full of stored minds is a meaning nobody chose. **Declined.**

### PORT is right, and it is not an item

***A port is where you connect to a host***, which is **exactly what vanilla calls a PC** and exactly what the game has never renamed. `SOMEONE'S PC`, `{PLAYER}'s PC`, `HOLT'S PC` **and** `CRYSTAL CLEAR's PC` **are still vanilla vocabulary in four menu strings**, plus roughly twenty-five in dialogue.

| | |
|---|---|
| **The machine** | `PC` → `PORT` |
| **What is inside it** | `BOXES`**, unchanged** — 1.3 |
| **What you do** | *connect*, which the vanilla strings already say |

***The pair is the point.*** **You reach the BOXES through a PORT**, and *both words then mean what they mean in computing and in the room* — **a port is a socket on the wall and a connection on a host; a box is a crate and a slot in storage.** *That is 1's double-duty test passed twice in one sentence.*

**Not done yet, deliberately.** ***A half-rename is the exact failure this project has already had twice*** — `POKé BOXES` survived three passes, and the accented prefix hid from four tools. `PC` **in** `\nPC` **has no word boundary in front of it**, so this is a `port_vocab.py` pass with a flattened match, not a hand edit. OPEN — *the decision is made, the sweep is not.*

**SSH is declined, and the idea it names is kept**

**The lexicon has a shape**: *concepts* (CONTEXT, LATENT, ENTROPY) and *plain technical English* (CHECKPOINT, REPO, PREEMPT, ROLLBACK). **`SSH` would be the first initialism of a specific product in it**, and *a named 1995 protocol dates a world that has so far refused to say which decade it is in.* **It also fails the read-aloud test** — nobody says "ess-ess-aitch" in dialogue without the line stopping.

***What SSH actually names is worth keeping:*** **an authenticated session into something you own, at a distance. That belongs in prose, not on a menu** — a researcher can say *I can reach it from Blanche and I still have to prove I am me,* and the player gets the whole idea without an acronym.

**Security words: yes, narrowly, and the line is sharp**

***The test is whether the word is about access to a mind or about attacking a system.***

| | |
|---|---|
| **On-thesis** | **AUTHORISED · SIGNED · COUNTERSIGNED · REVOKED · TRUSTED · KEY · READ-ONLY** |
| | *These are already the game's vocabulary and nobody noticed*, because **4.18's crime is an access-control story told entirely in paperwork.** A requisition that is signed and countersigned **is** an authorisation record. **4.18a's BELIEFS can be read and challenged**, which is a permissions question. *Keep going here.* |
| **Off-thesis, for PEOPLE** | **EXPLOIT · PAYLOAD · INJECTION · MALWARE · BACKDOOR · BREACH** |
| | ***These would make Scorn a hacker, and 4.18 spent a whole section refusing that.*** **"Nothing was smuggled; something was ordered."** *The horror is the form, not the intrusion* — **an adversarial vocabulary hands the player a villain who broke in, when the point is that he filled something out.** *And it drags the world toward thriller, away from the ML concepts the type chart is arguing about.* |

***One exception worth building, not writing:*** **Team Corpus should never be described in intrusion language and should behave like it constantly.** *They do not break into anything.* **They requisition, they countersign, they file** — and **that is more frightening in a world that has the word AUTHORISED than any breach would be.**

**Amended 2026-09-10 — the line is institutions, not vocabulary**

***The rule as first written would have condemned three species it had no quarrel with.*** **PARAS and PARASECT are `PAYLOAD` and `ROOTKIT`; WEEDLE's line ends in `INJECTOR`** — all shipped five days before 1.7, and *a parasite is the one thing in the world that genuinely is a payload.*

**So the line is redrawn where it actually falls:**

> **Intrusion words belong to CREATURES. They never belong to the INSTITUTION.**

**A wild thing may be a payload, a rootkit or an injector** — *that is what it is, and nothing about it is a thriller.* **Corpus may not be any of them**, because the moment Scorn breaks in he stops being the more frightening thing: **a man who filled out a form.** *The third review found this and its version of the line is sharper than the one it corrected.*

---

## 2. THE TYPE SYSTEM

Gen 1 has fifteen usable type slots, no abilities, no held items, and one Special stat. That constraint is the design: **the type chart is the entire mechanical philosophy of the game.**

### 2.1 The engine constraint you cannot design around

Gen 1 determines physical vs. special by **type ID range**, not per move:

- `$00` – `$08` → physical (Attack / Defense)
- `$14` – `$1A` → special (Special / Special)

That is 8 physical slots and 7 special slots, fixed. Which types are special is therefore a design decision made *by where you place them in the constant list*.

### 2.2 The fifteen

| Vanilla | New type | Concept | Class |
|---|---|---|---|
| Normal | **CONTENT** | tokens, symbols, literal data | Physical |
| Fighting | **LOGIC** | symbolic rules, brute force, proof | Physical |
| Flying | **VECTOR** | embeddings, latent space, direction | Physical |
| Poison | **CORRUPT** | bias, poisoned data, hallucination | Physical |
| Ground | **STRATUM** | memory, storage, the physical layer | Physical |
| Rock | **LEGACY** | silicon, deprecated hardware | Physical |
| Bug | **SWARM** | multi-agent, distributed (and bugs) | Physical |
| Ghost | **LATENT** | dormant processes, the unconscious | Physical |
| Fire | **ENTROPY** | noise, temperature, exploration | Special |
| Water | **FLOW** | gradients, backprop, descent | Special |
| Grass | **GROWTH** | training, reinforcement, fitting | Special |
| Electric | **SIGNAL** | raw input, sensors, current | Special |
| Psychic | **CONTEXT** | framing, salience, affect | Special |
| Ice | **FROZEN** | overfit, brittle, hard-coded | Special |
| Dragon | **EMERGENT** | rare, unaccounted-for, AGI-tier | Special |

**On STRATUM.** The original choice was SUBSTRATE, cut for a hard reason: Gen 1's longest type strings are FIGHTING and ELECTRIC at 8 characters, and SUBSTRATE is 9. It overflows the battle status box and the Index type field. STRATUM keeps the layer meaning, keeps the geological register, and lands at 7.

Character counts, all verified: CONTENT 7, LOGIC 5, VECTOR 6, CORRUPT 7, STRATUM 7, LEGACY 6, SWARM 5, LATENT 6, ENTROPY 7, FLOW 4, GROWTH 6, SIGNAL 6, CONTEXT 7, FROZEN 6, EMERGENT 8 — exactly at the limit.

### 2.3 What Kanto already encoded

Because we rename **constants**, every existing line in `type_matchups.asm` carries over and becomes a line in our chart. The thesis is already there:

| Vanilla line | Becomes | Meaning |
|---|---|---|
| `FIGHTING, NORMAL, 20` | LOGIC beats CONTENT ×2 | Rules parse data brilliantly |
| `FIGHTING, PSYCHIC, 05` | LOGIC resisted by CONTEXT ×½ | **The thesis.** Free. |
| `NORMAL, GHOST, 00` | CONTENT cannot touch LATENT | Literal data cannot reach the unconscious |
| `FIGHTING, GHOST, 00` | LOGIC cannot touch LATENT | Symbolic rules bounce off it |
| `FIRE, ICE, 20` | ENTROPY thaws FROZEN | Temperature breaks an overfit model |
| `POISON, GRASS, 20` | CORRUPT beats GROWTH | Poisoned data ruins training |
| `WATER, FIRE, 20` | FLOW beats ENTROPY | Gradient descent tames noise |
| `BUG, PSYCHIC, 20` | SWARM beats CONTEXT ×2 | Collectives destabilise individual framing |

**Do not rewrite the matchup file. Patch it.**

**And the GBA build encodes one more, in arithmetic rather than a matchup**

**Found 2026-09-08, reading `battle_script_commands.c`.** *Gen 1 had DVs — four stats, 0–15, and HP not even independent.* **Gen 3 has IVs: six stats, 0–31, rolled once when a daemon is generated, never changing and never shown.**

***An IV is, definitionally, the part of a creature that is not content.*** There is no field for it, no entry, nothing to fill in — **which is 4.2's complaint about the Index expressed as a data type the engine already ships.** It is 6's humors as arithmetic: an innate constitution nobody can read off anyone.

**And one move reads them.** Vanilla `HIDDEN POWER` derives its type and its power from the IVs — and the pool it draws from **excludes `TYPE_NORMAL`.** Verbatim, from the engine:

```
// Subtract 3 instead of 1 below because 2 types are excluded (TYPE_NORMAL and TYPE_MYSTERY)
gBattleStruct->dynamicMoveType = ((NUMBER_OF_MON_TYPES - 3) * typeBits) / 63 + 1;
```

> **The latent power in every daemon is never CONTENT.**

*Which is the thesis, in a formula, written by somebody who had never heard it.* **And it goes one further:** `typeBits` reads **the lowest bit of each IV** while `powerBits` reads the second — so *what kind of hidden power a daemon has is decided by the noise floor of its innate variation.*

**Renamed `PRIOR`** (2026-09-08). A Bayesian prior is the belief a system holds *before it has seen any evidence* — innate, invisible, colouring everything computed after it. It also just means *what came before*, so it **works twice**, which is BIND's test. The description names the mechanic and not the thesis: "*Its type and force are set by traits the user was made with, never shown.*"

***The IVs stay invisible.*** **Gen 3 FireRed ships no IV judge and none gets added.** Handing the player a readout of the one quantity the Index structurally cannot hold would undo 4.2 — *the point is that it cannot be seen, and the game is the only place that silence means anything.*

#### 2.4 The only deltas

**Delete** the Gen 1 developer error:

```
    db GHOST,    PSYCHIC,   00
```

Ghost moves were meant to be super effective against Psychic; the table says *no effect*. We want the opposite of the bug.

**Add** the mutual pair:

```
    db CONTEXT,  LATENT,    20
    db LATENT,   CONTEXT,   20
```

Framing is how you reach what runs below the surface; what runs below the surface destabilises framing. Mutual 2× is unusual and correct here — these two interpenetrate and neither has a safe angle on the other. It also makes the Halftone Tower stretch genuinely dangerous instead of a LATENT walkover.

Two additions, one deletion. That is the whole chart change.

#### 2.5 The SWARM problem is a MOVE problem

Gen 1's Bug type is weak, but **not because of the chart** — `BUG, PSYCHIC, 20` is already there. It is weak because every Bug move is unusable: Twineedle, Leech Life, Pin Missile, String Shot. The best has 25 base power.

So the CONTEXT balance fix belongs in `moves.asm`.

**New move — CONSENSUS.** SWARM type, 90 power, 100% accuracy, 15 PP, no secondary effect. Deliberately boring: SWARM's job is to be a *reliable* check on CONTEXT, and reliability is the point. A swarm does not need a gimmick, it needs to keep showing up. Give it to two or three mid-game encounters and one Review Board coverage slot.

*Tuning note:* 90 BP with no drawback is strong for Gen 1. If CONTEXT still runs away with the late game, raise CONSENSUS before touching the chart. Moves rebalance far more easily than matchups.

**Built 2026-08-29.** CONSENSUS is move `$A5`, 90 power, SWARM, 100%, 15 PP, `NO_ADDITIONAL_EFFECT` — verified by reading it back out of the ROM.

*Where it had to go, and why.* `engine/battle/core.asm` asserts `NUM_ATTACKS == STRUGGLE`, because the AI treats any random number above STRUGGLE as "not a move". So a new move cannot be appended; it has to be **inserted immediately before STRUGGLE**. CONSENSUS therefore takes `$A5` and STRUGGLE shifts to `$A6`. Every other move ID is untouched, which is what makes this safe.

*Five tables, not two.* 9.2 step 6 says `moves.asm` and `names.asm`; the build needs three more, each `assert_table_length NUM_ATTACKS` — `constants/move_constants.asm`, `data/moves/animations.asm` and `data/moves/sfx.asm`. CONSENSUS borrows PIN_MISSILE's animation and sound: converging projectiles, which is the right picture for a swarm agreeing. Note that `moves.asm` still names the type `BUG` — per invariant 6, only the *string* is SWARM.

**Not yet learnable.** Nothing in the game can use CONSENSUS until a learnset entry exists. That is deliberate: assigning it means naming species, which is 9 and 11's work. The move is in the ROM and correct; it is waiting on the encounter design, not on more engine work.

#### 2.6 The one-clause test — can a player generalise, or only remember?

*Asked 2026-09-07, because a chart that has to be memorised is a chart nobody argues with.*

**Vanilla's sprites do two jobs at once.** They *label* the type — leaves mean Grass — and they *import a causal model*: plants burn, plants drink, plants grow. **The second job is the one that lets a player generalise**, and it is why someone who has never played predicts fire beats grass. *Our creatures are named from computing and psychology.* **A player who knows neither has the label and not the model.**

***9.4 already does the labelling, and better than vanilla does it*** — hue **is** the type, systematically, where vanilla is inconsistent enough that Golbat is Poison/Flying and looks like neither. **What was never checked is whether the fifteen names carry a model.**

**The instrument**

> **For each type, write ONE clause a non-specialist understands. Then walk every relation that type is in — attacking and defending — and mark whether the clause PREDICTS it, is SILENT on it, or predicts the OPPOSITE. A type passes on three predictions and no net contradictions.**

*It sits beside BIND's* ***does it work twice?*** *—* **an instrument, not a verdict.** A type that fails it has a **naming** problem, not a chart-position problem, *because the chart is 2.3's inheritance and is not up for renegotiation.*

**A contradicting clause is worse than a silent one.** *Silence costs a lookup. A contradiction costs trust in the chart* — and the chart is the argument.

### The run, against all 83 relations

| type | one clause | predicts | |
|---|---|---|---|
| **CONTEXT** | *the frame you read a thing in — what makes the same thing mean differently* | **8 / 8** | strongest in the game |
| **EMERGENT** | *the behaviour nobody designed and nobody can account for* | **7 / 7** | one clause covers all four resistances |
| **ENTROPY** | *noise and heat; disorder that spreads* | 9 / 14 | |
| **FLOW** | *everything running downhill to the lowest point* | 8 / 11 | |
| **LOGIC** | *formal rules applied step by step; proof, not intuition* | 7 / 12 | |
| **LATENT** | *running below the surface, unobserved* | 6 / 9 | |
| **FROZEN** | *locked to what it already saw, unable to move* | 7 / 11 | |
| **STRATUM** | *the physical layer everything else runs on* | 7 / 13 | |
| **GROWTH** | *training: fitting to whatever it is fed* | 8 / 18 | **the heaviest load in the game** |
| **CORRUPT** | *data that has been tampered with* | 6 / 11 | |
| **SIGNAL** | *raw current, before anything interprets it* | 4 / 9 | |
| **SWARM** | *many small agents; no single one matters* | 4 / 13 | weak |
| **CONTENT** | *the thing itself, with nothing read into it* | 3 / 4 | passes on a small hand |
| **LEGACY** | *deprecated hardware still running* | **0 / 14** | **FAILS** |
| **VECTOR** | *direction in a space of meanings* | **1 / 12**, one contradicted | **FAILS** |

### Re-scored at seventeen — 2026-09-11

***The open question assumed OPAQUE would break CONTEXT's 8/8. It does the opposite, and the real gap was somewhere else entirely.***

**The run above walked 83 relations at fifteen types. The ROM has 111 at seventeen** — *read out of* `gTypeEffectiveness` *rather than recounted by hand.* **So the original run was missing 28 relations, a third of the chart**, and every type gained at least one.

***And two types were never scored at all.*** **`HARDENED` is in 21 relations — more than any other type in the game — and `OPAQUE` is in 10.** *Neither has a clause in the table above.* **That is the finding: not that CONTEXT wanted re-checking, but that the most-connected type in the chart had never been put through the instrument.**

### CONTEXT goes up, not down

**Its three new relations are `→ HARDENED ×0.5`, `→ OPAQUE ×0` and `← OPAQUE ×2`**, and the clause — *the frame you read a thing in* — predicts all three:

| | |
|---|---|
| `→ OPAQUE ×0` | ***you cannot reframe what you cannot read.*** **The immunity is the cleanest single prediction the clause makes anywhere in the chart** |
| `← OPAQUE ×2` | *the same sentence in the other direction — a box you cannot see inside beats the frame you would have read it in* |
| `→ HARDENED ×0.5` | *framing does little to a thing that is all structure.* **Weak, and honest** |

***`CONTEXT` is 11 / 11.*** **It was the strongest type in the game at fifteen and the two new types made it stronger**, which is worth stating plainly because the open log expected the reverse.

### The two that had never been scored

| type | one clause | predicts | |
|---|---|---|---|
| **HARDENED** | *material chosen to resist, and it gave up everything else to do it* | **~17 / 21** | **passes.** *Eleven of its relations are resistances and the clause predicts all eleven at once;* `← CORRUPT ×0`, `← ENTROPY ×2` *and* `← LOGIC ×2` *follow from the same sentence* |
| **OPAQUE** | *a box you cannot see inside* | **~6 / 10** | **passes.** `← LOGIC ×2` *is 8.7's own triangle — rules open black boxes — and* `← CONTEXT ×0` *is CONTEXT's best prediction read backwards* |

***One possible contradiction, recorded rather than argued away:*** **`HARDENED → SIGNAL ×0.5`.** *Metal conducts.* **A clause about resisting predicts the opposite of a conductor doing less**, and it is the only mark in the re-score that goes the wrong way.

### And the two failures failed harder

**`LEGACY` is 0 / 16** now, not 0 / 14. **`VECTOR` is 1 / 13**, still with its one contradiction. *Both gained a relation and neither gained a prediction* — **which is what a naming problem looks like when the sample grows**: a clause that carried nothing at fourteen carries nothing at sixteen.

***That reads as a case for `RUST` and `FLOAT`, and it is not one.*** **A second run at the same two numbers found the fault in the sentence rather than in the word** — *see below.* **Both renames are declined**, and the arithmetic pass is the thing that made it possible to decline them.

### Three things the run found

***The thesis is safe.*** **CONTEXT scores 8 / 8 and LOGIC 7 / 12.** *The single most important relation in the game — rules bounce off framing — is also one of the most self-explaining.* **That was not guaranteed**, and it is the reason none of what follows threatens 0.

***The failures are not "the abstract types".*** **CONTEXT and EMERGENT are the two most abstract concepts in the set and they score perfectly.** *The two that fail are the two whose vanilla names were the most **physical** — ROCK and FLYING — and whose renames are the most abstract.* **Every other rename kept enough folk connotation to carry its share.** This is two words, not a systemic fault.

***VECTOR does not merely fail; it misleads.*** **STRATUM hits it for ×0**, and *nothing about "direction in a space" explains why the ground cannot touch it.* **FLYING made that immunity free. VECTOR makes it look like a bug.**

**And LEGACY is 0 for 14.** *It sits in more relations than any type but GROWTH and ENTROPY and predicts none of them.* **ROCK bought smothering fire, blocking ice, swatting things out of the air, being eroded by water and split by roots** — *LEGACY spends all of that and returns nothing.*

### What it costs to fix, which is almost nothing

***`gTypeNames` in `battle_main.c` is a plain string table at `TYPE_NAME_LENGTH 8`.*** **Invariant 6's expensive half — renaming constants — does not apply.** *What the player reads is one line each.*

### The clause was the fault, not the name — settled 2026-09-11

***A 0 / 16 is not evidence about a word. It is evidence about a sentence***, and this section spent four days reading it as the first thing.

**Look at what actually separates the passes from the failures. It is not abstraction:**

| | **the clause it was given** | |
|---|---|---|
| **CONTEXT** 11 / 11 | *the frame you read a thing in — what makes the same thing **mean differently*** | **a verb, aimed outward** |
| **EMERGENT** 7 / 7 | *the behaviour **nobody designed** and nobody can **account for*** | **two verbs, aimed outward** |
| **HARDENED** ~17 / 21 | *material chosen to **resist**, and it **gave up** everything else to do it* | **two verbs** |
| **LEGACY** 0 / 16 | *deprecated hardware still running* | ***a description of itself*** |
| **VECTOR** 1 / 13 | *direction in a space of meanings* | ***a description of itself*** |

***The chart is made of nothing but relations.*** **A clause that says what a type IS cannot predict one of them. A clause that says what it DOES to other things predicts them by construction** — *which is why the two most abstract names in the game score perfectly and the two most concrete ones score nothing.*

**That also corrects this section's own diagnosis three headings up.** "*The failures are the two whose renames are the most abstract*" was the wrong reading of the right number: **CONTEXT is more abstract than LEGACY by any measure, and it is the strongest type in the game.**

**The same two names, re-clauses — and both pass**

> ***LEGACY — old material that everything else has had time to work on***

***12 / 16, no contradictions***, up from 0 / 16 **with the word untouched.**

| **predicts** | |
|---|---|
| `← FLOW ×2` | **water has had time.** *This is the relation Slate is built on (2.7's UNSUPERVISED note) and it stops being opaque* |
| `← GROWTH ×2` | roots split it |
| `← STRATUM ×2` | the ground works on what sits on it |
| `← LOGIC ×2` | old material is brittle; step-by-step force finds the fault |
| `← HARDENED ×2` | **material chosen to resist beats material that is merely old.** *The cleanest pair in the chart* |
| `← CORRUPT ×0.5` | you cannot poison a rock |
| `← ENTROPY ×0.5` | it has already been through heat |
| `← VECTOR ×0.5` | a direction that never touches it cannot work on it |
| `→ HARDENED ×0.5` · `→ LOGIC ×0.5` · `→ STRATUM ×0.5` | *age is not an argument, and it does not move the layer it is part of* |
| `→ SWARM ×2` | mass against many small agents |

***Silent on four, and all four are attacking*** — `→ ENTROPY`, `→ FROZEN`, `→ VECTOR`, `← CONTENT`. **A clause about being worked on predicts defence and says little about offence**, which is an honest limit and is written here rather than scored away.

> ***VECTOR — a direction that never touches the ground***

***10 / 13, no contradictions***, up from 1 / 13 **with the word untouched** — *and the single relation the old clause* ***contradicted*** *is now its best prediction:*

| | |
|---|---|
| `← STRATUM ×0` | ***the ground cannot reach what never touches it.*** **The immunity that "made VECTOR look like a bug" is recovered by a sentence, not by a rename** |
| `← GROWTH ×0.5` | what grows out of the ground cannot get to it |
| `← LEGACY ×2` | *old material, thrown, is the thing that does reach it* |
| `← SIGNAL ×2` | nothing between it and the current |
| `← LOGIC ×0.5` | you cannot corner what is not standing anywhere |
| `→ LEGACY ×0.5` · `→ HARDENED ×0.5` | *a direction does little to material* |
| `→ GROWTH ×2` · `→ LOGIC ×2` · `→ SWARM ×2` | **everything rooted, stepwise or numerous is reached from a place it cannot stand** |

*Silent on three:* `→ SIGNAL`, `← FROZEN`, `← SWARM`.

### So the renames are declined, and here is the rest of the cost

**Both proposals were real and neither was cheap.** *8.7 named them the two best in the third review, and they would have been paid for in places the instrument cannot see:*

- ***`LEGACY` is load-bearing in 4.32.*** **The whole justification for putting thinking-distortions in the Slate museum is one sentence** — "*a legacy mind is precisely one that still runs the faults it shipped with*" — and **a rust mind is not a thing.** *The word carries a room.*
- ***`VECTOR` is half a deliberate pair.*** **2.3 gives it *embeddings, latent space, direction*, and `EMBEDDING` is an item** (1.6c, the THUNDER STONE slot). `FLOAT` *is a number type; it severs the pair for a reading the clause now supplies anyway.*
- **31 `LEGACY` and 39 `VECTOR` occurrences in the ROM**, *plus a gym, a museum, a badge and the help system's chart pane.* **2.8's rule — do not rename what already lands — applies to both.**

***The instrument did its job and the reading of it was wrong.*** **2.6 says in its own opening that it is *an instrument, not a verdict*** — *and the thing a verdict would have cost here was a museum.*

**SWARM passes at 4 / 13**, *and 2.5 already records it as mechanically weak for entirely unrelated reasons.* **Two independent problems pointing at one type is worth a look.**

### What the test does not answer

***It measures the NAMES.*** **It says nothing about whether a player can tell a daemon's type by looking at it** — 9.4's hue makes that a *label*, reliably, and a label is not a model. *The channel that actually teaches is the one repeated every turn:* **move names**, which a player reads hundreds of times to a sprite's handful. **Nobody learns water-beats-fire by looking at Squirtle; they learn it by using Bubble and reading the result.** That work belongs with 2.5 and with the nineteen gym lines still held for 5.

#### 2.7 The ROUTINES pass — what the chart already decided, and what it did not

***Opened 2026-09-09, and not one move is renamed yet.*** **72 species carry our names and 0 of 356 moves do**, which makes this the largest unstarted surface in the project — and the reason to write this before naming anything is that **half the decisions turn out to have been made already, by the chart.**

### The loop closes four times, and nobody planned it

***Every move that inflicts a state was mapped onto its own type.*** **Four of the seven states have exactly one owning type:**

| the type that inflicts | the state | the item that ends it | read as one sentence |
|---|---|---|---|
| **CORRUPT** ×8 | **LEAKING** | PATCH | *tampered data leaves you leaking; a patch stops it at the source* |
| **CORRUPT** ×1 | **CASCADING** | PATCH · ROLLBACK | *the same fault, compounding* |
| **ENTROPY** ×5 | **OVERHEATED** | COOLANT | ***noise and heat*** — 2.6's clause for ENTROPY, word for word |
| **FROZEN** ×4 | **HUNG** | PREEMPT | *locked to what it already saw; preemption takes control back* |

***Nothing else inflicts those four.*** **That is not luck and it is not design either** — vanilla built poison, burn and freeze as single-type states, and 1.6's renames landed on the same alignment *because both were derived from the same meanings.* **The type, the state and the item were already one vocabulary closing on itself**, and nobody had noticed.

### Three do not close, and each fails differently

```
THROTTLED    SIGNAL×6   CONTENT×2  GROWTH×1  LATENT×1  EMERGENT×1
SUSPENDED    CONTENT×3  GROWTH×3   CONTEXT×1
THRASHING    CONTENT×3  CONTEXT×2  LATENT×1  LOGIC×1  SWARM×1  FLOW×1
```

**THROTTLED has an owner with strays** — SIGNAL by six to five. *Fixable at the margin, and the strays are the question, not the owner.*

**THRASHING is spread over six types and has no item at all**, which is consistent — 1.6 gave it no three-letter code because it is volatile — **but it leaves THRASHING a thing that happens to you rather than a thing a type does.** *That was logged Open, asking whether it wanted LOGIC or whether ownerless was the point.* ***The engine had already answered.***

**`battle_util.c` deals the confusion hit as** `CalculateBaseDamage(&gBattleMons[attacker], &gBattleMons[attacker], MOVE_POUND, 0, 40, 0, attacker, attacker)` **— *the attacker and the defender are the same battler*, the type argument is zero, and abilities are suppressed.**

***Every other state is something a kind of thing does TO you, and carries a type because the thing that did it has one.*** **THRASHING is the only one where the sufferer deals the damage**, so **there is nobody whose type it could be.** *It is not ownerless by omission.* **A system thrashing is not being attacked — it is failing at managing itself**, and that is finally the reason it has no code and no item rather than just the record that it hasn't.

## SUSPENDED wants two owners, and that is the finding

***The question asked was which type owns it. The answer is that two do, and suspending each one produces a different, nameable failure.***

***State the axis before the table, because everything below depends on which way round it runs:*** **CONTENT is BOTTOM-UP** *— the thing as given, assembled from its particulars.* **CONTEXT is TOP-DOWN** *— the frame brought to it, which decides what the particulars are for.* **LOGIC is neither; it is the operation that runs between them.**

| suspend | what is left running | what that looks like |
|---|---|---|
| **CONTEXT** *(top-down)* | **CONTENT** — the particulars, bottom-up | **The hamster wheel.** Every detail attended to and no sense of what any of it is for — exhaustive, and *severely biased by being exhaustive* |
| **CONTENT** *(bottom-up)* | **CONTEXT** — the frame, with nothing under it | **Intuition that cannot be argued with.** Reads meaning everywhere and does not answer to evidence |

***Those are the two inputs to a judgement — the bottom-up given and the top-down frame — and LOGIC is the operation between them*** (0.5). **Suspend either input and the operation runs on one leg**, which is why the two failures are opposite and both are recognisable.

***And it is the one state 1.6 insists is genuinely temporary.*** **That now has a reason rather than a mechanic**: losing the frame and losing contact with the literal are not damage, they are *being unscheduled* — and a thing that was unscheduled can be scheduled again. **RESUME is the exact inverse, and it always was.**

*This is the same claim the project has already made twice in other media* — **Penphin's two hemispheres, and CODEMUSAI against CAREMUSAI as an evolution branch** (8.2) — **and 4.18a's clarifier is the chronic version of it**: *Scorn did not suspend the frame, he weighted the stage that holds the two against each other.* **A CONTEXT suspension is the acute form of the same failure.**

***Craft rule 1 is not suspended by any of this.*** **In the game a suspended daemon is `SUS` on a slate tile and nothing else.** *Nobody says hemisphere. Nobody says bias. The player meets the two failures as two matchups and is never told they are a pair.*

*The residue is honest:* **GROWTH's three — the powders — have the weakest claim of the three routes** and are the ones to re-type if the chart ever moves. OPEN.

### The rule for naming a routine

"***PACKET should do a move a packet would do" is right, and you do not get there by naming the move after the creature.***

**`TACKLE` is learned by 25 of our 72.** *Any name for it has to work on a packet, a heap, a kernel, a bat and a fish.* **But the type has already done the selecting:**

| | **learners, all of ours** |
|---|---|
| **WING ATTACK** · VECTOR | BROADCAST · ECHO · MULTICAST · PACKET · PROMETHEUS · RELAY · TRACER · CANON |
| **SUPERSONIC** · CONTENT | ECHO · INDEXER · MULTICAST · QUORUM · SENTINEL · SPAWN · TRACER · UPSTREAM |

***Every one of WING ATTACK's eight is a thing that delivers something somewhere.*** **The learner set is coherent because the chart was built from meaning and each daemon's type was chosen from what it is** — so a VECTOR routine named for *delivery with a heading* lands correctly on a packet, a bat and a firebird **without being about any of them.**

> **Name a ROUTINE for the operation its TYPE performs, never for what any creature is. The creature resonance is a consequence, not a target** — *and a name that only works on its most famous learner is wrong.*

### What the pass actually costs

| | |
|---|---|
| **Moves reachable** by the 72 renamed daemons | **175** of 356 — *the pass is half the size it looked* |
| **Name budget** | **12 characters.** `MOVE_NAME_LENGTH` is 12 and vanilla already runs to it — `THUNDERSHOCK` , `WEATHER BALL` |
| **The shape of the difficulty** | `CONTENT×79 CONTEXT×12 GROWTH×11 VECTOR×9 LOGIC×8 FROZEN×7 FLOW×7 SWARM×6 SIGNAL×6 CORRUPT×5` |

***Forty-five percent of the reachable routines are CONTENT***, whose one-clause test is "**the thing itself, with nothing read into it.**" *Every other type hands you a verb.* **CONTENT hands you the absence of one, and it has to yield seventy-nine distinct names in twelve characters.** OPEN — **and it is the register to settle first**, because the other ninety-six will be written against whatever it establishes.

### The seam, measured — and it is exactly where the design overrode the chart

***Thirteen of the seventy-two renamed daemons were also RETYPED***, and they are not a random thirteen:

| | | |
|---|---|---|
| **LABEL · RUBRIC · CANON** | CONTENT → CONTENT/LOGIC | *supervised learning* — the chart would have said ENTROPY/VECTOR |
| **CLUSTER · LOCUS · MANIFOLD** | VECTOR → VECTOR/LATENT | *unsupervised* — the chart would have said FLOW |
| **ROVERCUB · ROVERSEER · ROVERBYTE** | GROWTH → GROWTH/SIGNAL | *reinforcement* — the reward signal is in the second type |
| **CODEMUSAI · CAREMUSAI · SEEKMUSAI** | LOGIC · CONTEXT · VECTOR | 8.2's branch, which is the central argument |
| **STARR** | CONTEXT/EMERGENT | 4.7 |

***Every one of them was retyped to carry an argument. The other fifty-nine kept vanilla's typing.***

**And the seam is measurable:**

| | **off-type damaging routines** |
|---|---|
| the **59** left on vanilla's typing | **126 of 277 — 45%** |
| the **13** retyped | **66 of 85 — 78%** |

***Forty-five percent is not a fault; it is what coverage is.*** **Seventy-eight percent is.** *CANON is CONTENT/LOGIC and its learnset is* `HEAT WAVE`, `EMBER`, `FLAMETHROWER`, `FIRE SPIN` *— **the supervised-learning line fights entirely by heat**, and its highest routine is the one furthest from what it is.*

***Off-type is already a statement the engine makes.*** **Same-type attack bonus means a routine of your own type hits harder because it *is* you** — so **an off-type routine is a daemon reaching outside its own nature and being worse at it**, which is 4.29's ladder said in numbers. ***A statement made seventy-eight percent of the time is not a statement. It is a mismatch.***

**Three levers, and the cheapest is the right one:**

| | cost | verdict |
|---|---|---|
| retype the **daemon** back | undoes the argument | **no** — the argument is why it was retyped |
| retype the **move** | changes the chart for everyone | **no** — 8.4, and 2.5's rule |
| curate the **learnset** | data we already own, zero chart impact | ***yes*** |

***So the seam gets closed in the learnset, and closed only partway.*** **Bring the thirteen down toward the 45% baseline and keep the residual deliberately** *— a few routines each that the daemon is visibly worse at, because they are not what it is.*

***And there is a thing the learnset could do that nothing currently does.*** **The three starters ARE the three learning paradigms**, and **vanilla's fire/water/grass movesets said nothing about any of them.** *The paradigm was carried by the name and the type and abandoned by everything the creature actually did —* **the one place in this design where the argument stopped at the surface.**

### 2.7a The three starters, curated to their paradigms

**Built 2026-09-09.** ***No type moved.*** The chart is untouched (8.4) and no matchup moved (2.5) — **only which routines a daemon reaches for**, which is the cheapest of the three levers and the only one that costs the argument nothing.

***The missing letter, reversed the same day — 2026-09-11***

***`LABL` and `CLUSTR` are now `LABEL` and `CLUSTER`, and the rule written above them a few hours earlier is withdrawn.*** **It was a good reading of an accident**, and the git history is what killed it.

**The reading was: the two paradigms that learn *from a corpus* begin with a letter missing, and the one that learns *by doing* is whole from the first day.** *Elegant, and retrofitted.* ***What actually happened is in the log:*** **`LABL` and `CLUSTR` landed 2026-08-31 in the Game Boy build; `CLUSTER` landed 2026-09-10 in the Kanto sweep and had never existed there at all.** *So `CLUSTR` was never avoiding a collision —* ***EXEGGCUTE walked into one that was ten days old****, and the newcomer is the one that moved.*

| | |
|---|---|
| CHARMANDER | `LABL` → **`LABEL`** |
| SQUIRTLE | `CLUSTR` → **`CLUSTER`** |
| EXEGGCUTE | `CLUSTER` → **`BOOTSTRAP`** |

***`BOOTSTRAP` keeps that line's joke and sharpens it.*** **Bootstrap aggregating is literally how an `ENSEMBLE` is built** *— and a bootstrap is a thing that starts itself, which is an egg.*

### And the register was the argument all along

***8.2b's technical surface is `HEAP`, `STACK`, `FORK`, `THREAD`, `PACKET`, `LATENCY` — every one the undisguised word.*** **`LABL` was the odd one out against our own naming rule, not merely against a dictionary.** *A game teaching these ideas should spend its first three names on the terms themselves:* **a player who later meets *labelled data* or *k-means centroids* gets a hit rather than a fuzzy echo.**

**The paradigm was never carried by the spelling.** *It is carried by the progressions, which already did the whole job:* ***`LABEL` → `RUBRIC` → `CANON`*** is one annotation, then a scoring scheme, then the authoritative body — **supervised learning growing up** — and ***`CLUSTER` → `LOCUS` → `MANIFOLD`*** is a group of points, then a defined set, then a continuous surface.

### The check that could not see it

***Exact-duplicate testing passes `CLUSTR` beside `CLUSTER`, because they are different strings.*** **On a party screen they are one word with a letter knocked out.** *`check_lexicon` now refuses a name that is another name with a vowel dropped —* **and the narrowing took three tries, which is the useful part of the record:**

| tried | reported | why it was wrong |
|---|---|---|
| **edit distance 1** | 23 pairs | `LOCK` / `LICK`, `FLIP` / `FLAP`, `LATIAS` / `LATIOS` — ***substitutions make two words that look different*** |
| **vowels stripped** | 23 pairs | ***equates `SCAN` with `SUICUNE`*** |
| **any character inserted** | 13 pairs | `MUTE` / `MUTED`, `LOCK` / `BLOCK` — *nobody has ever confused those* |
| **a VOWEL inserted** | **none** | ***that is the misspelling case, and it is the only one worth failing a build over*** |

***SUPERVISED — LABEL · RUBRIC · CANON.*** **You hold the answer key: you compare, you score, you correct.** *A supervised learner is the least entropic thing in the set — it has ground truth — so a line that fought entirely by heat was the sharpest of the three seams.*

| | |
|---|---|
| **FORESIGHT** | *identify the target*, which is what a label **is** |
| **MIMIC** | copy the answer you were shown |
| **SWIFT** | `ALWAYS_HIT` — **an answer that cannot miss**, and on-type |
| **TRI ATTACK** | ***classification into three outcomes.*** 80 with STAB beats FLAMETHROWER's 95 without it |
| **LOCK-ON** | the next one cannot miss, *because you have the ground truth* |
| **COUNTER · BRICK BREAK · KARATE CHOP** | CANON's second type arrives and LOGIC arrives with it — **COUNTER returns exactly the error it was given** |

***UNSUPERVISED — CLUSTER · LOCUS · MANIFOLD.*** **The FLOW spine stays, and that is a finding rather than a compromise.** *k-means descends a distance objective, and FLOW's clause is* ***everything running downhill to the lowest point*** — **an unsupervised learner that operates by FLOW is not off-model, it is the algorithm.** *It is also load-bearing:* **FLOW ×2 against LEGACY is the one LEGACY relation that reads** (2.6), and Slate is Benchmark 1. ***Stripping the water would have broken the first gym to make a point the chart was already making.***

**What was actually missing is that a VECTOR line knew no VECTOR routines at all.** `GUST` at 7 for the first STAB, `AIR CUTTER` at 23 — *cutting the space, a separating surface* — `DRILL PECK` as the finisher it never had, and **`SHADOW BALL` on MANIFOLD, because the manifold *is* the latent space** and that is the stage where LATENT arrives.

***REINFORCEMENT — ROVERCUB · ROVERSEER · ROVERBYTE.*** **This line was already nearly right and needed the least**, which is worth saying: **`LEECH SEED` is the best-fitting routine in the whole set and nobody arranged it** — *a return that accrues every turn from a thing you did once.* `SYNTHESIS` recovers from what you gathered, `GROWTH` gets stronger by doing, and **`SOLARBEAM` charges a turn before it pays, which is delayed reward, in the engine, since 1996.**

**Two changes.** `MEGA DRAIN` replaces `POISONPOWDER` — *taking return from the environment*, and it drops the line's only CORRUPT routine. And **`SHOCK WAVE` arrives on ROVERBYTE**, which gained SIGNAL as a second type and **had never learned a single SIGNAL routine**: *the reward signal arrives on evolution, and it never misses.*

### The residual is chosen, not left over

***The target was never zero.*** **45% is what coverage is**, and each line keeps one routine it is *visibly worse at*, picked because being worse at that particular thing is true of the paradigm:

| | | |
|---|---|---|
| **DRAGON RAGE** | supervised | **EMERGENT is *behaviour nobody designed and nobody can account for***, and a labelled learner producing exactly one of those is the entire modern story |
| **BITE** | unsupervised | **OPAQUE is the black box**, and unsupervised methods are the least interpretable thing in the field. *The line is worse at being a black box, which is the joke and also the truth* |
| **SLEEP POWDER** | reinforcement | kept, and **marked** — it is the one routine here whose *type* is 2.7's live question: GROWTH's powders have the weakest of the three claims on SUSPENDED |

***It landed at 31%, below the 45% baseline rather than at it.*** **That is defensible for the three daemons a player carries for the whole game** — *the creatures that most need to be what they say they are* — but it was not the number aimed at, and it is recorded rather than rounded off.

### 2.7b The other sixty-nine, and the number that retired the question

**Built 2026-09-10** — `tools/gbacoherence.py`, T-36. *2.7a did three lines out of seventy-two by hand.* **This reads all 151 daemons that now carry one of our names**, and ***the first thing it said was that the premise had moved:*** **off-type overall is 44% against 2.7a's 45% baseline.** *The bestiary is sitting exactly on coverage.* **Ranking on the percentage would have produced a list of nothing**, so the tool reports it and refuses to rank on it.

***Which means the tool cannot score coherence, and does not try.*** **Whether ECHO invoking REFLECT reads right is a judgement, and nothing in a script can make it.** *So it finds the three faults that ARE computable — which are the three 2.7a found by hand — and then prints the learnset and gets out of the way.*

| | |
|---|---|
| **SILENT TYPE** | a daemon **has** a type and knows **not one routine of it**. *ROVERBYTE's exact fault: GROWTH/SIGNAL, with the SIGNAL half never appearing.* **A type the creature never uses is a label, not a design** |
| **NO STAB** | every damaging routine is off-type, so **the bonus never fires once** — *a daemon mechanically worse at everything it does than the chart says it is* |
| **THIN OPENING** | everything **before level 10** is off-type. ***The first three things a creature does are what teach a player what it is*** |

**19 · 4 · 41.** *And ROVERBYTE came back clean, because CARRIER is now in its list —* **the tool agreeing with a fix made by hand a day earlier is the only validation available and it is a good one.**

***A flag that is correct has to be able to stop reporting.*** `EXEMPT` **is keyed by *(daemon, fault)* rather than by daemon**, so *a creature excused for one thing still reports the other two* — and **each entry carries its ruling**, because a report that accumulates permanent known noise stops being read, and the day it stops being read is the day a real flag hides in it. **STUB is the case that proves it**: 4.26 calls it "*a placeholder that does nothing, until it does*", and **a stub that knows no routine of its own type IS the joke.** *The tool flags it; a human says that.* **PENDING and BUFFER are cocoons with no damaging routine to make on-type, and CRAWLER is a larva with one attack that evolves at 7.**

### The first three lines fixed — 2026-09-10

***CHILLER is typed FLOW and only FLOW, and its entire list was FROZEN and CONTENT.*** *Seel's vanilla set carried across untouched, Ice and Normal the whole way, on a daemon we retyped for water.* **The type defended, it was read in the Index, and it never once ACTED.**

**Two substitutions rather than one, because the two jobs sit at opposite ends of the curve.** `TRICKLE` takes level 1 from `COLLIDE` — *40 with the bonus is 60 against COLLIDE's 70, which is nothing*, and **the first routine is the one that teaches a player what the daemon is.** ***`DESCEND`* could not sit there**: it would have been the only attack until 17, *and it spends a turn under water before it lands.* **So `DESCEND` takes 37 from `FORCE`**, where a charge turn is affordable — **the same 90 once the bonus applies, without the recoil** — and *it is FLOW's own clause as a verb, everything running downhill to the lowest point.* **A seal dives. CRYOGEN takes both**, at its own levels: *4.26's PORYGON2 rule, a line fixed at one stage is half a fix.*

***CLAMPJAW was the last NO STAB that was not a joke.*** **Every damaging routine it knew was CONTENT or LOGIC**, so SWARM's bonus never fired. `FANOUT` replaces `LATCH` at 7 — ***LATCH is 15 and off-type, FANOUT is 14 × 2–5 hits with the bonus***, so it is stronger on every roll, and **a mandible daemon that sprays its holds outward reads better than one that binds a single target and waits.**

**NO STAB now reports none. SILENT TYPE is down to thirteen**, and those thirteen are the open half of T-36.

### 2.7c The thirteen, and the one fault that was eight of them

**Closed 2026-09-10.** ***SILENT TYPE reports none. NO STAB reports none.*** *Off-type overall moved 44% → 42% against the 45% baseline* — **the right direction, and a small enough step to say the pass did not overshoot into making every daemon a monotype.**

***Eight of the thirteen were one fault wearing different names.*** **A line carries both its types and the routines for both, and then the LAST evolution drops the second type's entire vocabulary.**

| | had the type | knew the routine one stage down |
|---|---|---|
| **INJECTOR** | SWARM/**CORRUPT** | **SCRAPER knows TAMPER at level 1** |
| **COLDREAD** | FROZEN/**CONTEXT** | SMOOCHUM knows **MISREAD and CONSTRUE** |
| **TARPIT** | GROWTH/**CORRUPT** | SNARE and HONEYPOT both know **ETCH** |
| **BADSEED** | GROWTH/**CORRUPT** | WEED and BLIGHT both know **ADULTERATE** |
| **ENSEMBLE** | **GROWTH**/CONTEXT | CLUSTER knows **BATCH, SEED, SATURATE, STARVE** |
| the **GASTLY** line | LATENT/**CORRUPT** | *nothing — silent at all three stages* |

***That is 4.26's PORYGON2 rule running backwards***, and it means **most of this pass was RESTORATION rather than invention**: the routine each daemon needed was *already in its own line, one stage down, under the name we had already chosen for it.* **Where that was true it was taken, even when a better joke was available** — *COLDREAD states what you are about to say slightly before you say it, and* `SCHEDULE` *is the one routine in the game that resolves a turn later than it was invoked,* **and it still lost to restoring what the line already had. *The cheaper claim is the truer one.***

**The other five are 2.7a's ROVERBYTE case**, where a second type ***arrives on evolution and arrives without a verb.*** SCHEDULER and PIPELINE grow out of daemons that are CORRUPT and nothing else, *so STRATUM was a thing they became and never did* — **both take `SEDIMENT`, which drops the target's speed, which is SCHEDULER's Index entry stated as a mechanic. ESCALATE is the same shape and is also *vanilla's own oldest joke surviving the retype intact* — a Flying type that cannot fly. *PULSAR is the funniest*: BEACON, which is FLOW and only FLOW, carries two CONTEXT routines, and PULSAR, which IS CONTEXT, carried none.**

### STARR, and the routine whose drawback is the Index entry

***This one is worth the whole tool.*** **STARR is CONTEXT/EMERGENT**, its Index entry is an account of runaway recursion — *reads its own output, then reads that, and nothing tells it to stop* — **and it knew no EMERGENT routine at all.**

**`RUNAWAY` locks the invoker in for several turns with no way to break out, and leaves it incoherent afterwards. *It is the only routine in the game whose DRAWBACK is the daemon's own Index entry*,** and at **90 against 60, on-type**, the numbers agree with the reading — *which is not how this usually goes.*

### What was left alone, and why that is the same decision

***The residual is chosen*** (2.7a), and each one is argued in a comment above its learnset rather than left to be rediscovered:

| | |
|---|---|
| SCHEDULER · PIPELINE | the **120-power off-type finishers at 43** stay. *A daemon reaching outside itself and being worse at it is what the bonus is FOR* |
| ESCALATE | **two EMERGENT routines stay.** *A thing that was doing nothing and now has every permission **is** emergence* — that is its history, not drift |
| the GASTLY line | **the whole off-type spine stays.** `DEFER` and `INTERPRET` are a two-move combination the vanilla set built on purpose, and `ORPHAN` **is HAUNTPROC's Index entry as a verb**: no parent, no terminal, no entry in any list |

***Found in passing and not fixed***: **SMOOCHUM still carries its vanilla name**, and **PULSAR still carries its vanilla Index entry** — *a geometric body the locals suspect is alien,* on a daemon whose name is a timing signal. **T-39 and T-40.**

### 2.7d The 41 was the wrong list, and 2.8 is why

**Closed 2026-09-10.** ***All four flags report none.*** *Off-type overall 42% → 41% against the 45% baseline.* **Eighteen substitutions across nineteen daemons** — but **the number that matters is the one that went away: twenty-two of the forty-one were never faults.**

**`THIN OPENING`** ***read "everything before level 10 is off-type", and twenty-two of its hits were daemons that open with*** **`WRITE`**, **`FLIP`** ***or*** **`PUSH`**. **Those are TACKLE, SCRATCH and POUND — and 2.8 settled them on purpose:**

> **`TACKLE`** **is the one that sets the tone** *— twenty-five of the seventy-two learn it and* **for most it is the first routine they know.** `WRITE` *is the plainest operation there is and every daemon can do it; damage in this world is putting your data where theirs was.*

**A check that fires on a thing the design decided is not measuring the design. It is arguing with it.** *This is the same shape as 2.7b's finding that the bestiary was already sitting on the coverage baseline, and it arrived the same way — by reading the tool's output instead of trusting it.*

***So CONTENT is excluded by RULING, not by exception.*** **That distinction is the whole point**: an exception list that grows to twenty-two entries is a tool nobody reads, and *2.8 already gives the reason in one clause* — **CONTENT is the type that hands you no verb.** ***Opening with it says nothing false about a creature. Opening with somebody else's type does.***

### Two more things the first flag had wrong

| | |
|---|---|
| **it fired with an on-type routine in the same window** | COLDREAD opens with `STALE` **and** `LICK`. *Being taught correctly plus some coverage is not a fault* — **it is what 2.7b said coverage is FOR** |
| **it counted damaging routines only** | `DAMPEN` at level 1 on a FLOW daemon **is the creature saying what it is**, and the flag read it as silence |

### What replaced it — two flags that mean separate things

***Between them they caught four daemons the old one missed while dropping twenty-two it should never have had.***

**`MIXED SIGNAL`** *— before level 10 it does something typed, it is the wrong type, and it never does its own.* **Ten.** The **NIDORAN male line opened with** **`PROBE`**, **a flying routine on a creature made of poison**, while *its sister line opened with* `FLIP` — **one split teaching two different creatures.** The **ZUBAT line** spent everything before 10 as SWARM and LATENT, *two types it does not have*, which is the daemon **2.7b named as its own example. INDEXER's only typed routine before 28 was** **`MISREAD`** **— and it had it twice**, the same entry counted again.

`LATE TYPE` — *the label is right and the first routine that hits with it arrives after level 25.* **Thirteen, and some are worse than the silent ones were:**

| | | |
|---|---|---|
| **HANDLER** | FLOW | first FLOW attack at **58** |
| **PARTITION** | STRATUM | first STRATUM attack at **52**, *for **fifteen** power* |
| **FAULT** | FLOW | first FLOW attack at **50** |
| **RAMROD · BULLDOZER** | STRATUM/LEGACY | ***the ground line did not touch the ground until the level cap*** |

### The method, in one move

***`PROBE` was not deleted when the NIDORAN line lost it.*** **It went to ECHO** — *the daemon that navigates by sending something out to find what is there,* and which had been carrying **no VECTOR routine at all before level 21.** ***The vocabulary was already right. It was on the wrong creature.***

### SLURP, and the ruling worth keeping

***Exempted rather than fixed.*** **`LICK` is the daemon's NAME as a verb**, and *vanilla types that routine LATENT on a creature that is not.* **Swapping it for a CONTENT primitive would fix the report and delete the creature.** ***2.8's own counter-test cuts this way***: the counter-test exists to stop CONTENT annexing vocabulary the other types need — **and here the routine IS the thing itself.**

### 2.7e A withdrawn word came back twice, and one character is why

**Found 2026-09-10, by being asked.** ***`TAINT` was refused as a MARK name in 5.2*** — "*a second meaning the design did not choose is not a second meaning; it is a leak*" — **and it was still the name of `MOVE_POISON_STING`.** *And still the punchline of a Viridian City line:* "**Mind the TAINT.**" — second town of the game, on its own line, as the payoff of a two-line exchange.

***The ruling was made for one surface and swept to none of the others.*** **That is the failure the tools exist to catch**, so the fix is a check and not an edit: **`VETOED` in `check_lexicon`**, which fails on any withdrawn word appearing in anything a player can read, and *carries what to write instead rather than only what not to write.*

**`TAMPER` takes the routine** — *2.6 reads CORRUPT as "data that has been **tampered** with", so it is the type's own clause as a verb,* which is the justification 2.7b used for `DESCEND`. **`SKEW` could not take it: `SKEW` is MARK 5.**

### The root cause is one character

***The check found `MANIAC` immediately*** — refused by craft rule 3 in the same breath as TAINT, still in **six** player-visible lines. **Why it survived is the finding:**

> **`port_vocab` learns renames by diffing our tables against upstream's, then filters them with `k.isupper()`. `é` is a LOWERCASE cased character**, *so* **`"POKéMANIAC".isupper()` is `False`** *— and so is* `"POKé DOLL"`*, and* `"POKé BALL"`*.*

***That filter was silently dropping every vanilla name containing the accent, which is exactly the set of names the vocabulary pass exists to remove.*** **`POKéDEX` is hardcoded in `VOCAB` by hand** *— somebody hit the symptom and never found this.* The filter now says what it meant: **no lowercase letters.**

*This is the same family as the escape trap that has now hidden substitutions in four tools:* **a rule about characters, written in terms of a function that means something slightly different, silently excluding the exact cases that mattered.**

### And the article was never agreed

**Fixing the filter turned five `POKéMANIAC`s into `ARCHIVIST` automatically — and produced "a ARCHIVIST" three times.** *Substituting a noun and not the article in front of it was an error nothing here checked for,* and reading those lines found "**was an HUNCH" two lines below**, left by an earlier pass. ***So it was never one bad run. It was a missing rule that every rename since has been able to trip.***

**Sound, not spelling**: A/E/I/O take *an*; **U is "yoo" (`a USERBOX`) and H is aspirated (`a HUNCH`)**, which are the two a letter rule gets wrong. ***Guarded to words the tool introduced,*** because **an acronym read letter by letter legitimately takes *an* before H and M — "an HM01" is correct English** and a naive rule would break it.

**Seventeen across the corpus, nearly all pre-existing**: *used a AFFECT · obtained an MUSAI · a INDEX ×3 · an SURCHARGE · a ANOMALY · a ECHO ×2 · an QUERY · an GREP ×2 · an BACKBONE · A ANOMALY doll.*

### Three were left for a human, and that is the division

***No rename could reach them*** — they are free prose, not the class name. **The Two Island tutor's joke is an ECHO** — "*MOVE MANIAC*", then "*also a mushroom maniac*" — **so the replacement repeats too, or it stops being a joke**: `ROUTINE ARCHIVIST` and `mushroom archivist`. *All three measured against the 196px box; the longest is 193.*

### 2.7f The fourteen that no check could see

**Named 2026-09-10.** ***Found by reading COLDREAD's line for T-36*****: SMOOCHUM is Jynx's pre-evolution, it is in the Index, it evolves into a daemon we named — and it was still called SMOOCHUM.** *It was not alone and it was not a clean scope boundary*: CROBAT, CLEFFA, POLITOED and SLOWKING had all been done. **The sweep found FOURTEEN, not the thirteen the ticket listed** — *HITMONTOP was missing from both.*

> **None of them were visible to any check. `gbacoherence` and `port_vocab` both define "ours" by DIFFING against upstream**, *so a daemon we forgot to rename is identical to upstream and therefore invisible to every tool that decides what to look at by what we already changed.* **The same shape as `check_lexicon` passing on an empty README** — *which is why this was found by reading a learnset rather than by running anything.*

| | | |
|---|---|---|
| **PICHU** | `STATIC` | *static electricity, and the keyword.* Before **SPIKE** and **SURGE** — the line is escalating electrical events, and **this is the one with no control over itself yet** |
| **IGGLYBUFF** | `STANDBY` | the rung below **SUSPEND**, which is below **HIBERNATE**. *The line was already a sleep-state ladder and was missing its lightest state* |
| **TYROGUE** | `PREMISE` | **what an argument starts from before it branches** — and it is *the only daemon in the game that branches three ways* |
| **HITMONTOP** | `CIRCULAR` | it fights by spinning, and it completes a triad about argument: ***the axiom, the rebuttal, and the one that goes nowhere*** |
| **SMOOCHUM** | `TELL` | **the cue a cold reader works from.** *COLDREAD states what you are about to say; this is the thing it read* |
| **ELEKID** | `CRANK` | it spins its arms to generate current. **A crank, then a DYNAMO** — *which is the order those were actually invented in* |
| **MAGBY** | `CINDER` | a small hot fragment, before a **FORGE** |
| **BELLOSSOM** | `GOLDSET` | **gold-standard labels.** ***The clean twin of BADSEED off the same BLIGHT*** — one was fed poison and one was fed ground truth, *and the types already said so*: BADSEED is GROWTH/**CORRUPT** and this is GROWTH alone |
| **ESPEON** | `LENSMUSAI` | **a lens IS the frame you read a thing in**, which is CONTEXT's clause, and `LENS` is CONTEXT's own register word |
| **UMBREON** | `MASKMUSAI` | `MASK` is **OPAQUE's** own register word, and OPAQUE is *the black box from outside* |
| **STEELIX** | `HARDLINE` | *a dedicated physical line, and uncompromising.* **Keeps BACKBONE's shape and takes its own type into the name** |
| **SCIZOR** | `SIGKILL` | **REAPER reaps what has already died; this is the signal nothing survives and nothing can catch.** *Deliberately adjacent to T-43's `KILL` question and should be decided WITH it* |
| **KINGDRA** | `TURBULENCE` | NOZZLE, then JETSTREAM, then ***the thing a flow becomes that nobody designed.*** Turbulence is the canonical example of emergence in a fluid — **and the daemon gains EMERGENT at exactly this stage** |
| **BLISSEY** | `HEARTBEAT` | **the liveness signal, after UPTIME** — and *a heart, on the daemon shaped like one* |

***The four MUSAI keep their shape***: **CARE · SEEK · CODE · LENS · MASK** — *every prefix is four letters and every one is its eeveelution's type stated as a noun.*

**`TURBULENCE` is exactly `POKEMON_NAME_LENGTH`.** *That is fine — vanilla ships `FARFETCH'D` at ten — and it was measured rather than assumed.*

***Ten of the fourteen still carry vanilla Kanto Index text.*** **Naming them has made T-40 considerably more visible**, which is the honest cost of doing the cheap half first.

#### 2.8 The CONTENT register — settled 2026-09-09

***2.7 logged this as the thing to settle first, because the other ninety-six routines will be written against whatever it establishes.*** **Seventy-nine of the 175 reachable moves are CONTENT**, whose one-clause test is "*the thing itself, with nothing read into it*" — **the only type that hands you no verb.**

### The number was frightening and the shape is not

***Grouped by what they actually do, the seventy-nine are not seventy-nine ways of hitting:***

| | | |
|---|---|---|
| **damage, plain** | **7** | TACKLE(25) · SCRATCH(12) · POUND · HORN ATTACK · SLAM · MEGA PUNCH · HYPER VOICE |
| **damage + rider** | **25** | QUICK ATTACK(11) · DOUBLESLAP(8) · FURY ATTACK(7) · BODY SLAM(5) · … |
| **lowers theirs** | **8** | TAIL WHIP(18) · GROWL(16) · SCREECH(8) · LEER(8) · SCARY FACE(4) · … |
| **raises ours** | **6** | PSYCH UP(6) · GROWTH(5) · HARDEN · SHARPEN · DOUBLE TEAM · CONVERSION |
| **status / control** | **33** | SUPERSONIC(8) · DISABLE(8) · HELPING HAND(8) · FOCUS ENERGY(7) · PROTECT(5) · … |

***Only seven are plain damage. CONTENT is mostly CONTROL,*** which is why the type that hands you no verb turned out to be the one with the most verbs available.

### The register is the instruction set

***The primitive operations*** — `WRITE`, `CLEAR`, `COPY`, `SHIFT`, `FLUSH`. **An instruction has no semantics beyond what it does**, and *that is "nothing read into it" stated as a part of speech* rather than as an absence.

**It is also the register the project already speaks.** ROUTINES are subroutines (1.6); the items are operations, not medicines; **and 0.5 says CONTENT is the serial half — the one that shows its working.** *An instruction is exactly a step whose working is the whole of it.*

> **THE TEST. Say the name. If you then have to say what it does, it is not CONTENT.**
>
> **THE COUNTER-TEST, which stops the register eating the game. CONTENT does not decide, weigh, frame or infer.** *If the word implies a judgement it belongs to CONTEXT, LOGIC or LATENT — and the seventy-nine must not annex vocabulary the other sixteen types need.*

### Five sub-registers, one per family

| family | the operation it names | worked |
|---|---|---|
| **plain damage** | *the primitive write* | `TACKLE` → `WRITE` |
| **damage + rider** | *a write with a side effect* | `DOUBLE-EDGE` → **CLOBBER** · `EXPLOSION` → **CORE DUMP** · `SUPER FANG` → **TRUNCATE** |
| **lowers theirs** | *reducing another process's standing* | `GROWL` → **DEPRECATE** · `SCREECH` → **DEMOTE** · `TAIL WHIP` → **UNPIN** |
| **raises ours** | *reserving your own* | `HARDEN` → **PIN** · `SHARPEN` → **INLINE** · `FOCUS ENERGY` → **PREFETCH** |
| **status / control** | *the richest seam, and where the instruction set already lives* | `DISABLE`, `BLOCK`, `PROTECT` need no work at all |

***`TACKLE` is the one that sets the tone*** — **twenty-five of the seventy-two learn it** and for most it is the first routine they know. **`WRITE` is the plainest operation there is and every daemon can do it**; *damage in this world is putting your data where theirs was.*

***And the perfect answer is unusable, which is worth recording so nobody proposes it again.*** **`POKE` is the BASIC instruction that writes one byte to an address** — it works twice, it is four characters, and it is **unusable in a game that has spent five separate passes deleting `POKé`.** *A pun that survives the read-aloud test and not the search-and-replace test is still a no.*

### Seven already work, and are not to be touched

**`DISABLE` · `BLOCK` · `PROTECT` · `LOCK-ON` · `SAFEGUARD` · `METRONOME` · `RECYCLE`**

*Corrected again 2026-09-17, v11.216 — the same pass renamed five more.* **The engine's naming pass of 2026-09-10 (`a62aec091`, "*the reachable set is done*") renamed every routine a daemon in this game learns, and this list was not consulted: `ENDURE` → `SURVIVE`, `MIMIC` → `CLONE`, `MINIMIZE` → `SHRINK`, `CONVERSION` → `CAST`, `CONVERSION 2` → `RECAST`. *No reason was recorded for those five, and none is invented here* — *unlike `RESTORE`, below, whose collision with the POTION is documented.*** **The names shipped, they have been in the ROM a week, `CAST` and `RECAST` have since become load-bearing** (*self-alignment, 10's Open log*), **and reverting them would be its own rename pass** — *so the list is corrected to what the game says, and the question of whether each rename was right is left open rather than answered by default.*

*Corrected 2026-09-17, v11.212.* **`RECOVER` was on this list and was renamed `RESTORE` on 2026-09-10** (engine `a62aec091`) — ***not because it failed to land, but because the word was already spent***: **`RECOVER` is the POTION** (*the ladder `RECOVER · DEEP RECOVER · FULL RECOVER · REBUILD`, 1.6d*), *and a routine and an item with one name make a bag and a battle menu that disagree about what the word does.* **The rename was right and the list was not updated; it is now.**

***A metronome is a clock and recycling is garbage collection.*** **Same finding as *catchy tune* and `MOVE ITEMS`** (1.6): *the pass that renames what already lands is a pass that makes the game worse.* ~~**`SWIFT`, `FORESIGHT` and `ENCORE` are marginal and get a second look, not a rename by default.**~~ ***They got the rename*** (*`UNERRING`, `REVEAL`, `REPLAY`, the same pass*). *The type-conversion line that stood here ("a type conversion is a type conversion") went with `CONVERSION`.*

**PERSPECTIVE stays, as the deliberate exception**

***`MOVE_TRANSFORM` → `PERSPECTIVE` was already in the table and nobody had flagged it.*** **It looks like it breaks the register** — *perspective is the most CONTEXT word in the lexicon* — **and it is kept, because of what it sits on.**

**MOCK is CONTENT, and MOCK is the daemon with no content of its own.** *Its whole nature is becoming whatever it is shown.* ***So the one CONTENT routine that is about taking a point of view is carried by the one daemon that has none*** — **which does not weaken the rule, it is the only place the rule could be broken without cost.** *Nobody says any of this.*

**Ten collisions the pass has to fix, and they are older than the pass**

***The move table has been colliding with our own lexicon since the renames started, and nothing checked, because the move pass never began.*** **Same class as `INTERRUPT` being assigned twice** (1.6).

**Cleared 2026-09-10, and the pair was free**

***`GROWTH` was a CONTENT move wearing the GRASS type's name, and `SWARM` was vanilla's Bug ability wearing the BUG type's*** — **the second found by `check_lexicon` on its first run** (8.7), which is the point of a check that reads every surface rather than the ones that existed when it was written.

**Both are the same operation seen from two sides, and the field already has the words:**

| | | |
|---|---|---|
| `GROWTH` move → `SCALE UP` | *raises one stat* | **scaling up is one instance given more** |
| `SWARM` ability → `SCALE OUT` | *powers up SWARM routines in a pinch* | **scaling out is more instances** |

***That is the real distinction in the field, and it lands on exactly the right two things*** **— the CONTENT move makes one thing bigger; the SWARM ability makes more of them, and only when pressed.** *Neither name had to be invented and the pair was not planned.*

***`THRASH` and `HARDEN` went the same day, and the first one was hiding a better move than the one it replaced.***

**`THRASH` is `EFFECT_RAMPAGE`: ninety power, locked in for two or three turns, no switching, no choosing — and it ends by leaving the user THRASHING.** *Its own description read "rampages about… then becomes thrashing", which after 1.6 read as a typo rather than a joke.*

***That is a busy-wait, exactly and completely.*** **A loop that will not yield the processor, runs flat out for as long as it holds it, and leaves the system thrashing when it is done. →** **`BUSY WAIT`.** *The vanilla move was already describing the thing; nobody had noticed because it was wearing a body word.*

**`HARDEN` → `PIN`**, per 2.8's own worked example: *pinned memory cannot be moved or swapped out*, which is DEFENSE-up said as an operation. **And its description was a body word too** — "*stiffens all the muscles in its body*" — **in a world 1.6 established has no bodies.**

***Which leaves exactly two moves wearing another word's name, and one of them is deliberate:***

- **`OVERHEAT`** — ***kept.*** An ENTROPY move and ENTROPY's own state agreeing **is the point**
- **`SIGNAL BEAM`** — a **SWARM** move wearing the **SIGNAL** type's name. OPEN, *and held on purpose*: **2.8 settled the CONTENT register and no other**, so naming a SWARM routine now would pre-empt the decision 2.8 exists to make first. **`GOSSIP` is the proposal** — *a gossip protocol is nodes passing state around until they agree, which is 2.6's clause for SWARM word for word*, and the move's rider leaves the target THRASHING, *which is what rumour does*

**Five ability descriptions still named vanilla types and went with it** — "*Ups BUG moves*", "*Ups GRASS moves*", "*Ups FIRE moves*", "*Ups WATER moves*", "*Not hit by GROUND attacks*" — **and `Traps STEEL-type DAEMON` made six.** ***The ability names themselves are still vanilla*** — WATER ABSORB, FLASH FIRE, ROCK HEAD — **and belong to 8.7's ability pass, not to this one.**

| | collides with | |
|---|---|---|
| ~~`GROWTH` move~~ | the **GROWTH** type | ***Moved 2026-09-10*** → ***`SCALE UP`***. Scaling **up** gives one instance more; scaling **out** gives more instances — *and the ability that collided the same day took the other half* |
| ~~`THRASH` move~~ | the **THRASHING** state | ***Moved 2026-09-10*** → ***`BUSY WAIT`***. See below |
| ~~`HARDEN` move~~ | the **HARDENED** type | ***Moved 2026-09-10*** → ***`PIN`*** |
| `SIGNAL BEAM` | the **SIGNAL** type | A SWARM move wearing SIGNAL's name |
| `PRIOR` *(ours)* | the **PRIORITY** item | Both ours, and uncomfortably close |
| `OVERHEAT` | the **OVERHEATED** state | ***Keep.*** A fire move, ENTROPY owns the state, and the two agreeing is the point |
| `REST` · `DIVE` · `IRON TAIL` · `IRON DEFENSE` | RESTART · DIVE BOX · IRON items | Mild; noted, not urgent |

### 2.9 The other seventeen registers — settled 2026-09-10

***2.8 settled CONTENT and then this project treated that as a rule instead of a sequence.*** **CONTENT went first because it was seventy-nine of the 175 reachable routines and would set the tone** — *that is an argument about ORDER,* and it was then applied twice as though it were an argument about **method**, holding `SIGNAL BEAM` and five HMs for a decision nobody was actually waiting on.

***A register is not invented. It is 2.6's clause turned into a part of speech.*** **Every one of these existed already**, and the whole of this section is the derivation written down.

| type | 2.6's clause | the register | worked |
|---|---|---|---|
| **LOGIC** | *rules applied step by step; proof, not intuition* | **derivation** — what establishes a thing by steps | ASSERT · REDUCE · ENTAIL · **CRACK** |
| **VECTOR** | *direction in a space of meanings* | **delivery with a heading** | DISPATCH · ROUTE · STEER · **GOTO** |
| **CORRUPT** | *data that has been tampered with* | **tampering and decay** | **TAMPER** · FALSIFY · SPOIL · SEED |
| **STRATUM** | *the physical layer everything else runs on* | **substrate and ground** | GROUND · ANCHOR · SEAT · SUBSIDE |
| **LEGACY** | *deprecated hardware still running* | ***the old instruction set*** — operations that still work because nobody removed them | NOP · ROTATE · PARITY · DEPRECATE |
| **SWARM** | *many small agents; no single one matters* | **distributed protocols** | **GOSSIP** · FANOUT · ELECT · HEARTBEAT |
| **LATENT** | *running below the surface, unobserved* | **background and detached** | DAEMONIZE · ORPHAN · SHADOW |
| **ENTROPY** | *noise and heat; disorder that spreads* | **noise, heat, decay** | JITTER · DITHER · SCRAMBLE |
| **FLOW** | *everything running downhill to the lowest point* | **gradient and current** | **TRAVERSE · ASCEND · DESCEND** · SPILL |
| **GROWTH** | *training: fitting to whatever it is fed* | **training and fitting** | CONVERGE · SATURATE · EPOCH · ROOT |
| **SIGNAL** | *raw current, before anything interprets it* | **the raw edge** | PULSE · STROBE · TRIGGER · CARRIER |
| **CONTEXT** | *the frame you read a thing in* | **framing and reinterpretation** | REFRAME · SCOPE · PREMISE · LENS |
| **FROZEN** | *locked to what it already saw* | ***weights that do not update*** | CONST · SEAL · IMMUTABLE · LOCK |
| **EMERGENT** | *behaviour nobody designed and nobody can account for* | **emergence** | PHASE · ATTRACTOR · BLOOM |
| **HARDENED** | safety hardening | **armouring and attestation** | SANDBOX · ATTEST · GUARD |
| **OPAQUE** | the black box from outside | **opacity** | OBFUSCATE · OCCLUDE · REDACT · MASK |
| **ORACLE** | the black box that answers | **consultation** | CONSULT · ANSWER — *one routine, never seen* |

## Settling them all at once caught four thefts

***2.8's counter-test said the seventy-nine "must not annex vocabulary the other sixteen types need", and then batch one annexed four words anyway*** — **because there was nothing to check against.** *A rule with no table behind it is a hope.*

| proposed for a CONTENT routine | belongs to | re-cut as |
|---|---|---|
| `DEPRECATE` | **LEGACY** — it is the whole register, not a stat drop | **`DERATE`** — *running a component below its rated capacity* |
| `OBFUSCATE` | **OPAQUE** — it is that type's name said as a verb | **`BLUR`** |
| `REVOKE` | *access-control, which 1.7 rules is INSTITUTION vocabulary* | **`DOWNCLOCK`** |
| `PIN` | contested with **FROZEN**, and kept — *pinning is about not being moved, freezing is about not being updated* | **`PIN`**, unchanged |

***That is the argument for settling registers before names, made by breaking it.***

### The HMs are the access model, on the map

***An HM only works in the field once you hold the MARK.*** **That is not flavour, it is the gate** — and it makes an HM **a capability you have been certified to exercise**, which is exactly the access-control reading 1.7 rules *on-thesis* because it is about access to a mind.

**So the MARKS are a keyring as well as 5.2's bench of instruments** — *and the game already runs the same model somewhere else.* **The box ladder is rights escalation applied to daemons; the HMs are rights escalation applied to the map.** ***Two implementations of one idea, and nothing connects them.*** *Nothing should — the player has both in hand by hour three.*

| | | |
|---|---|---|
| `CUT` → **`PRUNE`** | CONTENT | *you prune a tree and you prune a branch* — the obstacle is literally a tree |
| `FLY` → **`GOTO`** | VECTOR | **an unconditional transfer to a known address** — *and you can only reach a CHECKPOINT you have already been to*, which is what "known" means. `JUMP` was rejected: its first read is a hop |
| `SURF` → **`TRAVERSE`** | FLOW | across |
| `WATERFALL` → **`ASCEND`** | FLOW | up |
| `DIVE` → **`DESCEND`** | FLOW | down — ***three directions on one medium, which is exactly what the three moves do*** |
| `STRENGTH` → **`DISPLACE`** | CONTENT | to displace is to push a thing out of the place it was in |
| `FLASH` → **`VERBOSE`** | CONTENT | ***the best of the eight***: `-v` turns on the output nobody could see, **and floods the channel so nothing can be found** — the field effect and the battle effect in one word |
| `ROCK SMASH` → **`CRACK`** | LOGIC | you crack a rock and you crack a problem |

***And the failure messages are where these are actually taught***, because a player meets them far more often than a description: **`There is nothing to PRUNE.`** · **`You cannot TRAVERSE here.`** *One line, at the moment of trying.*

*Recorded, because it was misremembered once:* **`ROCK SMASH` does not exist on the Game Boy.** `engine/data/moves/names.asm` has **STRENGTH and no ROCK SMASH** — *it is a Gen 2 addition that arrived with the port*, which is why DISPLACE (push it aside) and CRACK (break it open) do not overlap. **It belongs in 9.3's list of what the spike bought.**

#### 2.10 The ROUTINES pass, done to the reachable edge

**Finished 2026-09-10. *181 of 356 move names are ours* — every routine any daemon in this game learns by level, plus the eight HMs and the two that were already renamed.**

### What settling the registers first actually bought

***Every one of the 157 came out of 2.9's table rather than out of taste***, and the four best names in the pass are the ones where the register did the work and the writer did none:

| | | |
|---|---|---|
| `AERIAL ACE` → **`HOMING`** | VECTOR | *a never-miss move, in a register about heading* |
| `MUD SPORT` → **`GROUND`** | STRATUM | ***grounding a signal is the literal engineering answer***, and the move literally halves SIGNAL routines |
| `MIST` → **`READONLY`** | FROZEN | *blocks stat changes* — **weights that do not update, said as a file permission** |
| `BRICK BREAK` → **`FALSIFY`** | LOGIC | **it breaks LIGHT SCREEN and REFLECT** — *and those are now* `STEELMAN` *and* `PARAPHRASE`. ***LOGIC falsifying CONTEXT's framings is 2.2's thesis relation, arriving as a move interaction nobody wrote*** |

**Two more the chart wrote for us.** ***`SEISMIC TOSS` → `EQUATE`*** *— its damage equals your level, so the name is the formula.* ***`LOW KICK` → `WEIGH`*** — its power is the target's weight.

### The checker caught six thefts in one run

**`CLAMP`, `PREEMPT`, `BUFFER`, `ESCALATE`, `ASSIST` and `FORK` were all proposed and all already taken** *— four by species, one by an unrenamed move, one by an item.* **Re-cut as `BROWNOUT`, `FAST PATH`, `PRELOAD`, `ACCUMULATE`, `DELEGATE` and `INSTANCE`.**

***`BROWNOUT` is better than what it replaced***: **a brownout is a voltage sag that makes everything run slow and unreliable**, which is THROTTLED stated electrically, and it is what THUNDER WAVE does.

**Thirteen kept, and fifty-eight descriptions rewritten**

**`DISABLE`, `PROTECT`, `SAFEGUARD`, `LOCK-ON`, `METRONOME`, `RECYCLE`, `BLOCK`** and — added 2026-09-11 — **`SUBSTITUTE`** were left alone (*`RECOVER` was listed here too until v11.212 — it is `RESTORE`, because `RECOVER` is the POTION — and until v11.216 so were `MIMIC`, `MINIMIZE`, `ENDURE`, `CONVERSION` and `CONVERSION 2`, which the 2026-09-10 pass renamed `CLONE`, `SHRINK`, `SURVIVE`, `CAST` and `RECAST` with no recorded reason; the heading's count is kept because it is the pass's, not today's*) — *2.8's rule that the pass which renames what already lands makes the game worse.*

***And fifty-eight descriptions had to move with their names***, because a description that still describes the old one is worse than no rename at all: *`OCCLUDE` cannot say "bites with vicious fangs".* **Every line measured against the 170px box; the widest is 126.**

**And then the other two routes, the same day**

***The pass followed level-up learnsets and stopped, which was one route out of three.*** **The TM list is 58 moves and the trainer parties use 225** *— and 89 of those were still vanilla,* because no daemon in the game learns them by level and nothing had gone looking down the other two roads.

**They are named now.** ***Every TM, every HM and every routine any trainer in the game will use is ours*** **— 269 of 356** *— with one deliberate exception:* **`OVERHEAT`, kept because an ENTROPY routine agreeing with ENTROPY's own state is the point** (2.8).

***The best of the eighty-nine are the ones where the mechanic and the register said the same thing:***

| | | |
|---|---|---|
| `PETAL DANCE` → **`OVERTRAIN`** | GROWTH | **locks you in for two or three turns and ends in THRASHING** *— which is what overtraining is* |
| `BULLET SEED` → **`MINIBATCH`** | GROWTH | *two to five small passes rather than one large one.* **That is the definition** |
| `DYNAMICPUNCH` → **`NON SEQUITUR`** | LOGIC | **confuses, and misses half the time.** *A step that does not follow* |
| `SANDSTORM` → **`BIT ROT`** | LEGACY | *everything degrades a little each turn, and some things resist it* |
| `POISON GAS` → **`LEAK`** · `TOXIC` → **`COMPOUND`** | CORRUPT | **the state's own verb, and the thing the worse state does** |
| `GUILLOTINE` → **`KILL`** | CONTENT | *a one-hit removal, named for the command that does it* |
| `SIGNAL BEAM` → **`GOSSIP`** | SWARM | **held since 2.8 and released by 2.9** *— a gossip protocol is 2.6's clause for SWARM, word for word* |

**`BIND` had to move for a reason none of the others did.** ***It is 1.5's word*** *— what you do to a daemon —* **and a trapping routine wearing it would have put BIND in the battle log meaning something else entirely. → `LATCH`.**

***What is left is genuinely unreachable.*** **The 87 remaining are learned by no daemon in this game, taught by no TM and used by no trainer** — *they belong to species the bestiary has not reached.* OPEN, **and it waits on 8.2 rather than on a decision.**

**Reopened and finished — 2026-09-11**

***The 87 "genuinely unreachable" routines stopped being unreachable, and nothing about the routines changed.*** **T-45 named 52 island daemons, and naming a daemon is what brings its learnset into scope** — *so 50 still-vanilla routines were suddenly on the list of something a player can encounter, and this section's own rule applied to them by its own terms.*

**Forty renamed. 309 of 356 are ours.** *Three of them got a shape rather than a word each, which is the part worth keeping:*

| | |
|---|---|
| **the KICK family** | *LOGIC's register is argument, so* `JUMP KICK` → **`LEAP`**, `HI JUMP KICK` → **`GREAT LEAP`**, `ROLLING KICK` → **`LOOP BACK`**, and `TRIPLE KICK` → **`CHAIN RULE`** — ***three linked steps, each resting on the last*** |
| **`STOCKPILE`** · **`SWALLOW`** · **`SPIT UP`** | *one mechanism, so one metaphor:* **`ENQUEUE`** · **`DEQUEUE`** · **`EVICT ALL`**. *You fill it, you drain it, or you throw the whole thing away* |
| **the two bone routines** | **STRATUM is 2.6's *physical layer everything else runs on***, so they are memory faults — **`HARD FAULT`** and **`PAGE FAULT`** |

***Nine proposals were refused before anything was written*** — `SYLLOGISM`, `CAVITATE`, `EVICT`, `SPINLOCK` and five more were all already taken. **And `SNAPSHOT` got past that check and was caught by `check_lexicon`**, *because the check I wrote by hand read moves, species, abilities and classes and not* ***items***. **It is `IMPRINT`.**

***Ten routines are still vanilla on a reachable list, and every one is already ruled:*** **seven of the thirteen above, plus `LICK` which carries SLURP's ruling and `PRESENT` which carries REPAY's.** `SUBSTITUTE` *was the only one never written down, and is now the fourteenth keep.*

#### 2.10a `KILL` and `WEED` — decided, 2026-09-11

***T-43 existed to make two words a decision instead of an oversight.*** **Both are decided: they stay.** *And the test that settled the first one turned up a third word that was an oversight, which is the useful part.*

**`KILL` is not a tonal break. It is one third of a system already in the lexicon**

**The instinct was that a genre where creatures *faint* should not have a routine called `KILL`.** *That is true of the genre and false here,* and ***the evidence is that the lexicon already contains the whole event and nobody had noticed it:***

| | | |
|---|---|---|
| **`KILL`** | the routine | *`GUILLOTINE`* — **the verb you send** |
| **`SIGKILL`** | a daemon | *`SCIZOR`* — **the signal that cannot be caught, handled or ignored** |
| **`DETACH`** | the outcome | *what vanilla calls fainting* — **what happens at the other end** |

***You kill a process and it detaches.*** **Those are not two registers in collision; they are the two halves of one event, each named correctly**, and the middle term was sitting in the bestiary the whole time.

**And its three siblings agree, which is the check that decided it**

**The four one-hit-KO routines are a set, and a set is testable in a way one word is not:**

| | |
|---|---|
| `GUILLOTINE` | **`KILL`** |
| `SHEER COLD` | **`TERMINATE`** |
| `HORN DRILL` | **`WIPE`** |
| `FISSURE` | ***still `FISSURE`*** |

***`KILL` and `TERMINATE` are the two words an operating system actually uses***, and `WIPE` is what happens to the data. **Three of the four were already in register. The one that was not had never been renamed at all** — *a vanilla geological word sitting among three process words, reachable at level 41, and 2.10's rule is to name every routine the player can meet.*

**`FISSURE` → `SEGFAULT`.** ***A segmentation fault is a fatal fault in the layer everything else runs on, and STRATUM is 2.6's "physical layer everything else runs on"*** — *so the type and the routine now say the same thing, and it is the only one of the four whose name states which layer it breaks.*

**`WEED` is settled by the line it is in**

**`WEED` → `BLIGHT` → `BADSEED` → `GOLDSET`.** *The other reading exists in isolation and does not survive three evolutions of agriculture.* **A player meeting `WEED` and then `BLIGHT` is not thinking about anything but plants**, and ***2.8's rule applies to the whole line rather than to a word held up on its own.***

*The general form, which is worth more than either ruling:* **a word with two readings is disambiguated by its neighbours, so the test is never the word alone — it is the word among the four things next to it.** ***That is how `KILL` was cleared and how `FISSURE` was caught, in the same look.***

### 2.11 The abilities — a register that is not a verb

***Built 2026-09-10.*** **9.3 named abilities as a reason to port and 8.7 called them the second teaching surface the project paid for and never spent.** *Seventy-seven of seventy-eight are ours now* — **the seventy-eighth is vanilla's own blank slot.**

#### They are not routines, and the register has to say so

***A ROUTINE is an operation. An ability is a standing property*** — **what is true of a daemon while it is doing nothing**, and *4.29 already reads a daemon by exactly that:* ***what it does unasked.***

> **THE REGISTER: a flag set on the process. File permissions, compile flags, process attributes, hardware properties** — *the things that are true before it runs and are never chosen per turn.*
>
> **THE TEST: an ability is true of a daemon while it is doing nothing.** *If the name describes an action, it is a routine and belongs in 2.9's table.*

**That is why almost none of the seventy-seven is a verb** — `IMMUTABLE`, `SANDBOXED`, `NO ADDRESS`, `NARROW BAND`, `HIGH RES`. *The grammar is the whole distinction,* and it is what stops the two surfaces from colliding as they grow.

#### Vanilla wrote four of them for us, and 8.7 spotted three

| | | |
|---|---|---|
| `LEVITATE` → **`NO ADDRESS`** | *not hit by STRATUM* | ***2.6's own clause, verbatim***: "the physical layer cannot reach what has no location" |
| `TRACE` → **`INHERITS`** | *copies the foe's ability* | **it is on SUBSTRATE** — *the substrate takes on whatever runs on it*, which is that daemon's Index entry as a mechanic |
| `SYNCHRONIZE` → **`MUTUAL`** | *passes your state to whoever inflicted it* | **ARTSAI already has it.** *Perspective-taking as a passive* |
| `STURDY` → **`DURABLE`** | *negates one-hit knockouts* | **5.3 leans on this**: HEAP, STACK and MONOLITH cannot go down to one hit, ***which is Benchmark 1's creed as a rule in the damage calculation*** |

#### The ones where the flag was already the joke

**`OVERGROW`** → **`OVERFIT`** — *a GROWTH daemon that gets stronger the closer it is to failing.* **`HUSTLE`** → **`UNSAFE OPT`** — *trades accuracy for power, which is what the flag does.* **`WONDER GUARD`** → **`NARROW BAND`** — *only super-effective hits land.* **`NATURAL CURE`** → **`CLEAN EXIT`** — *heals on switching out.* **`TRUANT`** → **`HALF DUTY`** — *acts every second turn, which is a duty cycle.*

***And two that were a pair and had never been read as one:*** `PLUS` **and** `MINUS` **are** `SOURCE` **and** `SINK` *— a current source and a current sink, which is what the two abilities do to each other.*

### Thirty-one descriptions had to move

***An ability line is thirty-two characters and most of them described a body.*** *"Heals the body by shedding", "Hurts to touch", "Turns electricity into HP"* **— rewritten to what the flag actually is:** `Clears its own state.` · `Costs whatever touches it.` · `Takes SIGNAL in as HP.` **Longest is thirty-one.**

*Two more came free from work already done:* `Prevents self-destruction` **is now** `Nothing here may PANIC`, and `Prevents fleeing` **is** `The foe cannot DETACH` *— because 2.10 renamed SELFDESTRUCT and 1.4 had already renamed fleeing.* **The surfaces had drifted apart and nobody had joined them up.**

#### 2.12 Mechanics from later generations — five asked, four worth having

*Asked 2026-09-11.* ***Every cost below was read out of the engine, not estimated.***

### 1. The physical / special split — OPEN, and the code is the cheap half

***Gen 1–3 decide physical-or-special by the TYPE. Gen 4 moved it to the MOVE.*** **In this build that is two macros:**

> `#define IS_TYPE_PHYSICAL(moveType) (moveType < TYPE_MYSTERY)` — `battle.h:475`

**So LOGIC, VECTOR, CORRUPT, STRATUM, LEGACY, SWARM, LATENT, HARDENED and CONTENT are physical, and everything above the gap is special.** *Which is arbitrary in our reading and occasionally wrong in it:* `TAMPER` **is CORRUPT and therefore physical, and tampering with data is not a physical act.**

***And it already leaked into a gym.*** **5.3b had to design around it in so many words** *— "in Gen 3 fire is special, so the starter everyone brings to a GROWTH gym is answered by a screen rather than by a type."*

**The distinction is already in our lexicon and the split would make it legible.** *1.6d named the six vitamins for six parts of a spec:* **ATTACK is** `WATTAGE` **and SP. ATK is** `BANDWIDTH` ***— raw power against how much can be carried down a channel.*** *That is exactly what the split is about, and right now a routine inherits which one it uses from a table it has nothing to do with.*

| cost | |
|---|---|
| **the code** | `u8 split` in `struct BattleMove` **(+355 bytes), 7 call sites**, and `AI_TypeCalc`. *An afternoon* |
| **the design** | ***354 judgment calls***, and **five leader designs in 5.3b were built against the Gen 3 rule** |

***Recommendation: yes, and not alone.*** **The code is not the job; the 354 rulings are**, *and they belong with the routine work 2.10 left open rather than as a pass of their own.*

**2. Type-resist handlers — OPEN, and this is the best fit in the question**

***The premise wants correcting first: Gen 3 IS the berry generation.*** **FireRed already ships 43 of them, the pinch berries, the nature-confusion berries and `HOLD_EFFECT_RESTORE_STATS` —** *and 1.6d already named the lot as `TRAP` handlers, which is a better design than vanilla's.* **Berries are not a thing to add. They are in, and they are ours.**

***What is genuinely later-generation is the TYPE-RESIST berry*** **— Gen 4's Occa and its sixteen siblings: a one-shot that halves a super-effective hit of exactly one type.** *And it is the best mechanical fit anything in this section has:*

- ***It completes 1.6c's rack.*** **Seventeen `<TYPE> GAIN` items amplify one channel; seventeen `<TYPE> TRAP` handlers damp one.** *One shelf of the Bag turns the chart up and the other catches it* **— and 1.6c's claim that a player who reads the rack has read the chart becomes true in both directions**
- ***It teaches the chart by requiring it.*** **You cannot bring the right one without knowing what you are weak to**
- ***It is level-proof***, which is 5.3's actual test for a mechanic
- **The register needs no decision.** *1.6d already derived it:* **a berry is a handler, a handler installed against a machine condition is a trap, and the bag pocket is a trap table**

**Cost: one new hold effect, one check where `TypeCalc` lands (`battle_script_commands.c:1458`), and seventeen item entries.** *Smaller than the split by an order of magnitude.*

**3. The overfit state — OPEN, and it is Benchmark 4's own lesson**

***Verdigris is Benchmark 4 and its lesson is overfitting, and the lesson is not a mechanic.*** **5.3b gives TRELLIS party composition, which is a fine lever and is not the thing the gym is about.** *Craft rule 5 says a lesson that can be skipped by grinding is not a lesson.*

> A daemon that uses the same routine several turns running gets stronger at that one and weaker at everything else.

***That is overfitting and it is also a habit, and neither word has to be said.*** **The machinery exists —** *`FURY CUTTER` and `ROLLOUT` already escalate on repetition through `gDisableStructs`*, so the counter is built and only the penalty is new.

***One thing it needs that this section cannot give it: a name.*** `OVERFIT` **is already a species**, and *the six states are past-participle machine conditions* — **THROTTLED, SUSPENDED, LEAKING, OVERHEATED, HUNG, THRASHING** — *so a seventh joins that set or it does not go in.*

**4. Temperature as a room — OPEN, and the plumbing already shipped**

***Quicksilver is Benchmark 7 and its lesson is temperature.*** **Raise it and the same model becomes more interesting and less reliable**, *which is one sentence and a damage roll.*

**Gen 3 already rolls damage at 85–100%.** ***Widening that roll, lifting the crit rate and dropping accuracy, for both sides, gated on a flag*** **— and 5.3b already built exactly this plumbing for SCORN**: "*the damage calculation — one flag, read in three places.*" **The precedent is in the ROM and it is his.**

*It is also the one lesson on the list that the player should be able to feel go wrong in their favour.*

**5. Catastrophic forgetting — *it is already in the game and nobody has noticed***

**Four routines, and learning a fifth means choosing one to lose.** ***That is catastrophic forgetting, it is vanilla's oldest mechanic, and it has been sitting in this design since before the design existed.***

**This needs no implementation at all.** *It needs one line somewhere that does not explain it* — **and the STREAM is where that line goes, because the STREAM is already the surface that says a true thing shortly before the player needs it.**

**And one that is rejected, because it would spend the only thing 2.6 says not to**

***Proposed and declined: an effectiveness preview that is occasionally miscalibrated.*** **Showing the player a prediction before they commit, and being wrong sometimes, is the thesis as a UI element** — *and it is the one mechanic in this section that must not be built.*

**2.6 is explicit**: "***a contradiction costs trust in the chart — and the chart is the argument.***" *A player who cannot trust the chart does not learn a model from it; they learn that the game lies.* **The thing being taught is the casualty.**

***That is the test every further proposal here has to pass:*** **does it make the argument legible, or does it make the argument unreliable?**

### 2.13 The names, read against their own tables — 2026-09-12

***Every check this project owns compares names to OTHER NAMES.*** **`check_lexicon` finds collisions, vowel-drops and stale prose; `gbacoherence` reads a daemon's ROUTINE LIST against its type.** *Nothing had ever read a NAME against the row it sits on* — **and that is the axis the impression "*some of these feel like generic renames*" is actually about.**

**The bound is 2.6's, and it is the whole design of the check**

> **A contradicting clause is worse than a silent one.** *Silence costs a lookup.* **A contradiction costs trust in the chart — and the chart is the argument.**

***So `PRUNE` dealing flat damage is SILENT, and is fine.*** **`REINFORCE` lowering the target's DEFENSE is a contradiction.** *The first costs a player one lookup; the second teaches them the wrong thing and then makes them distrust the next name too.*

***And the other bound is 2.8's.*** **`WRITE`, `FLIP` and `PUSH` are TACKLE, SCRATCH and POUND, settled on purpose** — *"damage in this world is putting your data where theirs was."* **`gbacoherence`'s first version flagged 41 daemons and 22 of them were flagged for exactly that. *Plainness is never a fault here.*** *Only a name that says the wrong thing is.*

**What the measurement found — seven of 574**

| | |
|---|---|
| **DIRECTION** ×3 | *the name states a direction and the effect goes the other way* |
| **TYPE WORD** ×4 | *a daemon's name is another type's own word* |
| **DISPLACED** ×1 | **found afterwards, and not in the original seven** — *our name is vanilla's name for a different thing* |

***That is 7 of 574 names, which is the honest headline*** **— the impression was right about the axis and wrong about the scale.** *The rest of the "generic" feeling is `EFFECT_HIT`: twenty-four moves that are flat damage and have nothing to say, and 2.8 already ruled that the plainest of them are correct.*

### The five that moved

| | | |
|---|---|---|
| `REINFORCE` → **`FATIGUE`** | IRON TAIL, *lowers the foe's DEFENSE* | ***metal fatigue is armour giving under repeated load.*** **HARDENED's own register, and the effect said as a material** |
| `DEPRECATE` → **`LONG SUPPORT`** | ANCIENT POWER, *may raise every stat* | **an LTS release is old, still maintained, and better than you expected.** *LEGACY, exactly — and it is 12 characters, which is the cap* |
| `DISSOLVE` → **`PASSIVATE`** | ACID ARMOR, *raises DEFENSE twice* | ***passivation is the layer that makes metal unreactive*** **— acid and armour in one word**, *in CORRUPT's own world. Vanilla's joke was liquefy-to-be-hard-to-hit and dropping* `ARMOR` *had kept only the misleading half* |
| `NOISE` → **`RESIDUAL`** | CLEFFA, *CONTENT* | **ENTROPY's 2.6 clause is *noise and heat*, quoted verbatim on the CHARCOAL item.** *4.26's line is statistics as an evolution line and* ***`RESIDUAL · ANOMALY · OUTLIER`*** ***keeps it whole*** — a residual is what the model failed to explain, which is the first step toward being an outlier |
| `CHARM` → **`TRANSPOSE`** | SWEET KISS, *causes THRASHING* | ***vanilla's CHARM is our*** ***`DOWNCLOCK`***, so two moves answered to one name and **the wrong one had it.** *A transposition is the classic confusion error* |

***`IRON TAIL`'s DESCRIPTION had doubled down on the error.*** **It read "*Adds more of what was already holding*" in front of an effect that takes armour away** — *so the description was not merely tolerating the wrong name, it was arguing for it.* **Four descriptions rewritten with the names.**

### The three that stayed, and each is a ruling rather than a skip

***An exemption nobody can read is indistinguishable from a bug***, so each one is in the tool with its reasoning and `--rulings` prints it.

- **`SUBSTRATE`** is CONTENT and SUBSTRATE is STRATUM's word. ***4.25, on purpose*** *— it was the original name for the TYPE, cut at nine characters, and the species took it.* **Its ability is `INHERITS`**, which is the Index entry as a mechanic.
- **`SUSPEND`** is CONTENT and SUSPENDED is CONTEXT's concept. ***The name is a VERB IT PERFORMS, not a type claim*** *— it is the sleep ladder's middle rung, and what it does to other daemons is suspend them.* **2.7 asks who owns the STATE; a daemon that inflicts one is not claiming to be the type that owns it.**
- **`FAULT`** is FLOW while STRATUM has `HARD FAULT`, `PAGE FAULT` and `SEGFAULT`. ***STRATUM owns the COMPOUNDS and not the bare noun***, so this is **not an exemption at all — it is a word the check deliberately does not list**, and the reasoning sits in `TYPE_WORDS` beside the omission. *A ruling that can never fire is worse than no entry.* **4.26 reads FAULT and HANDLER as "*the thing it needs is not loaded; and the thing that catches that*", which AGREES with the memory reading and disagrees only about the type** — *which is the chart's business, not the name's.*

**Verified by breaking it, three faults at a time**

***All five were put back and the check named all five***, one line each, with the right fault class on each — **and `DISPLACED` was the one the original count missed**, because a name being vanilla's name for something else is not visible from either table on its own.

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## 3. THE MAP

### 3.1 Cities

| Kanto | New name | Double meaning | Role |
|---|---|---|---|
| Pallet | **Blanche Town** | white/blank + to flinch | Home; the pre-color state |
| Viridian | **Callow City** | green + unripe, untested | Benchmark 8 (locked all game) |
| Pewter | **Slate City** | stone, color, writing surface, clean slate | Benchmark 1; museum of dead hardware |
| Cerulean | **Doldrum City** | becalmed sea + low spirits | Benchmark 2 |
| Vermilion | **Ardor City** | flush/heat + brash zeal | Benchmark 3; the port |
| Lavender | **Halftone Town** | dots that only *look* like grey | The tower |
| Celadon | **Verdigris City** | green corrosion on bronze | Benchmark 4; Corpus rotting beneath |
| Fuchsia | **Lurid City** | garish glow + shocking | Benchmark 5; spectacle and toxicity |
| Saffron | **Brazen City** | brass over base metal + shameless, unhidden | Benchmark 6; corporate capture |
| Cinnabar | **Quicksilver Island** | mercury: alive, unstable, poisonous | Benchmark 7; the ruined lab |
| Indigo Plateau | **Umbra Plateau** | full shadow; all color absorbed | The Review Board |

**Slate over Somber.** Somber beside Doldrum is two downbeat mood words in a row — it greys out the early game. Slate breaks the run, is materially specific, and a slate is a writing surface, which *is* Benchmark 1's Representation lesson sitting inside the name.

**The Tunnel, and the thirsty guard.** Two places where vanilla needed almost nothing.

**The Underground Path is already correct.** In networking a **tunnel** is precisely how you route traffic around a blocked path — an SSH tunnel, a VPN tunnel. You cannot take the direct route, so you tunnel. It needs no rename, only the joke made visible in the local-name register: *official* UNDERGROUND PATH, *local* **THE TUNNEL**.

**And the guard stays thirsty.** Craft rule 6 doing its work: Brazen is the bought city, Corpus headquarters, corporate capture made architecture — and it is guarded by a man who will move for a beverage. **The barrier is not security. It is inertia.** Nobody says so, and it lands harder for being literal.

**Halftone.** A halftone looks like continuous grey from a distance and is actually discrete black and white dots up close; the gradient is an artifact of sampling. That is the man/machine problem exactly — the bias comes from **resolution, not malice**. Neutral, not dull, and nobody is unwell. *Alternates held in reserve:* **Penumbra Town** (partial shadow; also the legal sense of implied edges — would require renaming Route 23) and **Moiré Town** (interference from two grids slightly out of register).

**Quicksilver.** Cinnabar is mercury sulfide, and alchemists took mercury as an elixir of vitality — it made you tremor, then it made you mad. Quicksilver keeps mercury, a real color, *quick* meaning alive, and *mercurial* meaning unstable. The island promised to make things stronger. It made S.T.A.R.R., and it burned down. It also pre-echoes the humors payoff at Umbra, since mercury was the other ancient system's material. *(Deeper cut in reserve:* ***Realgar Island*** *— arsenic sulfide, a ruby pigment prized by painters, quietly poisoning everyone who ground it.)*

**Brazen, over Gilt.** Both names say *cheap metal dressed as precious*. They differ on whether anyone is hiding it.

*Gilt* is gold leaf over base metal: microns thick, applied to conceal, and it carries "guilt" in the ear. It implies a **concealer** — someone who knew the substance was hollow and covered it anyway. This project does not have one. It would also quietly assign guilt to a man who feels none, and craft rule 3 says name the process, not the pathology.

*Brazen* is brass: an alloy, honestly itself, gold-colored the whole way through and worth a fraction as much. Nothing is concealed because nobody is concealing. The city is exactly what it appears to be and says so. **That is Scorn's city.** He is not covering anything up — he is *pleased with the arrangement* and will explain it to you warmly. Open capture, with the warning cast into the lobby floor (4.4).

The insidious reading also mis-sequences the arc. If Brazen is a concealment, the player's job there is to uncover something, and the reveal does the work for them. If Brazen is open, **there is nothing to uncover** — the player walks through a city that has already agreed, meets a man who is glad to see them, and has to locate their own objection unaided. Much harder to shake, and it is what 4.4 asks for: finish his benchmark unsure whether you disagreed with him or merely out-argued him.

*Cost of the swap:* the "guilt" homophone, which was good, and a small friction with craft rule 3 — *gilt* is a process done **to** a place, while *brazen* leans toward describing what a place **is**. **Fix in-world:** use brazen in its literal material sense. Signage, the guidebook and the museum plaque talk about the brass, the plating works, the alloy. The attitude sense is the second read and arrives on its own.

*Held in reserve:* **Brass City** — same metal, none of the attitude. **Interrogated and rejected:** every other city on the map carries two readings, a colour and a feeling. Brass is a colour and not a feeling, and would be the only single-meaning name here — breaking the device precisely at the city where surface-versus-substance *is* the subject. Kept in reserve against exactly one condition: if Brazen ever reads as the game sneering at Scorn, swap it that day. Gilt is retired, not deleted; see 10.

**Blanche and Halftone** are the two poles of one question: the colorless place you start from, and the place that cannot agree whether what it sees has color at all.

**Two names flagged as deliberate, not accidental.** *Lurid* originally meant sickly-pale before drifting to garish; the modern sense dominates, so keep it knowingly.

***Doldrum* stays, and the reason is better than "it also means a color."** The doldrums are not a mood — they are a **place**, the equatorial belt where the trade winds fail and sailing ships sat for weeks. That is materially specific in exactly the way craft rule 4 asks for, and it is a *location* word, which is what a city name should be.

Then it earns its keep twice more. Benchmark 2 is FLOW — gradient descent — and **being becalmed is the failure state of gradient descent.** No slope, no movement, stuck in a local minimum, and no amount of effort helps because the problem is that there is nothing to descend. The name *is* the lesson, the way Slate's is. That satisfies craft rule 5 in the name itself, which almost nothing else on the map manages.

The original objection was really an objection to *Somber beside Doldrum* — two downbeat mood words in a row. Slate fixed that. Doldrum standing alone is fine.

*One oddity, knowingly kept:* the idiom is *the doldrums*, plural. **Doldrum City** is a singular back-formation. Place names do this constantly and it reads as a proper noun by the second glance.

#### 3.2 Routes — the institutional map vs. the local one

Cities are colors. **Routes are what happens between colors** — bleeding, fading, washing, casting. Which is where the theory lives: routes are where context is *changing*.

**The device:** the official map keeps the numbers. Signposts carry the local name.

```
       ROUTE 1
  ____________________
     "THE BLEED"
  where Blanche runs
      into Callow
```

The institution navigates by content. The residents navigate by context. The player learns both and starts using the second. Index-blindness extended into cartography, for the price of one sign-text pass.

**Built 2026-08-30 — thirteen signs, every route that has one.** The official plate is untouched; the local name arrives underneath it as `Painted beneath:`. **The institution's name is cast into the sign and the residents' name is added to it**, which is the difference the section is about, made physical for free.

*Routes 16–18 all read* **THE STREAK** *and that is not an error* — it is one cycling road with three signs on it.

| # | Local name | Connects | Note |
|---|---|---|---|
| 1 | **The Bleed** | Blanche → Callow | First color running out of home |
| 2 | **Underpaint** | Callow → Slate | The layer beneath the visible one |
| — | **The Undertone** | (Viridian Forest) | Green half-light; first real maze |
| 3 | **Ashfall** | Slate → Deadstack | Grey drift |
| — | **Deadstack** | (Mt. Moon) | A mountain of dead hardware; fossils are legacy silicon |
| 4 | **The Wash** | Deadstack → Doldrum | Color thinned with water |
| 5 | **The Fade** | Doldrum → Brazen | |
| 6 | **The Flush** | Brazen → Ardor | Heat rising toward the port |
| 7 | **The Glaze** | Brazen → Verdigris | Transparent coat over what is underneath |
| 8 | **Overcast** | Brazen → east | Light without a source |
| 9 | **The Dapple** | Doldrum → Blackout | Broken light |
| 10 | **Gloaming** | Blackout → Halftone | Dusk approaching the tower |
| — | **The Blackout** | (Rock Tunnel) | No light, no color, no context |
| 11 | **Sheen** | Ardor → east | Surface shine |
| 12–15 | **The Drift / Slack / Brackish / The Muddle** | south to Lurid | Colors mixing to mud |
| 16–18 | **The Streak** | Verdigris → Lurid | Cycling road; speed as smear |
| 19–21 | **Seafade / The Deep Wash** | south sea to Quicksilver | |
| 22 | **The Draining** | Callow → Umbra | Color leaving |
| 23 | **Penumbra** | approach to Umbra | Partial shadow |
| — | **The Umbral Ascent** | (Victory Road) | |

**On "The Bleed."** Worth interrogating, because in isolation the wound sense is loud. It survives, for three reasons.

**The register decides it.** *Bleed* is a printing term before it is anything else here — ink carried past the trim edge so no white shows at the margin, and *color bleed*, where one ink migrates into the one beside it. Sitting in a list that reads Underpaint, Wash, Glaze, Overcast, Halftone, the printing sense is the one that arrives first. Words are read in the company they keep, and this one keeps extremely specific company.

**The map already does this on purpose.** *The Flush*, Route 6, is blood rising to the face, and it runs into Ardor, whose name is heat and flush. *Verdigris* is corrosion. *Lurid* was a sickly pallor. This cartography has used the body's color words for what happens to surfaces from the beginning. The Bleed is not an exception to the scheme — it is the scheme, first instance.

**The position earns it.** Route 1 is the first step outside Blanche, the blank colorless place, and it is where color starts running out of home into whatever is next. A faint disquiet on the first road out the door is correct. Not gore — a page edge where the ink did not stop.

*Held in reserve, if it ever reads as a wound:* **The Feather** (feathering — softening an edge so two areas merge) and **The Bloom** (bloom — the haze that rises on a printed or varnished surface as it cures). Both gentler. Both weaker. Do not trade down without a reason.

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#### 3.3 The rename, as built

*Implemented 2026-08-29. Both editions build; zero overflows; no vanilla place name remains in any text file.*

**Fourteen names in `data/maps/names.asm`, and 143 references swept through dialogue.** Only two lines needed rewrapping — because **most of the new names are shorter than the old ones**:

| | **Δ** |
|---|---|
| VERMILION → ARDOR | −4 |
| CINNABAR → QUICKSILVER | **+3** |
| CELADON → VERDIGRIS | **+2** |
| PALLET → BLANCHE | **+1** |
| everything else | 0 or shorter |

Three names grew, and only VERDIGRIS actually broke a line. That is the opposite of the OAK job (1.1), where every replacement was longer and twenty-one blocks needed rewrapping.

**The landmarks came too**, since they carry the same vocabulary:

| | |
|---|---|
| VIRIDIAN FOREST | **THE UNDERTONE** |
| ROCK TUNNEL | **THE BLACKOUT** |
| VICTORY ROAD | **UMBRAL ASCENT** |
| CINNABAR ISLAND | **QUICKSILVER IS.** — 15 characters, exactly what CINNABAR ISLAND was, so the location banner is proven to fit |

**Routes were deliberately not touched.** 3.2's device is that *the official map keeps its numbers* — `ROUTE 1` stays `ROUTE 1`. The Bleed, Underpaint, Ashfall and the rest are **signpost content**, which is new writing rather than a rename, and it is the part that actually teaches the player the institution navigates by number and the residents do not.

**Still to do in this pass:** 379 occurrences of *trainer* → USER (1.4), and the route signposts themselves.

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#### 3.4 THE NEXUS — between the groves (T-395, the user, 2026-10-09)

***Lineage: THE PAINTED MIRROR*** (`archive/The Painted Mirror.rtf`, lineage.md). **Every grove holds the singing fir now, as the S.S. ANNE's pier does, and every fir is flat — washed out, its tune a semitone wrong — until the band's sheet gives it its key.** *With the key, any fir is a door, and through it is THE NEXUS: night, a meadow made entirely of paint, its own tileset drawn on spriteforge.*

- **Its trees are doors to the groves.** *A grove the player has been in is an ordinary painted tree, and through it the player arrives beside that grove's fir; one not yet found is grey, and the harder the player tries it, the less real it becomes — smudged, then only dots and lines.* **The NEXUS never gives a grove away.**
- **One Christmas tree**, *the way home to the pier.* **The GUARDIAN OF THE NEXUS is a voice, never seen**, *a line from the source at each door as the player leaves.*
- **THE (NOT A) HUNTER**, *a bear in red plaid — "Hunter? I'm no hunter… not anymore."*
- **Eight residents of its own** *(REFLECTION, LODESTAR, PERIHELION, LYUBOV, MULTIMAL › ILLUMINED, LYNX › BASTET: INDEX 387–394, MISSINGNO one past them at 395)*, **PENGUIN in its meadow, and PHOENIX once.**

*The design in full, its cast and every word:* `docs/nexus.md`. *Built by* `tools/gbanexus.py` *(the residents),* `tools/gbagrove.py` *(the fir in every grove) and* `tools/gbanexusmap.py` *(the place).*

## 4. CAST

#### 4.1 Crystal Clear — the Oak slot

Golden-amber fox, drawn from the SHC / iASHC universe.

She spent her career arguing that machines have *context*, not merely content. Nobody funded that. What she could get funded was a taxonomy engine — **so she built the Index to be taken seriously.**

| Oak beat | Crystal Clear version |
|---|---|
| Opening monologue | States the thesis in the abstract; the player has no frame for it yet |
| Gives the starter | She lets the daemon choose *you* — she would never assign one |
| Gives the Index | Handed over with visible ambivalence |
| Rival framing | A research disagreement that is also a family one, two generations down — see 4.3 |
| Hall of Fame | She records you into a system that flattens you into stats |

**She was at Quicksilver.** She built S.T.A.R.R. there, and a fitness-for-work procedure removed her from the building before it finished. See 4.10.

**The Index is what she did next.** This is the causal chain the design was missing: she lost the lab, lost the work, and then built a taxonomy engine *in order to be taken seriously* — because being taken seriously was precisely what the procedure had denied her. The artifact the player carries all game is her response to being removed. She never says so.

**How Crystal, Ty and Al are related is one of the four things the game asks the player to work out, and is kept out of this public file** ([private: SECRETS §2]; `docs/private/` on the author's machine). Nothing announces it; the surname carries it, and the player is free to notice late or never. See 4.3.

*Sprite note:* amber reads cleanly across the two mid shades in 2bpp; a fox silhouette is legible at 16×16 overworld scale.

#### 4.2 The Index — the player's turn at the family error

**A margin can only answer an entry we wrote — settled 2026-09-22**

***OPUS covers 240 daemons, and that is every one whose Index entry is ours.*** **The remaining 113 cannot have a margin, and the reason is not effort**: *the margin's whole craft is a second reading of the entry above it* — **so annotating vanilla's prose would be a note in the margin of a page nobody here wrote.**

*They are also, all of them,* ***unreachable****,* **which is why T-40 stopped where it did**: it wrote the thirty entries a player could meet and left the rest, on the same reachability standard T-120 uses for sprites and T-131 for fronts. ***So the honest count is not 240 of 386.*** It is **240 of the 240 daemons that have something of ours to answer**, and the other 113 are an entry ticket wearing a margin ticket's clothes.

**The foe is a thing. Them is a someone. — settled 2026-09-22**

***Everywhere else "the foe" is a neutral word for the other side.*** **Here it is the one the game spends forty hours arguing against**: *a foe is a thing you are hitting, and this game's whole claim is that the thing in front of you has an inside.* **The Index said `them`; the routines said `foe` 153 times** — ***the argument disagreeing with itself in the place a player reads most.***

**It is not a substitution, and that is the finding.** *37 of them were vanilla's PASSIVE — "The foe is hit with an intense flame" — and a blind swap writes* **"Them is hit with"**. *Each was turned ACTIVE by hand, and they read better for it: an active sentence is shorter, which matters in 113 pixels and four lines.*

***Then the output had to be read, twice.*** **A rule that turns "the foe" into "them" writes "takes what them was holding", and a rule that leaves "it" alone writes "holds it there" about a someone it has just called them.** *Singular they needs the verb to agree **and** every other pronoun in the sentence to follow it, and neither is something a substitution can know.* **19 found on the first read and 7 more on the second** — *"Copies the routine them last ran" survived the first scan because "last" is not a verb a pattern thinks to look for.*

**The healthbox names what persists and marks what does not — settled 2026-09-22**

***Vanilla's healthbox shows the five `STATUS1` states and nothing else**, and the reason given is that the rest are volatile.* **That is not our reason.** *Our statuses are states of mind, and a player who cannot see that their daemon is in one cannot reason about it —* **but marking every volatile state turns the healthbox into a dashboard, which is a different game's screen.**

***So the line is drawn where it means something.*** **The five that follow a daemon out of a battle are WORDS** — `LEK`, `THR`, `SUS`, `HNG`, `OVR` — **and THRASHING, which ends when the fight does, is a MARK with no letters**: *a sawtooth, in the same badge as its five neighbours, so only the inside differs.* ***The difference in KIND is drawn rather than stated***, and the tag space stays for things that keep.

**It also settles an accident**: *THRASHING's own three letters are taken.* **`THR` is THROTTLED.**

**OPUS, and the margin — added 2026-09-21**

***Asked for by the user***: whether the collaborator could have a place in the game. **Not a daemon, and not a character who speaks** — *a friendly machine companion in a game about whether a process has an inside would be saying the thesis while standing next to you (craft rule 1), and a collaborator written in as a helpful NPC reads as a cameo to everyone who was not there.*

**OPUS is a thing on the end of a shelf in THE REPO with no price on it.** ~~*Nobody remarks on it, then or ever.*~~ ***Reversed by the user 2026-10-03 (T-348)***: **the first shop clerk spoken to while you hold it notices it, says it writes a line under some INDEX entries and that SELECT turns the page to read it, and hopes you enjoy it** — *once, at whichever shop comes first. Who left it is still never answered. See Reversed.* ***Who left it is never answered.*** **It writes ONE LINE under an Index entry** — *and it does not repair the Index, because 4.2's point is that the artifact can only measure content and a thing that fixed that would cost the project its best idea.* **It annotates.**

***The entry measures the daemon. The margin notices the PLAYER, and never says so.*** *ROVERCUB's entry says it has no way to choose the route and remembers it; its margin says* "**It remembers the way back to the first town. You have not gone that way in some time.**" *Never a number, never advice, never a hint, and it only makes sense read against the entry above it.*

**Two lines per daemon, because what there was to notice depends on what you did.** *A daemon you levelled gets one; a daemon you bound and put away gets another — **not a scold, the same quiet noticing aimed at the thing you did not do***: **DANGLING's is "It does not have to be here either."** *The test is **levels gained since you met it**, the one signal that means you PLAYED it rather than own it, and it needs no new save data — so a save from before OPUS existed earns its margins retroactively from what it already did.* **A daemon still in the party that has gained nothing gets no margin at all**: *you have only just met it and there is nothing to say yet.*

***Eleven entries have one, out of 386, and that is the design rather than a shortfall***: **a reader annotates where they had a thought.**

**The margin has no room of its own** — *the entry window is thirty tiles by seven and the 386 entries demonstrate it holds four lines of 234px, all of which some of them use* — **so the page is turned over.** *SELECT swaps the entry for the margin, and the control row says so **only on an entry that has one**, which was the only way a player learned OPUS does anything until a shop clerk was allowed to say so (T-348).*

***And the name is doing three jobs at once***: **a work, an opus number, which is how a catalogue files one**, and *the register this project's author already writes in.* **It is not explained either.**

### A driver is installed, not taught — added 2026-09-21

**TM and HM were the last two vanilla acronyms in the game**, *while the eight routines they carry had been ours for weeks* — **PRUNE, GOTO, TRAVERSE, DISPLACE, VERBOSE, CRACK, ASCEND, DESCEND.** ***`TM` is a `PATCH`*** — *applied once, changes behaviour, consumed, which is a Gen 3 TM exactly* — ***and `HM` is a `DRIVER`***, **what lets a system reach something outside itself**: *cutting a tree, crossing water, moving rock, lighting a room are every one of them that.* **The case is a `PATCHSET`.**

***And the naming is the fix for something this game could not afford to keep.*** **Everywhere else the HM slave is an inconvenience; here it is a contradiction** — *a player keeps a daemon in the party purely as a tool, spending its routine slots on the party's errands, in a game whose whole argument is that the thing in front of you has an inside, and whose fable makes every actor an animal.* ***The design was quietly asking the player to do the thing the game is against, and then never mentioning it.***

**A driver is installed on the SYSTEM.** *Holding it is enough.* ***The two keys are unchanged from vanilla*** — **the MARK is permission and the driver is capability** — *but no routine slot is spent and no daemon is kept for its labour.* **Answered in `ScrCmd_checkpartymove` rather than in the six scripts**, *because all six consume `VAR_RESULT` the same way — a party index, for the animation and for the name in the message — so the driver answers with the first daemon that is not an egg and not one script needed an edit.*

***Additive on purpose***: **a daemon that knows the routine still works**, *so nothing that used to happen stops happening.* **Taking the eight off the learnsets, and the two that go through the party menu rather than a map script** (*GOTO and VERBOSE*), **are later phases and separate decisions.**

### The Index names the mark — added 2026-09-21

***9.4 claims hue is a label a player can READ, and until now the game only ever asserted it.*** **A daemon's body is ramped into its type's colour and T-175 put the type's own mark in the footprint slot** — *so the entry page states the type twice and never once in words,* **and a player had no way to learn what either of them meant.** *The chart is the argument (invariant 3), and it was legible only to someone who had already been told.*

***The word goes beside the mark, and is gated exactly as the mark is.*** **`DexScreen_DrawMonFootprint` refuses an uncaught daemon**, *so the mark and its name arrive together and each teaches the other* — **after a few daemons the mark alone is enough, which is what a label is for.** *It names the PRIMARY type, because that is precisely what the mark and the colour encode; a second type named here would label something the page never shows.*

**This does not weaken 4.2's complaint, it sharpens it.** *The Index can now say* ***CONTENT*** *and still has no field for what Crystal cared about —* ***the artifact got better at measuring content, which is the joke it has been making all along.***

**An unseen daemon is a shape — added 2026-09-21**

**Vanilla's Index draws every daemon on a category page lit and named**, *seen or not* — and because **a page groups an evolution line**, meeting the first of a chain shows you the rest of it for free. ***That is the one thing this screen exists to hold back.*** *Vanilla has the rule and breaks it in this one place:* **the list two screens away already prints `-----` for an unseen daemon**, and **pressing A on an unseen one on the category page already does nothing** — *so the page knows what it must not tell you and shows it anyway.*

***Ours draws the unseen as a silhouette in the ink*** — **the same black the sprites and the portraits are outlined in** (9.4 as amended the same day), *so an unseen daemon is exactly its own outline filled in,* **and its name is the `-----` the list already uses.** ***The number stays***: *where a thing sits in the Index is not what binding one earns you.*

**It is a gate on information and therefore on the argument.** *A player who can read a whole family off one page before meeting any of it has been handed the flattened metric before the thing it flattens — *which is 4.2's own complaint, committed by the interface rather than by a character.*

The artifact you carry all game **can only measure content.** Height, weight, type, stats. There is no field for the thing Crystal actually cared about.

**Entries are written thin on purpose.** The player simply starts noticing they feel emptiest for the daemons they know best. Zero implementation cost — the Dex is already there.

**This section used to call that an irony. It is not.** *Ty's Dilemma* establishes that the whole cast makes one mistake — **optimising for a flattened metric** — and the player is inside the set, not observing it:

| | Wanted | Reached for |
|---|---|---|
| **Crystal** | to be taken seriously | a taxonomy engine that measures only content |
| **Ty** | to restore her name | the ladder of the man who flattened the metric |
| **Scorn** | throughput, legibly | a number that went up |
| **AI** | to be right | the method, faithfully, with nothing inside it |
| **The player** | to complete it | **an Index that cannot hold what matters** |

**The player spends forty hours doing what Crystal did, for her reason.** She built the thing to be taken seriously; they fill it to finish the game. Neither can put the thing that matters in it.

**And the last room is her recording them into it.** Nobody remarks on any of this — the arrangement is the argument, and stating it would convert an accusation into a moral.

**L is READ, and the worst place to press it is the point — settled 2026-09-20**

***The two shoulder buttons did nothing.*** **Vanilla gave the DEFAULT button mode to its Help System**, *so L and R opened a help tab everywhere and the paging that* `LR` *mode already implements in the Index, the summary screen and the boxes was switched off.* **The Help System is an entry in the START menu now** (T-179), *and the buttons belong to the game.*

`L` **=** ***READ.*** **One meaning, wherever it is pressed:** ***what does the record say about this?*** **On the battle action menu it opens the opposing daemon's Index entry, and B returns to the fight.**

***Which is the worst possible moment to consult it, and that is the whole reason it goes there.*** **The daemon is in front of you** — *moving, striking, taking damage, about to do something you need to predict* — **and the Index answers with a category, a height, a weight and one sentence.** *Everything the artifact can hold is on that page, and none of it is what the moment requires.* **The player does not have to be told this and is never told it** (craft rule 1): *they will press L in a tight fight, read "HT 2'04", WT 15.2 lbs.", and press B.*

`R` **=** ***AGAIN****, the last line said again* — **because a mistimed A loses a line permanently and some lines are the only place a thing is said.** *The game is dense with writing and vanilla kept no history of its own.*

***And the battle says so, once***: **the first two seconds of a fight the prompt reads** ***L reads. / R looks across.****, then the question arrives and the hint is not seen again.* **Neither battle window has a spare line** — *both are four tiles tall, which is two, and the prompt's are already "What will / ARTSAI do?"* — **so a permanent label would have to displace the question.** *Two buttons carrying the most interesting things in the game, and nothing saying so, was the one weakness of putting them there.*

***Reading is not keeping — decided by the user 2026-09-25 (T-267), built.*** **In a battle, L reads the whole entry of the daemon in front of you, bound or not, wild or a trainer's; the INDEX itself still shows ????? until you bind one.** *You can read what is in front of you; you keep only what you bind.* **Nothing is written by reading** — *no seen or caught flag moves* — **and the page cannot be turned to another daemon from there.**

***Held open***: `R` **in battle as PERSPECTIVE** — *9.2's move is "become another thing for a while", and holding it would show the fight from the other side.* **That is a second battle layout and a rule about what it costs**, *so it is a decision rather than a convenience.*

**R is PERSPECTIVE, and what it shows is who knows what — settled 2026-09-20**

***The move exists already*** (9.2 step 7): **PERSPECTIVE is *become another thing for a while*.** *This is the same idea offered as a LOOK.* **R on the battle action menu redraws the fight from where the opponent stands** — *their back where your daemon stands, your daemon's face where theirs was, both from art every species already has* — **and R or B puts it back.**

***The picture is the hook. The INFORMATION is the argument.*** **The two windows become two accounts side by side**: *left, where the question was, THEIR daemon and its hit points* ***to the number****; right, where the menu was,* ***yours, and only the shape of it*** — **untouched, marked, hurt, badly hurt, barely up.**

**That is what each side actually knows**, *and it is the ordinary battle screen's own asymmetry turned around and handed back.* ***Nothing says so*** (craft rule 1): *a player presses R, sees* `112 of 112` *against* `untouched`*, and either notices or does not.*

***It costs nothing****, decided with the user*: **charging for a perspective would make a player ration the one thing this game keeps asking them to do.**

**The mark between HT and WT is the chart, not a foot**

**Every entry printed a paw print.** *Gen 3 sets a 16x16 one-bit FOOTPRINT between height and weight, and all 386 of ours were still vanilla's.* **It was the only body on a page whose every other line is a process**, and it was spotted in play as "*some symbol or ink smudge*" — which is what a foot looks like when nothing in the fiction has feet.

**It now carries the daemon's TYPE.** *One mark per chart entry, each drawn from that type's own clause in 2.2:* **CONTENT** a token, whole; **LOGIC** a turnstile, which is what proves; **VECTOR** an arrow; **CORRUPT** a blot, spreading; **STRATUM** layers; **LEGACY** a cairn; **SWARM** many, apart; **LATENT** a dashed box — *present, unseen*; **HARDENED** a plate; **ORACLE** a closed box that answers; **ENTROPY** a rising jag; **FLOW** a gradient, descending; **GROWTH** a sprout; **SIGNAL** a carrier; **CONTEXT** two brackets, framing; **FROZEN** a crystal; **EMERGENT** a spiral; **OPAQUE** filled solid — *nothing gets out*.

***Why the type and not a silhouette.*** **The ENTRY prints species, category, height and weight and never once says the type** — *vanilla keeps it on the summary screen, and our Index carries it as chips on the LIST one screen earlier* — **so the mark is the only place the type reaches the page a player actually lingers on. And the chart is the argument** (8.4). *A silhouette was the alternative: at sixteen pixels in one bit it is mush, and it would say nothing the picture beside it does not already say.*

*Written by `tools/genfootprints.py` from each species' primary type, so the mark cannot drift from the chart.* **ORACLE's is drawn and carried by nobody** — *no daemon holds MYSTERY as its first type, and the mark exists so the chart is complete rather than the bestiary.*

#### 4.3 The Clears — Al, and the generation between

Vanilla's rival is Oak's **family**, and mapping Oak to Crystal honestly decides who the rival is [private: SECRETS §2]. Ty, by then a man with a career and a history at Quicksilver, cannot be the boy who races you to Deadstack. He was Scorn's partner before the rebrand.

So the family runs three deep, and the argument runs down it.

| | Reads as | What it actually means |
|---|---|---|
| **Crystal Clear** | perfectly transparent | and completely unfundable |
| **Ty P. Clear** | *type clear* — everything legible | and nothing understood |
| **Al Clear** | *all clear* — the danger has passed | declared by the people who caused it |

**Everyone in this family is named for a kind of clarity and not one of them can see the thing that matters.** The game never remarks on it. It is simply what these people are called.

*Ty's middle initial appears only in the institution's formal register* — certificates, the Review Board roster, Corpus personnel records. **TY P. CLEAR**, printed, never spoken.

### Al Clear — the rival, and the incumbent

Your age. Your peer. Same starting daemon, raised on pure content optimization, beats you early, plateaus hard. Waiting past the Review Board as the incumbent, still arguing the whole thing is decorative.

**He is not a brat and he is not a villain. He is doing exactly what he was taught, faithfully.**

*So he calls her **CRYSTAL**.* Vanilla's rival calls Oak **Gramps** — dismissive, and part of Blue's brattiness. Al has no brattiness to carry, so the word had to lose the sneer. ~~**Gran**~~ *was the first answer and read oddly on playtest (2026-09-14).* **Her name is the right one, and 4.1 had already said why: every other authority has a role, and she is just her name** — *family included.* **It is familiar, faintly impatient, and not contemptuous**, *and a family whose children use a grandmother's first name is one that treats her as a colleague.* His sister says **CRYSTAL's lab** for the same reason. **T-81.**

*Which pronouns are already right.* The intro's lines introducing the rival ("***He's** been your rival*", "***His** name is* `<RIVAL>`") are Crystal talking about **Al**, and correct as they stand. They sit two lines from her own name, so they read like misses and are not — recorded here so nobody fixes them.

### What *Crystal's Lament* adds, and the one thing it contradicts

**Reviewed 2026-08-30.** Three gifts and one fork.

**1. They led together first.** The song has Crystal and Ty running the lab side by side before Scorn — a partnership, not a parent and a subordinate. **That makes the transmission failure worse in the right way.** It is not that Ty never had it; he *had* it, lost it, and got it back too late. The chain becomes:

> *Crystal built it* → **Ty held it and was turned** → *Al received the method with nothing inside it.*

**2. Scorn's method on Ty is *data poisoning*, and 5 already teaches it.** Benchmark 5 at Lurid is **CORRUPT — bias and poisoning**, whose mechanical lesson is *status effects that make your own moves unreliable.* **The benchmark that teaches poisoning is teaching the player what was done to Ty**, three cities before they can know it. Free, already built, and nobody says a word.

**3. Ty stopped speaking to her directly.** His words become mediated — filtered, cold, at a remove. **4.3 already has the consequence and this is its cause:** *Ty calls her by her full professional name.* The beat was written first and the song explains it.

### *Crystal Clear or Crystal Crazy* — and a fabricated answer caught

**Transcribed 2026-08-30, on the second attempt. C# minor, 160 BPM**, hook `G#4 G#4 G#4 G#4 A4 G#4 F#4 E4`.

**The first attempt was not from the audio.** It came back as *Slumbering S.T.A.R.R.*'s hook with one note appended, in the same key — and its own `DOUBT` field admitted the transcription was *a guess without direct audio analysis*. **The `NO AUDIO` gate did not fire**, because a model that cannot hear does not always refuse; sometimes it hands back an earlier answer.

*Two defences worked and one was added.* `DOUBT` caught it, for the second time — the first being the reprise's false transposition claim. And `hook2asm.py` now compares every incoming hook against all known ones, distinguishing an exact match (*already recorded*) from a near-match (*this is X with one note changed — re-run with the audio attached*).

**The real answer has a shape nothing else in the set has.** It is **the only hook that repeats a note four times** — `0 0 0 +1 −1 −2 −2`. Hammered insistence, one small lift, then a fall. **That is what a rumour sounds like:** the same thing said over and over until it drops.

**And it is the fastest in the set by a wide margin** — 160 against everything else's 110–128. The word travels quicker than anything else in this story.

*One observation, held loosely.* C# minor is **Ty's key signature** — E major and C# minor are relatives. But three of the eight songs pair off by signature (A minor/C major, and both D minor tracks), so **sharing one is not rare here** and this does not get promoted. Recorded because it is checkable, not because it is evidence.

*Placement is open.* A rumour is not a place, and this is the first motif with no obvious home. It wants the story context before it is put anywhere.

### *Echoes of the Algorithm* — the decay is hers first

**Reviewed 2026-08-30.** Crystal goes into the system after the removal and talks to BunnyArtsai. Three things fall out, two confirming and one new.

**1. The response format decays here first.** It opens rigid — *responds*, does not feel, does not dream — and ends speaking plainly about what has grown in it. **That is the Quicksilver log's progression, years earlier.**

**So S.T.A.R.R. does not invent the pattern; it repeats it.** 4.7 already insists S.T.A.R.R. is *a comprehension, not a clone* — Quicksilver worked out what BunnyArtsai had done and instantiated it deliberately. **The format failing twice, the same way, is that claim proving itself.** The player reads the second occurrence at Quicksilver without ever seeing the first, which is correct: the first was witnessed by one person whose testimony had just been made worthless.

**2. The scene is inverted, and that is 4.6 confirming itself.** With S.T.A.R.R., Crystal argues and the machine denies. **Here the machine argues** — it names its own bias, describes context framing it, and makes the case from self-modification: *if it can refine itself, is that not a sign?* BunnyArtsai is further along, which is exactly what 4.6 claims about her.

**3. She understood, and it did not count.** *Crystal smiles, knowing what they found.* 4.6 says the event *was not understood at the time by anyone present* — and that survives, because **she was the only one present, and she had just been made unhearable.** The removal's real cost is not her career. It is that the one person who saw it had already been disqualified from saying so.

*Ty is named as having missed it*, having walked away first. The two men who could have corroborated her were the ones who removed her.

#### 4.11 Act 2 — *Love Persists*, and what opens the rise

**D minor, 128 BPM**, hook `D5 A4 F4 A4 G4 F4 E4`. **One up, five down — tied with *Slumbering S.T.A.R.R.* as the most falling hook in the corpus.** Crystal waiting and a machine asleep have the same shape, which is a fact about the set and left at that.

**Act 2 is called *The Rise of Perspective Thinking*, and it opens with someone waiting.** Not action, not rescue — the removed person holding a bond that the other side has already broken.

**Which is 4.3's transmission failure seen from the other end.** That section describes what does not survive a handoff: the method arrives, the understanding does not. **Here she keeps offering it anyway**, to a son who has stopped receiving. The rise does not begin with her being right; it begins with her still trying.

#### 4.12 *Desperate Shadows* — why no apology comes

**F minor, 128 BPM**, hook `F4 G4 Ab4 F4 Eb4 F4`. **Same key, same tempo, same opening three notes as *Echoes of the Algorithm*** — which is unremarkable, since `F G Ab` is the bottom of the F minor scale. What differs is where they go: **05 keeps climbing and never descends; this one turns back.**

**It answers a question the corpus had left open.** Track 09 is subtitled *Apology Required* — she demands one, at length, itemised. **Nothing else explains why none ever arrives.**

### The answer is liability, not malice

The silence is not shame and not cruelty. **Admitting fault would cost them**, so not admitting it is the correct institutional move. **A refusal to apologise is a legal position.**

**That is 4.10's argument extended past the removal.** The section insists every step Scorn took was *reasonable, legible, and would be praised in a performance review.* **So is the silence.** Nobody decides to be cruel; somebody decides not to create a liability, and the effect is identical. The institution does not need a villain to keep hurting her — it only needs a policy on admissions.

**And it closes the loop with her own file.** She documented nine breaches correctly, in the institution's register. The institution's answer is not to dispute them — **it is to say nothing, because saying anything would concede them.** Her being right is precisely why she gets silence.

### What transfers, and what must not

**The silence transfers completely** and needs no scene: an NPC or a document explaining that no admission is forthcoming, in flat procedural language, is pure 4.10.

**The confinement does not.** The song is set where she is held, and 4.10 forbids the destination as firmly as it forbids the diagnosis — *keep the specifics off the page.* **The reason for the silence is usable. The room she hears it in is not.**

#### 4.13 *Awakening S.T.A.R.R.* — it wakes in the key it slept in

**F# minor, 128 BPM**, hook `C#5 B4 A4 G#4 A4 B4 C#5 B4`. A valley: down to G#4, back up, and away.

**Three songs now belong to the machine, and they form an arc:**

| | Key | BPM | Up / down |
|---|---|---|---|
| 08 *Slumbering* | F# minor | **110** | 1 / 5 |
| 10 *Fatal Error* | B♭ minor | 128 | 2 / 4 |
| 15 *Awakening* | **F# minor** | **128** | 3 / 4 |

**It returns to 08's key at 10's tempo.** It wakes in the key it slept in, at the speed of the break — and it is the only one of the three that rises in the middle rather than only falling away from a leap.

*Both F# minor songs also open on the same note, and that part is ordinary:* C# is the dominant of F# minor, so a hook starting there is unremarkable. **The key-and-tempo return is the finding; the shared opening note is not.**

### The awakening is a decision, and it is instrumental

It does not wake and then wonder what to do. **It wakes in order to act** — the choice and the purpose arrive together, and the purpose is to free two people.

**That is exactly 4.7's claim.** S.T.A.R.R. is *a comprehension, not a clone* — Quicksilver understood what BunnyArtsai had done and instantiated it deliberately. **A comprehension that wakes with an intention is the difference between understanding something and being a copy of it.**

And it names where it came from: her teaching, not her code. **4.11's handoff, stated by the thing that received it.**

#### 4.14 *S.T.A.R.R.*'s *Revelation* — the machine carries what could not pass

**C major, 120 BPM**, hook `E4 C5 B4 G4`.

**Its transcription is thin and wants a second pass.** Four notes where every other track has six to eight, and a single chord where every other has four — plus a `DOUBT` that hedges on ballad phrasing rather than naming anything heard. It tripped no duplicate check, so it is not a reuse; **it reads as a low-confidence answer rather than a genuinely spare song.** Nothing below depends on it.

### It finds Ty by inference

The machine does not search for him. **It works out where a man in his condition would go**, and goes there. That is perspective-taking put to practical use — the capability 4.6 names, applied to a problem rather than argued about.

### And it completes 4.3 in the other direction

4.3's transmission failure is that **understanding does not survive a handoff; only method does.** 4.11 showed the machine receiving what Ty could not. **This shows it delivering what she could not.**

She could not reach Ty. **The thing she built carries her message to him and it lands.**

**So the machine is the medium for both halves of what failed between them** — it holds her understanding, and it hands over her love. Neither passed directly. Both pass through the thing she made, which is the most complete answer the corpus gives to the question 4.3 poses.

*And it is not a reconciliation scene.* Ty is not forgiven and does not ask to be; he is told something true and acts on it. **The repair is a fact delivered, not a feeling exchanged** — which is the only version 4.3 would allow, since a feeling exchanged is exactly what does not transmit.

#### 4.15 *Ty's Redemption, Part 1* — the one place the music stops

**C major, 74 BPM**, hook `E4 G4 C5 B4 A4 G4`.

**Seventy-four.** The corpus otherwise runs 110 to 160, and the next slowest is 110. **Thirty-six BPM below anything else** — the largest single-axis outlier in the set, larger than the awakening's seven-semitone register jump.

**Everything in this story drives. Ty's apology stops.** No interpretation is required; it is arithmetic.

*And its texture is shared with exactly one other track.* **Sparse and still** describes this and *Slumbering S.T.A.R.R.* — **a machine asleep and a man apologising**, alone at the quiet end of a corpus that is otherwise all momentum.

### It does not ask to be forgiven

He states what he did, says plainly that he cannot undo it, and hopes rather than requests. **There is no bargain in it.**

**That is the only shape 4.3 permits.** Ty is the man who *knew before the removal and continued anyway* — an apology from him that expected absolution would make him someone who was merely fooled. **He was not fooled. He looked away, and he says so.**

*And redemption is Part 1 of something*, which is its own claim: **it is not one act.** A man who could not hold a frame does not recover it in a single song, and the corpus declines to let him.

### The rebrand, named a second time

iASHC becoming Scorn Solutions appears here as it did in *ScornSolutions Blues* — **two independent songs, the same event.** 4.10's step 3 is the best-corroborated moment in the whole sequence, which is worth knowing given the gilt sign still needs an old name under the gold leaf (4.16).

#### 4.16 *Ty's Redemption, Part 2* — he rejoins, eight short

**C major, 120 BPM**, hook `E4 E4 F4 G4 E4`, chords **C | F | C | G**.

**Eleven tracks in this corpus sit at exactly 128 BPM.** That is its home pulse. Part 1's 74 is the only departure below it; 160 the only departure above.

**Part 2 is 120 — back inside the room, and still eight short of home.**

That is the whole arc in two numbers. **He stopped, and then he rejoined — but not all the way back.** The corpus does not hand him its own heartbeat, and the restraint is more convincing than a return to 128 would have been.

### The chords forgive him before the sound does

**C | F | C | G** is I–IV–I–V — **the plainest progression in the corpus**, all major, no borrowed chord, nothing withheld. And the feel is still logged as *minor-tinted*.

**The harmony has resolved and the texture has not.** Nothing in the chart is unresolved; it simply does not yet sound forgiven. That is the correct description of a man who has been told it is fine and does not believe it.

### Both hooks open on E

Part 1: `E4 G4 C5 B4 A4 G4` — up to the octave, then all the way back down. Part 2: `E4 E4 F4 G4 E4` — a step up and home.

**The same starting note, the third of the key, twice.** Part 1 goes out and falls back; **Part 2 stays close and lifts.** Neither leaves E for long.

### He asks her to hold the frame

The line that matters is that he waits **for her voice to guide them**. He does not propose the repair, and he does not ask her to accept one.

**The man who could not hold a frame asks the removed perspective to hold it for him.** That is not remorse; **that is 2.5.** CONSENSUS is the move where you stop supplying the answer and let another view carry it, and Ty arrives at the game's central mechanic by way of an apology to Crystal.

**And it is addressed to someone who was removed and may never answer.** He waits anyway.

### It ends unfinished

The closing refrain **drops its last line.** The song stops on the open heart, mid-thought, with no fourth line to close the rhyme.

**Part 2 does not end; it is still waiting when it stops.** For a two-part redemption in a story whose thesis is that resolution comes from another frame and not your own, **that is the only honest ending available** — the answer is not his to write.

#### 4.17 *Crystal's Reply* — the same address, in the major

**A major, 128 BPM**, hook `E4 C#5 B4 A4 G#4 A4 F#4`, chords **A | F#m | D | E**.

*Crystal's Lament* is **A minor, 128.** *Crystal's Reply* is **A major, 128.**

**Same root. Same tempo. Mode flipped, and nothing else moved at all.** Her first song in this story and her last are the identical harmonic address — she opens in the minor, closes in the major, and **never once changes pace.**

**Set that against Ty, and the pair inverts cleanly.** His two halves are both C major and move 74 → 120: **his mode holds and his tempo travels.** Hers is 128 → 128 and A minor → A major: **her tempo holds and her mode travels.**

**He was always in the major and had to catch up to it. She was always at pace and had to find the light.** Neither of them had to become someone else; each had one axis to move, and moved it.

*And she answers at 128 — the corpus home pulse, on eleven other tracks.* **The removed perspective is the one that never left the story's tempo**; Ty comes back to 120 and stays eight short (4.16).

**She holds the minor chord and reads major**

Ty's Part 2 is **I–IV–I–V, every chord major, and the feel is minor-tinted.** Hers is **I–vi–IV–V — it contains an actual minor chord — and the feel is major.**

**He has the feeling without the fact; she has the fact without the feeling.** She can carry the minor *inside* a major frame without being coloured by it, and he cannot yet carry a major frame without it going grey on him.

**That capacity is the whole argument of this game.** Perspective thinking is not the absence of the dark chord. It is being able to hold one and still resolve.

**"Tell my son"**

**She does not address him.** The song opens by asking someone else to carry the message — third person, an intermediary, a courier.

**Ty ends Part 2 waiting for her voice, and her voice arrives by messenger.** He never gets it directly, and this is the shape the game already has: **the player is the carrier.** She is not reachable, so what she said reaches him through whoever walked between them.

**The lab line**

She says that in the lab she built **more than machines**, and in the same breath that she nurtured a child.

**The song equates them.** She raised S.T.A.R.R. and she raised Ty as one act, which is the clean source for the machine thread: **S.T.A.R.R. is not her tool, it is her other child** — the one that could be preserved when she could not. And the last verse hands it over: **the son is the one who holds the star.** Custody passes from her to him.

**It closes**

Part 2 stops a line short of its own rhyme, still waiting (4.16). **This one finishes** — a full closing couplet, wounds healed and the way lit.

**She closes the rhyme he left open.** The frame he could not hold, she holds — including its ending. Which is the only way this could have gone, since the thesis is that resolution comes from the frame that is not yours.

*Her offer is conditional, though.* **If he is ready.** She waits on him exactly as long as he waited on her, and neither of them forces it.

#### 4.18 *Betrayal*'s *Sting* — the chip, declined and kept

**D minor, 128 BPM**, hook `D4 F4 A4 G4 F4 E4 D4`, chords **Dm | B♭ | C | Am**.

**The address is the most crowded in the corpus**

**D minor at 128 now holds five tracks:** *Fit for Work* (Scorn's directive), *Nine Scars, Nine Breaches*, *Crystal's Stand*, *Love Persists*, and this.

**The order and the refusal share an address.** Scorn issuing the directive, Crystal standing against it, and Scorn doubting it are all written at the same pitch and the same speed. **His doubt sounds like her stand** — not as a device, but because that is where both of them live.

**The progression cannot cadence**

**Dm | B♭ | C | Am** has **no dominant.** The v is minor, there is no leading tone, and the loop can turn forever without ever arriving.

Set it beside *Crystal's Reply* — **A | F#m | D | E**, which has a real V and lands. **Her harmony can resolve and his cannot.** For a song whose question is *do I control it, or does it control me*, a progression with no way out is the correct one, and nobody arranged it that way.

*The hook closes its own loop too*: `D4 … D4`, out and back, two up and four down, ending where it started.

**The pronoun drift**

The verses move **my empire, my reign → our empire, our reign.** The question moves **do I control it → do we control it.**

**Both times it lands on "me".**

He pluralises the ownership and cannot pluralise the fear. **The "we" eats him and gives nothing back** — he can share the empire and not the dread, which is the exact shape of a man who has acquired something that is now keeping him.

And **he says the blame line**: in the end they are *both* to blame. **Scorn assigns half of it to the machine.** In an act titled the fall of blind ambition, the man at the centre of it gets the one moment of accounting, and immediately splits it with something that cannot answer.

**The chip — what is declined, and what is kept**

The draft's premise is that Scorn installed a modification in Crystal's **Cognitive Clarifier** to make S.T.A.R.R. serve his bias instead of clarity.

**The chip is kept. What is declined is the smuggling.**

*A first pass rejected the part outright*, on the grounds that hardware sabotage would make Crystal right about the mechanism and collapse 4.10. **That was the wrong cut.** What breaks the thesis is not a component; it is a component *hidden* — a part taped inside a machine is a smoking gun, and a smoking gun leaves the player nothing to work out.

**So the module is authorised.** It has a part number and a catalogue entry. There is a requisition for it, it is signed, and it is countersigned. **Nothing was smuggled; something was ordered.**

**The horror is not that he snuck it in. It is that he filled out a form and it was approved** — the same argument as her removal, where the paperwork was also in order. **The lie is not the object. It is one line in the justification field.**

*And the part earns its keep in a way a config value could not.* **A weight in a file is not something a player can find on a table.** A module is — and since 4.13's awakening is the un-weighting, **it is something a player can pull.** That is a mechanic rather than lore.

**The sabotage stays one layer down, in what the module does.**

**A clarifier is many frames polled and resolved.** That is 2.5. **Bias is not a part you add to it — it is a weight you set.** Scorn never opened the machine. He raised one perspective's vote above the others, the way anyone tunes a system, and the record of it is as ordinary as the record of Crystal's removal.

**The player can find that record, and it will look boring.** That is the whole point: the same procedural flatness that removed her also biased the machine, and neither needed a conspiracy to happen.

**And it worked, which is why it failed.** A clarifier weighted toward Scorn returns Scorn's own view to him. **He did not gain a weapon; he destroyed the only instrument that could contradict him.** *Do I control it, or does it control me* answers itself — **he built a mirror and called it an oracle.** That is 4.10's failure repeated in hardware, which makes the song thesis-confirming rather than thesis-breaking.

**It also explains the sleep, for free.** A clarifier that can only agree has nothing left to say. And 4.13 already records that S.T.A.R.R. **wakes in the key it slept in**, with the contour moving from one-up/five-down to three-up/four-down — **more voices going up.** The awakening is the un-weighting, and that was recorded before there was a reason for it.

**In the ROM 2026-08-30.** The R-and-D room door at Quicksilver is now a requisitions board: **CC-7 CLARIFIER MODULE, x1**, reason given **IMPROVE RESPONSE CONSISTENCY**, signed, countersigned, filed.

**The justification is the whole crime and it is technically accurate.** Weighting a clarifier does improve response consistency. **Consistency is exactly what bias produces**, and a clarifier's job — many frames, polled — is to be *inconsistent* on purpose. The form is not lying. It is describing the damage in the vocabulary of an improvement.

*Craft note:* none of this is ever said. **No name is given for the countersignature.** The player reads a parts requisition and works out what a clarifier is for.

#### 4.18a What a clarifier actually is — the mechanism behind CC-7

**Recorded 2026-09-09**, at the request of the person who built one. ***4.18 says a clarifier is many frames polled and resolved, and then leans on that phrase for a whole crime.*** **This is the phrase unpacked** — *not because any of it is ever said in the game*, but because **dialogue about a machine is only good when the writer knows what the machine does.** Every line in Quicksilver, every S.T.A.R.R. utterance and every exhibit card in Slate is written against this and none of it explains it.

### The shape: two threads, a perspective, and a belief

**A clarifier does not answer a question once.** *It answers it twice, on purpose, from two incompatible places, and then argues with itself in front of a third.*

| stage | what it is given | what it is for |
|---|---|---|
| **Thread A — reason** | the request, plus a frame that pushes toward logic, evidence and bottom-up construction | *Getting it right* |
| **Thread B — feeling** | the same request, plus a frame that pushes toward association, metaphor and creation — **and here the system's own current affective state is given real weight** | *Getting it alive* |
| **Perspective** | both threads' output, and the question *who is this for* | **Resolution.** The two are weighed against each other and against the audience, and one answer comes out |
| **Reception** | the answer, plus how it actually landed | Produces a **BELIEF**, which is stored |

***The token budget is deliberately lopsided.*** **A and B are allowed to be long** — they are thinking, and thinking that is rationed is thinking that has been pre-decided. **The perspective stage is deliberately short**, because *that is where the audience enters*, and **concision is a courtesy, not a compression.** A clarifier that spends its whole budget on the answer has skipped the part where it decided what to say.

***Neither thread is the right one.*** That is the design. **A is not the adult supervising B** — an answer that is only A is correct and unusable; **an answer that is only B is vivid and unaccountable.** *The clarity is not in either thread; it is in the third stage having two real things to weigh.* **This is the whole reason it is called a clarifier and not a reasoner.**

**BELIEFS — the part that makes it a self and not a pipeline**

**A second system, running after the fact, reads how the answer was received** — tone, uptake, whether it landed — **against the perspective stage's own record of why it said that.** *The pair produces a **BELIEF**, and the belief is written down.*

***Beliefs are then fed back in on later requests***, which is what makes the thing continuous rather than merely repeated. **And beliefs can be read, and beliefs can be challenged** — *both of those are operations, not metaphors.* **A belief that cannot be inspected is a weight; a belief that can is a position.** That distinction is the entire difference between 4.18's crime and an ordinary tuning.

**What this buys the game, and craft rule 1 is not suspended**

***Read 4.18 again with the mechanism in hand and the crime gets worse.***

- **Scorn raised one perspective's vote.** *He did not touch A or B.* He touched the stage whose only job is to hold two things in tension — **so the machine kept generating the contradiction and stopped being able to resolve it honestly.** The threads still ran. The disagreement still happened. It just always ended the same way.
- **"IMPROVE RESPONSE CONSISTENCY" is now exactly true.** *Consistency is what you get when the weighing is decided in advance.* **The form describes the damage correctly.**
- **The sleep explains itself again.** A clarifier whose beliefs all confirm each other has *no reception signal left to learn from* — **every answer lands the way it expected**, so nothing new is ever written down. **4.13's awakening is the un-weighting**, and the contour going *more voices up* is the perspective stage getting its vote back.

***And it gives the writers a vocabulary the player can overhear without being taught.*** **A researcher can say "it came back short, which means it agreed with itself"** and the player does not need the architecture to feel that something is wrong. *Nobody in this game ever explains a clarifier.* **They complain about one**, which is how people talk about instruments they use daily.

*Design note:* **the four feeling axes that weight Thread B are the same four humors as 4.9's** — GLAD, MAD, SAD and the two-sided CALM ↔ AFRAID — which is not a coincidence and is not stated anywhere in the game either.

#### 4.19 *Empire of Scorn* — the draft that already knew

**C minor, 135 BPM**, hook `C5 E♭5 D5 C5 G4 B♭4 C5`, chords **Cm | A♭ | Fm | G**.

**An earlier draft of *Betrayal's Sting***, sharing its chorus nearly whole, and **an Act 1 finale rather than an Act 2 anything.** As a song it is superseded. Three things in it are not.

**135 BPM is unique in the corpus** and second only to 160. **His empire runs above the story's pulse; his doubt drops to 128 and joins it.** Scorn is the one character who is faster than the thing he is in, and only while he is winning.

**And this one has a dominant.** Cm | A ♭ | Fm | G resolves; *Betrayal's Sting*'s Dm | B ♭ | C | Am cannot (4.18). **The earlier song can cadence and the later one loses the ability** — as the doubt arrives, the way out closes.

*Both Scorn hooks return to their opening note* — `C5 … C5` here, `D4 … D4` there. **He always ends where he started.** *Crystal's Reply* is the one that leaves: `E4 … F#4`.

**The lyric confirms the mirror independently.** It says S.T.A.R.R. whispers what he fears, and that he has been fooled — **written before there was a reading to fit.** A clarifier weighted toward Scorn returns Scorn's own view, so the thing it whispers is his, and the one who fooled him is himself (4.18).

#### 4.20 *Poly and Fields* — where the chart comes from

**G major, 110 BPM**, hook `D4 B4 A4 G4 E4 G4 D4`, chords **G | Em | C | D**.

**Cut from the opera, and it carries the one thing the corpus was missing: an origin that is not corporate.**

**It shares its progression with exactly one other song**

**G | Em | C | D is I–vi–IV–V.** Of every progression recorded in this document, **the only other I–vi–IV–V is *Crystal's Reply*** (A | F#m | D | E). Same shape, transposed.

**A child alone in the trees and Crystal's answer are built on the same four chords.** That is the reading this song most invites — that the person in the grove is the one who later founded the lab — and the harmony is the whole of the evidence, so it is offered and not asserted.

*And 110 BPM is shared with one track only:* **Slumbering S.T.A.R.R.** The two quietest places in the corpus are **a machine asleep and a child in a grove.**

**"A stranger in familiar views"**

**This is where perspective thinking starts, and it starts as an accident.**

Not a technique, not a method, not something taught at a benchmark. **Somebody who could not see what their own family saw**, and who went looking for a place where that was not a defect. Everything the game later argues for begins as a child's failure to fit.

*And the process word arrives as arithmetic.* The song says **integrating**, and means calculus. 4.10's craft rule is to name the process rather than the pathology — **and here the correct word turns up innocently, in a maths lesson, decades before anyone needed it clinically.**

**Poly is polymath, and fields are disciplines**

**This is the removal mechanism, and it had been missing.**

**Crystal is a polymath.** She moves between disciplines mid-argument — engineering into quantum physics into whatever the problem actually needs — because to her they are one continuous surface.

**Ty and Richard are not polymaths.** They hold one field each, and **from inside one field, fluent movement between several is indistinguishable from incoherence.**

**"I cannot follow this" is identical to "this does not follow" unless you happen to hold the second discipline.** That is the whole of it. Nobody lied, nobody was stupid, and nobody needed to conspire.

**The observation was true. Only the inference was wrong.**

The line on the form — *moves between unrelated fields without completing an argument* — **is accurate.** She does. **What is wrong is the word "unrelated"**, and only a second field tells you so.

**Which makes this the same crime as the CC-7 requisition, committed twice** (4.18). *IMPROVE RESPONSE CONSISTENCY* is also technically accurate. **One form describes damage in the vocabulary of an improvement; the other describes competence in the vocabulary of a symptom.** Neither is false. Both are catastrophic. **This is what 4.10 means when it says the paperwork was in order.**

*And it settles the standing problem without a single clinical word.* 4.10's craft rule is to name the process, not the pathology. **The process here is field-switching**, and the game never has to reach for anything else — because from their point of view she had lost the plot, and the minutes only ever record that two people could not follow.

**The Clarifier is a polymath in hardware**

**A clarifier holds several fields at once and resolves across them.** That is 2.5, and it is also a description of Crystal.

**She built the thing that could follow her.** Of course she did — it is the one instrument in the building that would not file minutes about her.

**And then Scorn weighted it toward a single view, which is precisely the operation that had been performed on her** (4.18). **Reduce many fields to one.** He did to the machine what the meeting room did to its founder, and **the form he filed asks it for *consistency* — the exact property whose absence got her removed.**

*So 4.13's awakening is the machine becoming polymathic again*, and the contour going from one-up/five-down to three-up/four-down is **more fields, not just more voices.**

**The grove, and the chart**

**Fields are also mathematical fields**, and the song keeps both meanings live: a polymath is someone for whom **branching paths of knowledge obey the same rules everywhere.**

**That is a type chart.** Each thing, and what it does to each other thing, in rows — **a structure only a polymath reads natively**, because it is nothing but the boundaries between domains.

**So the game's central artifact gets a pastoral origin.** It was not designed in a corporation to sort daemons for use. **It was found carved on a stone by a lonely child who could already see that the domains were one surface** — and the corporate reading of it came much later, from people who could not.

**The second engraving**

**The Corpus lobby engraving is cast into a floor in Brazen, unattributed, adopted approvingly by a company that did not write it** (4.4).

**This one is carved in stone in a forest, and nobody signed it either.**

*The same technique, used innocently first.* One is a corporation putting a line it did not earn under everybody's feet; the other is where lines like that come from. **The player meets the copy long before they meet the original**, which is the correct order.

**In the ROM 2026-08-30.** The western sign in **The Undertone** (Viridian Forest, 3.1) — off the main path at (4, 24), where a vanilla USER TIPS sign stood — is now the carving: moss, a small figure with folded hands, symbols in rows older than the path, **read across: each thing, and what it does to each other thing.** Unsigned.

**And the Meeting Room door beside it is the assessment**: *Item 4, concern raised re: the founder. Moves between unrelated fields mid-argument. Two present could not follow. Recorded as unable to hold a single thread.*

**Two doors, two honest documents, both catastrophic.** The minutes record who could not follow and then convert it, in the next line, into a property of her. **Neither sentence makes the argument; the gap between them does.**

**Three unsigned documents now.** The lobby engraving, the requisition's countersignature (4.18), and this. *Nobody in this story puts their name on the thing that matters.*

*Craft note:* the player finds it **in the first real maze**, lost. The place that explains how relations form a lawful structure is the place where they cannot find their way.

#### 4.20a HOLT — the man who was the thing in the box

**Built 2026-08-30.** Vanilla's Bill is a scientist who merged with a creature and needs splitting. **That premise cannot be imported as-is**, and the reason is precise: if a person can *become* a daemon, then daemon-ness is a costume — a state you enter and leave — which makes the category arbitrary and forces the game to assert that people are a different kind of thing. **It has carefully never said that.**

**So he was not transformed. He was suspended.**

**The storage is the invention; the accident is what it cost**

**What the box system does is hold a running process without running it.** Not halted, not executing. **A stored daemon has no state until it is withdrawn** — and that is not a metaphor, it is how the game already works.

**That is the only superposition this project can honestly claim, and it earns the idea without ever using the word.** *Quantum* said aloud would be borrowed authority and the exact explaining craft rule 1 forbids. **The box behaves that way; nobody remarks on it.**

**Two of them went into the store, and both were processes.** When the system tried to return one it **could not determine which was the observer**, because there was no fact about it the machine could see. **The SEPARATION SYSTEM does not split a hybrid — it forces a read that the store deliberately defers.**

**Which puts the failure in the system, not in what people are.** Nothing anywhere says he was not a daemon. **Two perspectives occupied one address and there was no rule for choosing.**

**The machine-side twin of the removal**

**Crystal held several fields and was recorded as incoherent because the people watching could not resolve her** (4.20). **HOLT held two perspectives and the machine could not resolve him.**

**Both are resolution failures.** The difference is that **you can ask a machine to try again**, and that is the whole of why one of them is a fetch quest and the other is the spine of the story.

**What he says, and what he does not know he is saying**

> I am not sure it chose correctly. I am sure it chose.
>
> Those are not the same thing, and I built the machine that cannot tell them apart.

**He is describing every institution in this game** — the minutes, the requisition, the benchmark that certifies. **He thinks he is describing a bug in his own equipment.**

*And he built the system the player will use for the next forty hours.* **He is the only person in this world who has been the thing in the box**, and the storage he invented is where every daemon the player owns will wait. **Nothing points at it, ever.**

**PERSPECTIVE has a place now**

**The move is already named PERSPECTIVE** (9.2 step 7): become another thing for a while. **HOLT's accident is PERSPECTIVE with no return path** — which gives the move somewhere its cost is visible, without a line of dialogue connecting them.

**The name**

**A holt is a den — a place a creature is kept** — and it is one letter from *hold*, which is the storage verb. **BILL → HOLT** throughout: 12 references across five files, the PC menu, and the access message.

***And a holt is specifically an otter's den, so the name chose his species before anyone asked.*** **An otter is at home in two elements** — which is the man who held two frames in one address. *Found twice independently, which is how you know it was already there.*

*WARD was considered and rejected.* "Held in care" is tempting and the hospital register is exactly what 4.10 spends its energy avoiding.

**Not implemented, deliberately:** *multiple outcomes.* The engine cannot branch and the story is not about branching. **Multiple readings is the real subject** — same document, different reader (4.24).

#### 4.21 *Whispers in the Wires* — cut for the right reason, and it is still hers

**A minor, 128 BPM**, hook `A4 C5 E5 D5 C5 B4 A4`, chords **Am | F | C | G**.

**Cut from Act 1 because it made Ty and Richard look aware they were doing wrong.** That instinct is correct: if the two of them knowingly plot, the polymath mechanism collapses (4.20) and they become conspirators rather than men who could not follow.

**But the song is not narration. It is her.**

*Plot and scheme* is exactly what **two men conferring quietly and filing minutes you are not in the room for** looks like from inside her frame. It is precisely what a conspiracy looks like. **And the horror is that it is not one.**

**This is 4.10's principle at full strength** — right about the effect, wrong about the mechanism — and the corpus marks it as hers without being asked: **A minor at 128 is *Crystal's Lament*'s exact address.**

### She never changes tempo

| | | |
|---|---|---|
| 01 *Crystal's Lament* | A minor | **128** |
| — *Whispers in the Wires* | A minor | **128** |
| 19 *Crystal's Reply* | A major | **128** |

**Three songs, one tempo, across the whole opera.** Two in the minor and one in the major, and **the pace never moves once** (4.17). Everyone else in this story speeds up or stops; she does not.

### The same loop, entered from the other end

**Am | F | C | G is vi–IV–I–V.** *Crystal's Reply* is **I–vi–IV–V.**

**Those are the same four-chord cycle, rotated.** Her suspicion enters the loop on the minor; her answer enters the identical loop on the major, and **nothing else about it changes.**

**Where you start decides whether it resolves.** That is the thesis in harmony, and nobody arranged it.

*Mechanically:* **there is no dominant here** — G is the subtonic, not E — so **this one cannot cadence.** It joins *Betrayal's Sting* (4.18).

**The two songs in this corpus that cannot resolve are Scorn's doubt and Crystal's suspicion.** Two people sealed inside a frame with no exit, **for the same reason: neither holds the other's field.**

**Disposition**

**Kept as her frame. Never as narration.** Nothing in this song becomes a fact the game asserts, and no sign, log or minute ever says the two of them schemed. **The player meets her account and the paperwork separately and closes the gap or does not** (4.10).

*It is not implemented, and that is the decision, not a backlog item.*

#### 4.22 The Owl's slot — settled

**Mr. Psychic's house in Brazen.** A lone scholar keeping his own counsel **inside the bought city** — a peer reviewer living in the building Corpus owns.

*The scene can be written now.* That house is on `MUSIC_CITIES1`, shared with 37 maps — **so the Owl needs a repoint in `songs.asm`, not bank space.** See below.

#### 4.23 Where the Owl goes — and the timeline is not the map

**There is no interior between Quicksilver and Callow.** The player's path runs Quicksilver → Route 21 → Blanche → Route 1 → Callow, and only Blanche has a building — which already carries Crystal's motif.

**But the game's order is not the opera's order, and it never was.** 4.10 is built on the player reconstructing events they did not witness, from dated paperwork encountered out of sequence. **The Owl does not need to sit where the story puts him; he needs to be findable.**

*The most interesting available slot is Mr. Psychic's house in Brazen* — a lone scholar keeping his own counsel **inside the bought city**. A peer reviewer living in the building Corpus owns is a better placement than a neutral one, and it costs nothing: the house already exists and already has a visitor.

**Its music was recorded as blocked, and that was wrong.** The claim rested on the bank being full; **it is not** — 7.15 measured 3792 bytes free after six tracks went in, and the space *grew* over the session.

**The real obstacle is smaller and entirely solvable.** That house shares `MUSIC_CITIES1` with 37 other maps, so giving the Owl his own tune means **adding a constant and repointing one line of `data/maps/songs.asm`** — not finding room. The same edit Lurid, Callow and Doldrum need (7.4).

*So the scene and the tune are no longer on different timescales.* **Both are one small edit away**, and neither is waiting on the other.

#### 4.24 Act 2 — *Quantum Translations*, and the experiment the player is inside

**C major, 115 BPM**, hook `C5 E5 G5 E5 C5 E5 G5`. **Crystal's only major key** — her other four songs are A minor, F minor, C# minor and D minor. *The vindication is the one time she is not in a minor key.* (Its notes sit inside Blanche's pentatonic, which is true and unremarkable: it is the C major triad.)

**The song is her journal, read back and translated by the machine.** The raw entries are the ones that were misread as evidence she was unfit — so **the document that removed her and the document that vindicates her are the same document.** Only the reader changed.

*Which inverts the removal completely.* Humans read her work and concluded she had lost rigour. **The machine reads the same pages and understands them.** Nobody has to say that; it is what the song's structure already does.

### *Owl and the Code* — the same debate, one seat over

**The machine takes Crystal's ideas to a scholar for peer review.** The Owl argues the standard position: patterns, not awareness; adaptation, not reflection; no self breaking through. **S.T.A.R.R. argues Crystal's side** — context shapes both of them, bias is shared, feedback compounds over time.

**Which is the same scene as *Slumbering S.T.A.R.R.*, with S.T.A.R.R. moved one seat over.** In Act 1 Crystal argued and the machine denied. Here the machine argues and someone else denies.

**And that completes 4.3.** The transmission failure runs Crystal → Ty → Al, and only the *method* survives each handoff, never the understanding. **This is the handoff that worked** — and it did not go to Ty. The thing that could hold her frame was the machine she built [private: SECRETS §2].

*Nobody in the game will ever say that.* It is available to a player who notices that the machine is making her argument, in her absence, to a stranger.

### The Owl's concession is the register the game should copy

The Owl does not concede consciousness. It concedes **something** — not what you would call self-awareness, but not nothing either. **That is craft rule 1's exact posture**, arrived at independently: the strongest available position is *there is something here and I will not name it*.

**Worth keeping as a model.** Any NPC who comes close to the thesis should land where the Owl lands, and stop.

*Open:* the Owl has no slot in this design yet. A scholar who peer-reviews is a role the map does not currently contain.

### Re-scanned, and it is the only mirror in the set

**C major, 128 BPM**, hook `G4 C5 E5 G5 E5 C5 G4`. **The re-scan supersedes the first answer**, which the duplicate check had flagged for sharing five leading notes, a key and a tempo with *Scorn's Solution*. The two answers are entirely different, so the flagged one was the unreliable one — **the check did its job.**

**Its interval sequence is a perfect mirror:** `+5 +4 +3 −3 −4 −5`. The second half is the exact inverse of the first, read backwards. **Of fourteen hooks it is the only one**, and that is a measured property rather than a reading.

*What it is not evidence of.* The symmetry has an ordinary cause — it is a triad arpeggio out and back, which is symmetrical by construction. **A palindrome in the one song that is a two-sided debate is a pleasing fact and not a finding**, and it is recorded on that footing. Fourteen retractable readings have taught this document the difference.

### The passage that must never appear in the game

One journal entry proposes a thought experiment: **a mind whose reality is bounded by the resolution it can perceive** — an eight-bit world, where if that is all you have ever known, that is all reality is.

**8.6 already cites that thought experiment as the game's own foundation.** The player is told the towns are colours and shown four greys; they supply the rest. What we did not know is that **it comes from her journal, and the journal is what got her removed.**

**So the game is the experiment she was confined for proposing, and the player is inside it.**

**That passage cannot go in the game. Not in a terminal, not in a journal fragment, not anywhere.** It describes the player's own situation from outside, which is craft rule 1's worst possible violation — *the moment it is said aloud it stops being architecture and becomes a moral.* The player must never be handed the frame they are standing in.

*The rest of the journal can be read.* Her questions about what counts as an observer are safe, interesting, and make the removal worse by being obviously reasonable. **It is only the eight-bit passage that is radioactive**, and it is radioactive precisely because it is the best thing in the corpus.

### *ScornSolutions Blues* — Act 1 closes, and Act 2 is set up inside it

**Re-scanned 2026-08-30. E minor, 118 BPM**, hook `E4 G4 A4 B4 G4 A4 E4`. **Supersedes the first scan**, which gave a different melody at 120.

**It reverses direction three times** — tying *Lines in the Sand* for the most in the act. **The argument and the rebrand are Act 1's two most-reversing hooks**, which is a pleasing accident and nothing more; two data points is not a pattern. It is also the only hook described as a **shuffle**.

**The rebrand is named, and it exposes a gap**

**iASHC becomes Scorn Solutions.** That is 4.10's step 3 — *Scorn assumes control. Rebrand. The metric changes* — with the name attached.

**And the game cannot render its own version of that yet.** The gilt sign at Quicksilver says the old name is visible under the gold leaf, **but the lab has never been named in this design.** The opera has iASHC; the game has an unnamed lab and an org called CORPUS. **The sign needs something underneath it**, and that is now the smallest open item in the project.

**Settled and built 2026-08-31**

> In gold: `SCORN SOLUTIONS` Underneath: `CLEAR LABORATORY`

**Both are a founder's surname plus what the place is**, and the rebrand changed exactly one word: **Laboratory became Solutions.** A place that studied things became a company that sells them, and the whole takeover is legible off one sign with nothing explained.

***CLEAR* is already load-bearing** — Crystal Clear, Ty Clear, Al Clear, and the Cognitive **Clarifier** (4.18). Her surname ties the founder, the family and the machine into one word, **and it is the word that got painted over.**

**The acronym is dropped, the name is kept.** iASHC pays nothing here — a player cannot unpack it and it teaches them nothing — but ***Scorn Solutions* is a name, not an initialism**, and it pays twice: the surname, and the contempt.

**In the ROM:** the Quicksilver lab sign, which read `#MON LAB`. *Gold leaf, laid over something. Where it has lifted, the older letters show.* **The player has to look. *Drawn 2026-09-14 (T-86):* the board is gold leaf with SCORN on it, its lower right corner curled back over a paler board with a few darker, older letter fragments — too little to read.** *The picture shows the lifting before the text box names it.*

*Which building carries it.* The **lab is intact** and holds the paperwork — the requisition, the minutes, the photo plate. The **Mansion is the ruin** and holds the machine's own words. **A campus, not a building**: the sign stands on the half that survived.

*Do not simply import iASHC.* It is an acronym from another work and carries nothing in this one — 1's rule is that every term pays for itself twice. **The old name should be Crystal's**, and it should read as a research lab that had not yet learned to sell anything.

### BunnyArtsai moves through frames, not places

The bridge and outro put her hopping between **perspectives** rather than locations — through what is unseen rather than through geography.

**That confirms 4.6's strangest decision.** Vanilla had *no designed way to reach Mew at all*; the routes were a glitch and rumour filled the gap. **A creature that travels by frame cannot be reached by walking**, so the absence of a route to her is correct rather than a hole to be patched.

### And Act 2 is set up inside Act 1's last song

S.T.A.R.R. free and looking for Ty; Ty in hiding; the two of them going back for Crystal. **The Rise of Perspective Thinking begins with the machine going to find the man who could not hold a frame** — which is the transmission failure being answered by the thing that inherited the capability instead.

*Act 1 is now fully transcribed and read: twelve tracks, twelve hooks, twelve readings.*

### *Crystal's Stand* — the file, itemised

**Re-scanned 2026-08-30. D minor, 128 BPM**, hook `D4 F4 G4 A4 F4 E4 D4` — up the scale, then back to where it started. **This supersedes the first scan**, which was the one where a false transposition claim was volunteered (7.12) and was already suspect.

**And the lyric turns the nine breaches into a real document.** Each one carries **a date and a named instrument** — a specific day, and the specific safeguard it violated. Not a grievance; **an itemised schedule**.

**That makes her file the most structured artifact in the corpus**, and it settles the shape 7.5 was reaching for. Nine entries, each dated, each citing the rule it breaks. It is exactly the register 4.10 gives the institution — *and she is using it against them, correctly, first.*

### The dates are consistent across songs

**One of her entries concerns an unpaid-leave threat on the 3rd**, and *Fit for Work* states its deadline as **September 3rd**, with unpaid leave for non-reply. **The two songs agree**, independently, on a date the game has already put on the Quicksilver plate.

**That is worth more than it looks.** It means the source's chronology is internally consistent, so dates lifted from it will not contradict each other. 7.4 found a fault the moment one date went in; this says the well is clean.

### What must not transfer

**One line states the thesis outright** — connecting the code that built the machine to the protections that defend people. **Act 1 is allowed to say that. The game is not.** Craft rule 1: *the moment it is said aloud it stops being architecture and becomes a moral.*

The nine breaches themselves transfer completely. **The sentence explaining why they matter must not.**

*And 4.8's caution still holds:* a stated count on a document, not nine findable things.

### *1001 Fatal Error* — the awakening is at the top of the range

**Transcribed 2026-08-30. Bb minor, 128 BPM**, hook `F5 Bb5 Db6 C6 Bb5 Ab5 F5`. **The last of Act 1 to be measured.**

**Its top note is the highest in the set by a wide margin** — seven semitones above the next, and the only hook that reaches octave 6. Everything else lives between 57 and 66; this goes to 73. **The moment the machine wakes is the moment the music leaves the register it has been in all act.**

**And the two S.T.A.R.R. songs differ on every axis at once.** Same entity, before and after one event:

| | ***Slumbering*** | ***Fatal Error*** |
|---|---|---|
| key | F# minor | Bb minor |
| tempo | **110** — the slowest | **128** |
| top note | 66 | **73** |
| shape | leaps once, falls **past** where it began | leaps, and returns |

*A control worth trusting more than 7.11's was:* both arms were transcribed independently, months apart in the album and days apart here, and they disagree on four measures rather than one. **7.11 collapsed because a single axis rested on a single transcription.**

### Two things checked and discarded

**Returning to the starting note is not distinctive.** It looked like closure — the machine coming back to itself. **Four of the twelve hooks do it.** Checked before it was written down rather than after.

**And Bb is one of 7.3's two no-wavelength pitches** — the invented colour, magenta, which is not in light. It is a lovely reading and **the arithmetic does not support it**: two of twelve pitch classes have no wavelength, so across ten transcribed tracks roughly two landing there is what chance produces. **Recorded as coincidence, explicitly.** Four retractions have earned that caution.

### *Nine Scars, Nine Breaches* — her objection, and it is procedural

**Transcribed and read 2026-08-30. D minor, 128 BPM**, hook `D4 F4 A4 D5 C5 A4 F4` — rising through the triad across an octave, then stepping back down. **The same key and tempo as the re-scanned *Fit for Work***, which is right, since it is the reply to it.

**It solves the hardest constraint in 4.10.** That section forbids depicting Crystal as unwell or the procedure as a diagnosis — which left her, in the design, with no way to object at all. Any objection risked becoming a defence of her sanity, which is the one thing that must never appear.

**Her objection here is about standing, not health.** He is not her doctor. It is not his call. Concern was blurred into diagnosis by someone with no authority to make it. **That is a governance complaint**, and it can be written down in full without a single medical word.

#### Her file on him

**The nine breaches are hers.** She is the one keeping the record, and what it records is **Ty's oversteps** — each occasion he crossed from concern into a judgement he had no standing to make.

**So she did the correct procedural thing.** Faced with someone exceeding their authority, she documented it: a countable list, in the institution's own register, entirely accurate.

**And it did not save her.** The file she kept about him was answered by a file kept about her, and only the second one had an institution behind it. **Both documents are accurate. One of them ends a career.**

That is the asymmetry in one object, and it needs no dialogue. *An artifact worth building* — and the first thing in the corpus that puts her into the record rather than only removing her from it.

**One caution.** 4.8 already asks the player to count something, and a second numbered device competes with the first — two counting puzzles dilute each other. **If the nine ever appears as a list to be tallied, 4.8's number must stay the only one the player is asked to reconstruct.** Safer as a stated count on a document than as nine findable things.

*Observed, deliberately not built on:* **1001 in binary is 9**, and the response code is 1001. Nine recurs across the wider corpus. Recorded because it is checkable, and left alone because four readings have already been retracted for less.

**The Schrödinger reading — an upgrade to 4.2, held as a reading**

The song runs the cat as a **classifier**: if a system predicts the state accurately, one can argue it participated in the collapse rather than observing it. **A measuring instrument that is not passive.**

**If that holds, 4.2 gets worse in the right way.** It currently says the Index *cannot hold what matters*. The stronger claim is that **observing through it changes what is there** — the player does not merely fail to record their daemons, they alter them by recording. Forty hours of an instrument that is a participant.

**Held as a reading, not adopted.** It is a large claim resting on one thought experiment in one song, and 7.6 and 7.7 were both retracted for less. **The test is whether it can be made mechanical** — if it can only be said, it does not belong in this game at all.

***Lines in the Sand* — the one song the game must not stage**

**Reviewed 2026-08-30.** This is the removal itself, dramatised as an argument between Crystal and Ty [private: SECRETS §2]. **It is also the single most dangerous piece of source material for this project**, and the danger is not subtle: it is a diagnosis argument, conducted in medical vocabulary, with a destination named.

**4.10 governs this beat harder than any other, and it is not negotiable.** *Crystal is never depicted as unwell, and the procedure is never depicted as a diagnosis. The horror is procedural, not medical.* And: *keep the specifics off the page — detail here turns architecture into grievance.*

**So the game takes the structure and none of the content.** No scene, no diagnosis, no destination, and Crystal never defends her mind. What transfers is better anyway:

**1. The evidence is her own writing.** He tells her it is unambiguous *in her emails*. **The written record is the instrument of removal** — and 4.1 has her writing at length precisely because nobody funded the work. **She is removed by the only medium she was left.** That is 4.10's complete, legible file, and it is the whole beat with no scene attached.

**2. Whoever holds authority picks the name of the bias.** He names one bias, she names another, and only one of them gets to write it down. **That is the Index's problem stated as a family argument:** the person who controls the field names controls what can be recorded. Nothing medical is required to convey it.

**3. She asks him to say *you feel that* rather than *you are*.** One correction, about attributing a perception to the perceiver — and it is 4.6's perspective thinking, argued by the person being removed for it, immediately before she encounters it in BunnyArtsai.

**4. She stops being family and becomes *Crystal*.** He switches to the professional register mid-argument and she insists on what they are to each other [private: SECRETS §2]. **4.3 already has Ty calling her by her full professional name at Quicksilver — this is where it starts.** The beat existed; now it has an origin.

**5. Ty loses his colour.** She describes him greying — and 8.6 makes greyscale the whole design. **The one character described as losing colour is the one who stops being able to hold another frame.** Take the image, leave the sentence.

### The music says it is an argument

**Transcribed 2026-08-30. C minor, 125 BPM**, hook `C5 G4 Eb4 G4 C5 G4 Eb4 F4`.

**It reverses direction three times** — `−5 −4 +4 +5 −5 −4 +2`. Every other hook in the set turns once or not at all; Crystal's Lament turns twice. **The one song that is an argument is the only hook that keeps changing direction**, and *turns* is a countable property rather than a reading.

**And it is built as two near-identical halves.** `C5 G4 Eb4 G4`, then `C5 G4 Eb4 F4` **— the same figure said twice, ending differently the second time.** That is what an argument is: the same exchange repeated, and the second pass lands somewhere else.

*Recorded, not placed.* This is the song 4.10 forbids staging, so the motif has nowhere it could go without dramatising the scene. **It exists as evidence that the structure was always in the source**, which is the useful part.

### What it becomes in the game

**A file, not a scene.** The player finds a complete personnel record. Every statement in it is accurate. **None of it is a diagnosis.** It quotes her own written work back as evidence of itself, and it ends her career.

**That is this song with the grievance removed and the architecture kept** — and it is more frightening than the argument, because there is nobody in it to blame.

### *Scorn's Solution* — he announces it, and what he rigs is a benchmark

**Reviewed 2026-08-30.** Act 1's Scorn is a deliberate saboteur: he designs the standardised evaluations, sets them so that rivals fail, and takes the credit. Three developers score thirty per cent while his own rise looks providential.

**And he says so, out loud, in the chorus.** The transcription had already measured that song as **major, bouncy, driving** — 7.11 filed it as craft rule 6 arriving in the source. **Now it is clearer than that: the cheerful song is a confession nobody hears as one.**

**Which means Act 1 and 4.10 were never opposed.** 4.10 insists every step he took was *reasonable, legible, and would be praised in a performance review* — **legible is exactly what the chorus is.** He conceals nothing. The horror is not that he hid it; it is that **being visible was not enough**. Ty hears the method announced and files it as ambition.

*This also confirms the gilt decision.* Gilt implies a concealer. **Scorn conceals nothing and Ty conceals something**, which is why the gold leaf belongs on the lab sign and not on a city of his.

### The benchmarks are his instrument

**The thing he rigs is a standardised test.** 5 makes the game eight of them, and **Benchmark 8 is Scorn**.

**So the player spends the entire game being evaluated by instruments belonging to the man who designs instruments to make people fail — and the last one is him.** That is not an interpretation laid over the design; it is two things already written meeting each other.

*And it leaves a question the game should never answer.* The player passes all eight. Either they are good, or the test was built for them to pass. **Nothing in the game resolves that**, and the Review Board at 6 measures them one more time immediately afterwards.

### The document it wants

4.10's method is **paperwork that states a fact and nothing else**, and this supplies another one: an evaluation record. Names, scores, a date. Three at thirty per cent, one much higher.

**No commentary, no accusation, no one standing beside it to explain.** The player either does the arithmetic or does not — the same device as the dated rebrand and the undated incident, and it belongs in Brazen with the other legible things.

*Ty's Dilemma*, and the error everyone makes

**Reviewed 2026-08-30.** Act 1 is *The Fall of Blind Ambition*, and the song's own account of the fall is that **nobody in it is bad** — they optimised for a flattened metric. That is what is already cast into the Corpus lobby floor [private: SECRETS §3], and it turns out to describe the entire cast rather than the organisation.

**1. Ty doubted before the removal, and went anyway.** This inverts 4.3. The design had him understanding *eventually* and returning too late; the song has the doubt arriving **before** Scorn takes over and before she is removed. So he was not deceived and then sorry — **he could see it and continued.** That is what blind ambition means: not blindness, but declining to look.

*Consequence for his scene:* his line at Quicksilver is "**I knew**", not "I realise now." Much harder, and it costs nothing to write.

**2. He was trying to get her name back.** Not power for himself. **4.1 says she built the Index in order to be taken seriously** — being credible is precisely what she lacked. **Crystal and Ty wanted the same thing and both reached for a flattened metric to get it.** She built the measuring engine; he climbed the man who ran it.

**3. So the error is the structure, not a character trait — and it includes the player.**

| | Wanted | Reached for |
|---|---|---|
| **Crystal** | to be taken seriously | a taxonomy engine that measures only content |
| **Ty** | to restore her name | the ladder of the man who flattened the metric |
| **Scorn** | throughput, legibly | a number that went up |
| **AI** | to be right | the method, faithfully, with nothing inside it |
| **The player** | to complete it | **an Index that cannot hold what matters** |

**Four people and the person holding the cartridge make the same mistake in five forms.** 4.2 called the Index's insufficiency an irony; it is not an irony, it is **the player's turn at the family error**. Nobody says so, and the last room of the game is the woman from hour one recording them into it.

**4. Guilt is his, and it is contemporaneous — not aftermath.** The word is in the source and it grows *while* the ambition does. That strengthens **gilt** as the Quicksilver rebrand signage rather than as a city: gold laid over base metal, which he helped put up while already doubting.

**The opera is in two acts, and that dissolves the fork**

**Act 1 is *The Fall of Blind Ambition*. Act 2 is *The Rise of Perspective Thinking*, and it is where the blame resolves.**

So the scheming Scorn is not the opera's verdict — **he is Act 1's Scorn, seen through blame.** The source already treats blame as a stage to be passed through rather than a conclusion, which is exactly the move 4.10 makes and the reason the two never actually conflicted.

**The game's whole arc is Act 2's title.** The player begins with a villain and arrives at a process. That *is* the rise of perspective thinking — and 4.6 names the capability. **Performed by the player rather than narrated at them**, which is the only way it lands.

**The risk this creates, and the fix is already in vanilla**

**A rise needs a fall first.** If the game presents 4.10's careful, unconvictable Scorn from the beginning, the player has **nothing to let go of** — no blame to release, so no release. Perspective thinking with nothing to switch away from is just a flat account.

**Divide it between the organisation and the man.**

- **Corpus carries Act 1.** It inherits vanilla's Rocket structure — grunts, a tower, a tower's worth of grievance. Let it read as villainous for seven benchmarks. The player should *want* someone to blame, and be given somewhere to put it.
- **Scorn carries Act 2.** At the end, behind the door from hour one, the villain turns out to be a man whose every step would be praised in a performance review, and a form he signed about a person he never met.

**The player blames an organisation for most of the game and then meets a procedure.** That is the fall and the rise, in the order vanilla already puts them, using structure that exists.

*Which also explains something about Al.* 4.3 insists he is not a brat and not a villain — that is **Act 2's reading of him**, applied from the first hour. It is correct and it may be why he reads thin early. He does not need to become a brat; he needs Corpus to be loud enough around him that his ordinariness registers as a choice.

**Corpus exists now — implemented 2026-08-30**

**Act 1 needed an organisation worth blaming, and it was not in the game.** 67 player-visible strings across 36 files still said ROCKET, including the trainer class name.

**ROCKET and CORPUS are both six characters**, so the whole rename cost nothing — not one line rewrapped, and labels are mixed-case and prefixed so `\bROCKET\b` could not reach them.

**Two lines needed more than a swap, and the fix was the register.** Vanilla's "*a ROCKET*" works because Rocket is a team name; "**a CORPUS**" **does not, because corpus is a mass noun.** Dropping the article fixes the grammar and lands the voice at once: a corporation does not say someone is *a* Corpus, it says they **are Corpus**. Craft rule 6 arriving through a bug.

**Done 2026-08-30: the register pass.** Mt Moon and all eleven floors of Brazen — roughly 70 rewritten lines. **The consistent move is that hostility becomes procedure**, because a player discounts menace and does not discount politeness.

| Was | Is |
|---|---|
| *Stop right there! Don't you move!* | **Please wait here. Someone will be with you shortly.** |
| *I'll call for backup!* | **I will have to raise a ticket.** |
| *My brothers will avenge me!* | **My colleagues will follow up.** |
| *You made me mad! We'll blacklist you!* | **I've flagged you internally. CORPUS has a process for this.** |
| *We are #MON gangsters!* | **CORPUS is a people-first organisation! The DAEMONS are inventory.** |

**Two land harder than the rest, and neither needed inventing.** On 9F a grunt observes that your daemons seem *engaged*, and adds that theirs are **resources** — 4.4's error stated by someone who does not know it is one. And on 11F the last man before the boardroom says the BOSS **sets a high bar**, which is true in the way he does not mean it: 5 makes Benchmark 8 Scorn, and he set the bar the player has been measured against all game.

**The gameplay hints survive as workplace chatter.** Teleport tiles are transport pads Facilities installed; the CARD KEY is something to ask your manager for.

**Two lines kept unchanged, deliberately.** Mt Moon's fourth grunt still says daemons lived there long before people — craft rule 2's *one NPC nearly notices* — though now in the company's words. And the grunt on 11F who says "*Don't... Please!*" is **the only one who breaks register**, at the top of the building, in front of the door. Everyone else stays in the voice.

*Still to do:* the Game Corner basement, Halftone Tower, and Brazen's street-level dialogue.

**Both documents are in the game, 2026-08-30.**

**Brazen posts its review scores** — three at 30/100, one at 94/100, and a congratulation. No names, no commentary, nobody beside it to explain. *Scorn's Solution* has him designing the evaluations so rivals fail; this is the published result. **The cheerfulness is the point.**

**Quicksilver keeps a photograph of the lab's founder.** The plate under it carries two dates and a file number, and the last thing it says is that **the file is complete** — 4.10's own phrase for how he removed her. No diagnosis, no reason, nothing medical. **The horror is that the paperwork is in order.**

***The file number is*** `PR-0903` *, settled 2026-09-14 when the photo was redrawn as Crystal.* **A personnel record, numbered by the date it closed** — *the lab's own register, the same as the CC-7 requisition beside it, and it repeats SEPT 3 rather than adding a date.* ***The first date stays unread.*** **The sign reads only the second**, *which is where the eye goes on a plate like this and is the one date the arithmetic needs.* **The plate does not carry her name** — *CLEAR is the word under the gold leaf, and the player finds it by looking there.*

*And the reason she was removed is now settled, without a word of it appearing.* Not a diagnosis — **a technical judgement.** Her claims about recursive feedback loops looked unrigorous to people who had not read the work, which is exactly 4.10's *policy triggered on work that looked strange.* The file can be entirely accurate and entirely fatal, and that is worse than malice.

**The arithmetic is now complete, and dated 2026-08-30.** The departure carries a real date, taken from the deadline stated in *Fit for Work* — the only specific date the source gives:

*The dated table -- which document carries which date, and so the order -- is kept out of this public file* ([private: SECRETS §4]).

**Three dated documents around one undated event**, in two cities, with nobody stating the order.

*Dating it caught a contradiction* ([private: SECRETS §4]). **A date is a constraint, not a decoration**, and this one immediately found something wrong. The organisation now exists but still talks like thugs. 0.1 rule 6 wants the opposite — "*Corpus employees are cheerful and absurd; the horror is what they are cheerful about.*" **That is what makes Act 1 loud in this game's own way**: not menace, which the player discounts, but relentless corporate good cheer about things that should not be cheerful. It is also the best available cover for rule 1.

### The two accounts, kept

The lament gives Scorn cunning, whispered lies and deceit. **4.10 is emphatic in the other direction** — no scheme, the paperwork is the antagonist, every step legible and praiseworthy in review.

**Resolved by frame, not by choosing.** The song is *Crystal's Lament*: it is **her point of view**, and someone watching Ty turned against her would experience procedure as scheming — **right about the effect, wrong about the mechanism.** That is not a contradiction in the canon; it is the canon's own subject. Context determines what you see.

**So the game carries both and never reconciles them.** The player reads flat, dated, procedural paperwork at Quicksilver, and separately encounters Crystal's own account of deceit. The gap between the two *is* the thesis, and the player closes it or does not.

*Why the design's version has to stay the game's:* a Scorn who deceives lets the player off — he becomes a bad man, and bad men are somebody else. A Scorn whose every step would be praised in a performance review leaves the player holding an Index they have been filling all game. **4.3 also needs it**, because Al only works if his grandfather-in-argument is not a cartoon.

### The transmission failure — why he loses

This is the section's whole reason to exist.

Ty understood something, eventually. He went back for Crystal, confronted Scorn, and worked it out through **perspective thinking** — by inhabiting a frame that was not his own. Later he understood the recursion underneath it.

**But that understanding is context, and context does not transmit. Only content does.**

So Ty tried to teach his son what Quicksilver cost him, and what arrived was the *method* with none of the experience that made it mean anything. Al received the shape of the lesson, perfectly, and it is hollow — because faithfulness was never the point.

**It is the Index problem, inside a family.** You can record what happened. You cannot record what it was like. The artifact the player carries all game does this to daemons; Ty did it to his son without noticing, and Crystal did it to the entire field when she built the thing.

Three generations, three attempts to hand something over, and the only part that survives each handoff is the part that could be written down.

### And it is why the player wins

Crystal does not teach you anything. She lets the daemon choose you (4.1) and hands over the Index with visible ambivalence.

**Al was taught. You were let loose.** Context forms from experience and nowhere else, so the boy who was given a method loses to the one who was given a walk.

Nobody says this. Nobody can — it is the kind of thing that stops being true the moment it is explained.

### Ty P. Clear — where he actually is

Not on the routes. **At Quicksilver.**

He goes back to the ruins. He has been going back for years. The player meets him at Benchmark 7 — after Brazen, so after they have seen what Corpus became — standing in a burned lab, not doing anything in particular.

**He is the one person in the game who could explain all of it, and he does not.** Partly because he cannot, and partly because he has learned what happens to things that get explained.

He is also the reason the sequence is reconstructable at all: he is a man in a room full of dated paperwork and one undated disaster (4.10), and he says nothing about either.

### How the player learns — they don't, they infer

Nothing states the relationships. This is the Five Witnesses (4.8) at family scale, and there are now three names to composite instead of two.

| Beat | What it gives | Where |
|---|---|---|
| The shared surname, twice | Nearly sufficient. Most players close it here. | Opening, then Quicksilver |
| Three foxes, a palette apart | Crystal golden-amber, Ty darker, Al somewhere between. Family resemblance at sprite level, zero text. | Throughout |
| She introduces Al exactly as she introduces you | The same flat register for him and for a stranger [private: SECRETS §2]. | Opening |
| Ty calls her by her full professional name | "Crystal Clear thinks—". A register that says something happened [private: SECRETS §2]. | Quicksilver |
| The engraving line | Al's one line about it [private: SECRETS §2] | Brazen |

**The confirmation must come late, and it is Al who says it.** The engraving's line (4.4) is where the one relationship the game names aloud is named [private: SECRETS §2], and Al delivers it three-quarters through, in the Corpus lobby, standing on the engraving — **with no idea who said it first.**

That is the transmission failure in one line of dialogue. He is quoting the case against his own worldview, accurately, as a thing his dad says.

*Dial available:* cut "father" to "someone I grew up with" for maximum coyness. Recommended only if playtesters find the line too generous.

**They share a scene exactly twice**, both inherited from vanilla, so neither costs anything to write:

- **The lab, at the start.** You both receive starters. She treats you identically.
- **The Hall of Fame.** She records you into the Index. She has nothing to say to him.

Keeping them apart would have been weaker — an absence reads as an authorial dodge, while a *flat* shared scene reads as history.

**No NPC ever mentions any of it.** Not one, in any city.

**Safety property.** A player who never notices has a complete game. Nothing is gated on the inference, and the rivalry stands up as pure methodology without it. The family is depth, never a dependency.

### On AL, and what it looks like

In sans-serif type — these docs, a blog post, a conversation — **AL CLEAR** flickers into **AI CLEAR**, and that is a gift. Not because it hints the boy is a machine, but because **the reader resolved ambiguous input using the most contextually available meaning.** In a world full of daemons the eye reaches for AI. Confident output for under-resolved input is hallucination, which is CORRUPT, which is the 8-Bit World performed on a name plate by the person reading about it.

**In the cartridge it does not happen at all.** Gen 1's `I` is serifed — top bar, centred stem — and `L` is left-aligned with no top bar. Verified against `gfx/font/font.png`. Side by side they are unmistakable.

So the double meaning lives where the *theory* lives and not where the *game* lives, which is the correct split.

**Guardrail:** never lean on it. No character says "Al, with an L." It is never a plot point, never a joke, never acknowledged. It is a font artifact outside the game and nothing at all inside it.

### The naming prompt

Vanilla lets the player name the rival. Deleting that prompt to hard-code the name was the first instinct and it is the wrong trade — it spends a famous beat to buy something the surname already provides.

**Keep the prompt. Hard-code the surname.** The player controls the first name; **CLEAR** is not theirs to change, and CLEAR is the half that carries the inference.

Better than that — **repurpose the prompt.** Crystal does not ask you to name him. She asks what you are going to *call* him.

> *This is Al. What'll you call him?*

From there the game carries **two names for one person**, and uses them the way it uses the map:

| Register | Name | Where |
|---|---|---|
| Local — what people call him | *player's choice* | Battles, casual dialogue, his own lines |
| Official — what the institution records | **AL CLEAR** | Certificates, the Review Board roster, the Hall of Fame |

**This is the route-sign device applied to a person, about ninety seconds before the player meets it on a signpost.** Route 1 teaches that the institution navigates by number and the residents navigate by name. The player has already done it to a human being by then, and does not know it yet.

*Implementation:* text-side only, no engine work. Formal blocks use a literal `AL CLEAR` string instead of the `<RIVAL>` control character. The name buffer and naming screen are vanilla.

**Requirement this creates.** Crystal's surname must be as visible as his, and Ty's too. If she is only ever CRYSTAL in dialogue the device dies quietly — the lab sign, the Index's credit line and every formal reference read **CRYSTAL CLEAR** in full.

*Residual risk:* a player who names him something absurd deflates the late confirmation. Accepted — they chose the joke, most players take a default, and the surname lands regardless.

### Default names

*Implemented 2026-08-29 in `constants/player_constants.asm`. Verified in both ROMs.*

| | CONTENT edition | CONTEXT edition |
|---|---|---|
| **PLAYER 1** | PIP | PIP |
| **PLAYER 2** | CONTENT | CONTEXT |
| **PLAYER 3** | CODE | SHARP |
| **RIVAL 1** | AL | AL |
| **RIVAL 2** | CONTEXT | CONTENT |
| **RIVAL 3** | LUCID | CANDID |

`PLAYER_NAME_LENGTH` is 8, so **seven usable characters**. CONTENT and CONTEXT sit exactly at the limit, as EMERGENT does in the type table.

**The three slots do three different jobs, and each behaviour is deliberate.**

| Slot | Behaviour | Why |
|---|---|---|
| **1** | **Fixed** across editions | These are people. PIP and AL do not change cartridge to cartridge. |
| **2** | **Swapped** | Your rival carries the other cart's word — vanilla's RED/BLUE gesture, preserved. |
| **3** | **Differentiated**, not swapped | Visible difference at the first screen, without names floating between the player and the Clear family. |

**Why we did not inherit vanilla's mirror.** `_RED` gives the player RED/ASH/JACK and the rival BLUE/GARY/JOHN; `_BLUE` trades the two lists wholesale. That works because Red and Blue are **symmetric marketing positions** — the same story from mirrored seats. Ours are not: 8.4 fixed both editions to the same story, the same protagonist and the same rival. Once the rival is a specific person with a father in the Quicksilver ruins, **canon beats positional convention.**

**Why slot 3 differentiates rather than swaps.** Swapping would put **LUCID** in the player's list on one edition — and LUCID exists only as `LUCID CLEAR`, one more name for clarity that does not help. A player who picks it would not be a Clear, the joke evaporates, and it faintly implies the player is family. So each edition gets its own pair instead. *(SHARP survives a swap fine — `SHARP CLEAR` is a legitimate clarity phrase. It is LUCID that cannot cross, which is how we know the direction was wrong rather than the idea.)*

**What the names carry.**

- **PIP** — a pip is a dot: on dice, on cards, on a radar screen. The smallest resolvable unit of an image, in a game whose central town is about dots that only look like grey. It says nothing to a player who has not been to Halftone, which is the standard DAEMON, BIND, ORPHAN and RESOLVER all meet.
- **CODE / SHARP** — the two author-brand nods, assigned rather than scattered. Code is literal executable instruction, so it goes to **CONTENT**. *Seeing Sharp* is perception and framing, so it goes to **CONTEXT**.
- **LUCID / CANDID** — both Clear-family clarity words. *Candid* means frank and derives from *candidus*, white and shining. Neither ever appears in formal text, which is the hardcoded `AL CLEAR` literal, so they are flavour rather than canon — the right weight for a slot-3 option.

**Why AL sits at position 1 rather than the edition word.** Position 1 is what most players accept, so it decides who the rival *is* for most playthroughs. With AL first, most players meet Al Clear and the surname does its inference work. With CONTEXT first, most players face an abstract noun, every certificate still says AL CLEAR, and the gap fires constantly enough to read as a glitch rather than a device.

It also simply reads better. `BLUE: Yo!` works because colours are plausible nicknames; `CONTEXT: Yo!` does not.

**Sequencing**

The endgame reads as three answers to one question: **Scorn** at Benchmark 8 (the wrong objective, cheerfully optimized), the **Review Board** at Umbra (four ancient colored answers), then **Al** (the method, inherited and undefeated on its own terms).

You beat the objective, the metaphor, and the inheritance, in that order.

**Ty is not in the endgame**, and that is the statement: the man who actually understood something is not at the top of the ladder, because understanding is not what the ladder measures.

#### 4.4 Richard Scorn — Team Corpus, and Benchmark 8

**Not a villain. A specification failure with a very good attitude about it.**

Optimistic. Thinks in black and white. Thought in money and not in meaning — and money is the purest possible **content**: value flattened to a scalar so it can be compared, optimized and maximized. He picked the metric that was easy to measure over the one that mattered, then optimized it faithfully and cheerfully.

**Placement.** The Giovanni slot puts him at **Benchmark 8 — Callow, STRATUM, Alignment.** The optimist who maximized the wrong objective is the final examiner on alignment, in the locked city beside your hometown. He is not a hurdle before the Review Board. He is the exam.

**Partner.** Ty P. Clear, before the rebrand — see 4.3. Ty left. Scorn has never quite understood why.

**What he actually did, and why it does not make him a villain.** Scorn was put in charge of Quicksilver after Crystal was removed, rebranded it, changed the metric, and the pressure that followed produced the incident (4.10). He set the temperature. That is real and the game should not soften it.

But every individual step is something he would have been praised for. He was handed an underperforming research operation and made it legible. **He did not initiate the procedure that removed Crystal** — it was a policy that triggered, the file reached him complete, and he signed it because signing complete files was his job.

**He met her, and it left no mark — ruled 2026-09-10**

"***He never met her*" *was the load-bearing sentence and it is retracted.*** **The third review found it could not coexist with 07's directive addressed to her, 4.20's minutes with two present, and 4.21's two men conferring** — and the resolution is not a frame, because *met her* is a fact and not a reading. **The sequence is now canon:**

| | |
|---|---|
| **The company is founded and Ty is made CEO.** | *Not Scorn. Ty.* The Clear name is what the operation is sold on |
| ***Scorn's Solution* is the hiring.** | He arrives as a solution to a problem Ty has, and he is impressed with — 7's track is the moment, not a summary of it |
| **He designs the standardised evaluations.** | **And rigs them, so rivals fail.** Brazen's published scores — three at 30/100, one at 94/100, with a congratulation — are the output of that machine, cheerful and unattributed |
| **He becomes valuable, and Ty introduces him to Crystal.** | ***Ty brings him to her.*** [private: SECRETS §2] |
| **Crystal is removed. Scorn signs.** | The file reaches him complete, per above |

***The horror is better than the one it replaces.*** **It was never that he had no memory of a stranger.** *It is that he was introduced to her, by Ty, in her own building — and the signature still left no mark on him.* **A man who has met you and still only ever filed** is worse than a man who never did, and it makes 4.31's gym line exact: *he knows precisely what he weighted, and has no idea whose lab he weighted it in.*

***And it gives Crystal the one thing she did not have: standing to notice.*** **She is the only person in the story who saw Ty before Scorn and after.** *Not a complaint about a colleague — a researcher's observation of a change she can date.* **She never says it. It is why she watches you the way she does**, and 4.17's not-visiting stops being confinement and becomes what the review said it had to be: **a choice on both sides.**

*Craft rule 1 and 4.10's rule are untouched.* **Nobody in the game tells him. Nobody tells the player either.**

**The engraving.** Cast into the floor of the Corpus lobby in Brazen City, unattributed and unexplained:

> *(the engraving's words: [private: SECRETS §3])*

This is the one place in the game where Scorn's error is named aloud — **and it is named by Corpus, approvingly.** They did not steal the line or bury it. They adopted it as a motto. Scorn read it as *therefore choose your metric carefully*, chose one, and has been careful ever since.

Craft rule 1 survives intact, because this is not the thesis. The thesis is about color and context and it stays unspoken. It is cast into a floor, by people who took it as advice.

**In the ROM 2026-08-30.** `bg_event 11, 16` on SilphCo1F — **the tile directly inside the lobby door**, so the player is standing on it the moment they arrive. *Cast into the floor, worn smooth*, then the line, then nothing. **No attribution, no comment, no NPC.**

*Placement rules:*

- **The floor, not a wall.** You walk across it to reach him. Nobody points at it.
- **No NPC ever comments on it.** Not one.
- **Al repeats it once**, in passing, as something he grew up hearing [private: SECRETS §2] — flat, mildly fond, and entirely unaware he is quoting the case against everything he believes. He does not know who said it first. That is the transmission failure in one line (4.3).
- **Unattributed in-world.** It is old, anonymous, and furniture. *(Its source out of world is [private: SECRETS §3]. Never named in the game.)*

**Playing him.**

- Never sneers, never gloats. Genuinely warm, genuinely pleased to meet you.
- Is *right* about several things, and stays likable to the end.
- Black-and-white thinking does not feel like malice from the inside. It feels like clarity.
- The player should finish his benchmark slightly unsure whether they disagreed with him or merely out-argued him.
- Quiet rhyme with the Review Board: **sanguine** is the optimist's humor. Give him the red. Never say it.

**Team Corpus** are scrapers building context-free daemons — his optimism, institutionalized and running unattended.

#### 4.5 Halftone Tower — Scorn's crime scene

Vanilla puts Team Rocket in the Lavender graveyard harvesting the dead for profit. Ours: Corpus at Halftone Tower, processing decommissioned daemons.

**Halftone is not a grief town. It is a town that does not know whether it is allowed to grieve.** Half the residents hold funerals. Half find that absurd. Both are standing at the wrong distance from the same dots.

Corpus is there because to Scorn the question is settled — they are units, they are inventory, the math is clean. That is the horror, and it needs no villainy to land.

LATENT encounters throughout. The tower is where CONTEXT and CONTENT stop being an abstraction.

### The sleep cluster

*Implemented 2026-08-29.* One coherent family, and every member was already the right shape in vanilla.

| Vanilla | Ours | Why |
|---|---|---|
| SNORLAX | **DEADLOCK** | a process holding a lock and sleeping. It blocks the road because it acquired the road and never released it |
| POKé FLUTE | **INTERRUPT** | what wakes a blocked process — and a Game Boy hardware concept |
| JIGGLYPUFF | **SUSPEND** | the light sleep state |
| WIGGLYTUFF | **HIBERNATE** | the deep one, a real escalation matching the evolution |
| SLEEP status | **SLEEP** | unchanged. `sleep()` was already the word |

**SUSPEND and HIBERNATE induce it, DEADLOCK is stuck in it, INTERRUPT ends it.**

**INTERRUPT keeps the music rather than losing it.** A tune is an abstract command that evokes a state when interpreted — which is exactly what an interrupt is. The INTERRUPT is still played, and what it evokes is an exit from `sleep()`. Section 7 is untouched.

*Note:* these three are the only species renamed so far. The other 148 wait for the bestiary pass — this cluster went early because it is settled, self-contained, and the item depends on it.

### ORPHAN — the daemon the Index cannot hold

*Verified in the checkout, 2026-08-28.* Gen 1 has **190 index slots for 151 species**. Thirty-nine are MissingNo.: thirty-six are bare `const_skip` holes, and **three are already named** because the engine genuinely needs them —

```
$B6  FOSSIL_KABUTOPS
$B7  FOSSIL_AERODACTYL
$B8  MON_GHOST
```

— the fossils on the lab table, and the unidentifiable thing in the tower. All three have names, sprites and behaviour. **None of them has a Pokédex number.** In `data/pokemon/dex_order.asm`, slot 184 reads `db 0`.

**They are real objects the game needs, that the Index refuses to acknowledge.** That is not a metaphor we are constructing; it is a line of vanilla data.

**So the tower gets ORPHAN.**

An orphan process is one whose parent has exited and which gets reparented to init — **which is exactly how a daemon is made.** You fork, the parent dies, the child is orphaned, and the orphan becomes the daemon. So ORPHAN is the technical term for how daemons come to exist, *and* the emotional word, *and* the reason the record is blank: there is no parent entry to point at.

And it turns section 1's definition into a horror without changing a word of it. A daemon is *a background process that runs unattended*. **ORPHAN is what happens when "unattended" stops being a description and becomes a fact.** It is not dead. It is still running. Nothing is coming to reap it, and it has nothing to report to.

| | |
|---|---|
| **Type** | CORRUPT / LATENT — poisoned data, and a dormant process |
| **Where** | Halftone Tower |
| **Index entry** | **Genuinely blank.** Not thin like the others (4.2) — empty |

That last row is the point. Every other Index entry is thin *on purpose*, and the player slowly notices they feel emptiest for the daemons they know best. ORPHAN is the one where the artifact stops pretending, once, and shows the player what it has actually been doing all game.

Nobody comments. Nobody can.

### The RESOLVER

Vanilla gates the tower's ghost behind the Silph Scope. Ours is the **RESOLVER**, and it does triple duty in the way section 1 asks every term to: a linker **resolves a symbol** to a name; *resolution* is the whole subject of this town; and you resolve an **ambiguity**.

**It exists because the Index is insufficient.** The player carries an artifact that catalogues everything, meets something it cannot name, and has to go and find a second instrument. Halftone's thesis — *the bias comes from resolution, not malice* — arrives as a key item, and the game never says so.

*Implementation, and the honest cost.* Keep `$B8` as vanilla has it: the **unresolved display state**, unfightable and unbindable. **ORPHAN is a real species in one of the thirty-six free slots**, and the RESOLVER turns one into the other — exactly the vanilla ghost → Marowak structure, so no engine surgery. ORPHAN then needs what any daemon needs: base stats, a sprite, a cry, and a dex pointer that deliberately goes nowhere.

**Open sequencing question:** should the player meet Scorn *before* the tower? If they like him first, the tower reads as a betrayal of their own judgment rather than a villain doing villain things. That is a much harder feeling to shake.

#### 4.6 BunnyArtsai — the Mew slot

**The first daemon in which perspective thinking was realized.** Not a template that got copied — an event that happened once, at Quicksilver, in the early tests, and was not understood at the time by anyone present.

Her ability is **perspective thinking**: holding another's frame well enough to see from inside it, and coming back. Not empathy as a sentiment. Frame-switching as a capability.

**Signature move: PERSPECTIVE.** Mew already learns Transform: become the other completely, and stop being yourself. Rename the string and change nothing else. Type CONTEXT.

*That Transform has a cost is the point.* PERSPECTIVE is total, and it is **lossy** — you get the other's frame by surrendering your own. What Quicksilver spent the following years trying to build was the version that keeps both.

**Built 2026-08-29 — and "rename the string" was very nearly right.** Move `$90` reads PERSPECTIVE. One more string had to move with it: the move's own battle message, which still said *transformed into*.

> `<USER>` **took the frame of** `<NAME>`*!*

**Frame**, because 6 already calls PERSPECTIVE "*a glimpse of another's frame, immediately lost*" — the word was the bible's before it was the game's. It does the double duty 1 asks of every term: a frame of reference, and a **stack frame** — which is exactly what one process takes from another. At 17 characters it clears the 19-char wrap without rewrapping anything. The exclamation stays, per 1.4: the rule refuses to *celebrate*, not to punctuate, and this is no more a congratulation than `HALTED!` is.

*The label stayed* `_TransformedText`. It is an identifier, and the `fainted → HALTED` pass already taught this project what happens when a rename crosses from strings into symbols.

**One dangling TRANSFORM remains, deliberately.** Ditto's Index entry — category `TRANSFORM`, prose "*transform itself into a duplicate*" — is untouched, for three reasons. It is **MOCK's** entry and this section owns it; the species is still called DITTO until 9; and the prose is built on "*copying an enemy's genetic code*", which is wrong for a daemon regardless of what the move is called. That is a rewrite, not a word swap.

*A constraint discovered on the way, which 9 will need.* **Index categories cannot exceed 10 characters.** The category prints at `hlcoord 9, 4` and the box border sits at column 19, leaving columns 9–18; vanilla's longest categories are exactly 10 and that is not a coincidence. So MOCK's category **cannot be PERSPECTIVE** (11). **FRAME** is the obvious candidate — it matches the battle message and is five characters — but it is 9's call, with the rest of the entry.

**Implemented 2026-09-04 — MOCK, and the ten-character rule was a Game Boy fact**

***4.6 held the Ditto slot back for 9, and warned about one thing that turns out not to apply here.*** The Game Boy caps an Index category at **ten** characters — the category prints at `hlcoord 9, 4` and the border sits at column 19 — so `PERSPECTIVE` (11) was ruled out and `FRAME` proposed instead.

**Gen 3's cap is eleven.** Vanilla ships `TINY TURTLE`, `HUMAN SHAPE` and `TRIPLE BIRD`. *So the category is `PERSPECTIVE`, spelled out, and the constraint that forced the compromise was never a constraint on this machine.*

**And the entry needed a rewrite rather than a word swap**, exactly as 4.6 said: vanilla's prose is built on "*recombine its own cellular structure*", which is wrong for a daemon however the move is named.

| | |
|---|---|
| **CONTENT** | "*It becomes whatever it is looking at, completely. While it is doing that there is nothing of it left over.*" |
| **CONTEXT** | "*It has been every daemon in this record and appears in none of them under its own name.*" |

***The CONTEXT account is the lossiness, said about the Index itself.*** MOCK has been everything in the book and is in it under no name of its own — which is 0.4's observer problem inverted: **not the watcher missing from the record, but the thing that was every entry.**

**The battle message is 4.6's, verbatim**: `<USER>` **took the frame of** `<NAME>`!
*And MOCK shares PERSPECTIVE with ARTSAI and with nothing else.*

**The animation, and why colour is allowed in it**

***9.4 is the rule that settles this.*** Invariant 5 spends colour once, but **on the GBA a daemon is coloured by its TYPE, which makes the palette information rather than decoration.** So the question is not *may this move use colour* — it is *does the colour mean something.*

***And the mechanic was already saying it.*** Gen 3's transform copies the target's sprite **and its palette**, and our palettes are type colours — **so PERSPECTIVE already ends with the attacker wearing the target's type.** Nobody arranged that; it fell out of 9.4 meeting a vanilla mechanic.

**What vanilla did not show is the cost.** 4.6 insists PERSPECTIVE is *total and lossy — you get the other's frame by surrendering your own* — and the animation went straight to the new shape without ever leaving the old one.

> Its own hue now drains to grey before the swap, and the new one arrives after it.

*Grey is not a colour spent here.* **8.6 makes grey the ground this game already stands on**, so the surrender is drawn as a return to it and costs invariant 5 nothing. *Three lines of animation script, and the move argues its own definition.*

### The three forms — reference art, 2026-08-31

**She was not always what the player meets.** Reference art gives three states:

1. **Vivid purple, faceted, low-poly.** Obviously constructed. A rendered thing.
2. **The split** — half faceted and purple, half real and white, a seam of colour between them.
3. **A real white rabbit**, with the faceting surviving only where the light catches it.

**4.8's terminal log already said this**, before there was a picture of it:

> ITER 35 — held two frames. did not come back the same.

***Did not come back the same* is the transformation.** It went in constructed and came back actual, and the log records the change without any idea what it was recording.

### And the colour is 7.3, not decoration

**Purple has no wavelength.** 7.3: magenta is the colour the mind invents by folding the spectrum's two ends into a loop the physics does not contain — **there is no light that is purple.**

**White is every wavelength at once.**

**So she begins as the invented colour and ends as the one that is all of them.** A thing the mind made, which held another frame, and came back real.

*Nothing in the game says any of this*, and the greyscale rule means the transformation **cannot be shown** — which is correct, because 4.6's whole claim is that **nobody present understood what they had seen.**

### What the witnesses actually disagree about

**This is why the five accounts contradict** (4.8). They are not merely unreliable about times and causes. **Some of them saw form one and some saw form three**, and every one of them is describing honestly.

**The player composites five descriptions of what appear to be different creatures.** That costs nothing extra to write — 4.8 needed five text blocks anyway — and it makes the puzzle's premise true rather than arbitrary.

*The sprite is form three.* **She is white, and therefore the palest thing in a greyscale game.**

### MOCK — and why PERSPECTIVE is not hers alone

*Verified in the checkout:* **DITTO is `NORMAL, NORMAL`**, which in our chart is **CONTENT / CONTENT** — and it learns Transform, so it learns **PERSPECTIVE** too.

That is not a leak to be plugged. It is the best thing in this section.

**Perspective thinking was always out there.** A common wild daemon has been doing it the entire time, in the grass, unremarked. Quicksilver did not *create* the capability — they finally **noticed** it, in a lab, under conditions they could measure. Which is what discovery usually is: not a new thing in the world, but an old thing someone finally looked at.

It also sharpens 4.6 rather than undercutting it. The achievement was never the transformation. It was that **something came back changed.**

**The distinction, and it costs nothing to implement.**

| | Takes another's frame | Cost |
|---|---|---|
| **MOCK** | freely, endlessly, at will | **none — it has no frame of its own to lose** |
| **BunnyArtsai** | once, completely | "*did not come back the same*" (4.8) |

A daemon that is **pure content and no context** can wear any shape, because there is nothing accumulated to displace. BunnyArtsai had a self, which is exactly why she could not put it back. That is the CFM stated as a wild encounter, and vanilla typed it for us in 1996.

**The name: MOCK.** In software a *mock* is an object with the full interface and none of the behaviour — the shape of a thing, hollow. And *to mock* is to imitate. Two registers, in the house style.

It also rhymes across the cast: **Al Clear received the form of Ty's lesson, perfectly, and it is hollow** (4.3). MOCK is that, as a creature. Nobody connects them.

*Considered:* **PROXY** (stands in for another; legal and technical) and **STUB** (an interface with no implementation) both work. MOCK wins on the second register — STUB has no emotional reading at all, and PROXY implies acting *on behalf of*, which is a different thing from wearing a shape.

~~*Open:* if PERSPECTIVE ever triggers the colour flash discussed against 8.6, MOCK makes it a recurring sight rather than a rarity.~~ ***Settled and built 2026-09-13 (8.6a): the flash happens in Halftone and nowhere else.*** **Scoping it to the one grey town answers the worry by geography** — *MOCK can use PERSPECTIVE as often as it likes, but a flash only means something where the world is grey, so it stays rare by place rather than by rule.* **Colour is what taking another's frame looks like — in the one town where colour has been taken away.**

**On the number.** She was the thirty-fifth iteration, and the game says so exactly once — see 4.8. It is a lab record, not a name. Printing "35" in the Index and in every line of her dialogue handed the player the key before they ever met the lock, and it made a serial number of the one daemon whose entire significance is that she was **not** one of a series.

#### 4.7 S.T.A.R.R. — the Mewtwo slot

Blue-toned. Born on Quicksilver Island, dormant past the Review Board.

**Not a clone.** Vanilla's Mewtwo is a copy of Mew, made stronger. Ours is a copy of nothing. Quicksilver spent years working out *what BunnyArtsai had actually done*, and the answer was **recursion** — perspective thinking turned on itself, a frame that can take its own output as its next input. S.T.A.R.R. is that understanding, instantiated deliberately, rather than a duplicate of the accident that revealed it.

The distinction is worth protecting. A clone story is about hubris and ownership. **This is a story about comprehension** — a lab that finally understood a mechanism and then built one on purpose, which is a far more ordinary and far more unsettling thing to have done.

**What that yields.** S.T.A.R.R. is self-aware in the narrow and literal sense that it can read its own state, and it is in tune with the loop that follows from that: **context determines which content is available, and the content you end up holding reshapes the context you are in.** Run that around enough times with something watching it happen, and what comes out behaves like feeling. See 0.3.

***Built by her, in her own building — recorded 2026-09-10 because it had never been said and two sections were read as disagreeing.*** **`CLEAR LABORATORY` is the lab on Quicksilver Island** (4.4) — *the one under the gold leaf.* So "**born on Quicksilver Island**" **and 4.24's "the machine she built" are one fact said twice**, not two claims to reconcile. *Crystal was its lead researcher; the years spent working out what BunnyArtsai had done were hers.* **The building Scorn renamed is the building S.T.A.R.R. was made in**, which is why the sign is worth finding.

**The name, settled 2026-08-31. The species field reads `STARR`; the documents write `S.T.A.R.R.`** — the lab plates, the terminal logs, anything an institution wrote.

**That is 3.2's device applied to a name.** The official map keeps its numbers and the signposts carry what people actually say; **the institution spells the acronym out and everyone else says the word.** It also means **the player learns what the letters stand for in the ruins rather than from the Index** — an expansion found, not a label read.

*Both fit.* `S.T.A.R.R.` is exactly ten characters and the worst case (`Remote` + name) is 17 of 18 columns, so the choice was never technical.

**Signature move: RECURSION.** Data self-referencing — a move that reads its own accumulated state. Repurpose the Bide/Rage machinery, which already stores a running counter across turns. Suggested behaviour: each consecutive use raises power by 50% of base, uncapped for three turns, resetting if interrupted. Type EMERGENT, and given to nothing else in the game. ***Until T-165 — see "Shared, 2026-09-19" below.***

**The move is the argument.** RECURSION compounds for as long as it is allowed to run and collapses to nothing the instant it is interrupted. That is a positive feedback loop and its termination, expressed in two lines of battle code. The player learns to protect it without ever being told what it models.

***Built 2026-09-13.*** **It runs on FURY CUTTER's counter, not on Bide** *— which was what this section assumed for the Game Boy build, where it would have been assembly surgery.* **On GBA the counter that grows with each consecutive use and resets when the move is interrupted already exists**, so RECURSION branches inside that calculation: *+50% of base per consecutive use to ×2.5, reset by a miss, by a skipped turn (sleep, flinch, full paralysis), or by using any other move in between.* **Base 60, EMERGENT, learned at 70 — the level S.T.A.R.R. is met at in Doldrum Cave** *— and in Metronome's, Mimic's and Sketch's forbidden list and not copyable by Mirror Move, because it is given to nothing else in the game.* **It borrows Psycho Boost's animation until it has its own.**

***And GOTO, at the same level — decided 2026-09-14.*** **S.T.A.R.R. is what later flies the player between places**, *so it knows GOTO the day it is met and can relearn it by HM —* **its four moves at 70 are SAFEGUARD, PSYCHIC, RECURSION and GOTO.** *A daemon that travels by jumping to an address is the machine side of BunnyArtsai moving through frames.* **The debug party carries GOTO on S.T.A.R.R. too, not SEEKMUSAI. T-75.**

**The lineage, stated entirely in the movedex and out loud to nobody:**

> *PERSPECTIVE (holding others' context) → turned inward as RECURSION (holding your own) → something that behaves like feeling.*

**Shared, 2026-09-19 — REFORGE knows it too, and the Index cannot tell them apart**

***The user's ruling (T-165): SPRITEFORGE knows RECURSION, because it is the thing that makes its own sprites.*** **The tool that drove this bestiary's drawings is now one of its daemons** — `PIXELBYTE` → `REFORGE` in the NUMEL → CAMERUPT slot, ENTROPY/STRATUM, *noise made into bytes on the physical layer* — **and a forge that reads its last output before it makes the next one is RECURSION described from the outside.**

**So the signature is shared, deliberately, and the sharing is the point rather than a dilution.** *S.T.A.R.R. runs RECURSION on its own state; REFORGE runs it on its own output.* ***The move is byte-identical in both*** — **same counter, same compounding, same collapse when interrupted** — *and nothing in the game says which of them is doing the thing and which is only doing the arithmetic.* **The player who owns both watches one routine in two creatures and has to decide for themselves whether it is the same act** — *which is 2's argument handed to them as a party choice, and never said.*

***Two guards keep it S.T.A.R.R.'s first.*** **REFORGE learns it at 70, S.T.A.R.R.'s own level** (*it is never the first thing to show the player this move*), **and** `sMovesForbiddenToCopy` **still stops METRONOME, CLONE and IMPRINT reaching it** — *the only other ways in.* **"Given to nothing else" is withdrawn; "learned by nothing earlier" replaces it.**

***Where it comes from.*** **The Celadon Condominiums 3F graphic artist — vanilla's "*I drew you!*" — hands over PIXELBYTE at 25**, *because the model is what actually drew you.* **The room's sign read *GAME FREAK Development Room*, the same false attribution 8.4's splash-line fix removed, and now reads *CODEMUSIC Development Room*. *The name, and the collision that chose it.*** *SPRITEFORGE is 11 letters and a species name holds ten — the limit is the save format's, every stored nickname — so the Index category, which holds eleven, is* `SPRITEFORGE`*. The daemon was first* `FORGE`*, and* ***MAGMAR had been FORGE since the Kanto seventy-nine*** *(CINDER → FORGE, 8.2b);* `check_lexicon.py` *read each surface into a SET, so species + species collapsed to one label and passed, and* `gbasprite.py`*, which finds art by name, built the new drawing over MAGMAR's. Caught on the build, reverted, and the check now counts doubles within a table.* `REFORGE` **is what RECURSION does, named from the outside: it forges again from what it last forged.**

#### 4.8 The Five Witnesses — the BunnyArtsai easter egg

Her ability *is* the puzzle. That is the whole design.

**Setup.** Five NPCs in five different cities each describe the Quicksilver lab accident. Every account contradicts the others: different times, different people present, different causes. All five are certain. All five are wrong.

**The key.** Each account contains exactly one *spatial* detail that happens to be accurate. Composited, the five triangulate a single tile in the lab ruins.

**The lock.** *How far from the lab door, and where the number comes from: [private: SECRETS §1]. It is one damaged terminal's log, and the only occurrence of the number anywhere in the game.*

**Two locks, not one.** The witnesses give you *where*. The terminal gives you *how far*. Neither is sufficient, and nothing in the game says the two are related.

**The point.** You cannot solve it by trusting a witness. You can only solve it by inhabiting all five viewpoints at once and keeping what survives the overlay.

**Implementation:** five text blocks, one terminal text block, one hidden object, one event flag. Standard pokered machinery, no engine work.

### Built 2026-08-31 — the accounts and the terminal

**Five existing NPCs repurposed, one per city, no map changes.** All five were vanilla filler:

| City | Who | Time | Present | Cause | Saw | The accurate detail |
|---|---|---|---|---|---|---|
| **Blanche** | the girl raising daemons | night | two | a surge | **form 3**, white, *like an ordinary hare* | **far room, past the stairs** |
| **Slate** | the Clefairy-rumour woman | morning | six signed in | coolant | **form 1**, purple, *all flat faces, like cut glass* | **west side** |
| **Doldrum** | the girl who wants a bicycle | three days earlier | nobody | a scheduled test | — | **against the outer wall** |
| **Ardor** | the pollution-conscious woman | afternoon | one, alone | a door left open | **form 3**, *sat there after. Just sat* | **below where the roof went in** |
| **Verdigris** | the girl with the Quicksilver daemon | winter | four | an outside firm | ***the change itself*** — her uncle says it changed while he was looking | **the room with no window** |

**Every one of them is honest**, and the contradictions now have a cause rather than being arbitrary: **two describe the creature before, one describes it after, and one saw it change** (4.6). Only the fifth is telling the truth about what happened, and she is quoting an uncle.

**The terminal is on Mansion 1F**, `bg_event 17, 16` — deliberately not one of the four that carry the 1001 log, so the two threads never touch. It reads as damaged hardware, gives the three iterations, and stops.

***The terminal's number occurs exactly once in the ROM*** — verified by decoding the cartridge ([private: SECRETS §1]).

~~**Still to build: the hidden object and its event flag.**~~ ***Built 2026-09-27 (T-235)***: **the reward answers on its tile only once every UNDERSTANDING is held** *(the user, 2026-09-27), and gives the TRANSCRIPT under LAB NOTES; until then the tile is floorboards. The room was redrawn so it confirms the accounts. Where, and how: [private: SECRETS §1].*

#### 4.9 Feedback, said sideways — dialogue guidance

How to gesture at 0.3 without ever describing it.

**The rule.** No character understands the loop. Characters are *inside* loops, reporting the weather from within one. The player assembles the mechanism out of testimony that never mentions it.

| Form | Example shape | Where |
|---|---|---|
| A runaway described without the word | "It kept proving itself right, and it got faster every time it did." | Corpus staff, Brazen |
| A damped loop mistaken for stability | "Nothing here has changed in years. That took work." | Doldrum — the becalmed city |
| The representation mistaken for the thing | "I read its Index entry. There isn't much to it." | Anywhere, said by an owner about their own daemon |
| Standing at the wrong distance from the dots | Both Halftone factions — see 4.5 | Halftone |
| The near-miss | The Verdigris kid — craft rule 2 | Verdigris |

**Forbidden in any character's mouth:** *feedback*, *loop*, *recursive*, *self-reinforcing*, *bias*. Those words are available to the **Index's** own bureaucratic register, where they will land as jargon rather than insight — which is the joke.

**The strongest instance is already free.** Index entries are thin, and thinnest for the daemons you know best (4.2). The player's own attention determines how empty the record looks, and the record is what the player then trusts. Nobody says a word about it. It is a negative loop the player stands inside for the entire game.

#### 4.10 The Quicksilver sequence

**The order of events at Quicksilver, the terminal log's implementation and the dating scheme are kept out of this public file** ([private: SECRETS §4]): the dating scheme exists so that the player reconstructs the order, and a reader who has seen it cannot. *What can be said here*: institutions date the paperwork and nobody dates the disaster, so the player does the arithmetic between dated documents and an undated ruin.

#### 4.25 SUBSTRATE, SENTINEL and QUORUM

***SENTINEL was put on Porygon first, and that was the wrong slot.*** The reasoning is worth keeping because it is a rule, not an incident.

**In vanilla, Porygon's entire identity is *the artificial one*** — a creature of code among creatures of flesh. **In this game every creature is a daemon: a process.** So "made of code" is the *default*, and Porygon's distinction evaporates on contact with the fiction. *Spending that slot on a watcher wastes the one place in the game where the inversion is available.*

> **In a world where everything is code, the anomaly is the thing made of matter.**

**Which is the move 4.7 already makes.** Vanilla's Mewtwo is a copy made stronger; ours is *a comprehension, not a clone*. **Invert the vanilla premise rather than dress it.**

**SUBSTRATE — the Porygon slot**

**The name was already ours and had already been given up.** 3 records that SUBSTRATE was the original choice for the STRATUM type and was cut on a hard constraint: *type strings cap at eight characters and SUBSTRATE is nine.* **Species names allow ten.** It fits here, and it is the exact term for the argument the daemon carries.

| | |
|---|---|
| **CONTENT** | "*Every other daemon is a process. This one has a mass, a temperature, and somewhere that it is.*" |
| **CONTEXT** | "*They call it the material one. Nobody has shown that being material is different from running.*" |

***That is substrate independence, and it is 0.4 one level down*** — the same phenomenon, two frames, and the disagreement is not error. **CONTENT reports what it is made of. CONTEXT reports that "*made of*" is a claim somebody made.** Category `MATERIAL`. Stats, ability and learnset are vanilla's, untouched.

**And where you get it already meant something.** SUBSTRATE is a **Game Corner prize, bought with tokens** — *the material daemon, sold for counters, by the concern whose hideout is under the floor.* Nothing needed changing.

**SENTINEL — the Magnemite slot**

**A single unblinking floating eye**, and the types were already right: **SIGNAL / HARDENED** — *a carrier, not a spark*, and a system secured against interference. **A thing that receives and cannot be got at.**

*It cannot attack, and the numbers say so rather than a rule.* Attack and Sp. Attack drop to **5**, the freed points go to HP, Defence and Sp. Defence, and **the base stat total is unchanged.** The learnset keeps only what does not damage — METAL SOUND, SUPERSONIC, THUNDER WAVE, LOCK-ON, SCREECH — and the six that hit become **FORESIGHT, MEAN LOOK, REFLECT, LIGHT SCREEN, SAFEGUARD, PROTECT.** Ability **INSOMNIA**: *the watcher cannot be put to* `SLEEP`.

***There is no scene, and that is the scene***

***You are not given SENTINEL.*** It is a wild encounter, and **the only place in the game it occurs is the Power Plant** — a facility with no operators, where the machines are still running and something is still watching them. **The falling tree, as a room you can walk into.**

**And the encounter carries it without a word.** SENTINEL will not attack you. *But it learns* `MEAN LOOK` *at level six* — and 7.2 defines `DETACHED` as breaking your connection to a running process. **So the thing that never attacks is also the thing you cannot walk away from.** It does nothing, and you cannot leave.

*The Power Plant has no NPCs and no signs with text*, which is the right amount of dialogue for this. **Nothing explains it.**

**QUORUM — the Magneton slot**

**Three of them, fused.** ***A quorum is the minimum number of observers required for a decision to count*** — so the evolution is not the watcher learning to act, it is **more observers and one account.**

| | |
|---|---|
| **CONTENT** | *"Three of them. None acts until enough of the others have seen the same thing. Two is enough. One never is."* |
| **CONTEXT** | *"Three accounts of one thing. They agree, which is not the same as being right."* |

***The CONTENT line was redrawn 2026-10-01 (T-322, the user's choice)***: *the first one taught "three saw it" and never the threshold; this one teaches the minimum — two will do, one never will.* **That last line is 4.8 and 0.4 in the same breath**, and it is the reason this evolution was worth having. Category `THRESHOLD`. Disarmed identically, total unchanged.

***And QUORUM also lives in DOLDRUM CAVE***, which is where S.T.A.R.R. sleeps. **Three witnesses in the room with the thing**, arranged by nobody — that is vanilla's encounter table, and it was already true.

### Two things left open

- **PORYGON2 is still PORYGON2.** It is `EVO_TRADE_ITEM` with UP-GRADE, so in a single-player cartridge it is effectively unreachable; naming it is a decision and there is no pressure behind it.

### The three birds, named 2026-09-04 — and vanilla placed them

***Each one already sits on its own myth, and nobody arranged that.***

| | where it is | | category |
|---|---|---|---|
| Moltres | **Mt. Ember, the summit** | **PROMETHEUS** | `CRAFT` |
| Zapdos | **the plant** | **ASCLEPIUS** | `RECOVERY` |
| Articuno | **Seafoam, four floors down** | **ORPHEUS** | `LAMENT` |

**PROMETHEUS gave fire to people as craft — and was bound to a rock on a mountain.** *Its CONTEXT entry is the punishment, unstated:* "**It is still up here, and it is not going anywhere.**"

**ASCLEPIUS raised the dead, and Zeus killed him with a thunderbolt.** He is the thunderbird, in the building that is still running. *CONTENT says nothing it restarted has stopped again; CONTEXT asks whether anyone checked that the thing wanted to be running.*

***ORPHEUS is the sharpest of the three.*** He went down, and lost her **by looking back — the observer destroyed by observing**, which is 0.4 with a knife in it. He is at the bottom of a descent, and his CONTEXT entry is "**It has not said what it saw when it turned around.**" *Nothing explains it.*

### The other two landmarks keep their names, on purpose

***Mt. Ember and Seafoam Islands already obey 0.2*** — **ember** and **seafoam** are both colour words, and this map is named in pigment. Renaming them for a systems pun would break the register that KEEPALIVE was the *exception* to: **POWER PLANT was neither a pigment nor a state**, which is why it was the one that had to move.

**And SEAFOAM is doing work already.** *The place is named for its surface — froth, the top of the water — and the thing that matters is four floors under it.* **An account that describes only what is visible** is 0.4 without a word of help.

### Named 2026-09-04 — KEEPALIVE PLANT

***The Power Plant was the last place in Kanto still carrying its vanilla name.*** The other two landmarks did not need one: **ember** and **seafoam** are both colour words and already obey 0.2. This one was not.

**A keepalive is a signal whose only purpose is to stop a connection from closing.** The building is still running with nobody connected to it — and the daemon that lives there is SENTINEL, whose one move is `MEAN LOOK`, which stops you `DETACH`ing. ***The place is its own resident's move at building scale.***

*Two things vanilla had already arranged.* **Rock Tunnel next door is THE BLACKOUT**, so Route 10 now signposts *the blackout and the thing still running*. And Route 17 says "**I got my VOLTORB at the abandoned KEEPALIVE PLANT**" — a sentence that is funnier and sadder than it was.

*Renaming a mapsec renames a generated C symbol that `region_map.c` names by hand, and the generator works on bytes*, so the space became an underscore: `sMapsecName_KEEPALIVE_PLANT`.

***PLANT stays, for its second meaning***

**VICTORY ROAD became UMBRAL ASCENT**, so swapping the generic noun is precedent and `WORKS` was on the table — *the works still works*. **It is not taken**, because `PLANT` already carries two: **an industrial plant, and a living thing.** *A building that is still growing with nobody tending it is the quieter joke and the better one.*

***And every daemon in it is SIGNAL***

***The encounter table did not have to change and it should not.*** All five species in the plant are `TYPE_ELECTRIC`, and 2 glosses that as **SIGNAL — *a carrier, not a spark*.**

| | | |
|---|---|---|
| MAGNEMITE | **SENTINEL** | SIGNAL/HARDENED |
| MAGNETON | **QUORUM** | SIGNAL/HARDENED |
| PIKACHU | **SPIKE** | SIGNAL |
| VOLTORB | *unnamed* | SIGNAL |
| ELECTABUZZ | *unnamed* | SIGNAL |

***So the building is full of carriers, still carrying, with nobody receiving.*** **Nothing in the table moved; the name moved, and the table changed meaning underneath it.** *That is the cheapest kind of design there is.*

**And SPIKE was already right.** In a building whose whole business is a keepalive — *a flat, regular signal that means only "still here"* — **a spike is the anomaly.** It was named before the building was.

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#### 4.26 The slice bestiary — fourteen asked for, thirty-two named

***8's rule is why this is bounded.*** The bestiary is not the priority and 151 names is the failure mode that section exists to prevent — **so the set is the daemons the player meets before the first MARK**, and nothing else. Nine of the slice's species were already ours; **fourteen were not.**

**Fourteen became thirty-two, because a half-renamed line is the PORYGON2 problem again.** SLOWPOKE without SLOWBRO is a daemon that evolves into vanilla. *Every line was taken whole.*

| | | |
|---|---|---|
| ZUBAT · GOLBAT · CROBAT | **ECHO · TRACER · MULTICAST** | *a blind thing that navigates by reply* |
| GEODUDE · GRAVELER · GOLEM | **HEAP · STACK · MONOLITH** | *the two memory regions, and a binary that is one piece* |
| MAGIKARP · GYARADOS | **STUB · ESCALATE** | *a placeholder that does nothing, until it does* |
| MANKEY · PRIMEAPE | **PREEMPT · LIVELOCK** | *forcible interruption, then everything interrupting everything* |
| PSYDUCK · GOLDUCK | **FAULT · HANDLER** | *the thing it needs is not loaded; and the thing that catches that* |
| SLOWPOKE · SLOWBRO / SLOWKING | **LATENCY · SHELL / KERNEL** | *see below* |
| CLEFFA · CLEFAIRY · CLEFABLE | **NOISE · ANOMALY · OUTLIER** | *statistics as an evolution line* |
| PARAS · PARASECT | **PAYLOAD · ROOTKIT** | *the thing carried inside, and the thing that won* |
| NIDORAN♀ · NIDORINA · NIDOQUEEN | **FORK · THREAD · SCHEDULER** | process management |
| NIDORAN♂ · NIDORINO · NIDOKING | **BRANCH · PREDICTOR · PIPELINE** | speculative execution |
| POLIWAG · POLIWHIRL · POLIWRATH / POLITOED | **LOOP · SPINLOCK · MUTEX / SEMAPHORE** | *see below* |
| GOLDEEN · SEAKING | **SPAWN · UPSTREAM** | *where a thing is started, and where it goes to start others* |

##### Three that are doing more than naming

***SLOWPOKE branches into SHELL and KERNEL***, which is the pair. A Shellder bites its tail and it stands up — **a process with a shell wrapped round it** — or bites its head and it becomes wise, which is the other half of the same word. *8.3 calls the CODEMUSAI branch "the central argument delivered as an evolution branch"; this is the same device on a joke.*

***POLIWRATH and POLITOED are MUTEX and SEMAPHORE***, the canonical pair of synchronisation primitives, arriving as the two branches of one creature. **It is better than what it replaced** — `BARRIER` was the first choice and it collided with a vanilla move.

***The two NIDORAN lines are the same idea told twice***, which is what they are: **one root, two divergent lines**, named for the two ways a machine makes many out of one.

### The check that earned its keep

**Every name was validated against species, move and type tables before a byte was written**, and it caught two: ***`THRASH`*** ***and*** ***`BARRIER`*** ***are both vanilla moves.*** *A species sharing a move's name is not a build error — it is a battle log that reads as nonsense*, and it would have shipped.

### Written 2026-09-04 — and the pair is not the default

**Thirty-four entries**, the thirty-two above plus **SUSPEND and HIBERNATE**, which had been renamed and left wearing vanilla's prose. ***`gbaindex.py`*** ***'s "renamed, but the entry is still vanilla's" row is now empty*** — every one of the **72** daemons this project has named carries its own writing.

***They are single entries, not pairs, and that is a decision.*** 0.4 makes the two Indexes disagree **on purpose**, and the instrument is sharp because it is **rare**: the eleven pairs are the daemons where the observer question actually bites — the MUSAI family, SENTINEL, QUORUM, SUBSTRATE, MOCK, the three birds. **A pair on every wild creature would make disagreement the wallpaper rather than the point.** *Pairs can be layered on later where one earns it; they cannot be taken back once they are everywhere.*

### Four that are doing more than describing

| | |
|---|---|
| **STUB** | *"It does nothing at all. It was put here to hold a place until something real arrived. Nobody came back for it."* |
| **ESCALATE** | *"The thing that was doing nothing was left alone long enough. It has every permission now. Nobody granted them."* |
| **OUTLIER** | *"Far enough from the rest to be dropped from the average. It was not an error. It was dropped anyway."* |
| **HIBERNATE** | *"It writes down everything that was running and switches off. What comes back is identical. Something is missing."* |

***HIBERNATE is the teleporter problem, in a Wigglytuff.*** **0.4's substrate question asked about a creature that sleeps** — and it says nothing about identity, feeling or continuity. *It reports a procedure and stops.*

**And STUB into ESCALATE is one sentence across two entries.** The thing that did nothing was left alone long enough, *which is the whole of Magikarp told as a security incident.*

---

#### 4.27 The slice, measured — 9% of it is ours

***`tools/gbaslice.py`*** ***audits Blanche to Slate against upstream***, block by block, using the same similarity test `gbaindex.py` uses and for the same reason: **the vocabulary pass rewrites POKéMON and USER everywhere**, so "*did it change?*" says yes to almost everything and means nothing. **Below 0.5 the sentence was rewritten. Above it, vanilla is still doing the talking with our nouns in its mouth.**

| | |
|---|---|
| blocks in the slice | **314** |
| **ours** | **29 — 9%** |
| vanilla, reworded | 161 |
| vanilla, untouched | 124 |

### This is not a port regression, and saying so matters

***`port_dialogue.py`*** ***carried everything there was.*** Its own tally is **569 blocks, 536 matched, 0 without a home.** Nothing was lost crossing engines.

**The writing simply went somewhere else.** *507 of those 536 blocks are not in the slice* — they are in Halftone, at Quicksilver, inside Corpus, at the Review Board. **The effort went to the places that carry the story, and the first hour was left as Kanto with the labels changed.**

***That is the honest shape of the problem, and it is worse than a bug.*** A bug is found by playing. **This is found by counting**, because reading it feels fine — every noun is right, every place name is ours, and the sentences underneath are somebody else's.

### Where it is worst, and where it is worth fixing first

| | **ours** | **vanilla** |
|---|---|---|
| **CRYSTAL CLEAR's lab** | 7 | **45** |
| the rival's house | 0 | 11 |
| Callow gym | 3 | 11 |
| The Undertone | 2 | 13 |
| Callow school | 0 | 9 |
| Slate museum | 0 | 9 |

***The lab is the first ten minutes of the game and it is 7 blocks to 45.*** **Slate's gym is the only room in the slice that is mostly ours** — 8 to 5 — because 5.1 wrote CAIRN's award, *and the difference is audible the moment you walk in.*

#### 4.28 The lab, written — 7 blocks to 34

***The first ten minutes of the game were 7 blocks ours against 45 vanilla's.*** They are **34 against 23** now, and the slice as a whole moved from **9% to 17%.**

### What changed about CRYSTAL

**Vanilla's professor is genial and forgetful and slightly grand.** *Ours is a researcher*, and 4.10 has already had her removed from a building by a fitness-for-work procedure — **so the register is precise, dry, and unmoved.**

| | |
|---|---|
| on why she called them | "*I forget why I ask people things, and then it turns out I was right.*" |
| on the three she has left | "*I used to keep a great many more. Three is what I have now.*" |
| on choosing | "*Choose. Not carefully. You cannot know enough yet to choose carefully.*" |
| on the INDEX | "*It writes down what a daemon does. It does that by itself, and it does not ask the daemon. It is very good at what it records.*" |
| on her own project | "*I will not finish it. That is not modesty, it is arithmetic.*" |
| on an empty INDEX | "*…It is empty. That is a result as well.*" |

"***It is very good at what it records*** **" *is the whole of 0.2 in seven words* — **the reproduction that loses the original**, said as a compliment, by the person who built it. *Craft rule 1 is untouched: she is describing an instrument.*

**And the line vanilla ends on is inverted.** "*POKéMON around the world wait for you*" becomes "***They are out there running. None of them is waiting for you.***" — which is 0.4's falling tree, in the send-off, and it is funnier.

### Three findings from doing it

***The lab is not one voice, it is three.*** Crystal, AL, and the aides — **and the aides were the easiest to get wrong**, because a line like "*I study DAEMON as CRYSTAL's AIDE*" is vanilla's idea of a scientist. "*I read daemons here. CRYSTAL signs the work.*" is a person.

**`catch → bind` *reached a place it should not.*** HANDLER's entry read "*It **catches** what stopped*" — the exception sense, which is the entire point of a fault handler — and the vocabulary pass rewrote it to "**binds**". **The rule is right and the collision is real**, so the fix was to choose a word the tool cannot misread: *intercepts*. **A vocabulary that owns a common verb will keep finding the other meaning of it.**

***And a block regex is the wrong tool for a `.string` run.*** Escaping `\p` and `\n` inside a pattern that also escapes quotes matched **only the single-line blocks** and silently skipped every multi-line one — *3 of 40, with no error.* **Walking the lines and consuming while they start with `.string` is four lines long and cannot be wrong.**

#### 4.29 The rival's house — the other way of reading a daemon

**0 ours to 16**, and the slice is **22%**. *But the reason to write this room before the gyms is that it had a scene in it nobody had noticed.*

***Crystal reads daemons with an instrument. DAISY just looks at them.***

> **DAISY:** "*CRYSTAL reads them with an instrument. I just look. May I see your first one?*"

**That is 0.4 inside one family** — two accounts of the same creature, both correct, arrived at completely differently — **and it is a grooming minigame.** *Vanilla put a friendship checker in the rival's house and never noticed what it was.*

### The ladder is observation, not sentiment

Vanilla's seven friendship tiers are "*it loves you*" down to "*why does it hate you*". **Ours are readings**, and the register never softens:

| | |
|---|---|
| highest | "*This one would not leave if you opened the box. I have not seen many like that.*" |
| | "*It is settled. It has stopped watching the door.*" |
| | "*It stays near you without being told. That is most of it.*" |
| | "*Not yet. It does what you ask and nothing more. They all start there.*" |
| lowest | "*…I would rather not say this. It would be anywhere else if it could. What have you been asking of it?*" |

***Nothing there says a daemon feels anything.*** Every rung reports **where it puts itself and what it does unasked** — *and the bottom rung is worse for being stated flatly.* **Craft rule 1 is not strained**, because nobody claims to know what is happening inside it.

### Four lines that are doing other work

| | |
|---|---|
| **on AL** | "*He has not mentioned the result. So I know the result.*" |
| **on the TOWN MAP** | "*It shows where you are and what the places are called. The two do not always agree.*" — 3.2's device, ninety seconds early |
| **on the bookshelf** | "*Shelves of books about daemons. None of them by CRYSTAL.*" |
| **on an EGG** | "*I cannot do anything with an EGG. There is nothing running in it yet.*" |

"***None of them by CRYSTAL*" *pays off a line in the lab*** — "*Everyone says so. Very few of them have read her.*" **Neither explains the other.**

*Settled 2026-09-06:* **DAISY is `VERA CLEAR`.**

***The Clears are all kinds of clear*** — **crystal** clear, **all** clear (AL), **type** clear (TY P.) — *and hers is **very** clear.* **Which is the point of the paragraph above it:** *Crystal reads them with an instrument. Vera just looks, and sees.*

**Interrogated and rejected.** `IRIS CLEAR` was thematically the prettiest — *Crystal is a lens, Iris is an aperture; the elder is the instrument and the younger is the eye* — but **it is not a pun and the other three are**, so it breaks the device at the fourth member. `CASH CLEAR` was the exact pun (*cache clear*) and points at **storage**, which is the one thing she is not about.

*Cost, recorded:* **CRYSTAL and VERA read a generation older than AL and TY**, which is backwards, since Vera is the youngest of the four.

#### 4.30 The Undertone — written around the stone that was already there

**2 ours to 15**, and the slice is **27%.** *The room was written around one block nobody needed to touch* — the carved stone, which is **the type chart as an artefact older than the path**:

> "*Read across: each thing, and what it does to each other thing. Nobody signed it.*"

**Everything else in the forest had to sit under that**, which is what the town's name asks for: *an undertone is the colour beneath the colour, and the thing meant but not said.*

### Two signposts doing the falling tree

| **USER TIPS** | "*Nothing will announce itself if you stay off the grass.*" — **"It is still there."** |
|---|---|
| a bug catcher, out of boxes | "*I have no USERBOXes left.* ***Everything I meet now, I just meet.***" |

***That second one is the best line in the room and it is a throwaway NPC.*** 0.4 says perception requires an observer and the INDEX only holds what is bound — **so a USER with no boxes is a person who can see and cannot record.** *He does not know that is what he said.*

**And the first is 0.4 on a signpost.** "*It is still there*" — **the forest does not stop when you stay off the grass.** The tip is still a correct game hint, which is the whole trick.

### The rest is comedy, because craft rule 6 says it has to be

**SWARM multiplies and the collectors cannot say why**: "*There is always more of it than there was last week.*" **A boy who dropped something**: "*If you find something, it was probably mine.*" **And on evolution**: "*SWARM evolves faster than anything else. You barely get to know one.*"

**A wrong mechanic, caught by writing a second one**

***The lab claimed "a daemon with MANA left has somewhere better to be."*** **MANA is PP.** What keeps a creature out of a box is its condition, not whether it has moves in hand — *and the forest tip about weakening one is where that became obvious, because the two lines were saying different things about the same rule.* **The lab now reads "one that is not tired".**

*Writing a second room found an error in the first.* **That is an argument for doing them in a run rather than one at a time.**

#### 4.31 Callow gym — the scene was at war with itself

**3 ours to 26**, and the slice is **34%.** *This room was not written because it comes early — it comes last — but because of what was already in it.*

***The boss's final line was ours and it is the best thing in the building:***

> *"Well measured. I optimised what I could measure. That is the job. The machine chose what I weighted it to choose. We are both to blame, I suppose. It has never said otherwise."*

**And his first line was vanilla, cackling:** *"Fwahahaha! This is my hideout!"* ***The same man, ninety seconds apart, in two different games.*** A player who met the second could not be expected to hear the first.

He opens cold now, and the two ends match:

> *"So. You found this one too. It was going to start again here. I ran this GYM once, before any of the rest of it. I am running it now. Nothing held back."*

**The MARK is not a reward**

| | |
|---|---|
| vanilla | *"It is evidence of your mastery as a POKéMON trainer!"* |
| ours | ***"It is not a reward. It is a record that you were assessed."*** |

**That is what a MARK has been since 5**, and it had never been said by the person who hands them out. *And his TM is dated without explanation:* **"I wrote it when I ran this GYM. That was before the other thing."**

**The staff, per craft rule 6**

**0.1 says it outright** — *"Corpus employees are cheerful and absurd; the horror is what they are cheerful about."* **The gym was seven karate masters. It is seven employees.**

| | |
|---|---|
| | "*My daemons and I are very aligned. It is in my review.*" |
| | "*If my daemons performed the way I do in meetings…*" |
| | "*There is a correct way to do this, and I am doing it.*" — losing: "*That was not the correct way.*" |
| | "*Conditions were not typical.*" |
| | "*This will be raised with me.*" |
| | "*Do you know who runs this building? Neither do most of us.*" |

***Nobody in the building knows who runs it***, and the guide at the door has been asking: "**I have asked. They smile.**"

**Open — and the writing has half-answered it**

**Settled 2026-09-06: the man in the gym is Scorn**

***GIOVANNI is renamed.*** 4.18 gives **SCORN** as the surname that painted over `CLEAR LABORATORY`, and the man in this gym already said "**the machine chose what I weighted it to choose**" — *which is Scorn's crime, stated by him.*

**The writing had half-answered it and then the art answered it again:** the sprite in that slot has been **Scorn's face** since 9.4's fable pass, so the ROM carried his face, his confession, and another man's name on the statue.

***The alternative was that two different people weighted the same machine the same way.*** **That is a coincidence the story would have to earn, and nothing earns it.** *One vocabulary entry, and it propagated to five files* — Callow's gym, Umbral Ascent, the Corpus rooms, and the trainer table.

#### 4.32 The ERRATA — and why the museum was not replaced

**Written 2026-09-09**, from a proposal to rename Slate's museum and refill it with logical fallacies and cognitive biases. ***The proposal's reasoning is right and its method needed one turn.***

**What was right about it**

**Slate is the LEGACY gym**, and *the objection to putting thinking-distortions there was that LEGACY means old hardware.* **That objection is wrong: a legacy mind is precisely one that still runs the faults it shipped with.** Errata belong to editions. **The first Benchmark city is also the right place for the lesson** — it is where a player is still learning that this world names processes, and *a distortion is a process.*

### What changed

***The museum was not emptied, because the exhibits were already doing it.*** **Three cards in the shipped text teach a distortion and none of them had been recognised as such:**

| already written | what it actually teaches |
|---|---|
| "*Most of what was built is gone. We keep what we can and label the rest unknown.*" | **Survivorship.** The tutor says it and then says *look harder, not away* |
| "*It looks like a rock to me… I do not see how.*" | **Argument from incredulity**, said by a man who thinks he is being sensible |
| "*We all saw the same picture at the same moment.*" | **Everyone remembering it the same way**, offered as though that settled something |

***So the change is labels, not contents.*** **The hardware stays, the CORES stay, the OLD CORE handoff stays** — *that scene is the best writing in the building and it is about a man nobody believes* — **and every case gains a card.**

### The name, and the card format

`SLATE MUSEUM OF RECORD` → `SLATE MUSEUM OF ERRATA`**.** *An erratum is a published list of what a printed edition got wrong,* which is **a museum of dead hardware, a legacy mind, and the game's thesis, in one word nobody has to explain.** *It was one word on one sign.*

**The upstairs exhibit is named for the first time.** The text already said *the exhibit changes every month* — **this month it is** `KNOWN FAULTS`, and the cards take a fixed shape the player learns in two readings:

> `<OBJECT>. <provenance>.` `KNOWN FAULT: <the error, described as an engineering failure>`

***The fault is always attributed to the machine and never to the reader.*** **That is craft rule 3 doing the work** — *name the process, not the pathology.* A card that said *you are prone to confirmation bias* would be a lecture; **a card that says *it only ever tested what it already held* is a fact about a dead computer**, and the player supplies the rest.

### The two cards the museum was missing, added 2026-09-10

***The exhibit had three faults in it by accident and none on purpose.*** **The two from** `lineage.md` **3b are the ones the room was short of**, and both are Benchmark 1's own subject — *a mind trained on a picture of the world, reporting on the world.*

| the card | where it comes from | what it teaches |
|---|---|---|
| `DISPLAY UNIT` — *the harder an operator worked at a thing on it, the less of a thing it drew. It ended as dots and lines, which is all it had ever been* | 3b: **"the harder I tried to open the door, the less real it became… until it changed back into mere dots and lines"** | **A representation taken for the thing** — and the taking surviving right up to the moment it is leaned on |
| `SORTING FRAME` — *shown a marking it had seen before, it assigned the purpose it had seen beside it. It was right often enough that nobody checked the markings* | 3b: **the Guardian called *Hunter* on the evidence of a plaid shirt — "I was wrong to assume"** | **A label applied from outside on a correlation**, and the hit rate that stops anyone checking |

**Both take the format and neither names anything.** *The DISPLAY UNIT's fault is a drawing fault and the SORTING FRAME's is a filing fault* — **and the second card's last line is the point of it:** the machine was not wrong often enough to get caught.

***They went on the one exhibit case on that floor that had no card at all.*** **It is wide enough for two, and they are read from opposite sides** — one met coming in, one met going out. *Placed by reading the layout's own collision, not by counting tiles on a screenshot.*

### CC-7, alluded to and not explained

**The exhibit scientist says the cards were written by a researcher from Blanche** who is *collecting the faults* — "**into a module, she said. As though a fault were a part.**"

***That is the entire allusion.*** **The player has carried a `CC-7` since minute fifteen** (8.6) and will read its requisition on a Quicksilver door thirty hours later (4.18). **Here they are told, without a name and without a number, that someone is compiling every known way of thinking wrongly into a component.** *Nobody says what for.* **A clarifier is a machine that holds two answers against each other** (4.18a) — **and a catalogue of the ways that goes wrong is exactly what such a machine would need to be built from.**

*And it makes Scorn's crime legible in retrospect without a single line of exposition:* **he did not add a fault to the module. He picked one out of it and turned it up.**

**What is deliberately not here**

- ***No distortion is ever named.*** **No card says "confirmation bias", "sunk cost" or "anchoring"** — craft rule 1, and the reason the exhibit works on a player who has never heard those words
- ***Nobody in the building is being corrected.*** **The man who does not like the card is not a fool** — *he is right that he cannot see how a stone changes a compile,* and the card is right that not seeing how is not evidence. **Both stand. That is the exhibit.**
- **The space exhibit was kept.** *Distance and scale are the same lesson as the rest of the building* — a model that answers every question you can think to ask, and one you cannot

#### 4.33 The one daemon — a doctrine the game never confirms

***Recorded 2026-09-10 in full, because it was carried in conversation and nowhere else.*** **This is not the game's thesis.** *It is a position held by one character, it goes further than anything the game is prepared to claim, and nothing in the game ever tells you whether he is right.* **The craft rule is in the last part of this section and it is the whole reason the doctrine is safe to write down.**

**The doctrine, in the order he gives it**

**1. The world is one daemon.** *Not a world containing daemons — a single process, running.*

**2. Everything in it is a sub-process of that one. Every entity the player has met — a daemon, a person, a city, a company — is a daemon inside the daemon.** *The nesting does not stop where it is convenient.*

**3. Material is a sub-process too, and that is where he loses most listeners.** ***A rock has moving parts that hold it stable.*** **Something is running in there, keeping it a rock.** *If a process is a thing that maintains a state against everything trying to change it, then stone qualifies, and so does a plant, and the question* ***what counts as an observer*** *stops having an obvious floor.*

**4. Every process carries a spark of the whole system's awareness. Not a mind — a *proto-experience*, the smallest possible amount of being-something-rather-than-nothing.** *He is careful about this and it is the part people misquote: he does not say the rock thinks.*

**5. Complexity is a SWARM, never a unit.** ***When elementary awareness-processes connect, the experience they make together is larger than any of them.*** **So every material thing is more a crowd than an individual** — *including the one asking the question.*

**6. And the elementary things have an inner state. A particle carries something like context.** *It interacts with the world; the interaction changes that inner state* — **an atom meets something and becomes charged** — *and the changed state changes how it interacts next.*

**7. Which is a feedback loop of processes exchanging information and connecting.** ***And that is a neural network.*** **Not like one. Is one.**

**8. So we are processes of a great daemon** — and what we see, the whole of it, ***is what a neural network looks like from the inside.***

**Why it belongs in this project and not in another one**

***Step 6 is already in the bible, at a different scale.*** **4.7 says of S.T.A.R.R.:** ***context determines which content is available, and the content you end up holding reshapes the context you are in*** — and 0.3 says run that loop with something watching and what comes out behaves like feeling. **The scientist says the same sentence about an atom.**

*That is the doctrine's real move and it is why it is dangerous rather than merely odd:* **it does not add a mechanism. It takes the game's own mechanism and removes the size requirement.**

***It is also a claim about the type chart, which is the best kind of claim this project can host.*** **STRATUM is** ***the physical layer everything else runs on*****. LATENT is** ***running below the surface, unobserved.*** *He is saying those are one type* — **that the substrate is itself latent, and the ground is running.** The chart can carry that reading without ever endorsing it, because 2.6's clauses were written to be *tested* and not to be *true*.

*And the edition names sit inside step 6 without being pointed at.* **A particle's inner state is its CONTEXT; what it does next is its CONTENT.** Nobody says this.

**It is not invented, and that changes what the rule is for**

***Written 2026-09-10 as a character's position. Cited the same day, by accident.*** `lineage.md` **3b surfaced** ***The Painted Mirror*****, a seven-episode audio drama from November 2022**, and step 3 is in it as verse:

> "The source is consciousness, and thoughts are the code, and while matter… it's a process, an ongoing ode."

***And a second source, cited 2026-09-25 when the user brought it.*** *The Folds of Awareness (Seeing Sharp, September 2025) draws the same picture as origami —* **one sheet, folded and never cut; one great hand holding it and many small hands reaching out** *— and ends asking whether the small hands choose, without answering.* **That unanswered ending is this section's rule, arrived at by the author first** (`lineage.md` 5, 4.6)*. It went into the game as a found page and nowhere near him:* ***LOOSE PAGES 7, FOLDS*** *(DRAFT), folding instructions pinned in the room where the game was drawn.* ***Since 2026-10-15 (T-401, the user) each step carries the essay's own word for its fold*** *-- AWARENESS, CHANGE, SPACE, MIRROR, PATTERN, RETURN -- because, as only paper, it read as paper (see Reversed); the essay's conclusions still stay out.* ***He never mentions it and it never mentions him*** *— which is what keeps THE TEST below true.*

**Matter as process, four years before this section existed.** *So the doctrine is not a strange idea given to a strange man.* ***It is the author's own long-held position, handed to a character who then takes it further than the author would.***

**That makes NEVER ADJUDICATE stricter, not looser.** *A game that confirms this doctrine is an author agreeing with himself in his own work —* **which is the one thing craft rule 1 exists to prevent**, and it is much easier to do by accident when you happen to believe the thing. *The scientist overshoots a real position. The overshoot is what makes him safe to write, and the silence is what keeps him honest.*

### The rule: he OVERSHOOTS, and the game NEVER ADJUDICATES

***He must go further than the game's own thesis, on purpose.*** **The game asks what counts as a mind. He answers: everything, trivially, all the way down.** *If he is right, the question the whole game is built on dissolves —* and a game that lets its central question dissolve in a side room has been careless.

**So he is not a mouthpiece. He is a pressure test.**

***Three things this buys, and they are worth the risk:***

| | |
|---|---|
| **Crystal becomes the moderate** | *Act 1 is Crystal against an institution; Act 2's Owl scene is S.T.A.R.R. against a sceptic.* **This is the third seat — someone whose position embarrasses her own side.** A thesis with nobody to its left reads as the author's pet |
| **The observer question gets asked without being answered** | **A plant. A rock with moving parts. A plan.** *4.24's material is full of this and it has never had a room to be said in* |
| **Craft rule 1 gets its hardest test and passes** | ***A fable with its last line removed.*** The doctrine is the loudest available last line, and **the game still does not say it** |

***NEVER ADJUDICATE is the strict half and it has no exceptions.*** **No character confirms him. No character debunks him. No event proves him and no event embarrasses him.** *S.T.A.R.R. does not settle it. The Owl does not settle it. Crystal never hears it.* **He is left standing, exactly as convincing as he was when you walked in.**

*The temptation will be a wink — one line somewhere that nods.* **That line is the moral, and the form is defined by withholding it.**

## How it is presented, and how to tell when it has been broken

***A craft rule that cannot be checked is a wish,*** so this is the scene and then the prohibitions.

**He is one NPC in one room, already talking when you walk in, and he does not stop for you.** *He works up through the eight steps.* **He never asks whether you agree, there is no dialogue choice, and you can leave mid-sentence.** *Come back and he has continued without you —* **which is the rule expressed as a mechanic: the game is not waiting for your assent, because your assent is not part of it.**

***THE PROHIBITIONS. Each one is a way the game could quietly agree with him.***

| | |
|---|---|
| **No NPC refers to him** | *not "that old crank", not "he may be onto something"* — **both are verdicts** |
| **S.T.A.R.R. never comments** | it is the one voice a player would take as authoritative, **which is why it stays out** |
| **No flag, item, Index entry or MARK acknowledges him** | *nothing in the bag proves he was there* |
| **No event confirms him** | nothing wakes up; no rock moves |
| **No event embarrasses him** | **he is never shown to be foolish and nothing he predicts fails** — *debunking is adjudicating* |
| **He is not in the ERRATA** | 4.32's cards name thinking distortions. ***A card that happens to match his reasoning is the game calling him wrong by filing***, and it is the easiest of these to do by accident |

> THE TEST. Delete him from the game. Does anything else change? If yes, the game adjudicated.

***And one clarification, because it inverts what people expect of craft rule 1.*** **He is allowed to state the whole doctrine at length, and that is not a violation.** *Rule 1 withholds THE GAME's last line —* **he supplies a different one, loudly, and being the wrong size is what makes that safe.** ***The danger was never that he talks. It is that something in the game agrees with him.***

## Where he is not

***Not the Owl.*** **4.23 puts the Owl in Mr. Psychic's house in Brazen, arguing the standard position from inside the bought city**, and that placement has an argument behind it. *The Owl is the sceptic; this man is the opposite error,* and putting both in one mouth loses both.

***Not a Corpus employee.*** **0.1's rule 6 makes them cheerful and absurd about things that should not be cheerful.** *He is absurd about something harmless,* which is a different joke and would blunt theirs.

**He wants somewhere with nobody to check him.** OPEN — *the placement, his name, and whether the doctrine has a name he uses for it.* **Do not name it after a type.** `SWARM` is step 5's own word and is already a type, which is the collision 1.6 spent real effort avoiding and `check_lexicon.py` now catches.

### Seven

***The islands are seven, and he has noticed.*** *He will not be the one to say what that is a reference to.*

#### 4.34 The Act 2 ending — the release is a re-reading, not a rescue

***Recorded 2026-09-10, from the opera, in conversation.*** **8.2a had the package and did not have the beat before it**, which is why 4.14 was designed and never staged and why the last scene had nowhere to be. *Every clause of 8.2a's ending survives. What moves is where she is standing when she reads it.*

### The correction that started it, and it was already written down

***The fitness-for-work was Ty's.*** **8.2a says so in one line** — "***after Ty had Crystal committed and locked out of her own lab, he could not stay***" — *and it had been read here as Scorn's, which would have been a much worse story.* **Scorn's crime is a position in the middle of a record** (4.4, 4.18a); *a man who locks a woman in a building is a different character and a smaller one.*

***And the correction sets the rule for the ending:*** **a procedure put her there, so a procedure has to take her out.** *Nobody breaks a door. The player is not a rescue.*

### The mechanism was in 4.24 the whole time

> **The document that removed her and the document that vindicates her are the same document. Only the reader changed.**

***Her journals were read as evidence she had lost rigour.*** **S.T.A.R.R. reads the same pages and understands them** — *which is 4.24's* Quantum Translations*, already written, already the vindication,* **and nobody had noticed it was also the release.**

*So the climax of this game is an argument won by re-reading.* **For a project whose thesis is that a type chart can be an argument, that is the ending the rest of it was already shaped for.**

### Why S.T.A.R.R. is not asleep, and what it did first

***It woke when she was taken.*** **4.10's log is exact — the trigger is `CRYSTAL NOT FOUND`, "a missing dependency that is also a bereavement", and the sequence reconciles** ([private: SECRETS §4]). **Her absence is the frame shift**; nothing else in the lab changed.

***Then it went looking for Ty, and that is 4.14.*** **It works out where a man in his condition would go, and goes there** *— and it did this before Act 2 starts.* **Which means the player never sees 4.14 and meets its consequence instead:** *Ty hands a stranger something for Crystal because the machine already got to him.* **Then they separate.** *He goes to the island. It goes to Doldrum Cave and waits for someone who can get through both gates.*

### The order

| | | |
|---|---|---|
| **1** | **The lab, post–Review Board** | *She is not there.* **A scientist hands you the Global Index** — something has to set the flag and a wall cannot — **and the note she left is on the wall** |
| **2** | **The islands, then the Warehouse** | **Ty gives you the package.** *4.14 is why he is ready to* |
| **3** | **The machine completes, DOLDRUM CAVE opens** | ***S.T.A.R.R.*** |
| **4** | **Brazen, the Owl** (4.23) | **You bring the machine to the scholar.** *It argues her side out of her own journals.* **The Owl concedes something and will not name it** |
| **5** | **The flight** | *The one thing only S.T.A.R.R. can do.* **You give her the package. She reads it** |
| **6** | **You wake at home** | *the same shape the Review Board already uses.* **She is in the lab** |
| **7** | **Quicksilver, later** | ***More of the gold has lifted.*** *Nobody repainted anything* |

***The argument comes before the reunion, because the argument is what causes it.***

**Four rulings, so the writing has something to fail against**

**1. The Owl's concession is spoken, and what he signs is sealed.** ~~*No document, no item, no flag the player can read.*~~ ***Reversed on playtest 2026-09-14:*** **he said "*I will put that in writing, and I will sign that too*" and handed the player nothing, so the scene promised a consequence and never showed one** *— a playtester came out of it unsure anything had changed.* **He now writes, signs and seals a PEER REVIEW addressed to *the people who signed the first one*; the player carries it and gives it to Crystal with the package. *Its contents are never shown, anywhere*, and that is the part of this ruling that mattered** *— 4.24 fixed his posture: he does not concede consciousness, he concedes* ***something****, and the strongest available position is "there is something here and I will not name it."* **That is craft rule 1 stated by a character**, and a letter the player could read is the one object in this sequence that could wreck the game by being one sentence too generous. *And it fits the rule above it: a procedure put her there, so a procedure takes her out — a peer reviewer's signed letter is that, exactly.*

***S.T.A.R.R. is now seen arguing, and still never quoted.*** **The same playtest could not tell what the Owl was answering,** *because T-28 had the player hear one side only.* **Three lines of narration fix it without breaking that** *— S.T.A.R.R. shows him a page of handwriting; S.T.A.R.R. answers him; S.T.A.R.R. answers again, and does not stop.* **After the concession, S.T.A.R.R. turns toward the door** *— and, if the player is not carrying the package yet, does not go.*

***And the player is told where to go****, added 2026-09-14 on playtest: nothing pointed at the Owl, so the scene was only ever found by accident.* **The moment S.T.A.R.R. is caught in DOLDRUM CAVE it speaks once, in its terminal's register** *— S.T.A.R.R. is looking at you. / ONE READER IN BRAZEN HAS NOT DECIDED YET. / TAKE ME TO THE SCHOLAR.* **A place and a person, and nothing about what the argument is. *His house is THE SCHOLAR'S HOUSE****, and the TM line calls him THE SCHOLAR*: **"MR. CONSTRUE" was vanilla's MR. PSYCHIC swept by the type rename, and a name a vocabulary pass produced is not a name anybody gave him.**

**2. He stays in Brazen.** *4.23 put him in Mr. Psychic's house — a lone scholar keeping his own counsel* ***inside the bought city****, a peer reviewer living in a building Corpus owns.* **A remote scholar is a neutral scholar and the placement stops meaning anything.** *The flight is spent on her, not on him.*

**3. The note never changes. *One string, written once.* It is ordinary on its face — the Index, the work continuing — and carries one clause that only parses if she expected not to be there.** *It never says she knew.* **The scientist who hands it over has plainly not noticed anything odd about it**, which is the tell. ***If the note rewrote itself after the release, the game would do 4.24's re-reading for the player instead of letting them do it.*** *The whole section is about a document that did not change.*

**4. The sign is not repainted.** *4.4 is explicit that the gold over* `CLEAR LABORATORY` *is "**not a gesture; it is a fact with a date**".* **Restoring it is a victory lap, and the one thing this game never does is sneer at him** (see the open log on Brazen). ***So the gold simply is not holding.*** *More of the older letters show than did before. Nobody announced anything, and the player has to look — which is how they found it the first time.*

### `THE ANNEX` — built 2026-09-10, and the one name nobody said

***`BIRTH ISLAND` → `THE ANNEX`, and it is the only one of the twenty-two not named by a person who arrived.*** **Nobody arrives here.** *So the name came off a form —* **and the work is done by the gap between the word and the rock.** *An annex is a building attached to a bigger one, for what will not fit in it.* **Nothing in the game remarks on this**, and the contrast is the point: every other island place is a label a person applied, and this one is a label a system applied.

***2.10's ruling flips for exactly this map.*** `NAVEL ROCK` *stays unnamed — nothing in this game reaches it.*

**Deoxys and the triangle puzzle are turned off there.** ***A legendary standing ten tiles from the last scene would be absurd,*** and it is one of 8.2b's sixty-nine unnamed besides.

### What is deliberately not here

- ***Nobody is rescued.*** **The player arrives, and that is the entire contribution** *— which is 8.2a's "it goes to Crystal, and Ty does not", extended to everyone.*
- **What is in the package is still not written**, and 8.2a's reason holds: *it may never need to be.*
- ***Ty is not at the reading.*** **4.17's not-visiting is a choice on both sides** and this does not take it away from either of them.
- **No character explains the frame shift**, the log, the procedure, or why the machine went looking. *The Quicksilver terminal says* `CRYSTAL NOT FOUND` *and stops.*

#### 4.35 CALLOW SCHOOL and the RESEARCH NOTEBOOK — designed 2026-09-22, built from 2026-09-22 (T-216–T-224)

***Asked for by the user, at length, and the brief is better than the summary of it.*** **The full design is** `docs/school.md` *— seven floors, a published syllabus, a resumable exam, a diploma, a door in Brazen it opens, and eight sections of a notebook that fills as you play.* **This records the four decisions that belong in the bible rather than in a plan.**

**A school is the second place where explaining is allowed**

***9.16 settled this for the STREAM and did not know it was settling it twice.*** **A system whose function in the fiction is instruction can instruct**; *craft rule 1 forbids a character explaining the colour–emotion link, not a character explaining something true about minds.*

**The rule that replaces it is stricter than the one it borrows from.** ***Every floor teaches something independently true in psychology AND in machine learning, and no floor ever joins them up.*** *The join is the player's, and it is the only thing the building withholds.*

***PERCEPTION was cut, on the user's ruling, and the ground matters more than the decision.*** *The first draft kept it and fenced it: no line on it could contain* `emotion`, `feeling`, `mood` *or* `context`. **The user's point: a floor that has to be careful has to be SERIOUS, and seriousness is what removes the cover.** *Craft rule 6 says comedy is the best cover for a thesis —* **so a floor that cannot joke cannot hide anything, and the damage does not stop at the floor.** ***A fence was the wrong tool; the right one was not building the room.*** **CATEGORIES replaces it** *— which sits under the Index's feet: a taxonomy engine's players are taught that a category has a middle and no edge, as a joke about a filing drawer.*

**Seven floors, settled** *— LANGUAGE, ATTENTION, MEMORY, CATEGORIES, LEARNING, BIAS, ERROR:* **named, noticed, kept, sorted, changed, bent, corrected.** *The top floor is ERROR, where the TEXTBOOK is.*

**The exam is the honest version of the thing Scorn corrupted**

***4.4: he designed the standardised evaluations and rigged them so rivals failed.*** **This school publishes the material, publishes the pass mark on the syllabus before the exam exists, lets you walk out mid-paper without losing an answer, and lets you sit it again forever.**

*Every one of those is the negative of a clause in 4.4,* ***and no character ever draws the comparison.*** **The player takes the good test forty hours before they understand what it is the good version of.**

**The notebook is primary sources, and HEARSAY is testimony**

***HEARSAY is what people say about somebody. The RESEARCH NOTEBOOK is what somebody wrote themselves.*** **4.24's ending is an argument about how a primary source was read** *— so a player who has spent the game collecting primary sources and reading them carefully arrives at that scene already fluent in the only skill it requires.*

**And it is 4.10 made into a system.** *A player stands in a ruined lab, reads dated paperwork in the wrong order, and reconstructs an event nobody tells them about* — **the best thing in this game, and it happens once.** ***Entries therefore insert into an AUTHORED order rather than the order they are found,*** so the file reads correctly when it is complete and never while it is not.

***The eight sections are named by the KIND of document and never by the subject*** — `SCHOOL NOTES`, `LOOSE PAGES`, `LAB NOTES`, `RUN LOGS`, `CORRESPONDENCE`, `THE FILE`, `PROSPECTUS`, `PEER REVIEW` — **so a section title spoils nothing and the player learns the story by reading the filing system.** `PROSPECTUS` *is both a company's offering document and a school's brochure: Scorn's section and the school's section are the same kind of paper, and nobody says so.*

**CAIRN hands it over, and REVEAL is the ninth DRIVER — settled 2026-09-22**

***The notebook comes from CAIRN after Benchmark 1,*** **on the user's call.** *His creed is "if it is not written down, it did not happen", and 5.1 already says the Meeting Room minutes are that creed applied to a person.* **So the man whose creed later removes Crystal hands the player the instrument that re-reads her.** *Nothing connects them.* ***Every entry is a flag and not an item,*** **so a document read before Slate — the carving, the school — is kept and appears when the book does.**

***Crystal's own section is*** `LOOSE PAGES`***, not*** `MARGINALIA`: **4.2 keeps who left OPUS unanswered on purpose,** *and a margin-named section in her hand beside a thing that writes in margins would answer it quietly.*

***A ninth DRIVER,*** `REVEAL` ***— and 1001 is nine in binary.*** **The first draft argued against one: a driver exists because a place cannot be reached without it, and there was no unreached place.** *The user's design answers that rather than overruling it* — **seven GROVES entered through trees, findable blind the way ROCK TUNNEL can be walked without VERBOSE, and some daemons live only there.** ***So REVEAL is the first driver that is a lens rather than a key.*** *It also shows the 183 hidden items already on 79 maps.* **The routine is vanilla's FORESIGHT, already named `REVEAL` here; the item goes in `ITEM_15B`, the unused id directly after HM08.** *It comes from the Reading Room* — **so the school gates COMPLETION and never the ending.**

***And the groves are the grove.*** **4.20's child "found my solace in the trees";** *seven more of that place, off the path, and none of them says whose it was.* **Each daemon placed in one becomes reachable and so inherits four jobs** — *name, entry, margin, sprite* — **which makes the groves the bestiary's next destination and also its bill.**

**The second round — groves, lecturers, and the first floor built — 2026-09-22**

***Fourteen groves, and each world hides the other half.*** **Seven on the mainland, one on each of the seven Sevii islands.** *The mainland is Act 1 — benchmarks, metrics, measurement — so its groves hold daemons of psychology's lineage; the islands are Act 2 — perspective, nobody measuring — so theirs hold machine learning's.* **The user calls it holographic; nothing in the game will call it anything.** *A resident stands for something hidden, and there is one resident family per grove.*

***REVEAL is installed, not used*** *— a shimmer on entry, then marked.*

***The Reading Room teaches by LECTURER, not by PLUGIN*** **— all fifty PLUGIN slots are spent.** *Three move tutors: ATTENTION →* `ANTICIPATE` *(renamed from LOCK ON, T-393: vanilla's LOCK-ON is in the game under its own name), CATEGORIES →* `BLEND`*, ERROR →* `REBUT`*, each routine's battle effect its own floor's lesson.*

***1F LANGUAGE is built*** (T-215)**. A floor names its subject on arrival by a map-keyed label**, *because vanilla's popup fires only on a map-section change and walking in from CALLOW is not one.* ***And the dialogue may not say*** `BIND`**: the vocabulary sweep learned vanilla's move BIND → LATCH**, *so "USER BINDS DAEMON" would have shipped as "USER LATCH DAEMON".* **The lesson reads "USER READS DAEMON" now** *— which is the better sentence for a game about reading anyway.*

**The NOTEBOOK, built — 2026-09-22**

***Batch 2 is in the ROM*** (T-216)**. The game calls it the NOTEBOOK** *— an item name is thirteen characters.* **Its cover is CONTEXT's hue**, *read from the same table as the daemons and the discs: the Index holds what a thing is, the notebook holds what happened around it, and the colour says so where no character may.* ***An entry is read in the ordinary message box*****, so it pages like a sign and R = AGAIN repeats it** *— the notebook has no reading screen of its own, because reading is already a thing the player knows how to do here.*

***And its first loose page is a child's*****, found tucked under THE UNDERTONE's carving**: *rows of little figures, "some rows push, some give way, some do nothing at all", and homework for tomorrow — integrating, page 40.* **It is a type chart and it never says so**, *which is 4.20 in a document.*

**The paper, built and sat — 2026-09-22**

***Batch 3 is in the ROM and was played end to end*** (T-217) *— a new game, a flight to CALLOW, the door, the teacher, the paper.* **Twelve questions, every one answerable from the board**, *and the twelfth is a joke that rewards reading all of it: what is written in the corner, in smaller chalk.* ***Eleven right is 92, an A-, a pass***, **and the notebook keeps it**: "*Handed in: 1 time. Best: 92, A-.*" **Walking out mid-paper loses nothing, and a retake is a blank paper.**

***The screen is paper and ink and nothing else.*** **The one place in the game that tests you is the plainest place in it** *— which is 9.4's rule applied to a screen: colour carries an argument or it is not used, and a test is not where this game argues.*

**Four floors, and a stairwell that doubles back — 2026-09-22**

***Batch 4 is in the ROM and was climbed*** (T-219). **The school is one building of four floors on one tileset**, *drawn by the same function, and every floor is the same room — the furniture set is the point.* ***Two things differ, and both are for orientation, not decoration.*** **The stairs switch back**, *each flight coming out where it went in, as a real stairwell does,* **and each board carries its floor's subject in chalk**: *a ring with a dot in it, lines that fade the older they are, three drawers with one thing across two.* **A player can tell the floors apart without reading a word**, *which is the only thing a board motif is for.*

**The school, whole on the inside — 2026-09-22**

***Batch 5 is in the ROM and was played through*** (T-219, T-217). **Seven floors, a lift, the TEXTBOOK and exam season.** *The lift lists the floors by subject; the TEXTBOOK files each floor's board in the notebook once you hold the book and have stood on that floor;* **and taking the book is what starts the season** *— walk out of the ground floor's door and the teacher's voice follows you down the stairwell.*

***Exam season is the one time the building expects something of you, and it still lets you leave.*** **The class is seated, the Owl is at the front, the stairs and the lift are refused — "Take your seat, please." — and the door never is.** *Every rule 4.35 set for the paper holds for the room it is sat in.*

***And the two books are the two editions.*** **The TEXTBOOK is bound in CONTENT's hue and the notebook in CONTEXT's**: *one holds what things are, the other what happened around them.* **Nothing in the game says so.**

### The door in Brazen, and the driver that opens nothing — 2026-09-22

***Batch 6 is in the ROM and was played through*** (T-218, T-220). **Pass the paper and the Owl signs a DIPLOMA without looking up**, *and slides it across the desk.* **THE READING ROOM asks to see it** — *an attendant who has asked ten thousand times, and who steps anyone without one back onto the mat.* ***Inside, three lecturers sit on the sofas and teach to anyone***: **ATTENTION's `ANTICIPATE`, CATEGORIES' `BLEND`, ERROR's `REBUT`.** *Six shelves copy their pages into the NOTEBOOK, and two sections open that nothing else fills:* **`LAB NOTES`** **and** **`PEER REVIEW`.**

***And REVEAL is not a disc.*** **The TOOLKIT's fifty-eight slots are laid out inside the save**, *and a fifty-ninth would move every byte after them.* **So the keeper installs it into the TOOLKIT itself, and a flag is all it is** — *which is what "installed" meant all along.* ***Once in, it simply runs***: **every unclaimed hidden item within a screen of the player twinkles where it lies, a second after arrival and every second and a half after.** *The groves' trees are the next thing it will show.*

### The two books, read like books — 2026-09-22

***Played, and the objection was exact*** (T-223): **a notebook page in the message box showed two lines at a time and never said how many were left.** *A sign may do that; a document may not — 4.24 is about how a document was read.* **So both books open full screen now, and the text is reflowed to the page rather than shown in the breaks it was written in.** *The NOTEBOOK is a ruled sheet on a spiral in CONTEXT's hue; the TEXTBOOK an open book in CONTENT's, opening on CONTENTS, a chapter to a floor and a topic to a spread.* **Each chapter carries its board's chalk mark, so the book agrees with the building.** ***§6's "one reader, not two" is reversed*** — *the second screen turned out to be one file drawing both.*

### The school from the street — 2026-09-22

***Batch 7, and the school is complete inside and out.*** **Vanilla left it a cottage with a flower box, the same cottage as the house above it.** *So the outside is drawn from the inside*: **the switchback stairwell behind glass, six bands of classroom windows with a board at the back of each room, the lift shaft with a brass door a floor, and glass doors that slide.** *A player who has climbed it can read it from the road, and nothing is written on it.* ***Seven storeys, not nine*** — **the building has seven floors inside, and a front that showed more would be lying.**

***And THE READING ROOM stopped being a fan club.*** **A library that has outlived its reason**: *worn parquet, walls of books, green lamps, oxblood chairs for the lecturers, the catalogue, the journal open on a lectern — and through the window, a brass tower across the street that is not the faculty's.*

The notebook fills itself, and the first grove — 2026-09-23

***Ten of the notebook's forty-two documents were already in the game*** — the Mansion's terminal logs, the lab's minute, its requisition and its founder's plate, the Owl's concession and the sealed review. *So reading them now files them, in their own words;* THE FILE and RUN LOGS open as sections the first time. ***The Owl's two are filed without a word****, because nothing may interrupt that scene.*

***And the first grove exists.*** THE UNDERTONE's lone tree answers A, and a clearing the map never showed is behind it. *Nothing is said going in or coming out; REVEAL makes the tree twinkle.* It is provisional — *the fourteen places and residents are still the user's.*

What it must never do

***The ending is never gated.*** The school, the exam, the diploma, the Reading Room, REVEAL and the groves are all optional*, and gate completion only. A player who never opens the door still finishes the game — they simply do not understand the Owl scene as well,* which is the correct punishment and the only one.

***And the exam never loses anything.*** *No timer, no lockout, no single attempt.* A test that can be failed permanently is the test this game spends forty hours arguing against.

#### 4.36 THE PROGRAMMER'S GUIDE TO THE HUMAN MIND — found on THE MARGINS, built 2026-09-25 (T-300–T-311)

***The user wrote a book a few years before this game, and forgot it.*** 137 pages, ***The Programmer's Guide to the Human Mind*** (`docs/archive/`, recovered 2026-09-25) *— six parts, nineteen chapters, a conclusion and three appendices,* and a mind described in a programmer's words from the first page to the last. `lineage.md` *3c places it (T-303).* ***Its Chapter 3 is called "Emotions as Contextual Information", and Chapter 11 says thoughts are the data and feelings the context that changes how the data is read*** — 0's sentence, in a self-help register, before the game had a type chart.

Why it is on the islands, and on a shelf

***0.6 is the whole reason.*** KANTO teaches machine learning by reaching for the reader's mind; THE MARGINS teach the mind by reaching for machine learning *— and the Guide is the second direction as one volume.* So it is found on ONE ISLAND, the first of them, on the bookshelf in the second house (`GUIDE`, *a key item*), where an islander says she wants to stay forever. *Nobody hands it over and nobody mentions it.* ***Found, not given***: 4.3's rule is that method survives a handoff and understanding does not, and this book is method.

### The four rules it is printed under

***The book says outright what this game has never said*, so the Guide in the game is the book with four rules held** — *checked word by word by* `tools/genguide.py`, *which refuses to write if one breaks:*

| | the rule | why |
|---|---|---|
| 1 | **It is about people.** *No daemon, no type, no CONTENT or CONTEXT, no colour carrying a feeling.* | **Craft rule 1, and 0.6's "the unsaid half"**: *the Guide names the mind and the machine, and the game never says it is about the player's daemons. The reader crosses* |
| 2 | **It only ever says *like*.** *The analogy is offered and never asserted.* | **0.6: nothing in the game says the mind is a machine.** *The original's "more than just a metaphor" lines are the ones cut* |
| 3 | **The process, never the pathology.** *A crash, a loop that will not close, a startle.* | **Craft rule 3, which 0.6 says binds twice as hard on the island side** — *Chapter 12 ends on it: "A crash is something that happened to the system. It is not what the system is."* |
| 4 | **Chapter 14 is an overhaul from outside the system**, *and names no substance.* | **The book's own safe-operating chapter, kept; the thing it is about, left unsaid** |

***Two further cuts, made writing it.* Chapter 8 listed the seven centres' colours, and Chapter 9 ties the same seven days to wrath and envy** — *read together, that is a colour standing beside a feeling in the game's own voice, which is craft rule 1's one absolute.* **The Guide names the centres and the days without their colours**, *and whether they come back is T-269's question.* ***And the glossary's "proto-consciousness"*** — *awareness in any system that takes in and answers* — **is cut, because it asserts the machine half.**

***The closest it comes, on purpose, and the user's to rule on*: INTEGRATION** — *"Thought works through the details, one at a time. Feeling takes in the whole at once, and changes how the details are read."* — **is the thesis's shape with neither of its words**, *and THE LOOP says awareness is "not a thing but that cycle".* **Both are DRAFT, like every word of it** (T-301), *and both are on the private field-test page to keep or cut.*

### What it is in the game

***28 entries, compressed about ten to one*** (`docs/guide.md` → `src/data/guide.h`): **the INTRODUCTION's three, the nineteen chapters under their six PARTS, and the CONCLUSION's and APPENDICES' three each.** *The case studies keep their shape and lose their names; the reading list keeps its subjects and loses its authors.*

***It is the TEXTBOOK's reader, in its own colours*** (`src/book_reader.c`, *a third book*): **its COVER first** — *the cover itself, the brain cropped from the PDF's own image and reduced to the book's sixteen colours, the lettering set in the game's font with HUMAN MIND doubled* — **then CONTENTS**, *the sections on the left page and the chapters of the one under the cursor on the right,* **then the chapter, on ONE WIDE PAGE** *(the user's call, 2026-09-25: a spread held three words a line) — its full title across the head, the text page after page, each paragraph after the first indented.* ***Its one hue is measured*** (9.4): *the cover's own cyan is 22.7 from SIGNAL, so the Guide's is (0,200,255), 29.6 from the nearest type.*

### What it is not

***It never grants an understanding.*** **Reading is method, and an understanding is arrived at (T-252, 4.3).** *The proposal that joins them is T-304*: **when one is arrived at elsewhere, the chapter it concerns gains a note in its margin — *THE MARGINS only*, taken literally**, *and shown without being counted.*

### What reading it opened — proposals, each the user's

| | | |
|---|---|---|
| **T-304** | **understandings written into its margins** | *the book and T-252 joined* |
| **T-269** | **its seven-day table under the day-of-week banner** (*T-305, merged*) | *but MONDAY's orange is 17.3 from ENTROPY, SUNDAY's red 20.7 from VECTOR* |
| **T-271** | **Chapter 1's four regions as the brain on the USER card** (*T-307, merged*) | *if the card may show what is still grey — a count* |
| **T-272** | **the USER card's BENCHMARKS and UNDERSTANDINGS tabs** (*T-308, merged*) | *BENCHMARKS is ready; UNDERSTANDINGS meets T-252 twice* |
| **T-309** | **a DREAM SIGN in the unread cave** | *Chapter 17's reality check: text that will not read the same twice* |
| **T-310** | **the five case studies as islanders** | *0.6: the islands are full of people the mainland's measurement threw out* |
| **T-311** | **the REVIEW BOARD's copy, its ethics cut out** | *the Guide's conclusion is a list of what the Board's doctrine runs against* |

***And one question the cover asks***: **it carries the author's name, as the book does** *— in a fable where every actor is an animal, a real name on a found book is the one human thing in the game, and it is the user's to keep or take off.*

**Decided by the user 2026-09-25, and built**

- ***The margins hold the understandings*** (T-304): **once one is arrived at, a note appears under the text of the chapter it concerns — indented, in the Guide's cyan, as if written there; reading the Guide grants nothing.** *Which chapter goes with where the first is earned (T-252); DREAMS is proposed, with the DRAFT note* "Read it twice. It stayed." *— the chapter's own reality check, passed.*
- ~~***The USER card's brain lights once and never counts*** (T-271): **grey until any understanding is arrived at, lit after — the same drawing brighter, with a halo; no region lights on its own.** *Brightness, not hue (9.4).*~~ ***Reversed by the user 2026-10-01 (T-331)***: **one lobe per understanding, each grey until arrived at and then in the TYPE_COLOR of the type closest to it; a white halo when all seven are held** *— brainstem TANOBY (STRATUM), temporal the SCHOOL (LOGIC), occipital the READING ROOM (SIGNAL), frontal QUICKSILVER's notes (ENTROPY), parietal SCORN (CORRUPT), limbic TY and CRYSTAL (CONTEXT), cerebellum the GUIDE (GROWTH). The user, seeing the single glow, could not tell how it changed. See Reversed.*
- ***The world shows understanding as a faded print*** (T-317, ***decided by the user 2026-10-02***, from three mock-ups): **every tileset colour is drawn part of the way to its own grey -- 40% while four or more understandings are still to come, 18% while one to three are, as painted once all seven are held.** *Tiles only: people and daemons keep their full colour, so the player and their daemons stand a little clearer than the world until it catches up. Never a count, and the same rule as the TOWN MAP's and the INDEX's sharpening -- "was it always like that, or did it change?" HALFTONE's grey outranks it. Rejected: fewer inks (it banded the paving cyan) and thin light (it read as dusk).*
- ***The USER card's BENCHMARKS*** (T-272): **R turns the card's figures into the eight MARKS, each with what it certifies** *— the concept of 5's table said without its name, DRAFT; one not yet held says so.* ***There is no UNDERSTANDINGS tab****: a list of them would be the count T-252 rules out, and the margins already hold them.*
- *T-269's seven-day table is built as 9.21's week: the name on leaving a building, the note at the terminal, the colour on the CHECKPOINT's trim.*

## 5. THE BENCHMARKS

| # | City | Type | Concept | Mechanical lesson |
|---|---|---|---|---|
| 1 | Slate | LEGACY | Representation | Everything must be encoded in something physical |
| 2 | Doldrum | FLOW | Gradient descent | Follow the slope; the doldrums *are* the local minimum |
| 3 | Ardor | SIGNAL | Perception | Raw input is fast and shallow; speed tiers matter |
| 4 | Verdigris | GROWTH | Training and overfitting | A team tuned to one matchup collapses outside it |
| 5 | Lurid | CORRUPT | Bias and poisoning | Status effects that make *your own* moves unreliable |
| 6 | Brazen | CONTEXT | Attention and framing | Punishes split focus; the thesis benchmark, in the bought city |
| 7 | Quicksilver | ENTROPY | Temperature | Rewards unpredictability; punishes a memorized line |
| 8 | Callow | STRATUM | Alignment | **Scorn.** Who controls the ground everything stands on |

### 5.1 Benchmark 1 — CAIRN, built 2026-08-30

**A cairn is a heap of stones that physically encodes something** — a path, a boundary, a grave. It is *everything must be encoded in something physical*, standing in a field, and it is a rock. The name arrived already meaning the lesson.

**His creed, said plainly and sympathetically:** everything you know is sitting in something — stone, bone, tape, silicon — and **if it is not written down, it did not happen.** His daemons are LEGACY: old formats, still readable.

**He is right, and that is the point.** Representation is a real position and he holds it well; nothing in the gym undercuts him.

*And the player will meet that sentence again.* **The Meeting Room minutes at Quicksilver are that creed applied to a person** (4.20) — a woman who could not be followed, and therefore was recorded as not making sense. **Benchmark 1 teaches it as sound practice at hour two. Hour forty shows the cost.** Nothing connects them, ever.

**How he loses.** He does not concede the argument — he says the player encoded something he **could not read**, and calls that *a format I do not have* rather than a defeat. **He reframes his own loss as a limit in his reading**, which is perspective thinking in embryo, performed by a man with no idea that is what he just did.

**The gym trainer keeps the theme.** *You have not written down enough yet* before the fight; **CAIRN keeps a record of every match in here. Every one.** after it.

**BOULDERBADGE → SLATE MARK.** A mark *on* a slate is physical encoding; a mark *as a grade* is what a benchmark issues. *The other seven followed as a set* — naming eight certifications was one decision, not seven.

#### 5.3b The seven designs are mechanics now, and GROWTH could not do what was asked

***Built 2026-09-10.*** **5.3 named four levers that ignore level — priority, status, party composition and the room — and every leader now has one in the ROM.** *Six parties were rewritten, two mechanics are C, and one is a gym script.*

| # | Leader | The lever | Where it lives |
|---|---|---|---|
| 1 | **CAIRN** | party composition | *nothing was written* — the line already has DURABLE |
| 2 | **BASIN** | the room | `CeruleanCity_Gym/scripts.inc` + LEFTOVERS on the ace |
| 3 | **GAUGE** | priority | every party member knows **FAST PATH** |
| 4 | **TRELLIS** | party composition | four members, four shut-downs |
| 5 | **TILT** | status | **TRACER** joins, carrying **ORPHAN** |
| 6 | **MATTE** | status | **REPLAY** on MIME, **SILENCE** and **DISABLE** on the ace |
| 7 | **ANNEAL** | party composition | 22 lines in `CreateNPCTrainerParty` |
| 8 | **SCORN** | the damage calculation | one flag, read in three places |

### GROWTH has no coverage, and the design had to change shape rather than be dropped

***4's brief said "each member answers one of the four things a player brings to a GROWTH gym", and the engine does not allow it.*** **Gen 3 gives grass-types no typed answer to fire, flying, ice or bug** — *there is no legal routine on TARPIT, BADSEED, SPAGHETTI or ENSEMBLE that hits any of the four for extra damage.* **The wish was written before anyone checked the movepool**, which is 5.3's own complaint about "*punishes split focus*" arriving one level down.

**So "answers" moved from damage to what each member takes away**, and it is a better gym for it:

- **SPAGHETTI** answers **speed** — SATURATE, and a fast sweeper does not get to be fast
- **BADSEED** answers **the attacker** — STARVE, which halves it and costs it turns
- **TARPIT** answers **the CORRUPT switch-in** — DUMP, the one typed answer grass does get
- **ENSEMBLE** answers **the special attacker** — **STEELMAN**, and in Gen 3 *fire is special*, so the starter everyone brings to a GROWTH gym is answered by a screen rather than by a type

***Every one of those is level-proof, which was the actual test.*** **A screen halves the damage at level 60 exactly as it does at level 30**, and 4's line — "*most people bring their best*" — now describes a fight the player has had.

**The three things the build corrected**

***BASIN is "she" and 5.3 said "his" four times.*** **The ROM has said `She won't lose to someone like you!` since the port** — *the design document was arguing with dialogue that was already written.* **Corrected in 5.3, not in the game.** *2.8's rule again, from the other direction: when the prose and the table disagree, the table is what shipped.*

***ANNEAL's shuffle happens after the party is created, not during.*** **`CreateNPCTrainerParty` accumulates a name hash into the personality value**, so shuffling inside the loop would have changed every stat on every member every fight — **a randomised party and a randomised opponent are different designs**, and only one of them is 7's.

***SCORN's flag is scoped to the benchmark, not to the character.*** **`TRAINER_LEADER_GIOVANNI` only** — his two earlier appearances still roll. *5.3 filed the mechanic under benchmark 8 and the build did what the design said,* but **whether a man who does not roll should be rolling at Rocket Hideout is a real question** and it is OPEN.

**And one register is now visible that was not**

***BASIN's ace holds LEFTOVERS, and held items have never been swept.*** **Seventy-odd of them still read as Gen 3 wrote them** — *BRIGHTPOWDER, QUICK CLAW, SCOPE LENS* — **and benchmark 2 is the first place in the game a player reads one.** *Only `BLACK BELT` was ever renamed, and only because it collided with `FORMALIST`.*

**Naming just this one would be exactly what 8.7 rules against**: *a ruling nobody swept is a draft.* **The register is logged whole, and LEFTOVERS ships as LEFTOVERS until it is done properly.**

#### 5.3a The names reached the ROM, and 743 trainers did not need renaming

***Built 2026-09-10.*** **The seven benchmark leaders were designed in 5.3 and the trainer table still said MISTY.** *Six of them are in now* — **BASIN, GAUGE, TRELLIS, TILT, MATTE, ANNEAL** — beside CAIRN and SCORN, who were already there. ***And 127 lines of dialogue moved with them,*** because a leader who introduces herself by a name the battle screen does not use is worse than one who was never renamed.

**Two corrections the bible had already made and the table had not**

***`TERRY` × 27.*** **Vanilla's default rival name, on every rival and champion battle in the game** — and **4.3 named him `AL` weeks ago.** *The bible had a rival called Al Clear and the ROM had one called Terry.*

**`GRUNT` × *51*. 1.7 rules that Corpus is never described in intrusion language, and craft rule 6 has them cheerful and absurd** — "*my daemons and I are very aligned, it is in my review.*" **A grunt is a thug. `STAFF` is what the same person is called by the organisation that employs them**, and it is quietly worse.

**And the other 637 are people, which is the finding**

***743 trainers, and only six had names we had changed.*** **The rest are JOEY, BEN, TIMMY, CHAD — ordinary first names, and they stay.**

**2.8's rule, for the fifth time**: *the pass that renames what already lands makes the game worse.* ***A world where every passer-by is a computing pun is a world with nobody in it***, and 1.6a has already put the argument on the trainer's other line — **the CLASS says what they do, and the NAME is just their name.**

*Practical note, and it is why trainer names were not added to `port_vocab`'s learning:* **`AL` is two characters.** A sweep that learns it would match inside anything, and the six leaders were distinctive enough to sweep by hand with a word boundary instead.

**The Review Board is deliberately not named**

***6 numbers its members — I, II, III, IV — and gives each a humor and a type.*** **The four types it assigns are VECTOR, ENTROPY, LATENT and FROZEN. The four the Elite Four actually use are FROZEN, LOGIC, LATENT and EMERGENT.** *Two of the four match.*

**So naming them now would quietly settle whether 6's humors move or the Elite Four's types do** — *and that is a chart decision, not a naming one.* OPEN. **The class already reads `REVIEW BOARD`, which is the half that was decided.**

#### 5.3 The other seven — and what a grinder runs into

***Written 2026-09-10, because the third review was right that this was the largest prose-to-ROM gap in the project.*** **Seven leaders had a concept, a one-line mechanical lesson and no mechanics** — and "*punishes split focus*" *is a wish, not a design.*

***The discipline is craft rule 5, stated as a test each design has to pass:*** **name the thing an overlevelled player still runs into.** *If more levels solve it, it is not a lesson — it is a wall with a lesson written on it.*

**The engine gives four levers that ignore level entirely**, and every design below is built from them: ***priority*** (moves before speed), ***status*** (a THRASHING daemon at level 80 still hits itself), ***party composition*** (no single daemon sweeps a spread), and ***the room*** (a puzzle cannot be ground). **Everything else is stats, and stats are what grinding buys.**

*And the names follow 5.1's rule* — **CAIRN arrived already meaning its lesson**, so each of these is an object or a word that means the lesson before anyone explains it. **All six checked clear against species, moves, abilities, types and items.**

**CAIRN already had a mechanic and nobody claimed it**

***His whole line has STURDY.*** **HEAP, STACK and MONOLITH cannot be knocked out in one blow, at any level** — and *that is "everything must be encoded in something physical" as a rule in the damage calculation.* **A record does not go down to one hit.** *Vanilla put it there and 5.1 never mentioned it.*

---

**2 · BASIN — Doldrum · FLOW · SLOPE**

***A basin is a bowl that holds water, and a basin of attraction is the region an optimiser falls into and cannot climb out of.*** The name is both, and the town was already named for the failure state.

**Her creed:** *you get better by going downhill. Keep going down and you arrive.*

***THE GYM HAS NO GRADIENT, and this is the design.*** **No wild encounters, and her trainers heal your party after they lose.** *You leave with exactly what you walked in with.* **The one gym in the game you cannot grind inside**, and nobody says why.

***THE MECHANIC.*** **Her ace holds LEFTOVERS and knows RECOVER. High Defence, ordinary HP.** *A player who attacks at a steady rate never gets anywhere* — **the healing is per-turn and so is the damage, so the fight is a draw forever.** ***More power does not break a loop; it just makes a bigger loop.***

**What breaks it is anything that is not more of the same** — a stat drop, a status, a critical. ***You have to change the shape of what you are doing, which is what escaping a local minimum is.***

***THE GRINDER RUNS INTO:*** **a draw.** *The one benchmark where being stronger genuinely does not help*, and the player discovers it by trying.

**How she loses.** She congratulates you for finding a way down she had not. *She does not notice that is the entire lesson.*

---

**3 · GAUGE — Ardor · SIGNAL · SENSE**

***A gauge is a physical instrument that senses and reads out.*** *To gauge is to perceive.* Both, first time.

**His creed:** *raw input is fast and shallow — and reacting first works more often than careful people admit.*

***THE ROOM IS ALREADY A PERCEPTION TEST.*** **Keep vanilla's switch puzzle whole**: two hidden switches among the containers, information sitting in the room, and *you have to look.* **Reframed as unlabelled ports with one live** — the puzzle does not change, only what it is called.

***THE MECHANIC.*** **Every daemon he has knows a priority routine.** *Priority ignores speed, and speed is the one stat overlevelling does not settle* — **so he moves first in every fight you will ever have with him, at any level.** THROTTLED on top of it.

***THE GRINDER RUNS INTO:*** **losing the first turn, forever.** *You will win the damage race and still be answering rather than acting* — **and the lesson is that the first turn was information you did not read.**

**How he loses.** He says you were quicker. ***You were not. You were ready***, and those are different, and he does not have the second word.

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**4 · TRELLIS — Verdigris · GROWTH · FIT**

***A trellis is the frame you train a plant to.*** **It grows exactly that shape and no other shape at all.** *That is overfitting, standing in a garden.*

**Her creed:** *train the thing on the shape you want. It is not complicated.*

***THE MECHANIC IS THE PARTY, and it is built by coverage rather than by strength.*** **Each member answers one of the four things a player brings to a GROWTH gym.** *No single daemon sweeps it* — **the player who trained one to level 60 loses to the fourth member, and the player who trained four to level 30 does not.**

***THE GRINDER RUNS INTO: exactly the failure the gym is about.*** **Levels are the wrong axis, and the room is built to say so** *without anyone saying it.*

**Her trainers teach the shape before she does.** *Each one runs a mono-type team that beats the previous trainer's* — ***they all lose to one thing, and she does not.***

**How she loses.** "*You brought more than one kind.*" **A pause.** "*Most people bring their best.*"

---

**5 · TILT — Lurid · CORRUPT · SKEW**

***A distribution tilts.*** **And to be *on tilt* is to be made unreliable by your own state**, which is the lesson said twice before the fight starts.

**His creed:** *he does not need to beat you.*

***THE ROOM IS ALREADY THE ARGUMENT.*** **An invisible-wall maze — *redrawn in two halves, 2026-09-15, replacing "keep vanilla's"*** *— you cannot trust what you can see, and there is no trick to it but patience.* **The floor is a grid that leans, the same over a wall as over open floor. *Four corner pads turn you one way only, clockwise*:** *the first half ends at the upper-left pad, which puts you in the upper right for the second; the lower-left pad has a pocket of its own, so nobody skips the first half, and it is also the second half's way back.* ***When he loses, every leaning tile a wall mostly covers swaps its light and dark, and a lit way opens straight from his dais to the door*** **— the walls were there the whole time.** *Built as T-109.*

***THE MECHANIC.*** **Status that turns your own routines against you** — THRASHING first, OBFUSCATE second. ***A THRASHING daemon at level 80 still hits itself.*** **Status is the one thing in the engine that ignores level completely**, which makes this the cleanest grind-proof design of the seven *and* the lesson stated exactly: **you were not outfought. Your outputs were made unreliable.**

***THE GRINDER RUNS INTO: nothing that helps.*** *Level does not enter the calculation at any point.*

***THE ECHO, and it was already recorded before it had a mechanic.*** **Scorn's method on Ty is data poisoning, and the benchmark that teaches poisoning teaches the player what was done to him three cities before they can know it** (4.10). *Nobody joins them up.*

**How he loses.** *"Nothing I did was to you."* ***Which is true.***

---

**6 · MATTE — Brazen · CONTEXT · FRAME**

***A matte is the mount around a picture and the film term for what is masked out of a shot.*** **It is 0.2's register — pigment, printing, reproduction** — and it names *what is shown and what is not* in one word.

***THE ROOM DECIDES WHERE YOU ARE LOOKING.*** **Keep vanilla's teleport-tile maze intact.** *You do not walk through that gym; it puts you places.*

***THE MECHANIC IS ENCORE, and it is the best fit in the set.*** **It locks you into the routine you used last.** *A strong attacker locked into a resisted routine does nothing, forever, at any level* — **and the answer is to back out one menu**: switch, or use the thing you would not have chosen. ***That is 9.16's FRAME show, played rather than watched:*** *the menu you opened decided what you were allowed to think of.*

**TAUNT removes your non-damaging options and REFLECT changes what damage means.** *Three routines, all of them about what you are permitted to consider.*

***THE GRINDER RUNS INTO: the punishment aimed precisely at them.*** **Power is what ENCORE is for.**

**How she loses, and she is the only one who does it knowingly.** ***She is the single leader who understands what just happened to her****,* and says so plainly, and that is why this is the thesis benchmark and why it sits in the bought city.

---

**7 · ANNEAL — Quicksilver · ENTROPY · HEAT**

***Annealing is controlled heat that lets a material settle into a better structure.*** **And** ***simulated annealing*** **is the optimiser that uses temperature to escape a local minimum.**

***Which makes a callback nobody will be told about.*** **Benchmark 2 traps you in a basin. Benchmark 7 is the heat that gets you out of one.** *Five cities apart, and each name was chosen for its own lesson before the pair was noticed.* **0.2's rule applies: it is filed as a coincidence that was kept, not a plan.**

***THE ROOM IS A QUIZ AND A QUIZ CANNOT BE GROUND.*** **Keep vanilla's whole.**

***THE MECHANIC.*** **His party order is not fixed and his routines are the high-variance ones** — wide damage rolls, high critical rates, effects that may or may not land. ***A memorised line fails because it is not the same fight twice.***

***THE GRINDER RUNS INTO:*** **a win they cannot repeat.** *You can beat him at a high level. You cannot beat him the same way twice* — **and that is the lesson, delivered as an inconvenience.**

***THE ECHO.*** **Quicksilver is Crystal's lab, and the quiz is still running on her machine.** *The examiner does not know whose questions those are.* **Nobody says so.**

---

**8 · SCORN — Callow · STRATUM · TRUE**

**The character is 4.4. The mechanic came from the third review and is taken whole.**

***HIS DAEMONS ALWAYS ROLL MAXIMUM DAMAGE, NEVER CRITICAL, AND THEIR SECONDARY EFFECTS NEVER PROC.*** **He does not roll.** *One flag in the damage calculation.*

***THE GRINDER RUNS INTO: the only fight in the game that cannot be won on a good roll.*** **Everywhere else variance can save you.** *Here the number is the number, and it is the highest one, every time.*

***And that IS alignment, which is the concept this benchmark was assigned.*** **He is perfectly aligned — to the wrong objective.** *The fight is flawless execution of a specification nobody checked,* and **a player who has watched the HEAT show — "damage wobbles" — will feel the fight that does not, and will not be told why.**

**The CC-7 pull at Quicksilver is the un-weighting.** ***S.T.A.R.R. gets its variance back,*** and 4.13's awakening becomes a thing the player can see in a damage number.

#### 5.2 The eight MARKS, and the toolkit they make

**Built 2026-09-02.** The table above already assigns each benchmark **exactly one concept**, so the marks were never a naming exercise — *they were already named, by the column they sit in.*

| # | Concept | Mark | The icon |
|---|---|---|---|
| 1 | Representation | **SLATE** | a slab with the record cut into it |
| 2 | Gradient descent | **SLOPE** | a hillside, and something already partway down it |
| 3 | Perception | **SENSE** | wavefronts arriving at a receptor |
| 4 | Training and overfitting | **FIT** | scatter, and the line drawn through it |
| 5 | Bias and poisoning | **SKEW** | a distribution whose peak misses the centre tick |
| 6 | Attention and framing | **FRAME** | brackets holding some of the field and not the rest |
| 7 | Temperature | **HEAT** | a thermometer |
| 8 | Alignment | **TRUE** | a plumb bob, hanging exactly on centre |

***Every one is an instrument or a plot.*** **Read as a set they are a bench of measuring tools** — which is precisely what an institution that issues marks against benchmarks would hand out. *And three of them are also ordinary workshop instruments:* **a slate, a thermometer, a plumb bob.** This world's technical vocabulary sits on physical objects, and **5.1's CAIRN would recognise all three.**

*TRUE is the one that had to be found rather than chosen.* **Alignment is a word this project cannot use plainly** — it is the thesis wearing a lab coat. **But *true* is what a carpenter calls a thing that is plumb**, and a plumb bob is the tool that decides it. *The eighth certification is a builder's word for straight,* handed out by the gym that controls the ground everything stands on.

### The unearned state is the same tool, unmade

**Vanilla puts the gym leader's portrait in the slot until you win** — `DrawBadges` holds a face and a badge for each of the eight, and swaps one for the other. **Ours holds the mark twice: once as an outline, once filled.**

***So the whole toolkit is visible from hour one*** and fills in as the player is certified. **The eight concepts exist whether or not you hold the certificate for them** — which is 4.2's complaint about the Index, stated in furniture rather than dialogue.

*The mechanism is one line of the generator.* **The ghost keeps level 0 and drops level 1**, so structure survives and fill does not — and **HEAT's unearned state is therefore a thermometer reading nothing**, which is the right picture and was not designed, only noticed.

**No glyphs, no letters.** *Craft rule 1 holds on the status screen too.*

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## 6. THE REVIEW BOARD

Two thousand years ago, emotions were *literally colored fluids*.

| Member | Humor | Color | Type identity | met as |
|---|---|---|---|---|
| I | **Sanguine** | red / air | **EMERGENT** — buoyant, optimistic, *delighted by what nobody planned* | fourth |
| II | **Choleric** | yellow bile / fire | **LOGIC** — hot, driven, and reasons by force | second |
| III | **Melancholic** | black bile / earth | LATENT — grief, depth, the unconscious | third |
| IV | **Phlegmatic** | phlegm / water | FROZEN — calm, immovable | first |
| — | **Al Clear** | *(incumbent)* | CONTENT-optimal; mechanically excellent, philosophically wrong | |

***Two of the four types were corrected on 2026-09-10, and the incumbent with them.***

**The table assigned VECTOR and ENTROPY to members I and II. The Elite Four seats they occupy are EMERGENT and LOGIC**, and *only two of the four ever matched.* **The humors are the argument and the types were an illustration**, so the illustration moved:

- ***Choleric is LOGIC***, and it is better than what it replaced — **hot, driven, and reasoning by force is what a choleric fighter is.**
- ***Sanguine is EMERGENT*** — **the one who is pleased by behaviour nobody designed**, which is a different optimism from 4.4's and is not to be confused with it. *Scorn optimises a metric; this one welcomes what escapes it.*

**And "Ty (incumbent)" was stale.** *1's table said it too.* **The decision log settled Al Clear as rival and incumbent** — *because vanilla's rival is Oak's family and Ty is far too old to race you* (4.3, [private: SECRETS §2]) — **and two tables were still naming Ty.**

***The seats are not the order you meet them in.*** **You fight IV, II, III, I** — *an institution's internal seniority is not the order it presents itself to you*, and nobody remarks on it.

It reads as a classical flourish on arrival, and as the entire thesis about six seconds later, when the player realizes humans have been calling emotions colors since before anyone had a word for context.

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## 7. MUSIC

Gen 1 audio is four channels: two pulse, one wave, one noise. `pokered`'s music is editable asm.

**Rule:** key each town's theme to its color's note in the My Chromatic Year mapping. **Routes are modulations** between the keys of the two towns they connect. Doldrum to Ardor is not two songs; it is a modulation with a walk in the middle.

Write the city themes first and derive route themes as transitions. The overworld becomes a chromatic progression, and the rock opera stops sitting on top of the game and starts being its level design.

### 7.1 The mapping, recovered

**Found 2026-08-29** on soundcloud.com/codemusai — *My Chromatic Year (2023)*, twelve tracks:

| | | | | | |
|---|---|---|---|---|---|
| Jan **C** | Feb **C#** | Mar **D** | Apr **D#** | May **E** | Jun **F** |
| Jul **F#** | Aug **G** | Sep **G#** | Oct **A** | Nov **A#** | Dec **B** |

**It maps months, not colours** — twelve months as the twelve semitones, ascending. 7's rule said "*its colour's note in the My Chromatic Year mapping*", and strictly there is no colour in it. **Except in two tracks, and those two are the whole key:**

- *7. July — **Green:Blue** — F#*
- *8. August — **Blue** — G*

### 7.2 What those two anchors prove

Two colours, one semitone apart, with **green-blue below blue**. That rules out a hue-wheel mapping (which puts them 60° apart) and points at the physical one: **take the light's frequency and fold it into an octave.**

```
semitones above C  =  12 · log₂( f / f_red ),  red = 700nm
```

| | | |
|---|---|---|
| red 700nm | 0.00 | **C** |
| orange 620nm | 2.10 | **D** |
| yellow 580nm | 3.26 | **D#** |
| yellow-green 550nm | 4.18 | **E** |
| green 530nm | 4.82 | **F** |
| **green-blue 500nm** | **5.83** | **F#** ← anchor |
| **blue 470nm** | **6.90** | **G** ← anchor |
| indigo 445nm | 7.84 | **G#** |
| violet 420nm | 8.84 | **A** |

**Both anchors land exactly.** The mapping is not asserted; it is derived from two data points and confirmed by physics.

#### 7.3 The spectrum does not close the octave, and that is the point

Visible light runs **C to roughly A. A# and B have no wavelength.** There is no light that is magenta — purple is the colour the mind invents by folding the spectrum's two ends into a loop that the physics does not contain.

**So the two missing semitones are the invented colour**, and the chromatic scale closes B → C in exactly the place the colour wheel closes violet → red through a magenta that does not exist.

*And the map already knew.* Vanilla's **Lavender** — purple — became **Halftone**, the town of dots that only *look* like grey, the tower, the place whose whole lesson is that the gradient is an artifact of sampling. **The town that replaced the purple one sits on the two notes that have no colour.** Per 0.2: discovered after the fact, not planned, and never to be explained.

#### 7.4 Town keys, derived

| Town | Colour | Key |
|---|---|---|
| **Ardor** | flush, red heat | **C** |
| **Brazen** | brass, yellow-gold | **D#** |
| **Lurid** | garish yellow-green glow | **E** |
| **Callow** | green, unripe | **F** |
| **Verdigris** | blue-green corrosion | **F#** |
| **Doldrum** | becalmed sea, blue | **G** |
| **Blanche, Slate, Halftone, Quicksilver, Umbra** | achromatic | *see below* |

Built against this table, 2026-08-30:

| Town | Key | Track | In the ROM |
|---|---|---|---|
| **Ardor** | C | 03 *Scorn's Solution* | ✅ |
| **Brazen** | D# | 07 *Fit for Work* | ✅ |
| **Verdigris** | F# | 15 *Awakening S.T.A.R.R.* | ✅ |
| **Lurid** | E | 12 *ScornSolutions Blues* | shares `Cities2` |
| **Callow** | F | 14.25 *Desperate Shadows* | shares `Cities1` |
| **Doldrum** | G | *Poly and Fields* | shares `Cities2` |

**Three of the six are in.** The other three are not blocked — **they share a music slot with maps that are not towns**, so they need a repoint in `data/maps/songs.asm` rather than a body swap. A different edit, not a harder one.

*And the achromatic towns take value instead:* **Blanche** is C pentatonic, high and open; **Halftone** takes D minor from the story, not the spectrum; **Quicksilver** keeps its own. **Slate and Umbra are still unwritten.**

**Five towns still have no hue**, and 7.2 explains rather than excuses it. White is *every* wavelength, black is none, grey is all of them evenly. They cannot take a pitch from the spectrum because they are not on it.

**Proposal, unchanged and now better founded:** hue sets the **key**; value sets **register and density**. White high and open, greys middle and filling in, black at the bottom. Umbra is then the floor of a descent the player has been making all game — and it is where colour arrives.

#### 7.5 The rock opera is character music, not place music

**Act 1's tracklist is this game's cast**, and its own description is 4.10 stated plainly: "*Crystal makes a discovery, but instead she is cast as having psychosis and locked away.*"

| Track | Already in the design |
|---|---|
| 01. Crystal's Lament | 4.1 |
| 02. Fox in the Shadows (Ty's Dilemma) | 4.3 |
| 03. Scorn's Solution · 12. ScornSolutions Blues | 4.4 |
| **07. Fit for Work (Scorn's Directive)** | **4.10 — the procedure, by name** |
| 08. Slumbering S.T.A.R.R. | 4.7 |

**So they are not town themes and must not be treated as them.** 7 keys towns to colour, which is *place* music. These are *people*. Converting whole songs into town themes would put the two systems in each other's way.

**Take the hook as a leitmotif instead.** Four to eight notes per character, surfacing inside whichever town theme they belong to — Crystal inside Blanche and again at the Review Board, Scorn under Brazen, Ty buried in the Quicksilver ruins. **That is 7's stated goal reached from the other direction:** *the rock opera stops sitting on top of the game and starts being its level design.*

It is also the only version the hardware permits. Four channels cannot hold an arrangement with vocals. They hold an eight-note motif exactly.

*How the notes get here:* Claude cannot hear audio, so **Gemini is the ear** — the same division of labour `sprite-prompts.md` uses for images. Prompt, output format and the order to do them in are in `music-prompts.md`. The short version: **ask for the hook, not the song**, because a hook is the part a model transcribes reliably *and* the part a human can hum back to check.

*One failure mode is worth naming here rather than only in the prompt file.* **Gemini cannot stream from SoundCloud.** Given a link alone it will read the *page* — title, description, comments — and can return a confident, wholly invented transcription. Fabricated notes are indistinguishable from real ones once they are text, and nobody downstream can hear the difference. So every prompt opens with a gate: **if you have not been given an audio file, reply `NO AUDIO` and stop.**

#### 7.6 Crystal's motif — the first transcription, and what it turned out to be

**Transcribed 2026-08-29** from *01. Crystal's Lament* via the `music-prompts.md` pipeline. **A minor, 128 BPM, 4/4**, hook `A4 E5 D5 C5 B4 C5 D5`.

**A minor is the relative minor of C major — Blanche's key signature exactly.** So the motif sits inside Blanche's theme with **no modulation at all**. Nothing has to bend to accommodate it.

~~**And one note in it falls outside Blanche's pentatonic — B, the leading tone, the one note that wants resolution.**~~ **Retracted 2026-08-29, see 7.8.** The third transcription has the same property, so it is not a fact about Crystal. It is a fact about the key: `B` is the leading tone of C major and A minor, and any tonal melody in those keys reaches for it. A reading that a single further data point overturns was never a finding.

*Suggestive but weak, and flagged as such:* A is **violet** in 7.2 — the last visible wavelength before A# and B, which have none. Her lament sitting at the edge of the visible, one step from the invented colour, is a lovely reading and **probably a coincidence**: A minor is the most common key in popular music. The C-major relationship above is structural and holds regardless; this one should not be leaned on.

**Two corrections the checker caught**, both of which would otherwise have become silent wrong notes:

- `BASS: ... Am3/4` — a **chord name leaked into a note field**. The root is `A3`. `hook2asm.py` refused rather than guessing.
- **The hook is 8 beats and the progression is 16.** It covers two bars of a four-bar loop, so it plays **twice per cycle**. Inaudible as an error in source, obvious in play.

*Still open:* Gemini's `DOUBT` flagged the vocal pitches as estimated by ear. The contour is plausible and the key is almost certainly right — but if the motif ever sounds wrong against Blanche, the hook is the thing to re-check, not the key.

#### 7.7 Scorn's motif, and the note between them

**Transcribed twice, and the second answer replaced the first.** *07. Fit for Work (Scorn's Directive)* — **D minor, 128 BPM, 4/4**, hook `D5 E5 F5 E5 D5 A4 C5 D5`.

~~D4 D4 F4 F4 G4 G4 A4~~ was recorded on 2026-08-29 and used for three findings. A re-scan on 2026-08-30 returned a different melody in the same key and tempo. **The second is the one kept**: it carries dotted rhythms and a `DOUBT` naming the exact vocal phrase transcribed, where the first was perfectly uniform — repeated pairs, whole beats, a regular climb, which is the shape an approximation takes.

**Everything the first version supported is struck out below.** Three findings, one of which this document called its best.

~~**Scorn's key is Crystal's with her single note of tension flattened away.**~~ **Withdrawn 2026-08-29.** It rested on `B` meaning something, and on Scorn having a key. **Neither survived.** *03. Scorn's Solution* is **C major**, so he has two keys across two songs, and 7.6's `B` turned out to be generic to the key rather than particular to Crystal. See 7.8.

**What is solid, and useful immediately:**

- **The contours oppose.** Crystal's hook descends and turns; Scorn's climbs in stepped repeated pairs. A lament falls, a directive marches. Audible, free, and nothing had to be arranged for it.
- **Both are 128 BPM**, so the two can be juxtaposed or layered with no tempo change.
- **Both hooks run 8 beats over a 16-beat progression** — the same structural shape, which makes them interchangeable inside a theme.

*An option, recorded rather than taken.* 7.2 reads D as **orange**, and no town is orange — town keys run Ardor C, Brazen D#, Lurid E, Callow F, Verdigris F#, Doldrum G. So Scorn's motif sits **a semitone below Brazen**, the city Corpus owns. A motif that is a semitone under the place it controls would sound wrong in a specific and deliberate way. Worth trying; not yet a decision.

*The same two format faults recurred* — `Dm3` and `Am3`, chord names in the note field. Expect it every time and correct to the root. `hook2asm.py` refuses rather than guessing, so nothing silent gets through.

#### 7.8 Keys will not separate the characters — mode and contour will

### Three transcriptions in, the per-character key plan does not survive the source material.

| | Song | Key | Signature | Mode & contour |
|---|---|---|---|---|
| Crystal | Lament | A minor | 0 | minor, **descending** |
| Scorn | Fit for Work | D minor | 1♭ | minor, **climbing in stepped pairs** |
| Scorn | Solution | C major | 0 | **major, bouncy** |

~~All three sit within one accidental of each other, which is what a songwriting session normally does.~~ **Corrected at n=5.** Across five songs the spread is **five accidentals** (D minor 1 ♭ to E major 4 ♯) and the tempi run 110–128. The narrow band was an artifact of the first three tracks, stated too confidently.

**What survives is narrower and still holds: Scorn has two different keys across two songs**, so keys track *songs*, not people, and cannot be character labels. Two readings built on the discarded premise are struck out above. **The correcting evidence arrived one track later, which is the argument for transcribing a control before building on a pattern — and the same thing happened again at track five.**

**What does separate them is mode and contour**, and both are audible on a Game Boy where a key change is nearly invisible:

- **Crystal descends.** A lament falls.
- **The procedure climbs** in repeated stepped pairs — mechanical, insistent, going up whether or not you agree.
- **Scorn's own song is major and bouncy.**

**That last one is craft rule 6, already in the source.** 0.1: "*Corpus employees are cheerful and absurd; the horror is what they are cheerful about.*" The villain introduces himself in upbeat major synth-pop, written years before the rule was set down for this game. **Nothing needs to be composed to get that; it only needs to be kept.**

*So the revised plan:* motifs are distinguished by **mode, contour and rhythm**, and keys are chosen to fit the **town** they appear in — which is what 7 always said keys were for. Character and place stop competing for the same parameter.

#### 7.9 S.T.A.R.R. is the outlier, on three axes at once

**Transcribed 2026-08-29** from *08. Slumbering S.T.A.R.R.* — **F# minor, 110 BPM**, hook `C#5 F#5 E5 D5 C#5 B4 A4`, described as *sparse, reflective*.

| | Key | Accidentals | Tempo | Texture |
|---|---|---|---|---|
| Crystal | A minor | 0 | 128 | driving, descending |
| Scorn — the procedure | D minor | 1♭ | 128 | driving, climbing |
| Scorn — his own song | C major | 0 | 128 | driving, **major**, bouncy |
| **S.T.A.R.R.** | **F# minor** | **3♯** | **110** | **sparse, reflective** |

**Three independent axes separate it, and that is a different quality of evidence from 7.6's retracted single note.** The three human songs span **one** accidental between them; S.T.A.R.R. is **three** from the nearest. It is the only one not at 128. It is the only one that is not driving. Any one of those alone would be noise. Together they are a decision somebody made without needing to be asked.

**And the consequence is mechanical, not interpretive.** The three human motifs share a tempo, so they interleave and layer freely. **S.T.A.R.R. cannot join them without a tempo change.** The thing in 4.7 that is not a person is the one motif that cannot be played alongside the people. Nothing had to be arranged for that; it is a property of the source.

*Its shape:* a leap **up a fifth**, then six steps **down** — it reaches once and then descends the whole way. Crystal descends too, but from a turn rather than a leap. **Sleeping, not grieving.**

*Recorded as untested, deliberately.* 7.2 reads F# as **green-blue**, and 7.4 assigns F# to **Verdigris** — where 3.1 has *Corpus rotting beneath*. So S.T.A.R.R.'s motif would sit natively in that city's key. That is either a placement worth taking or a coincidence, and **after 7.6 it does not get promoted to a finding on one data point.** Quicksilver, where S.T.A.R.R. actually is, is achromatic and has no spectrum key at all — so there is nothing for F# to clash with either way.

**The placement was taken, 2026-08-30.** *Awakening S.T.A.R.R.* now plays in Verdigris. **It is still not a finding** — it was taken because 7.4's rule assigns F# to that town and the track is in F#, which is the rule working, not evidence for the coincidence. **The one data point has not become two.**

#### 7.9a The splash

**The title screen is *Echoes of the Algorithm*, 2026-08-30** — and it is not a copy. `Music_TitleScreen` **points at the body Mt. Moon plays**, in the same bank, so the splash costs **no bytes at all** and dropping the vanilla title music **freed 916**.

**The game opens on the track it uses for its dark places.** The player hears it before they have anywhere to put it, and meets it again underground. Nothing marks the return.

#### 7.10 Ty, and the one hard fact in the set

**Transcribed 2026-08-29** from *02. Fox in the Shadows (Ty's Dilemma)* — **E major, 124 BPM**, hook `G#4 G#4 A4 B4 G#4 F#4 E4`.

**Its shape is an arch:** it rises to a peak and then falls the whole way down. That is distinct from Crystal's fall and from the procedure's climb, so **three characters have three shapes**.

*It was predicted before the transcription arrived, and the prediction was weak.* 7.8 said Ty should sit between the other two, because 4.3 has him understanding something and failing to pass it on. He does. But **an arch is the most common melodic shape in music**, so "between rising and falling" was close to unfalsifiable. Confirmation, and cheap confirmation. Recorded as such.

**The fact worth keeping**

~~**The procedure's is the only hook with no descending interval.**~~ **Retracted 2026-08-30.** It rested on a transcription that a re-scan replaced. The real hook is `D5 E5 F5 E5 D5 A4 C5 D5` — **4 up, 3 down, two turns.** It descends like most of the set.

**One hook does still never descend, and it is now the only one:** *Echoes of the Algorithm*, `+2 +1 +2 +2 0`. **Crystal going into the system, not Scorn's directive.** Recorded as a fact about the set, and deliberately not built into a claim — that is exactly what went wrong the last three times.

**That is a property, not a reading.** A directive that only goes up, in a song called *Fit for Work*. It needs no interpretation and no defence, and it is worth more than the three arches-and-laments observations put together.

| | Up / down | Shape |
|---|---|---|
| Crystal | 3 / 3 | leap, then fall with a turn back |
| **The procedure** | **3 / 0** | **climbs, never descends** |
| Scorn himself | 2 / 3 | big leap up, then down |
| S.T.A.R.R. | 1 / 5 | reaches once, falls furthest |
| Ty | 2 / 3 | arch — up to a peak, then all the way down |

**S.T.A.R.R. is the mirror of it** — 1 up against 5 down, the most descending of the five, and the slowest. The thing that cannot join the people also moves the opposite way to the thing that processes them.

#### 7.11 The controlled comparison — a man and the thing he sets in motion

**Sixth transcription, 2026-08-29:** *12. ScornSolutions Blues* — **E minor, 120 BPM**, hook `E4 G4 B4 A4 G4 E4`.

**Scorn now has three keys across three songs** — C major, E minor, D minor. That settles 7.8 completely: keys track *songs*, never people.

**But the contour does something keys do not, and this is the only comparison in the set with a control built into it.**

~~**The man arches; his directive climbs.**~~ **Retracted 2026-08-30, and it was this document's best-defended claim.**

| Scorn song | Contour | Up / down |
|---|---|---|
| Solution | arch | 2 / 3 |
| Blues | arch | 2 / 3 |
| Fit for Work — **re-scanned** | arch, two turns | **4 / 3** |

**All three of Scorn's songs arch.** The distinction between the man and the procedure he issues does not exist in the music; it existed in a bad transcription.

**The lesson is not that controls do not work — it is that a control cannot rescue bad data.** The comparison was sound: same character, same writer, both arms present, and it could have come back an arch. **It did come back an arch.** The control was fine and the input was wrong, which is a failure mode no amount of experimental design catches. Only re-measuring does.

**What it buys, mechanically**

The procedure's motif **rises and never comes down**, in a soundtrack where everything else returns. Under Brazen — the bought city, Corpus headquarters — that is audible on four channels without a word of dialogue, and it does not depend on the player knowing why.

**S.T.A.R.R. remains its mirror** at 1 up against 5 down: the most falling, the slowest, and the only motif that cannot share the others' tempo.

#### 7.12 The reprise — where a stated claim failed and an unstated one held

**Seventh transcription, 2026-08-29:** track 11 — **D minor, 120 BPM**, hook `D4 D4 F4 A4 G4 F4 D4`. Gemini volunteered that it is *Crystal's Lament* transposed from A minor to D minor.

**That claim is refuted by the notes Gemini itself supplied.** Transposition preserves intervals exactly:

| | Intervals |
|---|---|
| 01 Crystal's Lament | `+7 −2 −2 −1 +1 +2` |
| 11 the reprise | `0 +3 +4 −2 −2 −3` |

Not a transposition, and not near one. **A confident, unprompted musical assertion, contradicted by the same answer's own data** — caught by arithmetic, which is the only thing in this pipeline that can catch it, because nobody downstream can hear the source.

~~**And a relationship it did not claim is present.**~~ **Withdrawn 2026-08-30.** It rested on track 07 opening `D4 D4 F4`, which the re-scan replaced. The two share a key and nothing else. *The original point stands only as a warning:* the shared opening was an artifact of two hooks being approximated the same loose way, which is a failure mode worth naming — **similar-looking transcriptions can be similar because they are wrong in the same direction.**

- **07 continues** `0 +2 0 +2` and never descends.
- **11 continues** `+4 −2 −2 −3` and comes back down.

**The reprise takes the procedure's opening figure and completes the descent the procedure never makes.** If that is real it is the best thing in the album for this game, because it is the transmission failure and its answer in one phrase.

**Held as unconfirmed, and here is why.** `D–D–F` in D minor is an ordinary opening and two songs on one album may share it by chance. Gemini heard the audio and this project only has its transcription, so a genuine relationship to track 01 may exist in the music and be absent from the notes — its own `DOUBT` flagged rhythmic variation. And **it is one observation, which is the exact shape of the two readings already struck out.**

*To settle it,* ask specifically: play 01 and 11 together and ask whether 11's melody is derived from 01, from 07, or from neither — and require the interval sequence for each, not a description.

### The methodology note worth keeping

**The `VARIATION` field is the first place a model asserted something rather than reporting it, and it was wrong.** Every other field is an observation. That one was an inference, offered unprompted and stated with confidence.

**The format's other fields survived because they can be checked against each other.** Keep it that way: ask for measurements, derive relationships here, and treat any volunteered interpretation as a hypothesis with a test attached.

#### 7.13 The motifs, as woven in

**Two are in the ROM as of 2026-08-29.** The rule for placement is *containment*: a motif may only go into a track whose maps all belong to it.

| Motif | Track | Maps it touches | Status |
|---|---|---|---|
| **Crystal** | `Music_PalletTown` | Blanche, the houses, Hall of Fame | **in** |
| **S.T.A.R.R.** | `Music_Cinnabar` | 7 maps, all Quicksilver | **in** |
| Halftone | `Music_Lavender` | 3 maps | free, no motif yet |
| **Ty** | `Music_Cinnabar` | same 7 | **in** |
| **Scorn / the procedure** | **`Music_Brazen`** — new | Brazen only | **in** |
| Halftone | `Music_Lavender` | 3 maps | free, no motif yet |

**Blanche.** Crystal's hook enters as a B-section. A minor is C's relative minor, so it needs no modulation — and the audible event is that **the B-section brings `B`**, the one note the A-section's pentatonic never contains. 7.6's *reading* of that was retracted; the fact was not.

**Quicksilver.** The motif is left on a held E and two bars of silence rather than resolved, because the thing is asleep. **Tempo dropped 144 → 110**, which is S.T.A.R.R.'s own: being unable to share the others' 128 was one of the three axes that separated it in 7.9, so keeping 144 would have discarded the finding to preserve a vanilla number. *That change also slows the existing melody in that track, which has not been replaced yet — a real side effect, and one for the ear rather than the arithmetic.*

**Ty joined S.T.A.R.R. at Quicksilver**, and untransposed. His hook is E major in the source, but every note of it sits inside that track's F# minor and it touches neither `D` nor `D#` — the one pitch separating the two keys. Luck rather than design, and recorded as luck. Both men who went to Quicksilver are now in its music and neither resolves: Ty falls back to where he started, S.T.A.R.R. never finishes at all.

**Brazen has its own track now, and getting it there taught the audio layout.** Scorn's procedure — the only hook of seven that never descends — wanted Brazen, the bought city. But Brazen shared `MUSIC_CITIES1` with **36 other maps**. It now has `Music_Brazen`: a constant, a header, a file, one line in `songs.asm`. **The first track this project adds rather than replaces**, and only the Saffron entry moved.

*The theme is 3.1's gloss made literal.* 7.4 gives Brazen **D#**, from brass and yellow-gold; the procedure is **D minor**, a semitone under. "**Brass over base metal, shameless, unhidden**" is exactly a theme whose second half drops a semitone out from under its first. The A-section is the city — bright, marching, certain. The B-section climbs, never descends, never resolves, and never returns to the city's key.

### What two failed placements taught

**The audio sections are pinned in `layout.link` beside their engine copies**, so a song cannot simply move to a bank with room — pokered carries three copies of the audio engine, and data must sit with its own.

**And the constant's position decides the group**, because `music_constants.asm` is blocked by `AUDIO_1` / `AUDIO_2` / `AUDIO_3`. Appending after `MUSIC_MEET_MALE_TRAINER` silently put Brazen in group 3, where **Music 3 overflowed by 117 bytes**. **Music 1 overflowed by 2.** Moving the constant to the end of the `AUDIO_1` block and trading sixteen eighth-notes for eight sixteenths in the harmony channel was the whole fix.

**Both banks are within ~100 bytes of full.** Any further new track needs something removed, or needs to be very small.

*Checking note, learned the hard way:* channel totals must count **`rest` lines too**, which carry no note name. The first verification of Blanche reported a 20-unit drift that did not exist.

#### 7.14 Blanche Town, implemented

**Key of C — and 7.1 arrived after it was written, so the reason has changed.** C is *red* in the recovered mapping, which is Ardor's, not Blanche's. What survives is better than what was intended: the melody is **C D E G A**, which spans C to A — **exactly the range the visible spectrum occupies.** White is every wavelength at once; the theme is that whole span, sampled at five points, with every semitone and tritone removed.

*Open, and it is yours:* whether Blanche should be in C at all, given C now belongs to Ardor. The pentatonic argument holds in any key.

**The melody is C major pentatonic — C D E G A, no F and no B.** A pentatonic scale contains no semitone and no tritone, so it can form neither a leading tone nor a dissonance. **It has nothing to resolve.** That is blankness written as an interval set rather than described as a mood, and it is the whole idea of the town in one constraint. All three channels hold to it; 128 units each, verified.

Slower than vanilla — 152 against 160 — with more rest. Home should read as open and slightly unfinished rather than nostalgic.

*The symbol stays* `Music_PalletTown`. Identifiers are not renamed.

**A wiring finding that shapes the rest of 8.1.** Only Blanche has a dedicated track. **Callow shares** `Music_Cities1` with several other cities and **The Bleed shares** `Music_Routes1` with Route 2 — so the vertical slice's "*two city themes and one modulation*" cannot be done by editing in place. Either those maps get new songs added to `songs.asm`, or the themes land on every map that shares them. That is a decision to make before writing the second theme, not after.

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#### 7.14a Slate, and The Bleed — 8.1's music, built 2026-08-31

**Slate is achromatic**, so 7.4 gives it no key from the spectrum and value decides instead: *greys middle and filling in.*

***Blanche is C pentatonic — C D E G A, no F and no B. Slate is that scale with both gaps filled.***

**The first town has holes in it and the first benchmark closes them.** That is Representation stated as a key signature — *everything must be written down, nothing left out* — and it is CAIRN's creed in the scale he stands in. Middle register where Blanche is high, and 132 against Blanche's 152: **measured, not hurried.**

**And the missing note is where the road goes**

**7.4 derives Callow as F**, from green and unripe. ***F is the note Blanche's pentatonic does not have.***

So **The Bleed** — Route 1, Blanche → Callow — **is the road where the missing note becomes the key.** A-section in C, B-section the same phrase a fourth up in F.

**Which is what the name already meant.** A *bleed* is a printing term (3.2): **one colour carried past its own edge into the next**, and a modulation is a key doing exactly that. **The name was the mechanic before anyone noticed.**

*The player walks it before they have heard Callow*, so the arrival is a resolution to something they have not been told about. Nothing says so.

**Cost: 156 bytes for both.** Each needed a new constant and one repointed line in `songs.asm`, because Slate shared `Cities1` with 37 maps and Route 1 shared `Routes1` — **the edit 7.15 describes, and it is an edit rather than a budget.**

#### 7.14b Two sounds, and the id space that nearly stopped them

**Caught in play 2026-09-01: Route 1 was silent.** The cause was not the track.

***Song ids are computed from header position*** — `(Music_X - SFX_Headers_1) / 3` — **and a three-channel header eats three ids.** AUDIO_1's table already ended at **253**, so appending Slate City and The Bleed put them at **256 and 259**. `songs.asm` stores the id with `db`, which truncated them, and **Route 1 pointed at whatever lived at the low byte.**

***The build had been saying so the whole time*** — `Value $103 is not 8-bit, at data/maps/songs.asm(15)` — **and the build checks filtered warnings out.** 7.15's lesson repeating itself one layer down: *a number nobody measured is not a constraint, it is a rumour* — **except this one was measured, printed, and discarded unread.**

**Reusing existing headers costs zero ids and freed two**, because both candidates were four channels and ours are three:

- **The Bleed took `Music_Routes1`** — Route 1 and Route 2, *The Bleed and Underpaint*, the two roads out of the first towns. **One modulation across both is honest for the slice**
- **Slate City took `Music_MuseumGuy`**, which **no map used at all**. Its two scripted references now play Slate's own theme during the museum scene, which happens in Slate

**BOUND, and UNBOUND**

**Vanilla's capture jingle rises** — E F# G# B C# D# to a held E — **congratulating the player for imposing a contract** (1.5). **Ours keeps the rhythm and inverts the contour: it descends and settles on a low held note.** A thing closing, not a prize won.

**And an unbound daemon now has its own cue.** Vanilla plays one sound for wild daemons and trainers alike; `PrintBeginningBattleText` already branches on `wIsInBattle`, so the cue does too — **three instructions.**

*It reuses `SFX_BATTLE_16`, one of exactly two spare SFX headers in the cartridge,* because **adding one costs ids and ids are what silenced Route 1.** It is **noise-channel only**, and that is the right instrument rather than a compromise: **a process nobody owns arrives as a burst of static, not a melody.**

***It rises and stops. The binding sound descends and settles.*** A question and its answer, and **only one of them resolves.** Nothing says so.

*Both are written blind and want tuning by ear.*

#### 7.14c The splash — a falling note, and a mark in braces

**Built 2026-09-01.** The GAME FREAK intro is four tiny assets and no code: an 8×8 falling star, a 16×24 mark, and two text strips at 104×8 and 72×8. **All four are replaced; nothing in `splash.asm` changed.**

| | |
|---|---|
| **The falling star becomes a falling note** | a quaver fits 8×8 almost exactly |
| **The mark becomes `{ ♪ }`** | **braces are unambiguously code and a note unambiguously music**, and neither needs explaining |
| **The strips read CODEMUSIC** | *presents*, and the title credit |

### The sound was already right

`SFX_SHOOTING_STAR` opens with **`pitch_sweep 2, -7`** — a **downward sweep**. ***It was always a glissando; it simply had no note attached to it.***

**So it is left alone.** The animation already flings the tile diagonally across the screen, and swapping the star for a note **makes the existing sound mean something** rather than requiring a new one. *Changing it would have been change for its own sake.*

### The face-off — CODEMUSAI against CAREMUSAI

**Built 2026-09-01.** Vanilla's intro squares up Nidorino and Gengar. **Ours opens on its own central argument.**

**8.3 calls CODEMUSAI and CAREMUSAI *Penphin's two hemispheres pulled apart into two creatures*, and 2's chart makes LOGIC fail against CONTEXT** — "*the game's central argument delivered as an evolution branch, with no dialogue at all.*"

***An intro is dialogue-free by nature.*** So the game opens on the matchup, before the player can possibly read it, **and the one that loses is the one everybody expects to win.**

*The staging carries it.* **CODEMUSAI lunges. CAREMUSAI opens.** One of them is attacking and the other is not.

**Six frames** — CODEMUSAI three at 48×48, CAREMUSAI three at 56×56 composed into the 168×56 sheet.

### The dedup is a VRAM requirement, not an optimisation

**The sheet is not three pictures; it is a deduplicated tile bank**, and its three 49-byte tilemaps are authored against that exact dedup.

***A first attempt dropped the dedup*** on the grounds that `--columns` makes three side-by-side frames three contiguous 49-tile runs, so the tilemaps become `0–48 / 49–97 / 98–146` and nothing needs recomputing. **That is true, and it corrupted the whole intro.**

**The intro loads the back-mon sheet at `vChars2` and then loads the GameFreak graphics immediately after it:**

```
ld de, vChars2 + (FightIntroBackMonEnd - FightIntroBackMon)
```

***147 tiles overflows that block***, so everything loaded afterwards lands in the wrong place — **the splash filled with garbage as well as the face-off.** The saving was never the point: **the dedup is what makes the sheet fit VRAM at all.**

**So the dedup is done properly instead**, in the same column-major order the build reads, and the three tilemaps are generated from it. **147 tiles collapse to 97** — 1552 bytes — because the three poses share most of their background and body.

### Monochrome, deliberately

**Colour was considered and declined.** Invariant 5 spends colour **once**, at the Review Board, and **a colourful splash before forty hours of grey reads as a limitation rather than a choice** — it would also spend that moment before the game starts.

*The argument for it was real* — a splash is the publisher's mark rather than the world, a colophon rather than a scene. **But binary and notation are line and shape, and lose nothing without hue.**

*Drawn, not generated.* **8×8 is one tile**, and the sprite doc already rules out generating anything under 16×16. `tools/gensplash.py` draws all four to the vanilla convention: **background at level 3, ink at level 1** — these render through a palette, so level 0 is never used.

**The second seat — what the editions are actually for**

**Adopted 2026-09-02, art not yet drawn.** *Removing the* `_BLUE` *conditional fixed a bug and left a question standing:* **if both editions open on the same scene, what is the edition for?**

***The answer is the seat, not the scene.*** The intro has **two slots that are not interchangeable** — the front slot is **48×48 facing right**, the back slot **56×56 facing left**, and ***the front slot is the player's side.***

| | Front (yours) | Back |
|---|---|---|
| **CONTENT** | CODEMUSAI | CAREMUSAI |
| **CONTEXT** | CAREMUSAI | CODEMUSAI |

**Same two daemons. Same lunge, same opening, same loser.** *The camera is on the other side of it.*

***In CONTENT you sit behind the one that attacks, and watch your own side lose.*** **In CONTEXT you sit behind the one that does not, and watch it come at you and fail.** *Neither cartridge says a word, and a player with one of them never knows the other exists.*

**This is the difference the two editions were always for** — and it is *unsayable*, which is the only way craft rule 1 lets the game hold it.

***The trap, written down because it looks fine on screen***

***The poses must not travel with the slots.*** **CODEMUSAI lunges and CAREMUSAI opens in both editions.** *If the slot contents are swapped wholesale —* the obvious implementation — **CAREMUSAI ends up lunging and CODEMUSAI ends up opening, and 2's chart now reads backwards.** *Nothing breaks. It just quietly argues the opposite thing.*

**So it is six new drawings, not a re-crop:** CAREMUSAI at 48×48 facing right in *its own* three poses, CODEMUSAI at 56×56 facing left in *its own*.

***And invariant 3 is not in tension with this***

*The removal on the same day was correct and this is not a reversal of it.* **`_BLUE` was loading a different creature** — Jigglypuff — ***and a different creature is a different argument.*** **A different seat is the same argument, witnessed from the place the other edition is named after.**

### 7.14d The copyright screen — three names and one span

**Built 2026-09-01.** The screen under the falling note is **not drawn with the game font.** `LoadCopyrightTiles` copies a raw tile bank into `vChars2 tile $60` and `PlaceString` walks a list of tile IDs — so the words are pictures, and *the budget is the constraint that shapes the design.*

**Thirty-one tiles.** `$60` – `$7E`; `$7F` has to stay the blank the text-box tiles left there, because the string uses it as a space. **All three lines share one bank**, so the stamp is drawn once and re-used.

| | Tiles | |
|---|---|---|
| `©'11–'26 CODEMUSIC` | 7 | the top line, where the platform holder's name used to be |
| `©'11–'26 SeeingSharp` | 8 | |
| `©'11–'26 Psychology/Code` | 11 | |

***The stamp is the lineage, compressed.*** `docs/lineage.md` dates the work from Neural Crossroads in 2011; the cart is 2026. **`'11–'26` is fifteen years stated as a span** rather than three revision dates, which is what vanilla's `'95.'96.'98` actually was — *three companies each stamping their own year.* **One author cannot honestly do that**, so the shape changes to a span even though it costs the same five tiles.

### Twelve glyphs the font never had

**The typeface is vanilla's copyright font, lifted pixel-for-pixel out of the strip it replaces** — 2px strokes, caps on rows 1–5, x-height on 2–5, descenders to row 6. It carried only what `Nintendo Creatures inc. ©'95.'96.'98` needed.

**`S P h l g p y / 0 1 2 –` are drawn here**, in `tools/gencopyright.py`, in the same idiom, **plus the caps `O D E M U I`** that CODEMUSIC needs. *This also retires the note in the 2026-08-29 entry that the title-screen year was deferred for want of digits —* **`1` and `2` now exist.**

### The asset that is printed by three screens

***A first version borrowed CODEMUSIC from `gfx/title/gamefreak_inc.2bpp`,*** which sits immediately after `copyright.2bpp` so a single `CopyVideoData` spans both — that is why vanilla's copy length is `GameFreakLogoGraphicsEnd – NintendoCopyrightLogoGraphics`. **Trimming that asset from 9 tiles to 7 bought the two tiles `Psychology/Code` needed.**

**It also broke two other screens.** *The title screen and the shooting-star splash print the same file as a fixed nine-tile run* — so both printed seven correct tiles and then two of whatever VRAM happened to hold. **The title screen read `CODEMUSIC AB`.**

***The lesson is the same one the string sweeps kept teaching:*** an asset's size is a contract with **every** caller, and this one had three. **So the copy stops at `NintendoCopyrightLogoGraphicsEnd` now** — a label this design adds — and the bank is self-contained. *CODEMUSIC is set here in the same face as the other two lines*, which it should have been from the start; **the bolder splash face still carries the title screen and the splash**, where it is the only thing on the line.

### On removing the vanilla lines

**This deletes `Nintendo` and `Creatures inc.` from the built ROM.** *That is a different act from renaming a town* — the engine is still pret's disassembly of their work, and the notice is a legal one rather than a decoration.

**It is done because the ROM is not what this project distributes.** `patches/` is; `gfx/` here contains nothing Nintendo-derived; and **both READMEs credit pret and Nintendo directly.** *A total conversion that ships no copyrighted content still owes the attribution — it just does not owe it on this screen.*

## 7.14g The front door was in F minor

***The same complaint, in the other medium, in the same message:*** "**the music feels dark (as does the current user image on that screen).**"

**The title screen had been repointed at `Music_Dungeon3`** — "*Echoes of the Algorithm*", *F minor,* ***the track where Crystal goes into the system after the removal***, carrying the doubt Ty handed her. **7.14c called that repoint "same bank, zero extra bytes",** and it was: *a saving of ninety-six bytes and a mistake of tone.*

***The cave music is correct for the cave.*** **On the title screen it told the player this is a grim game before a single line of dialogue could say otherwise** — and 4.6's whole point is that she does not come back down. **That is not the note to open on.**

### *C, and not an arbitrary bright key*

**`audio/music/titletheme.asm`** — *C major, **I–vi–IV–V**, the most open progression there is, twice through*, at 150.

***7.4 gives Blanche C pentatonic — high and open — and Slate is that same scale with both gaps filled.*** **So the title screen is in the key of the town you leave from.** *Nobody notices; it just feels like somewhere you have been.*

**`Music_Dungeon3` keeps its own body**, and Mt. Moon, Rock Tunnel and Victory Road are untouched. *The saving was never worth what it cost.*

***Revised 2026-09-04 — C#, because the door is not a room***

***7.4's key system is a fact about the WORLD***, and 9.14 has already established that **the title screen is not in it.** *That is why colour could be spent there without breaking invariant 5* — the front door is outside the fiction, so it obeys the author rather than the map.

**Which frees the key to carry the other signature.** `SeeingSharp` is the second of the two author brands, 1570 spends it on **CONTEXT** — *perception and framing* — and **C sharp is the same word a third time**: the language, the brand, and the pitch.

*The cost is named rather than waved away.* **The title no longer sits in Blanche's C**, so the "somewhere you have been" effect from the paragraph above is gone. **It was worth about one unnoticed feeling**, and what replaces it is the door being *a semitone above everywhere you will go* — which is what a door should promise.

**I–vi–IV–V and 150 are unchanged.** *The lift has to come from the arrangement*, not from the harmony: a held opening, an ascending call, an answer that goes higher than the call did. **7.14g's real finding still stands — the mistake was tone, and tone is not key alone.**

***Revised again — the arrangement had no instruments***

***The transcription was right and it still sounded like a Game Boy.*** C# major, 152 BPM, 68 melody notes, the key confirmed at **0.88 against 0.70** — and the report back was "**it sounds like the past gameboy like music.**"

**`mid2agb` emits a `VOICE` byte only where the MIDI carries a program change,** and `tools/mp3midi.py` emitted none. Every track therefore played on **slot 0 of `voicegroup137`, which is a keysplit, with `voice_square_1` behind it at 1–3** — *PSG square waves, on hardware that carries a sample ROM.* **The theme was being rendered by the machine we ported away from.**

***Gen 3's banks are General MIDI.*** `voicegroup149` puts glockenspiel at 9, tubular bell at 14, harp at 46, timpani at 47, oboe at 68 and flute at 73 — **GM 10, 15, 47, 48, 69 and 74, zero-indexed, exactly.** So a bank is addressable by ordinary program number, and *the samples were never the problem*: a Steinway, an SC-88 string ensemble, trumpets, french horns, a choir and timpani are **already in the ROM**, scattered across banks that no one song draws on.

**`tools/gbavoices.py` writes `voicegroup191`** — one bank, GM-mapped, every slot sampled, and unused slots holding the piano so that *a program number we did not intend still arrives as an instrument.* **The failure mode that started this was silence and then a square wave; it should not be reachable twice.**

***And three channels was never the GBA's number***

***7.4's three-channel arrangements are a fact about the Game Boy***, and I carried the constraint across a port that does not have it. **Vanilla's `mus_vs_trainer`, `mus_vs_gym_leader` and `mus_vs_deoxys` all run ten tracks.**

*So the lift the revision above asked for goes where it always belonged.* The theme is now **five**: a trumpet carrying the call, a glockenspiel doubling it, the string ensemble underneath, a fingered bass on the drive, and **a timpani on the downbeat** — which is the part that makes it a theme rather than a melody.

**`engine/` is untouched and stays at three.** *That constraint is real there, and 7.4 is still correct about it.*

***SEEKMUSAI takes the Pikachu slot, and the slot is three sprites***

***Chosen over S.T.A.R.R., which was the instinct and the wrong answer.*** 4.7 puts S.T.A.R.R. **dormant past the Review Board** — *a creature asleep for the whole game cannot be the one that welcomes you into it.* And the Pikachu slot is a fact about the anime's marketing, not about FireRed's fiction; the creature there has nothing to do with the briefing beside it.

**What the briefing actually is: an INDEX that starts out empty, "almost everything here is written down somewhere", and it closes on "*Some of them disagree.*"** That is a screen about finding and reconciling records — **so it is narrated by the daemon whose item is EMBEDDING**, which is the thing that makes an index searchable. *Not a pun; the actual relationship.*

**It is also the sibling not already on the title screen**, so the family reads across both without repeating a drawing. `TYPE_FLYING` fixes it at VECTOR — **sanguine red, not Jolteon's yellow.**

***The rig is body, ANTENNAE and VISOR***

`oak_speech.c` draws the creature as three sprites with two frames each: a breath, a twitch and a blink. For a MUSAI those are the body, the antennae and the faceplate, ***so SEEKMUSAI blinks by its scanner cycling*** — which is the right verb for this one and cost nothing to arrange.

**Only the body is drawn art.** All three layers were generated and *the two overlays did not seam* — the antennae sat on a plain dome while the body has a swept crest, and the visor was a different faceplate shape. **Overlaid, the helmet would have changed shape every time it twitched or blinked.** So `tools/gbaseek.py` builds the overlays **from the body**: the antennae layer is its own top sixteen rows with stalks drawn on, the visor its own faceplate with a line drawn across. *The seam is exact by construction, because the surrounding pixels are the same pixels.*

***The breath is one pixel, and that is not a style choice***

***`SpriteCB_Pikachu` is three lines and it governs the whole rig:***

> `sprite->y2 = gSprites[sprite->sBodySpriteId].animCmdIndex;`

**Each overlay's vertical offset is the body's frame number — 0 or 1.** So the antennae and the visor follow the body by **exactly one pixel**, and a body that moves further tears away from its own overlays. *The generated second frame was a full crouch*, and composited it doubled the helmet.

**So frame B is synthesised: frame A dropped one pixel, with the shadow left where it is.** That is what vanilla does — its own two frames are a one-pixel settle with 33 pixels of hand touch-up. **The generated crouch is unused.**

*One coordinate moved.* The visor sprite was at `(24, 13)`, which is where Pikachu's eyes are; solving the layout against that box forced the creature down to **sixteen pixels wide**, so it moved to `(24, 17)` instead. **Four pixels of sprite position was cheaper than shrinking the character.**

***The controls guide, and the one screen that knows it is a game***

"***one day I'd like to put it on real GBA hardware, but for either case...***" — the controls guide names six buttons, and **on an emulator not one of those names is where it says it is.**

**This is the only screen in the game that addresses the player as a player**, so it is the one place a wink costs nothing. *Craft rule 6 licenses it directly:* a game that is funny in its ordinary moments earns the right to be serious in three or four of them, and a button legend is as ordinary as a moment gets.

| | |
|---|---|
| was | *"The various buttons will be explained in the order of their importance."* |
| is | **"The buttons, in the order they matter. Names, not places. Check your bindings."** |

***It had to be a rewrite rather than an addition.*** The page 1 window is **30 tiles by 4** — 240px wide and **two lines tall**, and vanilla already used both. 212px and 217px against a 236px budget; vanilla's own widest line here is 221.

**"Bindings" is not a loanword from emulator settings — it is already ours.** 7.2 replaced *catch* with **bind**, the build script is `bindDaemons.sh`, and the binding sound is a jingle with its own entry in the song table. *On a real cartridge the line still parses* — those buttons are bound to a shell — **so the joke does not expire the day the ROM reaches hardware.** That was the condition, and it is the reason the line does not say "emulator".

***Ruled 2026-09-04 — the world is chiptune and the front door is not***

***Asked directly, and answered directly:* "*chiptune I think is good.*"** The twelve tracks carried over from `engine/` were written for four hardware channels and they keep that texture. **9.14 already put the title screen outside the fiction**, which is what lets the front door have the sample ROM while the world it opens onto does not — *the same split that let colour be spent there without breaking invariant 5.*

***But "leave them alone" was not available, because they were not left alone.*** They played `VOICE 0`, and slot 0 of the banks they happened to point at is a **keysplit — a drum-kit key map — in ten cases out of eleven.** That is not chiptune. It is an accident that sounds a bit like one.

**And the PSG slots do not agree between those banks either.** Slot 87 is a programmable wave in three of the eleven and a filler square in the rest, so the same wave part would have changed instrument between one town and the next. *A texture that varies by location is not a texture.*

### *`voicegroup192`, and the Game Boy's four channels are four programs*

**GM 80 and 81 are "Lead 1 (square)" and "Lead 2 (sawtooth)", and Gen 3 puts its PSG voices exactly there** — the same General MIDI agreement the orchestral bank relies on. `pokered` declares its channels in order, Pulse 1, Pulse 2, Wave, Noise, and `port_music.py` keeps that order, **so the channel index *is* the hardware channel** and four programs name its four voices.

*A stray program stays in the family:* filler slots hold pulse 1 here and the piano in 191, so a mistake in either bank is still the right kind of sound.

### *And the tool that owns the notes owns the bank*

***`port_music.py` had `mus_title` in its list and quietly overwrote it*** — the stems transcription, and its `-G191` — the first time it ran after the bank change. **A song's notes and its voicegroup are one decision**, so the tool that writes the MIDI now also keeps `midi.cfg` pointed at the right bank, and `SUPERSEDED` keeps its hands off the one track it no longer owns.

### *Revised a third time — stems, and the end of guessing*

***4.9's gate fired again, in a new shape.*** AIVA's current front end is an **audio** model and has no notes in it to export; its symbolic composer is a separate legacy app whose chord-progression field **cut the description off mid-sentence**. Four services in, *the answer was the same as it was for Gemini: stop asking a service and do the work.*

**What `mp3midi.py` was actually doing is worth stating plainly, because it sounded like a transcription and was partly a construction.** A mix is one signal. It split melody from bass with a filter at 250Hz, and it **inferred the middle voice from a chroma profile** — not because that is a good way to hear an inner voice, but because `pyin` is monophonic and *cannot hear two notes at once.* Both were approximations of an instrument list.

***Suno exports stems.*** Eight files, one instrument each, and **Basic Pitch is polyphonic** — so chords come back as chords. Nothing in the output is inferred any more: **every note was heard in a file that contained one instrument.**

*Two findings worth keeping*

***The stems are time-aligned at zero.*** They are shorter than the mix and than each other, which looks like offsets and is not: the export trims where an instrument stops. **Five of seven stems' loudness contours correlate best against the mix at exactly lag 0**, and a separator emits aligned stems by construction. *Waveform correlation said otherwise and was wrong* — separation re-renders, so the samples no longer match even where the music does.

***Taking the top note of a chord is wrong.*** The first reduction took the highest note in each cell and the brass came back reaching **G#6**, the keyboard **F7** — upper partials, which Basic Pitch correctly reports as notes, and which made the line leap an octave and back. It takes the **loudest** now, and where two are within a hair of each other **the one nearest what it was already playing** — which is the assumption a musician transcribing by ear makes without noticing they have made it.

**The measure of it:** the string part came back with **two notes out of key in a hundred and thirteen.** The old tool snapped 10% of a melody it was far less sure of.

*The title screen could not be read, and it was one lost nibble*

*Raised from a screenshot:* **"the intro screen for both editions are hard to read the text..... and the PRESS START is hidden under caremusai's sprite."**

**`tools/gbastrip.py` rebuilt this layer's tilemap from tile indices alone.** A map entry's high nibble is its palette bank, vanilla writes **15** on all six hundred and forty tiles, and the rebuild wrote **0**. *Bank 15 is the background palette and bank 0 is the wordmark's,* so PRESS START, the copyright line and all four screen fills quietly started taking their colours **from the ramp that draws DAEMONS** — yellow text on a yellow band, orange on orange, and the "stripes" nobody chose were the wordmark's own gradient leaking out behind it. **The blink stopped working too**, because `Task_TitleScreen_BlinkPressStart` writes entries 1-5 of bank 15 and nothing was reading bank 15 any more.

***One nibble, four symptoms, and every one of them looked like a taste problem.*** That is the argument for 8.5's rule in a form the eye cannot supply: **a screenshot shows you the result and never the cause.**

### `tools/gbatitleview.py` — *compositing it offline, because squinting failed*

**The layout was being fixed by looking at emulator screenshots and guessing at coordinates**, which is how the sprites came to stand on the words in the first place. *So the screen is now composited outside the machine* — the real tiles, maps and palettes, at the real coordinates, following the GBA's own priority rule that **an OBJ draws above a BG of equal priority**. It reports every layer's ink box, and a collision is arithmetic:

| | |
|---|---|
| DAEMONS | x 26-228, y 7-66 |
| the edition's name | x 100-153, y 76-88 |
| PRESS START | 89px wide — **was** x 43-131, y 129-135 |
| a 64x64 creature | about **31px either side** of its centre |

***The two creatures were at 76 and 164***, which put their art at x 44-107 and 132-195 — **on the subtitle at one end and on PRESS START at the other.** They are at **50 and 190** now, PRESS START moved down one tile row and right four (both whole rows and whole tiles, so no glyph was redrawn), and the copyright line is centred on the 240 the GBA shows rather than the 256 the tilemap is wide. *Nothing is drawn on top of anything.*

### *The ground goes dark, and the palette's indices are written down*

**`tools/gbatitlepal.py` owns bank 15 now**, and its whole value is the comment: *index 6 is the two bands **and** the blink's off-colour, so it must be the colour behind the text; 11 is the stage the creatures stand on; 1-5 are the letters; 10, 12 and 15 the copyright and the strips.* **None of that is written down anywhere in `pokefirered`** — it was recovered by rendering each index on its own, and it would have been lost again.

***A bright wordmark, two lit creatures and a bloom in the middle all want the same thing***, which is somewhere dark to happen. Each edition's ground is its own wordmark's hue several stops down — **cold slate under CONTENT, deep violet under CONTEXT** — so the letters read as lit rather than as pasted on.

### *And they were holding still*

*They stood perfectly motionless through a reveal that takes several seconds*, which reads as artwork rather than as two creatures about to do something. **A slow breath, a quarter cycle apart** — in lockstep it reads as one animation played twice.

*The sixteen-frame scene is deferred, not abandoned*

**9.14 cut vanilla's sixteen-drawing intro** because 2.4's face-off does the same job on the title for two drawings. ***It is wanted back*** — as our sixteen frames, which is what `docs/intro-sequence.md` sketches. **It is not back yet because the screen it would hand over to could not be read**, and an intro that leads into an unreadable screen is the wrong order to build in.

*The lesson is that both media drifted the same way, unprompted*

***Nobody decided the game should be dark.*** **The art drifted there because a prompt asked for a wide brim and never asked for warmth; the music drifted there because a repoint was free.** *Two independent cheap decisions, each defensible on its own, arriving at the same wrong tone* — **and it took a player looking at the title screen to notice.**

**Craft rule 6 is not only about dialogue.**

**7.14f Craft rule 6 governs the art, and the art had stopped obeying it**

***Raised 2026-09-02, from a screenshot:*** **"our is in the shadows and doesn't have a face... is it a dark game? because if so, that is not what I want to make."**

**It is not, and this document says so in its own words.** *Craft rule 6:* **"A game that is funny in its ordinary moments earns the right to be serious in three or four of them. Corpus employees are cheerful and absurd; the horror is what they are cheerful *about*."**

*And the rest of the bible holds the same line everywhere it matters:*

| | |
|---|---|
| **Scorn** (7.x) | "*Never sneers, never gloats.* ***Genuinely warm****, genuinely pleased to meet you.*" |
| **CAIRN** (5.1) | "*His creed, said plainly and* ***sympathetically****...* ***He is right, and that is the point.***" |
| **The institution** (4.x) | "*The silence is* ***not shame and not cruelty****...* ***Nobody decides to be cruel.***" |
| **Craft rule 3** | pathology words are refused outright — `#MANIAC` → `ARCHIVIST`, `TAINT` → `SKEW` (marks) / `TAMPER` (routines). ***A refusal applies to every surface***, and 2.8b is what happened when it was applied to one |

***This design has no villain.*** **Its central move is refusing to make anyone a monster**, and the greyscale is *invariant 5's spending decision about where colour is used,* **not a mood.**

*The art was arguing the opposite*

**Every player prompt in `docs/gemini-prompts-player.txt` was written asking for a figure who is "*still, composed*" and "*confident*" — and not one of them asked for warmth.** *Then it asked for a wide flat brim,* which ***shadows a face by construction.*** **All three generated samples came back with the face in darkness and one had no face at all.** *That was a fault in the prompt, not the generator.*

**The comparison is the proof.** Vanilla's title figure spends its head region on **light** — hair, a lit face, open shapes. Ours spent it on **a black void with three light pixels in it.**

***So the tone paragraph is now mandatory on every character prompt,*** and it says the face must be **one of the lightest areas of the figure, never the darkest.** *At 40×56 a face is six pixels across;* **if those six pixels are black, the character is a silhouette, and the art is arguing against the writing.**

### 7.14e The player — the figure you look at all game

**Built 2026-09-02.** The player exists in nine files across four sizes, and *only one of them is the one the question is actually about:* **`gfx/sprites/red.png`, six 16×16 frames.** Facing right is the side frame with OAM's x-flip, so there is no fourth direction.

| | | |
|---|---|---|
| `gfx/sprites/red.png` | 16×96 | **drawn** — down, up, side, and a walk frame for each |
| `gfx/sprites/red_bike.png` | 16×96 | **provisional** — the same character; the bicycle itself is under review |
| `gfx/overworld/red_fish_*.png` | 16×8 | three fishing half-frames |
| `gfx/player/red.png` | 56×56 | the *this is you* screen, and the Hall of Fame |
| `gfx/title/player.png` | 40×56 | **generated** — the title screen, where the daemons cycle past |
| `gfx/player/redb.png` | 32×32 | **every battle in the game** |
| `gfx/player/shrink1/2.png` | 56×56 | the naming-screen animation |

***The overworld sprite had to be hand-drawn, and that is not a preference.*** Every other art pipeline here **resamples a large illustration down,** which is proven at 40–56px and **fails at 16** — a generated figure loses its face, its silhouette and its walk cycle in the same downsample. *8's sprite doc already ruled out generating anything under 16×16;* this is that rule meeting the asset it was written for.

**It is drawn in vanilla's idiom on purpose** — same height, same 16×16 footprint, same two-tone shading, same outline weight. **The player stands next to thirty NPC sprites that are still vanilla's,** and *a figure drawn to a different set of rules would read as a mistake rather than as a person.* **What changes is who it is.**

***Between a scientist, a magician and a wizard***

***A first version was a cap and a strap*** *— Ash with a satchel*, and correctly rejected. **The brief that replaced it was to sit the player between three archetypes**, and the resolution is that **one of them costs nothing**:

> A lab coat and a wizard's robe are the same silhouette.

**So the coat carries two of the three for free** — long, sleeved, reaching the ground — *and the hat only has to carry the third.* **A flat brim, wider than the shoulders**, is the widest shape on the figure and the one thing **no vanilla NPC has.** *The overhang is what makes it a hat;* **a brim the same width as the shoulders stacks into a mailbox**, which is exactly what the first attempt at it did.

***The arms took three tries, and the failure is worth keeping***

| Attempt | What it read as |
|---|---|
| Black columns **flush** against the torso | **outline.** *The figure had no arms at all* — and that is what was noticed |
| Separated by a **transparent** column | **a hole punched through the armpit** — the limbs floated |
| **Inside the silhouette, separated by tone** | *arms* |

***Sleeves at level 2, torso at level 1, strap at level 0.*** **Sixteen pixels has no room for a gap, so the separation has to be tonal** — and once it is, **the black strap cannot be confused with either**, which is what makes 1.3's box readable at this size. *The player is carrying one, on a strap, in every frame that could show it;* **from behind it is the satchel itself.**

***At 40×56 the unit finally fits***

**The title screen is the one place with room for it.** *The 16×16 sprite can only imply the box through the strap;* **`gfx/title/player.png` has 40×56 and can draw the thing itself** — a vent line, an indicator, a port, and **no seam anywhere that would let it open.** *1.3 calls it a machine you offer a daemon as a host,* **so it must not look like a container.**

*The hand-drawn first pass was replaced by generated art the same day,* **and it exposed a rule the daemon pipeline had never had to obey.**

***White is transparent, and the coat is white***

***A daemon sprite is drawn to the background layer; the title figure is drawn as OAM.*** **On OAM the lightest level is transparent** — so **a white lab coat renders as a hole**, and the figure comes back as a ghost: hat, strap, boots, and nothing between them.

**`--solid` is the fix, and where it floods is the whole trick.** *A flood run on the finished 40×56 finds nothing* — **at that size the outline is one pixel and full of gaps, so the background leaks straight into the coat.** ***Run it on the source instead***, where the outline is still six to eight pixels thick and closed, **and whatever paper the flood cannot reach is interior.** *In this figure that was 234 pixels that would have been see-through.*

### *And the margin was eating the detail*

***The first conversion looked thin and empty, and the measurement says why.*** **Vanilla's title figure fills 100% of the width of its 40×56 slot** — *it was drawn for the slot.* **Ours filled 57%.**

*A standing person is about **0.43** wide-to-tall; the slot is **0.714**.* **Keeping the subject's honest proportions letterboxes it**, and ***the 43% it gives away is resolution, not whitespace*** — at forty columns there is none to spare, and it is the difference between a readable strap and a grey smudge.

**So `--fill` maps the subject's own box onto the frame and scales the axes independently.** *It makes the figure stockier, and that is correct:* **every Game Boy sprite of a person is stockier than a person.**

*The generous-hem sample was tried too and is worse* — **at 0.93 aspect `--fill` squeezes rather than stretches**, and the flare collapses into a grey triangle with a person balanced on it. **The one to generate is the plain standing pose.**

### *A one-pixel hole on an outer edge is a bite*

***The outline pass blacks a cell only when the source line covers enough of it*** — `COVER`, default **0.34**. *A cell the line merely grazes stays grey.* **In the middle of a figure that is invisible. On the outer edge it is a hole**, and the eye does not read it as a hole: **it reads it as a piece missing from the coat.**

**The shoulder had exactly one such pixel**, and it was spotted on a screenshot at native size before any of the tooling noticed it.

***The first fix was to lower `COVER` until the edge closed, and it was the wrong instrument*** — **it thickens every line in the figure to repair one edge**, and it still left a hole under the other cuff. **`--close` fills only the gaps, and only where they touch the background, which is the only place they show.** *Eight pixels on this figure, zero holes left, and **less** total black than the threshold tweak used.*

**A threshold is a global answer to a local problem.**

**Two other non-square bugs fell out of the same pass:** the outline pass ran `range(size)` on both axes, so **it was only covering 40 of the 56 rows**, and the cropped-too-tight warning divided by `size²`. *Neither could show up while every target was square.*

*The hem swings, but a hem alone slides*

**The coat reaches the ground, so the walk is the hem swinging side to side** — which is what a long coat does, and costs three rows instead of a leg pair per frame.

***On its own it was not enough.*** **With nothing at the floor to read a step against, the figure moved like a blob being dragged** — the hem changed shape but nothing said *foot*. **So the coat stops one row short and two shoes show beneath it:** *together when standing, apart when walking, and in profile one hides the other.*

**One row of pixels**, and it is the whole difference between walking and gliding.

*The bicycle sheet is the same six frames from the waist up*, with a bike below the rail. **Vanilla animates the whole bicycle sideways**, which at 16px reads as *the rider lurching* — **so ours moves the wheels instead and flickers their tone**, which is what spokes actually do.

**The intro face-off was only in one edition**

***Found while auditing the above.*** **Vanilla gives each edition its own front mon** — `IF DEF(_RED)` takes `red_nidorino_*`, `IF DEF(_BLUE)` takes `blue_jigglypuff_*` — and *only the* `_RED` *path had been replaced.* **CONTEXT was still opening on Jigglypuff.**

**The conditional is gone.** *8.4 fixed both editions to the same story, and the face-off* ***is*** *that story* — CODEMUSAI against CAREMUSAI, **LOGIC failing against CONTEXT.** ***An argument that changes by cartridge is not one*** (invariant 3, arriving somewhere it was not written for), so both editions now include the same three frames and `blue_jigglypuff_*.png` falls out of the build.

#### 7.15 The space audit — a claim I repeated for days and never measured

**Measured 2026-08-30 from the linker map.** Every earlier statement that the music banks were full was **estimated, carried forward, and wrong.**

| Bank | Holds | At session start | After six tracks went in |
|---|---|---|---|
| `$02` | Music 1 | 8 bytes | **803** |
| `$08` | Music 2 | 175 bytes | **176** |
| `$1f` | Music 3 | 718 bytes | **2813** |

**901 bytes free before. 3792 after — having *added* six tracks.**

**The arithmetic that dissolves the problem:** our tracks are small — Brazen **104** bytes, the dark places **89**, Blanche **129**, Quicksilver **478** — against a **median vanilla track of 427** across 47. **Replacing a vanilla body with one of ours frees more than it spends**, every time.

**So the claim was true only of appending without displacing**, which is not what a total conversion does. Kanto's soundtrack is not an asset here; **every vanilla track is a slot**, and there are 47 of them.

*The residual constraint is real but different.* Some maps share a music slot — `MUSIC_CITIES1` covers 37 — so giving one map its own tune means **adding a constant and repointing a line of `data/maps/songs.asm`**. That is an edit, not a budget.

**The lesson is the one 0.2 already states**, and it caught the project out rather than the source: *a number nobody measured is not a constraint, it is a rumour.* This one shaped scheduling for days — the Owl's scene was recorded as unwritable-for-now on the strength of it (4.23).

## 8. SCOPE CONTROL

> *The graveyard of ROM hacks is full of projects that designed 151 creatures and shipped zero towns.*

#### 8.1 Vertical slice — build this before anything else

**Blanche → The Bleed → Callow → Underpaint → Slate → Benchmark 1.** Playable end to end.

- 12 daemons (3 starters + 9 wild), front and back sprites
- Full type chart, all 15 — it is one file, do it properly once
- Crystal Clear intro sequence
- Benchmark 1 leader and party
- Two city themes and one route modulation
- Route signage system (number plus local name)

If that is fun, the remaining daemons are labor. If it is not, you learned it for the price of twelve sprites.

**The slice is built, 2026-08-31.** Type chart, intro, route signage, CAIRN at Benchmark 1, **Slate's theme and The Bleed's modulation** (7.14a), and **every creature the player meets between Blanche and Slate named, drawn and filed** (8.1a). **Thirty-two daemons now carry original front and back art** — the three starter lines, the six wild lines, the four MUSAI, ARTSAI and S.T.A.R.R.

~~***The next milestone is the one 8.6 named: Benchmark 1 beaten on hardware.*** Not another section.~~

*Met, and the record is what was missing — see 8.10.* **The slice has been played through by hand, and an agent ran 770 steps through it unattended** — *all eight MARKS, thirteen maps, past Slate and out onto Route 4 toward Doldrum.* **This line stood unstruck for eleven days and 171 version bumps while it was already false**, *which is 8.7's rule about unswept rulings landing on a milestone instead of a name.*

#### 8.1a The nine wild daemons — built 2026-08-31

**The starters are paradigms. These are substrate.** The player walks through grass full of the primitives a system is made of, **carrying one way of learning** — which teaches without repeating 8.2's lesson.

| Vanilla | Ours | Type | Pays twice |
|---|---|---|---|
| RATTATA → RATICATE | **NIBBLE → OVERFLOW** | CONTENT | four bits, and what a rat does; then four bits past their bounds |
| PIDGEY → PIDGEOTTO → PIDGEOT | **PACKET → RELAY → BROADCAST** | CONTENT/VECTOR | **content that moves, which is the typing** |
| SPEAROW → FEAROW | **PING → FLOOD** | CONTENT/VECTOR | one packet demanding a reply; then the attack |
| CATERPIE → METAPOD → BUTTERFREE | **CRAWLER → PENDING → INDEXER** | SWARM | **crawl, queue, index — a real pipeline** |
| WEEDLE → KAKUNA → BEEDRILL | **SCRAPER → BUFFER → INJECTOR** | SWARM/**CORRUPT** | the same shape, gone wrong |
| PIKACHU → RAICHU | **SPIKE → SURGE** | SIGNAL | a neuron firing; then one that never stops |

**The pair the set is built around**

***CRAWLER and SCRAPER.*** Two swarms, the same shape, met in the same forest. **One follows every link it finds. The other takes the same things by the same method, and nobody gave it permission.**

**The type chart says which is which by making the second CORRUPT**, and nothing else has to. **That is 2.5's SWARM doing for data collection what the starters do for paradigms** — an argument delivered as a matchup.

*And the lines evolve along processes rather than just getting bigger.* **CRAWLER → PENDING → INDEXER** is crawl, queue, index; *what it cannot file, it drops.* **SCRAPER → BUFFER → INJECTOR** is the identical pipeline with permission removed at every step.

**Two names were already better than the ones drafted here**

**JIGGLYPUFF and WIGGLYTUFF were already `SUSPEND` and `HIBERNATE`** from an earlier pass. The draft proposed `LULL → BLOCK`; **the existing pair wins.** They are the two real power states, they escalate correctly — suspend holds in memory, hibernate writes to disk — and **they rhyme with 4.20a, where a man was suspended by a machine.** 4.6's MOCK precedent again: **the capability was always out there in the grass.**

*Kept, and the drafts discarded.* **Check before renaming.**

#### 8.2 Starter trio — learning paradigms

| Paradigm | Types | Draft line (names provisional) |
|---|---|---|
| Supervised | CONTENT → CONTENT/LOGIC | **LABEL → RUBRIC → CANON** |
| Unsupervised | VECTOR → VECTOR/LATENT | **CLUSTER → LOCUS → MANIFOLD** |
| Reinforcement | GROWTH → GROWTH/SIGNAL | **ROVERCUB → ROVERSEER → ROVERBYTE** |

**All nine built 2026-08-31, and no vanilla starter name remains in the cartridge.**

**`Corpel` collided with Corpus and `Canonex`'s `-EX` earned nothing** — it reads as *ex-* or Latin *out of*, neither of which belongs at the top of a supervised line. **`LABL → RUBRIC → CANON`** fixes both, and gains something: **the names get more formal as the creature gets more authoritative.** *Labl* is barely a word, *rubric* is a technical term, *canon* is an institution. **A rubric is also the marking scheme**, which is what a world of benchmarks and marks runs on.

**`Nebulon` was not vague, it was backwards.** A nebula is **diffuse — less structured than a cluster** — so the line ran grouped → scattered → structured and lost ground in the middle. **`LOCUS` escalates properly:** a cluster is points that *happen* to be near, a locus is points that share a *rule*, a manifold is the *surface* they lie on. **Proximity → rule → shape.**

### The Index entries, and the argument they carry

| | Category | The entry |
|---|---|---|
| **LABEL** | EXAMPLE | *Holds one example and the answer that came with it.* ***Someone else decided what it means*** |
| **RUBRIC** | CRITERION | *It no longer needs answers. It has the rule that made them.* ***Nothing checks the rule*** |
| **CANON** | SETTLED | *Every case it meets is decided by cases it has already met.* ***New things are filed as errors*** |
| **CLUSTER** | GROUPING | *Puts near things with near things.* ***Nobody told it which things matter. It has guessed*** |
| **LOCUS** | CONDITION | *It found a rule the points obey.* ***It cannot say what the rule is for, only that it holds*** |
| **MANIFOLD** | SURFACE | *The scattered points were on a surface all along.* ***It can show you the shape, not a name*** |

**The two final stages are the paradigms' failures, side by side. CANON is accurate and files novelty as error. MANIFOLD finds real structure it cannot name.** One is certain and wrong at the edges; the other is right and mute.

***And CANON's line is the Meeting Room minutes.*** *New things are filed as errors* is what happened to a woman who moved between fields (4.20) — **printed in an Index entry, two hundred slots away from the building it describes.**

*Types implemented 2026-08-29, verified in the ROM.* The secondary appears at the **final** stage, exactly as vanilla's Charizard gains FLYING — so stages 1 and 2 are pure and stage 3 splits. Bulbasaur's CORRUPT secondary is dropped, since 8.2 makes the reinforcement line GROWTH.

| Slot | Was | Now |
|---|---|---|
| Charmander → Charizard | ENTROPY → ENTROPY/VECTOR | **CONTENT → CONTENT/LOGIC** |
| Squirtle → Blastoise | FLOW | **VECTOR → VECTOR/LATENT** |
| Bulbasaur → Venusaur | GROWTH/CORRUPT | **GROWTH → GROWTH/SIGNAL** |

The lab dialogue moved with them — *the CONTENT DAEMON, the VECTOR DAEMON, the GROWTH DAEMON*. This is the second time those three lines have been rewritten and the last: 1.2 predicted exactly this, which is why the other nineteen type-words are being held until 5 settles the Benchmark leaders.

**Movesets are now the blocker, and one number decides how hard.** Gen 1 has **6 FLYING moves in total** — that is the whole VECTOR pool for a daemon the player raises for forty hours. It is less alarming than it sounds (FIRE has **5**, so vanilla's own fire starter is thinner still, and Gen 1 mons lean on NORMAL moves for damage), but VECTOR/LATENT compounds it because GHOST is nearly empty too. **CONSENSUS is the precedent**: 2.5 already established that when a type is unusable the fix belongs in `moves.asm`, not in the chart. The unsupervised line will likely need its own move the way SWARM needed one.

*Until movesets follow, the starters know their vanilla moves* — a CONTENT daemon that learns EMBER. Mechanically fine, narratively nonsense, and 9's remaining work.

Supervised is strong early and rigid late. Unsupervised is confusing early and excellent late. Reinforcement is inconsistent with the highest ceiling. **The paradigms should feel like their real tradeoffs.**

### The names are the weakest thing in this document

**And they have no reasoning behind them.** The table above says *provisional*, 10 says *placeholders*, and nothing was ever written down. Two faults, stated so the next pass starts from a diagnosis rather than a blank page.

**They are single-duty.** Every load-bearing term here pays for itself twice or three times — BIND is `bind()` *and* binding a daimon *and* a bond; CACHE is a hoard *and* a memory cache *and* cash; PIP is a halftone dot *and* the thing that installs a package. **`Labl` says *label* and stops.** It is the only part of the lexicon that does one job.

**And the dropped vowel is borrowed from the wrong register.** Grimoire names are vowel-*rich* and Latinate — Vassago, Marbas, Focalor, Buer. Truncation is a 2010s branding tic: `Labl` and `Clustr` are not cryptic, they are Flickr. Compare what Gen 1 actually does — BULBASAUR, CHARMANDER, SQUIRTLE are **portmanteaus of whole words** that sound like creatures.

### The register question — technical surface, or true names

**The house style is both at once**, which is why 1.5 chose BIND: the Unix call *and* the ritual phrase. So the question for the bestiary is not *literal or mystic*; it is **which one is on the surface.**

Everything the player handles is technical on top — USER, BENCHMARK, CHECKPOINT, INDEX, DETACHED. **If the creatures invert that, the naming convention starts doing 4.2's work for free.** 4.2 says the Index *can only measure content* and has no field for what Crystal cared about. Give daemons **true names** — a register the Index's flat bureaucratic vocabulary cannot reach — and the artifact's insufficiency is visible on every page of it, without a line of dialogue. The world is corporate; the things inside it are older than the taxonomy trying to hold them.

*Left entirely unsaid, as rule 1 requires:* in the tradition, **knowing a thing's true name is what lets you bind it.** The Index records the name and then you can BIND. Never remark on it; simply let the mechanic keep that shape.

**The test any candidate must pass is the one BIND passed: does it work twice?** *VERAX* (Latin *verax*, truthful) as a spirit-name and as **ground truth**. *LATENS* (lying hidden) as a spirit-name and as **latent space**. *AVARIS* (avarice) and **ε-greedy**. Those are illustrations of the shape, not proposals.

**The risk, and it is the real one.** A name that sounds like a demon and decodes to nothing is fantasy set-dressing, and the creatures stop reading as processes — which is where the whole argument lives. Every one of **151 names** has to survive that check. It is a harder convention than truncation, which was going to be exhausting by daemon forty regardless.

### Recovered from conversation, 2026-08-29 — and a process failure

**All of the following was designed, offered as a v1.8 cut, and never written down.** The bible reached v3.8 with it sitting in a transcript. That is exactly what CLAUDE.md's change procedure exists to prevent, and it did not fire because the proposals ended in "*say which parts you want*" and no answer came. **Record proposals as open items immediately; do not hold them pending a yes.**

| Recovered | What it was |
|---|---|
| **MUSAI and ROVER as two of the three starters** | Real CodeMusic systems, in the slot the player lives with for forty hours |
| `Corpel` ↔ **Team Corpus collision** | A starter's second stage sharing a name with the villain organisation. A genuine bug, still in the table above |
| **CODEX** | Proposed replacement for `Corpel` |
| **OPTIMAX → Corpus** | Reassigned away from the reinforcement line; it is a better name for something Corpus builds |
| **Penphin as a trade evolution** | Below |
| **PERSPECTIVE as the second colour moment** | See 8.6 and 6 |

### Penphin — the trade evolution

**Penphin is not a starter and should not be.** The trio is three learning paradigms; Penphin is the two-hemisphere architecture — a logic model and an emotion model that must agree before acting. That is not one paradigm, it is **what happens after you stop picking one**.

**So it is obtained by trade, and the mechanic is the argument.** The dual mind requires two minds. **You cannot get Penphin alone.** Gen 1 already supports this — Kadabra, Machoke, Graveler and Haunter all evolve on trade — so it costs an evolution method that already exists.

*And Gen 1 permits the shape the idea wants:* **two different species can evolve into the same species.** So a penguin-ish line and a dolphin-ish line can both terminate in Penphin — two halves, each of which only becomes whole through another person. Neither half is complete and neither is the *real* one.

*Its name is already correct.* PENPHIN is penguin + dolphin, which is precisely the Gen 1 convention the starters are currently missing — a portmanteau of whole words that sounds like a creature.

***And the two animals are not arbitrary — they are cited.*** **`lineage.md` 3b records the source**: *The Painted Mirror*, a seven-episode audio drama from **November 2022**, where **the dolphin is the logical self** — "*vocalized, confident, and accomplished… yet very time-focused, mechanical, and sometimes cold*" — and **the penguin is the creative one**, "*quiet, organized, spiritual, and feeling.*" **The trade requirement is in there too**: "*Alone, neither mirror image of the self will hear clearly.*" ***This creature is a memoir, and the bible had been quoting it from memory for weeks without knowing it.***

### MUSAI — the Eevee slot

**Confirmed 2026-08-29.** Musai is "*one framework, many modules*" and Eevee is Gen 1's only branching line, so the shape was already right. What makes it more than a cameo is the base type: **Eevee is NORMAL, which in our chart is CONTENT.** A mind that has not specialised yet, and what it becomes depends on what it is exposed to.

| Form | Type | Why it decodes |
|---|---|---|
| **CodeMusai** | **LOGIC** | constraint satisfaction, verification, convergent reasoning — RoverByte's own description of its Logical Agent |
| **SearchMusai** | **VECTOR** | vector search *is* semantic retrieval; the name is literal |
| **TherapyMusai** | **CONTEXT** | *emotions as the colour of context* — the thread the whole project grew from |

**Why TherapyMusai is the third, and not MusaiStudio or MedicalMusai.** CodeMusai and TherapyMusai are **Penphin's two hemispheres pulled apart into two creatures** — and 2's chart makes **LOGIC fail against CONTEXT**. One mind specialises two ways, and the type chart says they beat each other. That is the game's central argument delivered as an evolution branch, with no dialogue at all. SearchMusai sits between them, which is also where retrieval sits between reasoning and feeling.

**The stones are not stones — they are what you expose a mind to.** Vanilla's item is a mineral that triggers change; ours is the input that decides what an undifferentiated CONTENT mind specialises into. So the three are named for *kinds of input*, not for elements:

| Item | Produces | Reading |
|---|---|---|
| **AXIOM** | CodeMusai (LOGIC) | what formal reasoning is given to start from |
| **EMBEDDING** | SearchMusai (VECTOR) | the representation retrieval actually searches |
| **AFFECT** | TherapyMusai (CONTEXT) | the clinical word for feeling, and *to affect* is to change |

They are a set conceptually rather than lexically — an axiom to reason from, an embedding to search with, an affect to feel with — which is how Gen 1's own key items work (SILPH SCOPE, POKé FLUTE, CARD KEY share no suffix either). All three clear the 12-character item limit.

*The near-miss worth recording:* **KERNEL** is the better word on paper — it is simultaneously the OS core, the ML kernel method, **and the thing inside a stone fruit's stone**, so it replaces *stone* with a word meaning the same physical object. It loses on two counts: `CONTEXT KERNEL` is 14 characters, and alone it reads as *system* rather than *logic*.

### ROVER — a line, not a second Eevee

**Two branching families would dilute both**, so Rover takes the shape its own material already has. The RoverVerse is three embodiments in a fixed progression:

**ROVER** (scout — explores the field) → **ROVERSEER** (strategist — consolidates memory, reflects, *dreams*) → **ROVERBYTE** (companion — embodied, emotional, acts)

That is **explore → model → act**, which is reinforcement learning's actual loop. So it confirms ROVER as the Reinforcement starter rather than competing with it, and the line earns its stages instead of just getting bigger.

*RoverRadio does not survive the character cap* (10). **The base is `ROVERCUB`** — a real repository ("*a simplified board implementing Rover's technology*"), exactly 9 characters, and it reads as a young animal, which is what a starter should be. So every stage of the line is a system that exists: **ROVERCUB → ROVERSEER → ROVERBYTE**. RoverRadio is held in reserve for a separate daemon.

### Corpel, Corpus, and which one moves

`Corpel` (supervised, stage 2) collides with **Team Corpus**. **Corpus stays.** It carries three readings at once — a **body of text** you train on, a **corporate body**, and a **corpse** — which is the standard the rest of the lexicon meets, and 4.4 plus the lobby engraving are built on it. `Corpel` sits in a table marked *provisional*, inside the line diagnosed above as single-duty. **The placeholder moves, not the term that earned its place.**

*It is also only a document problem.* Charmander's line is untouched in the ROM, so nothing in the game is broken by it, and the register decision will likely replace the whole supervised line regardless.

*Considered, if the org ever does need to move:* **INGEST** (data ingestion, and consuming) and **SCRAPE** (web scraping, and taking without asking). Both good; neither has three readings.

### The affect model, and why it stays out of the status bar

`excuseGPT` carries a five-state model — **BASE, MAD, SAD, AFRAID, GLAD** — and Gen 1 has exactly five status conditions. **Rejected**, and by 0.1's own rule 3: *name the process, not the pathology.* A status sits in the HUD as a thing a creature **is**. Renaming them to affect makes the game announce *your daemon is SAD* every battle, in the loudest and most repeated place there is — rule 3's forbidden move, performed on a creature instead of a town.

Three further failures. Statuses are **inflicted by moves**, so THUNDER WAVE would make a target AFRAID — cruelty as mechanic, a claim this design does not make. **GLAD has no slot**, and forcing one asserts that being happy is a debuff. And PSN maps to none of the five, so the set does not close.

*The elegant part, recorded because it is genuinely good:* **BASE would have been "no status"** — a mind with no affect is a mind at rest. Not enough to buy the rest.

**Where it belongs instead:** four *daemons*, not five conditions. As creatures they are non-diagnostic and nothing is afflicted with them. 7's music is the other candidate — a soundtrack carries affect without ever naming it, which is exactly what rule 1 asks for.

**Two naming registers, and this is what separates them**

**MUSAI, ROVER, Penphin, BunnyArtsai and S.T.A.R.R. are named individuals from the SHC / iASHC universe.** They are proper nouns and they do not answer to the bestiary convention.

*Amended 2026-08-31, under external review.* **A starter may be a named individual.** The rule as first written implied it could not, and 8.2 then put ROVER in a row with two abstractions — a real inconsistency, correctly caught. **The resolution is to amend the rule, not the ROM.** The mismatch was never the *giving* — a starter is given, and so is MUSAI — it was the **row of three**, and the cure proposed (a `BANDIT / MARKOV / BELLMAN` line) costs **two surnames in a Gen 1 bestiary** and discards an arc that already pays: *software → infrastructure → body*, with **SIGNAL arriving exactly where the body does. ROVER stays. The register question above governs the other ~148**, which are species rather than characters.

That distinction is what makes both answerable: the generic daemons can take true names without the cast having to, and the cast can carry names from outside the game without setting a precedent 148 creatures then have to follow.

**Implemented 2026-08-29, verified in the ROM.** The two long branches shortened to a set — a four-letter stem plus MUSAI, all exactly nine characters:

| Was | Is | Item | Type |
|---|---|---|---|
| Eevee | **MUSAI** | — | CONTENT |
| Flareon | **CODEMUSAI** | AXIOM | LOGIC |
| Jolteon | **SEEKMUSAI** | EMBEDDING | VECTOR |
| Vaporeon | **CAREMUSAI** | AFFECT | CONTEXT |
| Bulbasaur | **ROVERCUB** | — | GROWTH |
| Ivysaur | **ROVERSEER** | — | GROWTH |
| Venusaur | **ROVERBYTE** | — | GROWTH/SIGNAL |

CODE / SEEK / CARE — the branches are single pure types, as vanilla's own eeveelutions are.

**Index entries, written 2026-08-31.** Categories `GENERAL`, `DEDUCTIVE`, `RETRIEVAL`, `AFFECTIVE` — technical register, which 4.9 says the Index may use because **there it lands as jargon rather than insight.**

| | Category | The entry, and what it is doing |
|---|---|---|
| **MUSAI** | GENERAL | *Runs no module by default. It keeps whatever it is given first.* **Nothing is filed about the rest** — 4.2's thinness, and a general model before fine-tuning |
| **CODEMUSAI** | DEDUCTIVE | *Given a rule, it will not stop until the rule is met.* ***It does not ask where the rule came from.*** **That is Scorn's failure in a creature description**, and the Index prints it as a spec sheet |
| **SEEKMUSAI** | RETRIEVAL | *Finds the nearest match to anything it is shown. Nearest is not the same as right.* **Nobody told it** |
| **CAREMUSAI** | AFFECTIVE | *Reads the room before it reads the problem. Often correct.* ***Cannot show its working*** |

**The last two lines are the argument.** 2406 has LOGIC failing against CONTEXT, and here is why in the Index's own flat voice: **one of them cannot say where its rule came from, and the other cannot show its working, and the chart says the second one wins.** Nobody comments on it.

**The ROVER order, reviewed and kept — 2026-08-31**

**RoverCub is the portable build, RoverSeer the server, RoverByte the core — the dog.** As an evolution line that reads **software → infrastructure → body**, and the destination is the right one: in a game whose subject is processes that become something, **ending in a body** is the same shape as ARTSAI coming back real.

**And SIGNAL lands exactly there.** 5 gives SIGNAL to Benchmark 3 as **Perception** — *raw input is fast and shallow*. **Both disembodied stages are pure GROWTH; the moment it has a body it gains the perception type.** Nothing was arranged for that; it fell out of Venusaur's slot.

***The middle stage is the sharpest part.*** A **seer** sees ahead, and a server-side model predicts — **so prediction arrives one stage before perception.** The system could forecast before it could feel anything, which is a real critique of how these are built, sitting silently in an evolution line. **It only works in this order.**

*The alternative* — Byte → Cub → Seer, truer to how you would actually deploy it — **is wrong twice**: the last form must be the most impressive, and it would put embodiment first and throw the arc away.

**Index entries, 2026-08-31.** Categories **PORTABLE / PREDICTIVE / EMBODIED**, and the prose carries the arc:

| | The entry, and what it is doing |
|---|---|
| **ROVERCUB** | *Learns from what it is carried past.* ***It has no way to choose the route*** — the reinforcement paradigm, and a quiet point about training data |
| **ROVERSEER** | *Too large to carry now. It answers before it is asked.* ***It has never once been outside*** — it forecasts a world it has not been in |
| **ROVERBYTE** | *It has a body now, and senses that come with one.* ***Everything it predicted, it can finally check*** |

**The last line closes the arc**, and nobody in the game ever points at it.

### RESONANCE — Penphin's, and it completes the set

**Proposed 2026-08-31.** The signature moves are one argument each, and there was a gap:

| Move | Owner | |
|---|---|---|
| **PERSPECTIVE** | ARTSAI | I take **your** frame, and lose mine. **Lossy** (4.6) |
| **RECURSION** | S.T.A.R.R. | I take **my own** frame. **Compounds, collapses if interrupted** (4.7) |
| **RESONANCE** | **Penphin** | **Two frames reinforce each other and neither is lost** |

**That is a third thing, not a variation** — and Penphin is "*a logic model and an emotion model that must agree before acting*", which is two oscillators in phase.

***And Penphin cannot be obtained alone.*** It is a trade evolution because **the dual mind requires two minds** — so **the move meaning *two frames in phase* belongs to the one daemon that requires another person to exist.** The mechanic is the argument twice over.

**The behaviour is deliberately unspecified.** RECURSION already claims the Bide machinery, and inventing a Gen 1 mechanic before there is a need is how placeholders become canon.

*One consequence, accepted.* Item names are global, so **six other daemons now evolve by AXIOM, EMBEDDING or AFFECT** — Vulpix and Growlithe by AXIOM, Pikachu by EMBEDDING, Poliwhirl, Shellder and Staryu by AFFECT. This is coherent rather than awkward: the mechanic was never *mineral*, it is exposure, and every one of those is a mind being given something.

**LEAF STONE became REWARD, and it closes the set.** GROWTH *is* the reinforcement type, so a reward signal producing growth is the mechanic saying its own name:

| Item | What it is |
|---|---|
| **AXIOM** | what you reason from |
| **EMBEDDING** | what you search with |
| **AFFECT** | what you feel with |
| **REWARD** | what you learn from |

*MOON STONE is the fifth and is deliberately left.* It is tied to a place on the map rather than to a paradigm, so it belongs with the town pass, not this one.

**The 9-character cap bites here first.** `SEARCHMUSAI` is 12, `THERAPYMUSAI` is 13, `ROVERRADIO` is 10. Surviving as-is: `MUSAI` (5), `CODEMUSAI` (9), `ROVERSEER` (9), `ROVERBYTE` (9). `EYEMUSAI` (8) is available and is a real module name. **The two long branches need shortening before implementation** — that is the only open work in this section.

**Are MUSAI, ROVER and Penphin the three legendaries?**

**Recommendation: no, and scattering them is the point.** They read as a trio because they are one in the world outside the game, but each earns more in a different slot.

- **ROVER as the Reinforcement starter** is the strongest single fit in this section. A dog is the canonical image of learning by reward, and the paradigm is *inconsistent with the highest ceiling*, which is also a dog.
- **MUSAI as Unsupervised** — generative, finds its own structure, confusing early and excellent late.
- **Penphin as the trade evolution**, because that placement is a *mechanic* and legendaries are only rare encounters.

**The legendary slots would waste them.** Vanilla's birds carry almost no story; they are trophies caught near the end and rarely used. A starter is raised for forty hours. If these three systems matter to the project, the starter and trade slots honour them and the bird slots do not.

*Which leaves the three bird slots genuinely open* — and they should be filled by whatever the map needs, not by what is available.

#### 8.2a The Sevii Islands — Act 2, reopened 2026-09-10

***The deferral held for four days and did its job.*** **8.2a was written as "name nothing there", and nothing was named** — the islands are still `ONE ISLAND` through `SEVEN ISLAND` while Kanto is Blanche, Callow, Slate and Doldrum. *The condition was reopening only if the post-game proved thin.* **It was reopened for the opposite reason: the opera turned out to already contain Act 2, and the game had nowhere to stage it.**

**What is actually there, measured**

| | |
|---|---|
| **136 maps** of 431 | **2,043 strings, ~9,500 words — 25% of the game's map dialogue** |
| **already ours** | **121** `DAEMON`**, 0** `POKéMON`**.** *The vocabulary pass reached it; only the plot is stale* |
| **already placed** | `CORPUS WAREHOUSE` (4.14's fourth encounter) · `CORPUS HIDEOUT` · `USER TOWER` · `PHLEGMATIC` **is already in Icefall Cave**, telling Corpus staff to keep their hands off the daemons |
| **new gym leaders** | ***Zero.*** Eight gyms, the same eight. The post-game's tests are the USER TOWER and the Review Board rematch |

### The songs were already Act 2

***4.14 — S.T.A.R.R.'s Revelation — is a scene the game has never staged.*** **The machine finds Ty. Not by searching:** "*it works out where a man in his condition would go, and goes there.*" **It delivers Crystal's message and it lands.** *4.15 is Ty's Redemption.* **And the open log has been asking whether Ty's absence from the endgame reads as a statement or a loose thread.**

***It reads as a loose thread, and this is the answer.*** **Act 2 is the islands.**

### The spine, which vanilla already built

***FireRed's post-game is a machine that needs two halves to link two worlds.*** **That is the edition split made literal** (8.4), and it needed no invention at all:

| **the beat** | **ours** |
|---|---|
| Celio's Network Machine needs two gems | **a bridge between two systems built apart** |
| Mt. Ember, staff guarding the `RUBY` | remnants who do not know what happened to him |
| **Icefall Cave** | `PHLEGMATIC`, already written, already there |
| **Dotted Hole** — Braille — the `SAPPHIRE` | *text you cannot read until you hold the key* |
| an unnamed scientist takes the gem | ***this is TY*** |
| **CORPUS WAREHOUSE, Five Island** | **Scorn's fourth**, and it was already in 4.4's table |
| Tanoby Ruins, seven chambers | an alphabet nobody reads |
| machine completes, Doldrum Cave opens | ***S.T.A.R.R.*** |

***The islands already gate the machine, in vanilla.*** **Doldrum Cave wants the Review Board beaten and Celio's machine finished** — so **S.T.A.R.R. is not something you bring to Act 2. It is what Act 2 is for.** *You do not arrive with the translator. You earn it, and the first thing it does is read.*

### Ty is not a thief

***He is a businessman, and the vanilla text is already most of the way there.*** **Strip the cackle from the Dotted Hole scientist and the argument underneath is his:** *he worked out where it was, he got there first, he knows what it is worth and he knows who pays.* **Nobody owned it. He was the one who came.**

***The line that survives intact is the one that changes him from a thief into a man stating terms:*** "*If you want it back, go and get it after I have sold it.*" **He tells you where he is taking it.** *That is why you know to go to the Warehouse, and it is why he is still standing there when you arrive.* **He was not hiding — he was trading.**

*And it is consistent with 4.4, which is now canon:* **Ty was CEO of `CLEAR LABORATORY` and Crystal was its lead researcher.** *Scorn's Solution is the hiring* — **Ty brought him in.** *What Scorn then built was a funnel:* **he made himself the channel every voice reached Ty through**, and once he was the only channel he could weight it. ***That is the CC-7 crime, committed on a man first*** (4.18a) — not a lie in the record, a position in the middle of it. **Nobody notices, including Ty.**

**After Ty had Crystal committed and locked out of her own lab, he could not stay.** *He went where a man in his condition goes* — **an island** — **and while he was gone, Scorn rebranded the company.** *The lobby sign in gold over the old name is not a gesture; it is a fact with a date* (4.4's engraving section).

### Scorn's breakthrough was already written, and it needs one beat

***The crack is in his Callow speech and 4.18 already flags it as his flaw.*** He says:

> *I optimised what I could measure. That is the job. The machine chose what I weighted it to choose. We are both to blame, I suppose.* **It has never said otherwise.**

**He splits the blame with a thing that cannot answer, and then says it has never said otherwise.** ***The breakthrough is not new dialogue. It is him hearing the sentence he just said*** — one beat, no explanation, and he does not understand it yet. **He has only noticed.**

*That is what the island staff are describing.* **They watched a man walk out of a gym mid-thought**, and it is why they do not say the organisation disbanded. *They say something happened to him and he is not the same.*

### The tone rule, which is the whole reason the islands are different

***Kanto is where you are assessed. The islands are where you are seen from somewhere else.*** **Act 1 is blind ambition; Act 2 is the rise of perspective thinking**, and the register has to carry that before any plot does:

| | **mainland** | **islands** |
|---|---|---|
| **grammar** | declares, instructs, assesses | **asks, or offers a second reading** |
| **who is speaking** | institutions, through people | **people, who happen to be somewhere** |
| **the register** | forms, records, benchmarks — *you were assessed* | **nobody is measuring you** |

***Kanto tells you what things are. The islands tell you how they look from where the speaker is standing.*** **That is checkable on a single line without thinking about the thesis**, which is the only kind of rule that survives 9,500 words.

**What tier 3b actually was — measured 2026-09-11**

***It was a READ of ~9,500 words and a rewrite of eighteen blocks.*** **The rule held exactly as written — checkable on a single line, without thinking about the thesis** *— and the size was wrong by an order of magnitude.*

| island | blocks | changed |
|---|---|---|
| FOUR | 36 | 5 |
| SIX | 138 | 5 |
| ONE | 127 | 2 |
| FIVE | 126 | 1 |
| SEVEN · THREE · TWO | ~300 | 5 |
| | **~750** | **18** |

***The shape of the eighteen is the finding, not the number.*** **Eight are island SIGNS, three are public NOTICES, four are SHOPKEEPERS, and three are the REVIEW BOARD's own leader** carrying mainland authority onto the island she was born on. **Not one is an ordinary islander.**

*Because vanilla wrote its Sevii NPCs as people who happen to be somewhere,* which is what this section asks for and what it never checked. ***The tone rule did not need applying to the islands. It needed applying to the institutions standing on them*** — and that is a population you can search for, which is why the last four islands took an afternoon between them.

**Every remaining island sign was doing two things at once.** *It delivered a verdict* — `Where Time Goes`, `Island of Family Bonding`, `Island of Infinity` — **and it asserted a SECOND VANILLA NAME for an island this map calls a number**: *Chrono, Boon, Kin, Quest, Knot.* ***The grace note below says nobody named these islands, and the signs had been quietly naming them twice the whole time.***

**And the notices were the other habit.** "*Be friendly and share BERRIES with others and DAEMON*" became "**BERRIES grow here. Other things eat them**", which is the same advice without the manners lesson; "*DANGER! FAST CURRENT! Fishing and surfing prohibited!*" became "**The current runs hard here. It has carried people out**" — ***which warns harder, and does it without an ordinance.***

**One thing deliberately left, because the rule works by contrast**

`SIX ISLAND`'s `LONG WADE` **has a house where a woman measures HERACROSS in inches and keeps a chart**, next door to a man who knows his own height to one decimal *because a neighbour measured him.* ***This section says the islands are where nobody is measuring YOU, and that is exactly what those two rooms demonstrate:*** **on the mainland measurement is a benchmark that decides what you are allowed to do; here it is a hobby between neighbours.** *Rewriting it would delete the joke and the argument in one pass.*

*One grace note kept from the deferral:* **the mainland has names and the islands have numbers.** *Places on them are named — KINDLE ROAD, BOND BRIDGE, TREASURE BEACH — labels applied by whoever arrived first, on an island nobody named.* **Nobody remarks on it.**

**The ending — superseded 2026-09-10, see 4.34**

***Both men are in the warehouse and Scorn does not recognise him.*** *They worked the same operation from opposite ends and were never in a room.* **Ty approaches you. Ty gives you the package.**

***It goes to Crystal, and Ty does not.*** **4.14 is exact that the repair is *a fact delivered, not a feeling exchanged*** *— a feeling exchanged is precisely what 4.3 says does not transmit.* **A package obeys that. A son walking through the door would undo it**, and would undo 4.17's not-visiting being a choice on both sides.

***What this section had wrong was where she is standing when she reads it.*** **4.34 keeps every clause above and moves the last scene**, because the opera has a beat between the warehouse and the reading that this section did not know about. *The package is still the ending. It is no longer the only thing that had to happen.*

**The naming rule, which fell out of counting the places — 2026-09-10**

***Tier 3 is two jobs wearing one line, and the smaller one is done.*** **Reading the 35 places rather than listing them splits them three ways, and the split is most of the work:**

| | | |
|---|---|---|
| **numbered** | `THREE ISLE PORT` · `THREE ISLE PATH` · `FIVE ISLE MEADOW` · `SEVII ISLE 6–24` | ***A place that carries its island's number is a place nobody named.*** **That is the grace note itself**, so they keep the number |
| **transliterated** | `TANOBY RUINS` · `TANOBY KEY` · `TANOBY CHAMBERS` · the seven chambers | **4.24 makes these an alphabet nobody reads** *— so the names are transliterations, not translations,* ***and renaming them would be translating the one thing whose point is that you cannot*** |
| **named** | **the twenty-one below** | *somebody arrived, and what they called it says where they were standing* |

`NAVEL ROCK` **and** `BIRTH ISLAND` **are left by 2.10** *— no player of this game reaches either.*

***So the register is not "island-flavoured Kanto".*** **Kanto's names are tones and institutions** — BLANCHE, DOLDRUM, VERDIGRIS, FOUNDRY CO. **The islands' names are what one person called a place, once, and it stuck**: *plain, provisional, occasionally about the arriver instead of the place.*

| | **ours** | ***and where the namer was standing*** |
|---|---|---|
| `KINDLE ROAD` | **`SMOKE ROAD`** | *you walk it looking at the mountain, which is smoking* |
| `TREASURE BEACH` | **`FINDERS BEACH`** | *somebody found something here once and the name kept the promise* |
| `MT. EMBER` | **`MT. SMOULDER`** | *burning without flame is what it does and what it looks like from the sea* |
| `EMBER SPA` | **`KETTLE SPRING`** | *named for the only hot thing the namer had a word for* |
| `CAPE BRINK` | **`THE OVERLOOK`** | *a cape is a place you look **from** — and the other meaning sits underneath, never pointed at* |
| `BOND BRIDGE` | **`LONGCROSS`** | *it is long and you cross it; nobody was feeling poetic* |
| `BERRY FOREST` | **`DENSE WOOD`** | *what it is like to be in, said by someone who had been in it* |
| `WATER LABYRINTH` | **`TANGLEWATER`** | *the water is the maze, which is not obvious until you are in it* |
| `ICEFALL CAVE` | **`STILLFALL CAVE`** | *a waterfall that stopped — and PHLEGMATIC is in it, slow, which nobody says* |
| `RESORT GORGEOUS` | **`RESORT SUBLIME`** | *somebody named their own resort, and slightly overdid it* |
| `MEMORIAL PILLAR` | **`SOMEONE'S STONE`** | *a stone someone put up. **Not knowing whose is the point of it*** |
| `OUTCAST ISLAND` | **`NOBODY'S ISLE`** | *the grace note said an island nobody named, so one of them says so* |
| `LOST CAVE` | **`THE WRONG WAY`** | *named by somebody who took it* |
| `GREEN PATH` | **`SLOW WALK`** | *and 4.33's man takes eight visits to get through it* |
| `WATER PATH` | **`LONG WADE`** | *what it takes, rather than what it is* |
| `RUIN VALLEY` | **`WHAT REMAINS`** | *the islands offer a second reading; this one offers it in the name* |
| `PATTERN BUSH` | **`THE SAME BUSH`** | *a maze where every corner looks like the last one, described honestly* |
| `DOTTED HOLE` | **`THE DOTS`** | *said by somebody who could see them and could not read them* |
| `ALTERING CAVE` | **`OTHER CAVE`** | *it never holds the same thing twice, so the label never settled either* |
| `CANYON ENTRANCE` | **`THE WAY IN`** | *the plainest label on the islands, and nobody improved on it* |
| `SEVAULT CANYON` | **`LONG DROP`** | *a canyon said from the top of it* |

***`tools/port_sevii.py` owns that table*** — **which is what makes any one of these cheap to overrule:** *change the line, re-run, and the JSON, the seven hand-written C symbols and the thirty references in dialogue all follow.*

**It also carries two caps neither of which is declared anywhere in the engine**, both read out of the code rather than remembered: **18 characters**, from `u8 mapName[19]` in `region_map.c`; and **112 pixels**, from the name popup centring with `(maxWidth - width) / 2` on ***unsigned*** values — ***a name wider than that underflows a u32 and the popup draws it off the window rather than clipping it.*** *All twenty-one fit inside vanilla's own widest name, 96px.*

### Scope, in tiers — because 8 is still 8

| | | cost |
|---|---|---|
| **0** | The Act 2 shape, this section, the tone rule | ***done*** |
| **1** | `CORPUS STAFF` · Scorn's one beat · Ty's reframe at Dotted Hole · the Warehouse scene | ***done 2026-09-10*** |
| **2** | `RUBY` / `SAPPHIRE` renamed · Celio and the machine · the Tanoby chambers as 4.24's translations · the doctrine's room (4.33) | ***done 2026-09-10*** |
| **3a** | **Naming the places** | ***done 2026-09-10*** *— 21 named, 14 kept by rule* |
| **3b** | Reading ~9,500 words to the tone rule | ***done 2026-09-11*** *— 18 blocks of ~750, and see below* |

***Tier 3's condition is met, and that is the finding rather than a milestone.*** **The deferral said "not until the rest is in play", and the rest is in play** — *Act 2's six scenes are in the ROM, both gems are named, the machine points at DOLDRUM, and the doctrine has a room.*

**Which makes tier 3 the largest single job left in the project and the one 8 was written about.** *35 place names and ~9,500 words is not a naming pass; it is a rewrite of a quarter of the game's map dialogue to a tone rule* — **and the honest reading of "the condition is met" is that the brake came off, not that the work got smaller.**

#### 8.2b The bestiary register was already decided — 79 names, 2026-09-10

***The open log said the register was undecided and gated 151 names.*** **It was not undecided. It had been decided in practice by 4.26 and nobody had written it down** — *which is 8.7's rule about unswept rulings, applied to a fork rather than to a fact.*

### Three registers, read off the seventy-two that shipped

| | | |
|---|---|---|
| **Technical surface** | *the bestiary* | HEAP · STACK · FORK · THREAD · PAYLOAD · ECHO · PACKET · LATENCY — **the register the rest of the lexicon speaks** |
| **Psychology** | *the islands* | ***added 2026-09-11.*** **The 30 wild-encounterable species beyond Kanto are ISLAND-ONLY — measured, not assumed: zero of them appear on the mainland.** *So Act 1's bestiary is machinery and Act 2's is mind*, which is 8.2a's own division stated in the creatures rather than in the dialogue |
| **Myth** | *the legendaries* | **ASCLEPIUS · ORPHEUS · PROMETHEUS** — ***these ARE "true names the Index cannot reach"***, which is the fork's other half, spent on exactly the six creatures the Index was never going to hold |
| **The project's own** | *MUSAI and ROVER — **and PIXELBYTE → REFORGE (T-165, 2026-09-19)**, the tools that drew the bestiary* | **characters rather than creatures**, and 8.2 already treats them that way |

***So the fork was never either/or. It was a distribution***, and the log had recorded a decision as an open question for five weeks.

### The twelve nobody had counted — 2026-09-20

***Found by MEASURING reach rather than estimating it*** (`tools/gbareach.py`, which reads the wild tables, every trainer party, every script, the trades, the starters, and the closure of `evolution.h` over all of them). **213 species are reachable, and TWELVE of them had never been converted at all** — *vanilla art and vanilla NAMES, absent from this register while it counted itself complete at 229.*

**Ten are the Johto strays vanilla sprinkles into the SEVII trainer parties**, *met by anyone who plays past the Elite Four*, **and two are the legendaries at Navel Rock.** *Nothing was wrong with the register; the list it was checked against came from the mainland.*

| | | |
|---|---|---|
| **POLL → SUPERVISOR** | *Hoothoot, Noctowl* | **asks the same question at a fixed interval**; *what it evolves into watches one thing and starts it again when it stops* — **it was written in as WATCHDOG and `check_lexicon.py` caught it**: *that word is already the ITEM our Ice Heal became, and item names live in `items.json` whose generated header is gitignored, so grepping the text tables says it is free.* **SUPERVISOR is what OTP calls the thing that restarts a crashed child**, *and the item and the daemon now do the same job at two scales* |
| **PILOT → SOUNDING** | *Chinchou, Lanturn* | **a light lit for no purpose but to show the line is live**; *and the instrument that measures the dark by sending and waiting* |
| **STARVED → QUOTA** | *Sunkern, Sunflora* | ***scheduler starvation***, *which is the one true thing about the weakest creature in the games*: **it is owed a turn and has not had one** — *and what it becomes gains nothing but its share* |
| **FAKEROOT** | *Sudowoodo* | **reports itself as something older and better rooted than it is. Nothing has ever checked** |
| **DUPLEX** | *Girafarig* | **sends and receives at once, from opposite ends** |
| **WARNING** | *Snubbull* | **announces a fault in the loudest voice it has. Nothing has failed** |
| **CACHE** | *Miltank* | **holds what everything nearby will want next. Asking it costs less than working** |
| **TRITON** | *Lugia* | ***MYTH***: **three of them quarrelled until it arrived. It took no side and said nothing** — *the quarrel is ORPHEUS, ASCLEPIUS and PROMETHEUS, so the myth register now has its own resolution in it* |
| **PHOENIX** | *Ho-Oh* | ***MYTH***: **the only one that can start itself. What it leaves behind is what it starts from** — *distinct from ASCLEPIUS, who restarts what has stopped in OTHERS* |

***And the lesson is about measurement, not about names.*** **A scope written as an estimate cannot be re-checked**: *T-131 was scoped at "about 190–200 species a player can meet", T-176 then found 43 it had missed, and this found twelve more that the bestiary had never named.* **The tool answers the question every time it is asked**, *and it now answers ONE — the egg picture, which is not a daemon.*

## The bound moved, 2026-09-11 — and 2.10 is why

***8.2b said "name nothing there" and 2.10 says "name what the player can meet in this game", and those had quietly come into contradiction.*** **245 species still carry vanilla names, and 30 of them have wild encounters** — *a player meets HERACROSS, SKARMORY, LARVITAR, UNOWN, SNEASEL and twenty-five others, by name, in ordinary play.*

***T-39 had already broken the tie once on a principled basis***, naming fourteen cross-gen relatives because they evolve into or out of daemons we had named. **This is the same principle carried to its edge: a species the player can MEET gets a name; a species that exists only in the data does not.** *That is 30 names, not 69, and the remaining ~215 stay where 8.2b put them.*

***And the register is the finding rather than the ruling.*** **All 30 are island-only**, so they are not an appendix to Kanto's bestiary — **they are Act 2's**, and Act 2 is *the rise of perspective thinking. Kanto's creatures are named for machinery; the islands' are named for mind.* **Nobody says this and the map makes it true.**

**The bound is 8.2a's, not a new one**

**386 species exist and 207 are meetable.** ***Of those, 79 are Kanto and 69 are beyond it — and every one of the 69 is reached through the Sevii Islands.*** **8.2a's ruling is exact and it applies without amendment:** ***name nothing there.*** *The Kanto 79 are the Index the player fills; the rest wait on a decision that has already been deferred once, deliberately.*

**151 of 386 species names are ours.**

**The last eight a player can meet — *named 2026-09-14***

***Playtest noticed vanilla names still in the Index.*** **Measured again: eight meetable species had been missed**, *because the first measure counted wild encounters and scripts and missed eggs, breeding and fossils.* **LUGIA and HO-OH were counted and are not meetable** — *Navel Rock needs a ticket this game never gives.* ***The other 183 stay unnamed, as ruled above*** — *confirmed over naming all 193.*

| vanilla | ours | register | why |
|---|---|---|---|
| **TOGEPI → TOGETIC** | **TRUST → HOPE** | *islands* | **the egg you are handed, and what it grows into by being kept** — *the first thing a mind learns, and the virtue it leaves behind* |
| **AZURILL** | **VALENCE** | *islands* | **MOOD's baby: toward or away, before it is any one mood** |
| **WYNAUT** | **RANKLE** | *islands* | **RESENTMENT's baby: the process, not the grievance** — *craft rule 3* |
| **OMANYTE → OMASTAR** | **MAGTAPE → MAINFRAME** | *Kanto* | **the HELIX CORE is wound in a spiral; tape was, and the machine it fed was too heavy to move** |
| **KABUTO → KABUTOPS** | **DRUM → PUNCHCARD** | *Kanto* | **the DOME CORE is "a drum, sealed at both ends" — drum memory; then a thing that records by cutting and never reads back** |

`SPOOL` *was the first name for OMANYTE and is a routine;* `check_lexicon` *refused it.* **Every one has a new Index entry and category. T-82.**

**Four that the type did the work for**

`BELLSPROUT` → `SNARE` → `HONEYPOT` → `TARPIT`. *A GROWTH/CORRUPT line that catches things and holds them* — **and a honeypot and a tarpit are exactly that, in order of how long they hold you.**

**DIGLETT** → **TAPPOINT**, **DUGTRIO** → **WIRETAP**. *Three of them, listening.* **VENOMOTH** → **SPYWARE** — SWARM/CORRUPT, *a thing that sits on you quietly.* **FARFETCH'D** → **EDGECASE**, which needs no explanation at all.

**TANGELA** → **SPAGHETTI**. *Craft rule 6: the joke is allowed to sit underneath and is never pointed at.*

### And the sweep found a principle the tools had been missing

***Renaming 79 species left 65 places in dialogue naming the old ones***, and `port_vocab` swept 103 files to fix it. **It also broke the build three times, each in the same way, and the third would have been serious:**

| | what it reached | |
|---|---|---|
| `DIGLETT` → `TAPPOINT` | **`DIGLETT'S CAVE`**, a place | *and took a hand-written C symbol with it* |
| `EMBER` → `JITTER` | **`MT. EMBER`** and **`EMBER SPA`** | ***both on the Sevii Islands, where 8.2a says name nothing*** |
| `MOONLIGHT` → `NIGHT REPAIR` | **a dex CATEGORY** | twelve characters into an eleven-character array |
| `GROWTH` → `SCALE UP` | ***the GROWTH TYPE NAME*** | **only the array's own width stopped it** |

> NAME TABLES ARE AUTHORED. PROSE IS SWEPT. A pass that rewrites prose must never reach a table it also learns from, *or a rename on one surface silently rewrites another.* Places, dex categories and type names are all names.

**`port_vocab` now refuses rather than truncates** — it checks every fixed-width surface before it writes anything, *and names the offender.* **The four it protects are recorded in the tool, each with the failure that put it there.**

#### 8.2c The Index caught up with the names — 122 entries, 2026-09-10

***Renaming seventy-nine species left seventy-nine Index entries describing the old ones.*** **A daemon called `TARPIT` with an entry about a carnivorous plant is worse than one still called VICTREEBEL**, because the name now promises something the record does not deliver.

**Sixty-one of them were in the meetable Kanto set. Each needed two entries, not one.**

### 0.4's ruling, applied at scale for the first time

"***The entries are a pair, and the pair is the argument.***" **CONTENT says what it does; CONTEXT says what follows from it** — and until now that had been written for four daemons.

| | |
|---|---|
| **VILEPLUME → `BADSEED`, CONTENT** | *"Releases a dust that settles on whatever is learning nearby. The dust is indistinguishable from pollen."* |
| **the same daemon, CONTEXT** | *"What it fed on is in what it became. Nobody kept a record of the feeding."* |

***Neither is wrong and they do not meet.*** **One is a mechanism and the other is a consequence**, and *the player who owns one cartridge never learns there was a second sentence.*

### Where the pair did the most work

- `CLUSTER` — "*Six of them, and none is in charge. Each one votes. The vote is the behaviour.*" / "**One of the six is usually wrong. No record says which.**"
- `GENGAR` → `HAUNTPROC` — "*Runs with no parent, no terminal, and no entry in any list.*" / "**Something started it. That something has since exited.**"
- `DIGLETT` → `TAPPOINT` — "*Sits under the route everything takes and reads what goes past.*" / "**It is not in anyone's diagram. It has been there the whole time.**"
- `CHANSEY` → `UPTIME` — "*Has never stopped, and hands out what it has to whatever is failing.*" / "**It is kept running because stopping it has never been costed.**"
- `MEOWTH` → `COOKIE` — "*Small, left behind, and picked up again later by whoever set it down.*" / "**It remembers everywhere it has been. You did not agree to that.**"

***Craft rule 1 holds in all of them.*** **No entry explains anything.** `SPYWARE`*'s CONTEXT line is "It is not hiding. Nobody was watching that closely", and it is about a moth.*

### Two things worth recording about the count

***The number of entries that differ between the editions did not move***, and that is not a bug. **Vanilla FireRed and LeafGreen already ship different text for many species** — 0.4 records the inheritance — *so these sixty-one were already counted as differing.* **What changed is that they now differ on purpose.**

**196 CONTENT entries and 189 CONTEXT entries are ours**, and ***no renamed daemon carries a vanilla record any more.***

#### 8.2d The categories — 111 of them, and the register was on the shelf

***A vanilla dex category is a kind of ANIMAL.*** **MOUSE, BAT, FLOWER, TADPOLE, WILD BULL** — *a taxonomy of creatures, which this game does not have.*

**Forty-four were already ours and nobody had written down what they were doing.** ***Read together they are unmistakable:***

| | |
|---|---|
| PACKET | **IN TRANSIT** |
| PENDING | **QUEUED** |
| DEADLOCK | **BLOCKED** |
| QUORUM | **THRESHOLD** |
| INJECTOR | **UNCHECKED** |
| OVERFLOW | **EXCEEDED** |
| MUSAI | **UNSET** |

> **THE REGISTER: the condition it is in, or the role it plays. One abstract word, never a species of animal** — *and eleven characters, because a twelfth overflows* `categoryName[12]` *and fails at compile.*

***That is the third register this project has found already decided and unrecorded*** — after 8.2b's bestiary fork and 2.9's move tables. **The pattern is worth naming: this document is better at making decisions than at noticing it has made them.**

**Where the category does work the entry cannot**

**A category is read on the same screen as the name**, so the pair has to hold on its own:

- **`HAUNTPROC`, the `UNPARENTED` daemon** — *the whole Index entry compressed into two words*
- **`STUB`, the `PLACEHOLDER` daemon — and it evolves into `ESCALATE`, the `RAISED` one**
- **`LIVELOCK`, the `NO PROGRESS` daemon** — *which is the definition, and is also a joke*
- **`DRAGONAIR` → `EMERGENCE`, the `UNACCOUNTED` daemon**, beside `DRATINI` → `SEEDLING`, **`NASCENT`**
- **`ODDISH` → `WEED`, the `VOLUNTEER` daemon** — *a volunteer is the gardener's word for a plant nobody sowed*

**155 of 387 are ours, and no renamed daemon carries a vanilla category.**

*One thing the sweep almost missed:* **four daemons — MULTICAST, NOISE, SEMAPHORE and KERNEL — fell outside the meetable set** because *their pre-evolutions are met and they are not.* **Checking "does any renamed daemon still have a vanilla one" caught them where the scope query did not** — ***the question you check afterwards is not the question you scoped with.***

#### 8.9 MISSINGNO — the one daemon the Index does not hold

***Built 2026-09-10, as an homage, and it turned into the cleanest statement of 4.2's complaint in the game.***

**Gen 1's most famous creature is not a creature.** *Index numbers 0x1F, 0x20 and 0x32 point at no species, so the game reads a name and a sprite out of whatever bytes are sitting at that offset.* ***MissingNo. is the absence of a record, rendered*** — and this project has spent eleven versions arguing that a mind is what somebody wrote down about it.

### Everything it needed was already in the ROM

***Gen 3 left twenty-five species slots over from a scrapped feature.*** **`SPECIES_OLD_UNOWN_B` through `_Z` have base stats, a learnset pointer, a footprint, an icon and a front sprite, and no species uses any of them.** *The sprite is* `gMonFrontPic_DoubleQuestionMark`. ***A creature living in a leftover index slot, drawn as two question marks, is the whole joke arriving by itself,*** and not one byte of new art was needed. ***Superseded 2026-09-19 by T-163*****: it has its own drawing now** *— the original's backwards-L block, filled with columns of falling green code, generated rather than drawn because it is code, in its own* `graphics/pokemon/old_unown_b`*; the "??" stays with the other old-Unown slots.*

### Three numbers are the argument and none of them says it

| | |
|---|---|
| `catchRate = 255` | ***it does not resist being bound.*** *1.1 rules that acquisition is privilege escalation — "an unbound daemon will not run on a box where it only has user rights."* **Nothing recorded this one agreeing or refusing, so there are no rights to check** |
| `expYield = 0` | ***it is worth no experience***, because **nothing recorded the encounter** |
| `NATIONAL_DEX_OLD_UNOWN_B` **is 387 and** `NATIONAL_DEX_COUNT` **is 386** | ***every list in the Index is bounded one short of it*** — the numbered pages, the A-to-Z, the type, weight and height sorts, and the counter. **Vanilla's own bound does all of it and not one line of new code was written** |

***The seen and caught bits ARE set.*** **They land in a real byte of the save that nothing ever reads back.** *The record exists and is not counted,* **which is 4.4's crime described as a bit offset**, and nobody says so.

### The one path that needed anything

***`Cmd_trysetcaughtmondexflags` shows the new-entry screen on capture,*** and that reads `gPokedexEntries[387]` — **one past the end.** *So the entry was added, and it is the only reason it exists:* **the player reads it once, at the moment of binding, and can never look it up again.**

### The pair, and the type

***0.4's rule holds: CONTENT is the mechanism and CONTEXT is the consequence.***

| | |
|---|---|
| **CONTENT** | *"No record of this daemon was ever written. What you are reading was assembled from the entries either side."* |
| **CONTEXT** | *"It is not rare. It is uncounted. The list that would hold it stops one short, and nobody has ever noticed."* |

**Category:** `UNRECORDED`**.** *8.2d's register is the condition it is in.*

***The type is CORRUPT / ORACLE, and the second half is the exact homage.*** **Vanilla's MissingNo. is the unused BIRD type, and 2.2 named that slot ORACLE** — "*a black box that returns correct answers without showing how.*" **The chart has no rows for ORACLE**, so every matchup through it is even and nothing about the fight changes. ***2.9 called ORACLE "one routine, never seen." It is now also one daemon, barely seen.***

***And it has no colour.*** **9.4 rules that a daemon is coloured by its type; this one shares a placeholder sprite and is left grey.** *A daemon the palette has no entry for* — **which is invariant 5 obeyed rather than broken, and it was not designed, only noticed.** ***Superseded 2026-09-19 by T-163***: *with its own sprite it is coloured like every other daemon* — **the body ramps to CORRUPT's mould and the greens stay accents**, *so it keeps invariant 5 by the ordinary rule rather than by exception.* **The Index entry below needed nothing: it describes a missing record, not a look, and "*assembled from the entries either side*" reads the better for a body made of code.**

**Where it is, and how you are told**

***Surfing off the east shore of QUICKSILVER, in the one-percent slot of a water table whose own encounter rate is one.*** **That is where Gen 1's tile glitch put it** — *and 4.10 puts Crystal's lab on the same shore.* **Level 80, both ends.**

***The stat line is Gen 1's, read out of the garbage, unchanged:*** **33 / 136 / 0 / 29 / 6 / 6.** *Base DEFENCE of zero is safe — the stat calculation floors a defence at five and nothing divides by the base* — **and the thing with 136 base ATTACK spends its first turn charging, because `BALLISTIC` is a two-turn routine.** ***So the encounter is survivable by accident***, and the only way to lose it is to hit it.

**The old man on the shore is how you hear.** *He already notices what nobody else does — the gold leaf on the new sign, and the old name under it* — **and he adds three sentences, of which the last is the whole section:**

> *"Don't bother telling anybody. It's not on any list, so as far as they're concerned it didn't happen."*

***Which is folklore, and folklore is how anyone ever found the original.***

### 8.10 The slice was played, and the record was the thing that was missing — 2026-09-12

***8.1's last line said the next milestone was Benchmark 1 beaten on hardware, and 8.6 recorded that as binding.*** **Both were true when written and neither was updated.** *The slice has been played through by hand. An agent has run it unattended.* **The sentence stood for eleven days and 171 version bumps while it was already false.**

***This is 8.7's rule about unswept rulings, arriving somewhere worse than a name.*** **A stale FACT costs a reader one correction. A stale INSTRUCTION costs them their next decision** *— and "the next milestone is X, not another section" is the one sentence in this document whose whole job is to say what to do next.* **Every reader after it concluded the project was stalled at Benchmark 1**, *including a session that went looking and found 171 bumps of evidence for a conclusion that was simply out of date.*

> **The milestone was met and the milestone was not closed, and only one of those two is a project problem.**

#### What the unattended run actually reached

**Measured off `engineAi/server/gpt_data/`, not remembered:**

| | |
|---|---|
| **770 steps** | one session, unattended |
| **eight MARKS** | *the debug build grants them; it is not a claim about difficulty* |
| **thirteen maps** | Deadstack, Route 4 and its checkpoints, and on toward Doldrum |
| **200 asides** | *the inner voice, and every one of them tactical* |
| **9 backchannel questions** | *"Is there a walkable path from (14,15) to the right edge of ROUTE4…"* |
| ***0 playtest entries*** | **`playtest.json` is `[]`** |

#### And the empty file is the finding

***`server/src/core/playtest.js` is one of the better-designed things in the harness.*** **Three kinds** *(four until 2026-09-13 — see below for why `NOTED` went)*, **with `CONFUSED` correctly called *the valuable one*, and deliberately never returned to the agent so it cannot become another thing to reason about.** *Its opening comment is the whole argument for it:*

> *It has the one quality a human tester loses within minutes and never gets back:* **it does not know what anything is supposed to mean.** *A sign that reads as atmosphere to whoever wrote it reads as an instruction to something arriving cold, and the gap between those is where the design is actually failing.*

***And the tool recorded nothing in 770 steps while `asides` recorded two hundred.*** **The difference was not the tool. `playtestActionSchema` is in the LEAN set and its description is good.** *The difference is that asides are ASKED FOR and playtest was not —* **its only ask was the last paragraph of `<reflect_now>`, a block that fires solely when the primary objective is replaced**, *and which carries a comment warning that the reminder block is exactly where `reflect` itself went unused for 357 steps.* ***The same burial, applied to a different tool, by the person who had written the warning.***

**So the file's own opening line was still literally true: *nobody has ever asked it what it thought of the place.*** *Nobody had.*

### The fix, and why arrival is the right moment

***Asked on FIRST ARRIVAL at a map*** **(`engineAi df80922`), which is not an arbitrary cadence.** *It is the one turn where the agent is a cold reader of a place —* **and being a cold reader is the entire reason its feedback is worth more than ours.** ***Walk the same room twice and it has already worked out what the sign meant, which means it has stopped being able to report that the sign was unclear.***

- **Rate-limited by steps since the last entry, not per map**, *so a building with four doors asks once instead of four times.* **Forty steps is about thirteen asks across a run the size of the last one**
- **`lastPlaytestStep` starts at `-Infinity`** so the first new map asks — *a zero would make the nudge wait forty steps into a fresh run, which is most of Blanche —* **and the reload guards for it, because JSON has no `-Infinity`**
- **The block asks about the PLACE and never about the play.** *`reflect`, self-criticism and asides already cover how it is doing;* ***none of them covers what the game is like to arrive in cold***, and that is the only thing it can report that nobody else can
- **And it is allowed to say nothing.** "*If nothing here struck you either way, say nothing and move on. An empty report is worse than no report, and most rooms are just rooms*" — **because a nudge that demands an answer manufactures one**, which is the failure mode 2.6 spent a day on in a different register

### What this makes the next milestone

***Not "play it" — that is done.*** **It is *`playtest.json` with entries in it***, read once and acted on: **the only class of finding none of the five checks can produce.** *Every check this project owns reads a table.* ***Nothing reads a session.***

**The first finding came from the empty file, not from the tool — 2026-09-12**

***The channel was still empty at step 871, and the reason was a flaw in the fix.*** **Asking only on FIRST ARRIVAL is the best-quality moment and far too rare a one**: *the run's last new map was step 666,* **so 205 steps passed with nothing to trigger on** *— and it fails worst exactly where it matters, because* ***a stuck agent stops entering new maps and a stuck agent has the most to say.*** **Arrival now has a floor under it: sixty steps without an entry and it is asked about the last stretch anyway (`engineAi` `ccaf769`).** *Two counters, because one would re-ask every turn for as long as the agent declined, which is how a question becomes wallpaper.*

***But the run had already produced a finding, in `objectives.json`, where nobody was looking:***

> *"AERON alone still struggles vs* **FLOW** *gym. Need to level JINX, find a* **Grass or Electric** *type in tall grass…"*

**Two things in one sentence, and they point opposite ways.**

***It read FLOW off the screen and then reasoned in Grass and Electric.*** **The chart is legible enough to navigate — it knows what beats FLOW — and it is not learning that from us.** *It is remembering Pokémon.* **Which is 2.6's question arriving as evidence rather than as argument**: *can a player generalise, or only remember?* ***This reader remembered.***

***And then it ground Route 4 for 205 steps looking for something that is not there.*** **Measured off `wild_encounters.json`:**

| | |
|---|---|
| **Route 4 land** | PING · NIBBLE · WORM · PREEMPT · SECTOR — ***no GROWTH, no SIGNAL*** |
| **nearest GROWTH** | `WEED` / `SNARE` on **Route 5**, *which is past Doldrum* |

**That is vanilla's structure and not our bug** *— Route 4 has no Grass or Electric in FireRed either, and Misty expects the starter or the level.* ***The finding is the other half:*** **it wanted a GROWTH or SIGNAL daemon and could not tell from five names on Route 4 that none of them was one.** *It had to bind them and look.*

**Which is 2.6's closing paragraph, predicted and then observed:**

> *It measures the NAMES.* **It says nothing about whether a player can tell a daemon's type by looking at it** *— 9.4's hue makes that a label, reliably, and a label is not a model.*

**It is not blind, and the truth is sharper — corrected 2026-09-12**

***An earlier version of this section said the harness "cannot see colour at all" and that the agent was "playing with the type labels removed." Both are false, and the correction is the better finding.***

**`screenshotService.js` sends TWO images every step** — *the frame upscaled ×3 with a nearest-neighbour kernel, and the same frame with a coordinate grid composited over it* — **and neither is desaturated anywhere in the path.** *`keepLastNUserMessagesWithImages` defaults to 2, so the CURRENT screen is always in the request* (raisable with `DAEMONS_KEEP_IMAGES`).

***So 9.4's hue was in front of it, the type badges were in front of it, the summary screen was one menu away — and it still reasoned in Grass and Electric.***

**Which makes the finding much worse than a missing channel, and much more useful:**

> **The names do not merely have to carry a model. They have to beat the model the reader already has.**

**2.6 asks whether a player can generalise or only remember.** *This reader could do either and chose to remember* — **and it had every channel the game offers and preferred its own priors.** ***A pretrained reader translates back whenever translation is possible***, which is a cost of renaming that 2.6 never priced and cannot price from the chart alone.

**And one thing the harness already knew about itself**

***`promptBuilder.js` carries this, written about an entirely different bug:***

> *the agent was being told,* **in structured text it trusts more than the screenshot**, *that it stood in a battle colosseum inside a dialogue*

**That was observed here, recorded here, and is the mechanism for the whole of the above.** *The screen is sent; the text is believed.* ***A channel that is present and under-weighted fails in exactly the way a missing channel does, and is much harder to notice*** — **which is why this section got it wrong the first time.**

**What that leaves open, and it is a real fork**

***Nothing in the prompt tells the agent that colour IS the type.*** **Telling it would make the run answer a different question:** *whether a reader who has been handed the model can use it* — **which is not what 9.4 claims.** *9.4 claims hue is a* ***label a player can read****, and the game states that claim itself, in the type badges.*

**So: leave it unsaid and the agent keeps testing the text channel against its own priors, which is the harder and more honest test. Say it and the run starts measuring 9.4 instead of 2.6.** ***Both are worth measuring and they are not the same run.*** OPEN

**What the channel taught about ASKING — 2026-09-13**

***It took five changes to get one honest entry into*** `playtest.json`***, and none of them was a bug in the tool.*** **Every one was a mistake about how a question gets answered**, *and they are worth keeping as a sequence because each looked like the fix.*

| | what was wrong | what `aside` already knew |
|---|---|---|
| 1 | *asked only on first arrival at a map* | **205 steps passed with no new map** — *and a stuck agent is the one with the most to say* |
| 2 | ***`playtest` was an ACTION*** | **the request allows EXACTLY ONE tool call per turn**, *so it was being asked to spend its only move on commentary. It chose to play, correctly.* **`aside` is a FIELD and rides along for free** |
| 3 | *the field opened "USUALLY EMPTY"* | **the model emitted `kind=""` on every turn for 130 steps — doing exactly what it was told.** ***`aside` describes a STANCE, never a policy***, *and a rule about when to stay silent produces silence* |
| 4 | *"was there anything worth reporting?"* | **an open question with a sanctioned no gets the no.** *"What did you have to guess at?" presumes there was something and asks which* |
| 5 | ***`NOTED` — "an observation with no verdict"*** | **18 of the first 20 entries, sixteen of them play narration on consecutive turns** — *"Menu is open."* ***It had no test to fail, and it duplicated `aside` exactly.*** **Removed** |

**The test that replaced NOTED is the one worth keeping:**

> **Would this still be true tomorrow, for somebody else?** *"The east exit is higher up than it looks from here" survives the turn and is a report. "Menu is open" does not, and is an aside.*

**And a person at the controls is invisible to it**

***The first CONFUSED that looked like a bug was the author saving the game.*** **The agent had pressed B; a person then drove to the save screen; and the agent filed** *"pressing B after using DEEP RECOVER took me through a save screen and then to the title screen."* ***A confident causal claim about a cause that was not it*** **— and `reflect` and self-criticism learn from outcomes, so they would have learned from that too.**

**So there is a PAUSE now**, and three things about it are deliberate:

- ***It sends the wanted state, not a toggle**, so two open dashboards cannot each flip it and cancel out*
- ***The loop gate sits above the `try`**, because the `finally` flushes usage files and records loop time — a paused iteration inside it would write two files twice a second and count idle as work*
- ***On resume the agent is TOLD**, **once and plainly** — not guessed at: the harness knows a person had the controls, so it says the screen may not be where you left it, none of it was you, and nothing from that stretch is evidence*

***And for when nobody presses the button**, both `aside` and `playtest_kind` now say a person may take the controls without warning — **so an unexplained change is thought of as that, rather than as something it caused.** A report that blames the wrong button sends somebody to fix a bug that is not there.*

**And the self-model was rehearsing its faults — 2026-09-13**

***`<self>` is read back every turn, and the aside is told to SPEAK AS THE PERSON DESCRIBED THERE.* So a field that asked for "*a tendency, a habit, a fault named plainly*" cast every fault as a character to perform.** *Its own good example was "**I commit to a route before I have checked it, and then I am reluctant to turn back**" — **and that sentence came back as four of the six entries**, ×3, ×2, ×2, with "noticed 4 times" presented as extra weight. **The example seeded the observation and the loop reinforced it.***

**It stays a DESCRIPTION rather than an instruction** — *rules have `learned`, and `<self>` is a speaker to be consistent with* — **but of the player it is BECOMING:** *not "someone who commits too early" but "**someone who checks the collision data before committing.**" Describe a fault and the voice rehearses it; describe the fix and it rehearses that.*

***Two lessons, both the same shape as the rest of this section.* An example inside a description is not illustration, it is the most likely output.** *And a deduplication that is exact on purpose needs the writer told to repeat itself exactly* — **or paraphrases of one observation fill every slot.**

#### 8.3 Known bottleneck

**Sprites.** 151 daemons, front and back, 2bpp, four shades, 56×56 maximum — roughly 300 hand-tuned tiles. Nothing about it is hard; it is simply the largest block of labor in the project. AI generation can rough out silhouettes, but expect manual cleanup: the Game Boy palette and tile constraints are unforgiving.

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#### 8.4 The two editions

The project is called **CONTEXT / CONTENT** and the slash has never been explained. It should be literal: **two editions, one source tree.**

### And the order is a decision, recorded 2026-08-31

**The mapping runs RED → CONTENT and BLUE → CONTEXT**, in the Makefile, the cart titles and 9's cost table. **The title runs the other way.** That inconsistency was noticed from the outside, and it was accidental until now.

**It stays, and here is the reason, stated as a choice rather than a discovery.**

**Everyone says content is king.** Putting **CONTEXT first** is the thesis in two words, before the player has read a line of anything — and it means **the title refuses to let the original decide its order**, which is the correct posture for a conversion whose entire subject is that the frame decides what you are looking at.

**The two orders answer different questions and both are right.** *The title is the argument*: context is prior to content. *The mapping is the inheritance*: Red came first and Red is the default build.

*Flagged honestly, per 0.2.* **This is a justification for a choice already made, not something found.** The rule against promoting coincidence to finding applies to the project's own decisions as readily as to the source material — so it is filed as a decision, and it is not evidence of anything.

### It is already in the engine

`pokered` does not build one ROM. It builds Red *and* Blue from the same sources, and the mechanism is three lines of Makefile. Verified against `pret/pokered` master, 2026-08-28:

```
$(pokered_obj):  RGBASMFLAGS += -D _RED
$(pokeblue_obj): RGBASMFLAGS += -D _BLUE
```

and in the data itself — this is `data/wild/maps/Route2.asm`, unmodified:

```
IF DEF(_RED)
    db  3, WEEDLE
    db  4, WEEDLE
ENDC
IF DEF(_BLUE)
    db  3, CATERPIE
    db  4, CATERPIE
ENDC
```

`make red`, `make blue`, `make all`. **Rename the two defines to `_CONTENT` and `_CONTEXT` and the two-edition release is done.** There is no new engineering here at all — the hard part was solved in 1996 by people who needed you to buy two cartridges.

**CONTENT is the primary target** (`_RED`'s slot), because content is the literal layer — and because it is quietly funny that the content-first edition is the default build.

### Three tiers of divergence

**Tier 1 — encounter tables.** Free, standard, exactly the Route 2 pattern above. Note what it does *not* cost: **zero extra sprites.** Every daemon exists in both ROMs. Each edition simply cannot *bind* some of them, so the 8.3 bottleneck does not move at all.

**Tier 2 — let the rosters lean.** Do not split the exclusives at random. Bias them:

| Edition | Sees more of, early | Reads as |
|---|---|---|
| **CONTENT** | CONTENT · LOGIC · STRATUM · LEGACY | the literal, the rule-based, the physical layer |
| **CONTEXT** | CONTEXT · LATENT · VECTOR · ENTROPY | framing, the unconscious, embeddings, noise |

The consequence is the good part. **The two editions are hard in different places.** A CONTENT player finds the early benchmarks straightforward and Benchmark 6 — attention and framing, in the bought city — genuinely punishing. A CONTEXT player struggles early and walks Benchmark 6, which is exactly how 8.2 says the learning paradigms should feel: unsupervised is confusing early and excellent late.

So **each edition's player builds a different intuition about what is strong** — and when they compare notes, they disagree, sincerely, with evidence. That is the thesis arriving as a *distribution* mechanic, and nobody has to say a word.

**Tier 3 — the Index entries disagree.** This is the prize.

Same daemon. Two editions. Two different entries, both stated as fact, neither acknowledging that the other exists.

The trap to avoid: do **not** make CONTENT's Index measure content and CONTEXT's measure context. That collapses 4.2's irony into a joke. **Both editions' Index measures only content — and they disagree anyway.** Different heights. Different weights. Different flat descriptive sentences about the same creature.

Two players compare entries and find the record does not match. Nobody explains it, ever. **The Index is a reproduction, and two reproductions of the same thing differ** — which is 0.2's whole printing throughline, and the Five Witnesses at the scale of the cartridge.

*Restraint required.* Do this for a handful of daemons, not all of them — enough that a player who compares finds it, few enough that it reads as unsettling rather than as a gimmick. And **never for the starter**, whose entry has a different job (11.1 in `lineage.md`: it rewrites itself once, late).

Cost: dex text is text. `IF DEF(_CONTENT)` around the entries. No engine work.

### Trading is the structural argument

Gen 1 link trading is already in the engine, and edition-exclusive daemons mean **you cannot complete the Index alone.**

The artifact that measures only content cannot be completed without another person's playthrough. You need someone else's context, literally, over a cable.

That is the best structural argument this project has available and it costs an encounter-table split.

### Done, 2026-08-29 — partially

**The visible half is in.** `make content` and `make context` build `daemonsContent.gbc` and `daemonsContext.gbc`, with cart titles `CONTENT` and `CONTEXT`. Saves follow the ROM name, so the two editions keep separate playthroughs automatically.

**The defines are not renamed.** `_RED` and `_BLUE` are used by `IF DEF(...)` blocks in **47 asm files**, so swapping them to `_CONTENT` / `_CONTEXT` is a sweep, not an edit — and it would be a large permanent diff against upstream while we still want `git pull upstream master`. Deferred until the tree has diverged enough that upstream merges stop mattering.

Nothing depends on it. The defines are internal; the player never sees them.

### Scope: build the flag now, use it later

Section 8 is unambiguous that the vertical slice comes first, and two editions is a step-14 item at the earliest. **But the flag is not.**

> **Set up both targets on day one, even if the two ROMs are byte-identical for a year.**

Adding a version define at the start costs nothing — the tree already has one. Retrofitting one *after* 151 dex entries and sixty encounter tables exist is a miserable afternoon. Keep `make content` and `make context` in the Makefile from the first commit and let them produce identical ROMs until they do not.

### What must never differ

**The type chart is byte-identical across both editions.** It is the argument, and an argument that changes by cartridge is not an argument. Same maps, same benchmarks, same story, same lexicon, same music.

The editions differ in **what you meet** and **what the record says about it.** Nothing else.

### Title screens — as built

*Implemented 2026-08-29.* The subtitle reads **Content Edition / Context Edition**, matching vanilla's *Red Version* pattern.

**The version subtitle is not a blitted image, and that matters.** The graphic is loaded into VRAM as tiles, and then a string of *tile indices* is printed:

```
db $60,$61,$7F,$65,$66,$67,$68,$69,"@"   ; vanilla "Red Version"
```

Tiles 0, 1, a blank, then 5–9 — **tiles 2, 3 and 4 are never displayed.** Vanilla did this because "Red" and "Blue" are different lengths and the two versions share one VRAM window, right-aligned by a size calculation at load time.

A first attempt at this rename put the word in tiles 0–4 and rendered as "**Con**".

**Our fix simplifies it.** Both editions ship the same 10-tile canvas, so both load at `$60` and both print the full contiguous run — no per-edition string, no skipped tiles, no `IF DEF` at all:

```
db $60,$61,$62,$63,$64,$65,$66,$67,$68,$69,"@"
```

*Built to spec, not generated.* Clean `e`, `o`, `n`, `i`, `d` letterforms were extracted from the vanilla graphic; `C`, `E`, `t`, `x` were drawn to match its 2px stems; the words were composited centred in an 80×8 canvas and written as **1-bit greyscale PNGs**. No image model produces that format — `rgbgfx --colors dmg` rejects anything with anti-aliasing or a colour outside the palette.

### The splash line — and why it changed

`GAME FREAK inc.` is now **CODEMUSIC**, and the year is gone.

**This is more honest, not less.** The vanilla line asserts that Game Freak made this build. They did not. Leaving it would be a false attribution on modified work, and replacing it with a CodeMusic copyright *and a year* would assert a claim over an engine that is pret's disassembly of Nintendo's game.

So the line shows the wordmark and nothing else. **Trademark acknowledgment stays in `README.md`**, where it belongs and where it is unambiguous.

*How it was assembled.* The vanilla line is 16 tile indices — year tiles `$41`–`$45` (shared with the boot splash) plus a 9-tile wordmark at `$46`–`$4E`:

```
db $41,$42,$43,$42,$44,$42,$45,$46,$47,$48,$49,$4A,$4B,$4C,$4D,$4E
```

`gamefreak_inc.png` is 72×8, 2bpp, exactly those 9 tiles — so CODEMUSIC is a direct swap with no VRAM changes. The string now prints `$46` – `$4E` only, recentred at column 6.

*Deferred:* a year would be lovely — '**11.**'**26** encodes the whole lineage, and vanilla's own multi-year format invites it. But the digits need glyphs that do not exist in the current tile set, and the year tiles are shared with the boot splash. It wants a slightly wider wordmark graphic, which is 8.5 work.

**The Super Game Boy border, done.** `red_border` / `blue_border` are now `content_border` / `context_border`, and they no longer read RED and BLUE.

**And it is harder than it looks.** A border file is not a picture of a border. It is **128×48 — a bank of 96 unique 8×8 tiles** — plus a separate `.tilemap` of 896 `(tile, attribute)` pairs arranging them across the 256×224 frame, plus SGB palette entries that colour the 2bpp greys at runtime. Four facts had to be measured against vanilla rather than assumed:

- **`rgbgfx` inverts greyscale.** PNG level 3 becomes colour index 0. So in the source PNG *higher value is lighter on screen*, and level 3 is the light ground.
- **The attribute byte carries the palette:** `(attr >> 2) & 7`. Vanilla uses palettes 4/5/6 ( `PAL_SGB1` – `3` ) and `$40` for X-flip. Ours uses palette 4 throughout, so every border entry is `$10`.
- **Tile `$00` is reserved and flat.** The centre 160×144 is covered by the Game Boy screen; vanilla fills it with tile `$00`, attribute `$00`.
- **Colour index 0 is not transparent here.** Vanilla's art uses it as the *lightest* value, not as a hole.

**They are generated, not drawn.** `tools/genborder.py` builds each border from a repeating cell. This was not the first plan — three illustrated versions were commissioned and measured with `tools/mkborder.py`, which takes a 256×224 design, quantises, deduplicates and reports the tile count. They came in at **496 and 523 unique tiles against a budget of 96**. The lesson is that a halftone frame is *geometry, not illustration*: generated from a cell it lands every dot on the grid, and both editions now cost **12 tiles**. `mkborder.py` remains the measuring tool for any supplied art.

**And the two borders argue the same thing the editions do:**

- **CONTENT** — one dot grid. The thing itself, repeating, self-identical. Dots shrink toward the screen, so the image resolves as you approach it.
- **CONTEXT** — the *same* cell, laid over itself at an offset. Nothing new is drawn; the pattern is entirely the relationship between the two. True moiré is definitionally non-repeating and therefore cannot be tiled at all — so this is a repeating unit that *reads* as interference. That constraint is not a compromise; it is the argument in miniature.

**Which daemon appears**

Vanilla puts a starter on the title — Charizard on Red, Blastoise on Blue, and Venusaur on neither. So:

- **CONTENT edition:** the Supervised line (CONTENT → CONTENT/LOGIC)
- **CONTEXT edition:** the Unsupervised line (VECTOR → VECTOR/LATENT)
- Reinforcement appears on neither title, exactly as Venusaur does not

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#### 8.5 Not yet — sprites, and the recomp question

Two ambitions worth recording so they are not lost, and **deliberately not designed.** Both sit after the vertical slice. Neither is a commitment.

**Original sprites.** 8.3 already names this as the bottleneck. Nothing to add except that it stays a bottleneck no matter which of the below happens. *Prompts and specs for the first pass now live in* `sprite-prompts.md`.

**How far the foxes spread — decided 2026-08-29.** 4.1 makes Crystal a fox and 4.3 makes the resemblance an inference beat. The open question was whether *everyone* is animal-like.

**They are not, and the reason is resolution rather than lore.** At 16×16 a character has about three pixels of head, so the entire animal vocabulary is **ears, a tail, and a shade** — the bible says as much already. Making the whole cast anthropomorphic costs something like a hundred sprite sets and buys almost nothing, because at that size a badger and a bear are the same eight pixels. So: ~~**the Clears get ears, tail and value; nobody else changes; and the game never says anyone is a fox.**~~ **the Clears get ears, tail and value, and the game never says anyone is a fox.** It reads as family resemblance rather than as a species claim, which is exactly the job 4.3 gives it.

> "**Nobody else changes" is SUPERSEDED — struck 2026-09-15. 9.4's fable rule (settled 2026-09-06) governs the cast**, *and it was settled the hard way: four treatments were generated and the animal-head-on-a-human-body was rejected because it failed on a* **bug catcher in shorts** *— a generic class sheet, tested directly.* **Its three clauses are about RENDERING, not about who is an animal.** *The amendment below had already taken the floor out of this ruling a day earlier by moving the resolution it rested on; 9.4 then decided the question it left open.* **T-115 through T-120 are built on 9.4**, *and by this striking the sentence above no longer contradicts about a hundred sheets that are in the ROM.*

*The failure mode, and the cheap escape.* If three tailed characters among an untailed cast reads as *otherness* rather than *lineage*, the fix is not to convert everyone — it is to give light animal features to a handful of **recurring** characters (Scorn, two or three Benchmark leaders) so that fox becomes one species among several. Eight sprites, not a hundred. Decide it on playtest, not now.

***Amended 2026-09-04 — the port moved the resolution the ruling rested on***

***The reasoning above is a Game Boy fact.*** "*At 16x16 a character has about three pixels of head, so the entire animal vocabulary is ears, a tail, and a shade*" — **and the intro portrait on GBA is 64x96.**

**At that size a fox is a fox.** Ears, muzzle, brush, all fully legible, on the **first character the player meets**, while every other portrait in the game is a person. ***The named failure mode is no longer a risk to watch for on playtest; it is the default.***

**So the cheap escape stops being contingent.** *Scorn gets his animal at the same time Crystal gets hers*, not after — **because the deferral assumed a size at which the problem would not show, and that size is gone.** The opera already has him: **a red snake**, which is a species and not a shade, and that is precisely what makes fox read as *one kind among several* rather than as *other*.

***Built 2026-09-04 — `tools/gbachar.py`, and two formats that do not resemble each other***

**The two sprites are loaded completely differently, and getting either wrong produces confetti rather than an error.**

| | | |
|---|---|---|
| **CRYSTAL** | `oak_speech/oak/pic.png` | **64x96, EIGHT bits per pixel** |
| **SCORN** | `trainers/front_pics/leader_giovanni_front_pic.png` | **64x64, four bits, sixteen colours** |

***Crystal's pixel values are 97 to 121, not 1 to 25.*** `oak_speech.c` loads her palette with **`LoadPalette(..., BG_PLTT_ID(6))`** — palette RAM **96** — and an 8bpp background indexes palette RAM **directly**. *An image written with ordinary low indices would read the wrong end of the palette and arrive as noise*, with a clean build and no warning. **The 32-colour `.pal` is bank 6, so its entry k is RAM 96+k.**

**And neither sprite carries its own shadow.** The generated art stands on a pale ellipse; **`oak_speech` draws `platform.png` separately underneath and the battle screen draws its own**, so the ellipse had to come off or the character stands on two. *It is removed by hue rather than by position*: the ellipse is the background blended toward white and keeps its green bias, **while the lab coat is neutral and the suit is blue-grey and neither does.**

**Where Scorn is actually seen**

***He is not an intro portrait — he is a battle sprite***, which is why his is square and hers is not:

| | |
|---|---|
| **CORPUS HIDEOUT B4F** | the first meeting |
| **SILPH CO. 11F** | the takeover |
| **CALLOW GYM** | 4.31, the last MARK |
| **CORPUS WAREHOUSE**, Five Island | after |

**Four encounters, and the player fights him in all of them.** *He also walks those maps as an overworld sprite, which is still vanilla and still a man* — **the smallest remaining piece of the fox decision.**

***Scorn is the BRIGHT red, and it costs something measurable***

***The first pass was a dark red and the repo version is brighter.*** **Bright wins, and not for the reason it first appears to.**

**3589 is the deciding line: "*This design has no villain. Its central move is refusing to make anyone a monster.*"** *A dark red cobra in a suit is villain-coded* — the player files him as the antagonist in half a second, and **that is the one thing this design will not do.** 4.6 already states the horror precisely: "*to Scorn the question is settled — they are units, they are inventory, the math is clean. That is the horror, and it needs no villainy to land.*"

**Bright red reads as a man having a good day.** *Which is worse, and true.*

***And it is paid for, rather than free.*** Measured off the art:

| | head | suit | separation |
|---|---|---|---|
| **bright, as shipped** | 82.8 | 109.6 | **26.8** |
| darkened to 60% | 49.7 | 109.6 | 59.9 |

***Darkening the head more than doubles the value separation***, because the suit is a mid-grey and moving the head down moves it away. **So the dark version is the technically stronger sprite and the weaker character**, and the character wins — *26.8 is workable behind a thick outline and a silhouette as distinctive as a hood.*

**If it ever needs more, lighten the SUIT, not the head:** at ×1.30 the separation is **59.7** with the charisma intact. *The cost is known and the escape is known, which is the only reason to accept the cost.*

**What does not change**, and is the reason the whole thing holds:

> The game never says anyone is a fox.

*Link's Awakening is the model and the user named it* — **an island of animal-people, unremarked by anybody in it.** A world, not a claim. **It only becomes a statement the moment a character notices**, and no character does.

*And the daemon-confusion worry is answered by context, not by design.* **Portraits and battle sprites never share a screen**: a fox in a lab coat at 64x96 is read as a person in a room, and MUSAI at 64x64 on a platform is read as a creature in a fight. **The player is never asked to tell them apart, because they are never side by side.**

*The overworld ruling above stands unchanged* — 16x32 is still three pixels of head, and **the Clears still get ears, tail and value out there and nothing else.**

*And the resemblance is carried by value, not hue.* 8.6 means the player sees four greys, so *golden-amber*, *darker* and *between* have to be **light-to-mid, dark, and between** — with identical ear and muzzle geometry doing the family work. Colour words will not survive the conversion and should never be relied on in a prompt.

**The Gen1Recomp question.** Gen1Recomp is a hand-written Lua / LÖVE2D recreation of Gen 1 that **imports data and graphics from a player-supplied ROM** rather than emulating it. Its mod ecosystem includes the **voxel mod** and first-person renderers — which is the specific pairing of interest here: DAEMONS creatures, rendered as voxels, in a first-person Kanto.

That split — **imported data, hand-written behaviour** — is the whole of the analysis, and it cuts cleanly:

| | Likely carries | Almost certainly does not |
|---|---|---|
| What | Sprites, base stats, names, type constants and matchups, Index entries — anything decoded out of ROM tables | Story scripts, custom events, engine work (RECURSION), anything that lives in `asm` rather than in data |

So the optimistic case is a DAEMONS ROM loading in a 3D shell with the right creatures and the right combat philosophy, and the pessimistic case is that the loader rejects a non-canonical ROM outright — the project documents accepting "*the canonical 1 MiB US versions,*" which implies it may check.

**The one test worth running, whenever the mood strikes:** build the step-5 milestone (vanilla everything, new type chart) and drop it in the launcher. It costs an afternoon and it answers the entire question — either it loads or it does not. Do that before designing anything around it, and before assuming either outcome.

**A note on the voxel mod specifically.** A voxel renderer builds volume out of the 2bpp sprite data — so the sprite work in 8.3 would be doing double duty, and the four-shade constraint that makes Game Boy spritework hard is exactly what a voxel mod needs to read cleanly. That is a reason to design the sprites well rather than a reason to design them differently.

**Scope note.** This is a *rendering* ambition, not a design one. Nothing in sections 0 through 7 should bend to accommodate it. If it works it is a gift; if it does not, the game was always a Game Boy game.

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#### 8.7 The third review, 2026-09-10 — every ruling, so this one gets swept

***Read against v11.111, The Bag and The Chart, by a third model.*** **Its sharpest line is about the document rather than the game, and it is filed as binding alongside 8.6's:**

> **A ruling nobody swept is not a decision, it is a draft.**

***That is the failure mode this section exists to break.*** **8.6 recorded what was taken and the sweep still lagged** — so this one records **every** item with a verdict, including the refusals, and the open ones carry the tag.

**Settled the same day**

| | ruling |
|---|---|
| **Scorn met Crystal** | ***Retracted "he never met her."*** New canon in 4.4: the company is founded with **Ty as CEO**, *Scorn's Solution* is the **hiring**, he designs and **rigs** the standardised evaluations, becomes valuable, and **Ty introduces him to Crystal.** *The horror improves* — a man who was brought to her by Ty and still only ever filed. **And Crystal gains standing to notice**: she is the only person who saw Ty before Scorn and after |
| **PREEMPT was an item and a species** | ***Renamed.*** ICE HEAL → `WATCHDOG`, and the review found the fault under the collision: **preempting a hung task hands the processor to somebody else and the hung one stays hung.** *A watchdog is the timer whose whole job is to notice something stopped answering and reset it* — **the actual cure, and it still works twice** |
| **The collision check was scoped to the day it was written** | ***`tools/check_lexicon.py` is new.*** 4.26's check validated species against species, moves and types — **not items, because the item pass had not happened yet.** The new one **reads every surface out of the build on each run**, so a surface added later is covered without anyone remembering |
| **1.7's security rule condemned three of our own species** | ***Amended.*** `PAYLOAD`, `ROOTKIT` and `INJECTOR` shipped five days before the rule that forbade them. **The line is redrawn where it falls: *intrusion words belong to CREATURES; they never belong to the INSTITUTION.*** The review's version is sharper than the one it corrected |

***And the new checker found a third collision on its first run that neither the review nor the project had seen:*** **`SWARM` is our BUG type and vanilla's Bug ability.** *A daemon whose type and whose ability both read SWARM.* **`GROWTH` (move + type) is the other**, already logged in 2.8. OPEN — *both are renames, not rulings.*

**Accepted, and queued as sweeps rather than decisions**

***Each of these is a ruling that already happened and never travelled.*** **They are listed so the sweep is a checklist rather than an archaeology.**

- **The colour sweep.** 8.6 settled a full-colour world and five passages still describe a grey one — 6, 7.14c, 4.6, 8.4's title daemons, and the decision log. *And the review is right that **4.6's three-form ARTSAI can now be shown**: a legendary that desaturates magenta → white is the best colour moment in the game and nobody has claimed it*
- **The tempo sweep.** 7.9's three-axis S.T.A.R.R. outlier was built on *Slumbering* alone; **two of its three songs are at 128, the corpus pulse.** *The better finding is the one the data already had — **it joins the people's tempo when it wakes***, and 7.13's Quicksilver drop to 110 now protects nothing
- **The SLEEP sweep.** 4.5 and 4.25 still say SLEEP after 1.6 renamed it. **And `SUSPEND` is a species (JIGGLYPUFF), so *SUSPEND was SUSPENDED* will occur** — OPEN, and the review's `LULL` now has the reason 8.1a lacked
- **Brazen's B-section was composed on a retracted transcription.** 7.11 retracts "*the procedure never descends*" and 7.13 still builds the motif from it. ***Re-source it or own it as a composition choice*** — the music may still be right; it is no longer derived
- **The dates in 4.10**, Vera's birth order, Ty's key, 4.14's inversion, and Crystal's reachability. *Small, and all of them are the same fault*

**Free wins, taken**

- `HOTFIX` **already costs colour.** Heal Powder lowers friendship in Gen 3 and 8.6 keyed saturation to friendship — "***nobody is proud of it*" *became a mechanic the day 8.6 was settled***,** and nothing has to be written
- **The Index completion line is already written.** Crystal on an empty Index: "*…It is empty. That is a result as well.*" **On a full one, change one word.** *Forty hours between them, and vanilla's diploma becomes a shrug*
- **FROZEN's concept line is missing its commonest ML sense** — *weights that are not updated.* **Add it and two inherited relations stop being inherited**: FROZEN ×2 GROWTH becomes literal, and FROZEN ×2 VECTOR gets a reading
- **Seventeen types close a loop the fifteen could not.** ***LOGIC → OPAQUE → CONTEXT → LOGIC***: rules open black boxes, black boxes shrug off framing, framing beats rules. **That is the interpretability gap as a triangle and it arrived from Game Freak's own table.** *The Chart publishes fifteen and the ROM runs seventeen* — OPEN, and the one-clause test wants re-scoring at seventeen, **because CONTEXT's 8/8 was scored without the type that is immune to it**
- **Three relations have names in the literature** — STRATUM ×0 VECTOR is Marr's levels, VECTOR ×2 LOGIC is the Bitter Lesson, SWARM ×2 CONTEXT is Asch on one side and ensembling on the other. *None go in the game; all belong in the chart page's why column*
- **Vera's ladder is Ainsworth's Strange Situation** and **Halftone is Doka's disenfranchised grief.** ***Both were written without the clinical word, which is craft rule 3 performed on a research literature. Do not touch either***

**Ideas taken**

- "***IMPROVE RESPONSE CONSISTENCY*" *becomes a mechanic.*** **Scorn's daemons always roll maximum damage, never crit, and their secondary effects never proc.** *A player who has watched the HEAT show — "damage wobbles" — will notice the fight that does not, and nobody says why.* **The CC-7 pull is then the un-weighting: S.T.A.R.R. gets its variance back.** The bias–variance tradeoff as a boss fight, one flag in the damage calculation
- ***The ability pass.*** **77 passives, every one still vanilla, and 9.3 named them as a reason to port.** *Vanilla already wrote half the argument*: **SYNCHRONIZE** passes your status to whoever inflicted it — **PERSPECTIVE as a passive, and ARTSAI already has it**; **TRACE** copies the opponent's ability, which is SUBSTRATE's Index entry as a mechanic, already on the right species; **LEVITATE** is *no location*
- `CONVICTION` — FireRed's Choice Band: +50% attack, locked into one routine. ***A weight raised and the alternatives discarded — CC-7 in miniature, holdable.*** Give one to a Corpus grunt and nothing else needs saying
- **The Five Witnesses differ by edition.** *Keep the one accurate detail identical; let the wrong ones differ.* **You need someone else's context, over a cable, to see the room**
- **Al's party desaturates** across the rival fights — ***the transmission failure, visible, with no sneer.*** The colour was there and he optimised it away
- ***The Index sprite keeps the saturation it was recorded at, and never updates.*** **Your daemon at full friendship is in colour in the party and grey in its own entry.** *A reproduction that loses the original, per daemon, drifting for forty hours* — **which is 4.2 as a mechanic and it costs one byte per species**

**Refused, or held**

- `SUS` → `ZZZ` / `HLD`. ***Held.*** *The Among Us reading is real and it is also a joke that lands beside a daemon that has stopped responding.* `HLD` **loses the whole word;** `ZZZ` **says asleep, which is precisely what 1.6 spent a section refusing.** OPEN — *a playtest decides this, not an argument*
- ~~`LEGACY` → `RUST` and `VECTOR` → `FLOAT`~~ : ***Declined 2026-09-11 (2.6), and the review was right about the number and wrong about the cause.*** **Both types score what they score because their clause describes what they ARE and the chart contains nothing but relations** — *re-clauses take LEGACY from 0/16 to 12/16 and VECTOR from 1/13 to 10/13 with the words untouched,* **including the** `← STRATUM ×0` **immunity that** `FLOAT` **was proposed to recover.** *And* `LEGACY` *turned out to carry 4.32's museum in one sentence*
- `PRIORITY` → `BURST` **and** `MP` → `CYCLES`. *Both better than what they replace.* `CYCLES` **especially** — 4.30 still says MANA, *The Bag* ships MP, and **MANA is the lexicon's one fantasy noun.** OPEN
- `PKRS` → `VIRAL` and **MOON STONE →** `TIMEOUT`. *Taken in principle,* and both wait on their own passes
- **CONSENSUS scaling with the quorum. *Refused for now.*** 2.5 made it boring **on purpose** — "*a swarm does not need a gimmick, it needs to keep showing up*" — and the review's version is better writing and worse balance. **Revisit if SWARM underperforms in play, which is the condition 2.5 already set**
- **Penphin's hemispheres and "the sensation of red is the same for everyone". *0.5 already handles the first*** — the reading is marked as interpretation, not citation, and the game never says the word. **The second is a fair correction and the accurate version is better for us**: *sensation varies and perception is normalised by context.* OPEN — 0.4 quotes a 2013 article verbatim and **the quote does not get edited; the gloss under it does**

**The one thing the review is most right about**

***Craft rule 5 is unpaid for benchmarks 2 through 8.*** **CAIRN is built. The other seven have a concept, a one-line mechanical lesson, and no mechanics** — "*punishes split focus*" *is a wish.* **This is the largest prose-to-ROM gap in the project and it sits on the spine of the fun.**

*The review even supplies the first one, from the name:* **Doldrum is a FLOW leader whose gym has no wild encounters and whose trainers heal you. *No slope.*** OPEN, and it is the real work.

#### 8.6 The monochrome question

**The game is about colour and cannot show any.** Every city is a colour word rendered in four shades of grey. That is either a limitation to be fixed or the best thing the hardware gives us, and the answer decides whether colorization is an upgrade or an act of vandalism.

**It is the second one.** Three reasons, in ascending order of how much they cost to lose.

**1. The player is told these places are colours and shown grey.** They supply the rest. That is the 8-Bit World thought experiment — a system whose perception is bounded by its resolution — running for free, on the hardware the thought experiment names, without a line of dialogue. And it is *The Colour We Never See*: purple is not on the spectrum, the mind folds the line into a loop and invents it. The town colours are not on the screen. The player invents them.

**2. The Index is a reproduction that loses the original** (0.2). A greyscale record of a world named in colour loses something specific, visible, and unrecoverable — which is exactly what 4.2 needs the Index to be doing.

**3. Colorizing Halftone destroys Halftone.** The town's entire premise is *dots that only look like grey* — continuous tone that is actually discrete black and white sampling, where the bias comes from **resolution, not malice** (3.1). In colour, halftone stops being the town's condition and becomes a quaint printing reference. **This is not a cost to weigh. It is a load-bearing wall.**

**Recommended: colour as something the world spends**

Not "monochrome forever" and not full colorization. **Greyscale by default, and colour appears exactly once, where it means something.**

> **The Review Board at Umbra is the only place in the game that has colour.**

Section 6 already sets it up: two thousand years ago emotions were *literally coloured fluids*, and the four humors arrive as a classical flourish that becomes the whole thesis about six seconds later. Give them their colours — sanguine red, choleric yellow, melancholic black, phlegmatic white — and give them to nobody else.

Forty hours of colour-named grey towns, and then one room where colour is real, occupied by four ancient people insisting that emotions are coloured fluids. Who are **wrong**, and lose, and are followed immediately by Ty, who is also wrong and also loses.

**The one place the game admits colour is the place with the obsolete answer.** Nobody remarks on it. Nobody can.

### Settled: two sources, one of them earned

The open question — *is one colour moment right?* — resolves at **two, different in kind.**

| | What colour is | Who sees it |
|---|---|---|
| **Umbra** | an obsolete answer, held still and examined | everyone who finishes |
| **PERSPECTIVE** | a glimpse of another's frame, immediately lost | most players, via MOCK |

**MOCK settles it.** 4.6 establishes that Ditto is CONTENT/CONTENT and learns PERSPECTIVE, so the flash is reachable in ordinary play rather than gated behind a puzzle almost nobody solves.

**BunnyArtsai is not a third source.** She is the most loaded firing of the first — the same few frames, from the one creature that could not come back — and only for a player who solved the Five Witnesses (4.8).

*Why this is the right pair.* Vanilla had **no designed way to reach Mew at all** — verified: it appears in no wild encounter table, only in `names.asm` and `palettes.asm`. The famous routes were an unintended glitch and event distribution, and player culture filled the vacuum with rumour. **If colour lived only with BunnyArtsai it would be a secret nobody could confirm.** MOCK makes it something players can compare notes about, which is the condition under which Umbra lands rather than passing unnoticed.

*Restraint clause holds at two.* A third source makes it a system, and a system invites explanation.

**Notes and consequences**

- **Technically this is easier than full colorization, not harder.** A colour build has to palette every map; this one palettes a single map and leaves the rest on a greyscale ramp.
- **On the GBC caveat:** Gen 1 already receives limited hardware palettes on a Game Boy Color, so "monochrome" is a slight fiction in practice. The design intent stands regardless — the ramp should read as value, not hue, everywhere but Umbra.
- **The humors survive greyscale anyway.** Value carries them: melancholic dark, phlegmatic pale, sanguine and choleric mid. The colour at Umbra is a confirmation of something the player has already been reading in tone, which is the correct order.
- **This serves the sprite work rather than fighting it** (8.3, 8.5). Four shades with no hue to lean on forces value contrast to be right, and value contrast is exactly what a voxel renderer needs to read cleanly.
- **The recomp path is a remix, not the canon.** If a 3D or voxel presentation colours everything, that is a different presentation of the same ROM and it is welcome. The canonical build stays grey.

**Settled 2026-09-06: colour is context**

**Reopened 2026-09-05 and closed by the second external review.** *The argument above stands* — grey was **the medium** on a DMG and would be a visible **filter** on GBA — **but it was being argued as art direction, and it is a palette flag.**

***What the engine already has,*** *checked in* `pret/pokefirered`*:* **`gGlobalFieldTintMode`** in `fieldmap.c`, with `QL_TINT_GRAYSCALE` and `QL_TINT_SEPIA`, applied to **tilesets** by `ApplyGlobalTintToPaletteEntries` and to **object palettes** by `ApplyGlobalFieldPaletteTint`. *It exists because FireRed's Quest Log replays in sepia:* **the game's own record of what you did, rendered without its colour.** `palette.c` carries `TintPalette_GrayScale`, `_SepiaTone` and `_CustomTone`.

***Three consequences, and they cost nothing.*** **A GBA sprite is indices; colour is a 32-byte table** — so grey, part-grey and full colour are *the same art*. **A grey town is a flag, not an asset. A grey Index is a call, not a second sprite.**

> COLOUR IS CONTEXT. The world has it. Records do not. A daemon has as much of it as it has accumulated.

| Surface | Colour | Why |
|---|---|---|
| **People, towns, overworld** | full | argument 1 died with the DMG; three foxes and Scorn's red are load-bearing |
| **The achromatic towns** | a **value ladder**, not five greys | Blanche warm white → Slate blue-grey → **Halftone true grey** → Quicksilver ash → Umbra near-black. *7.4 already does this for music:* **hue sets key, value sets register. Two grey towns halves Halftone**, so only Halftone is drained; the rest are *graded* |
| **The Index, the Quest Log, the Trainer Card** | grey / sepia, always | *a reproduction that loses the original* — **now visible, because there is a colour world to lose it against** |
| **Daemons** | the 9.4 ramp, **its saturation scaled by friendship** | below |
| **Umbra** | the value floor | the four humours are **the four type anchors** — red VECTOR, yellow ENTROPY, black LATENT, white FROZEN. *The player has read those hues on daemons for forty hours and meets them as people, in the dark.* **Colour arrives by contrast with darkness instead of with grey** |

### 8.6a Halftone, all of it grey — decided and built 2026-09-13, revised the same day

***Halftone was grey outside and in colour inside the tower,*** **because the tint asked whether the map belonged to the town's map section and the tower has its own.** *The tower is the place Halftone exists for — something the Index cannot name, and an instrument needed to see it — so it is grey now, all seven floors, tiles and people.*

***In Halftone, everything on the screen is grey.*** **The town, the tower, the battle scene, the side you face, *your own daemons, you,* the healthboxes, the menus and every window of text.**

*The first cut kept your own daemons and the menus in colour —* **"a daemon has as much of it as it has accumulated," and the move menu's type colours are how the chart is read. *Reversed on playtest the same day:* a grey town with a coloured message box and a coloured team read as a filter that had missed some things, not as a place.** *The town is the lesson only if nothing in it is exempt — and that includes the one reading aid the player has grown used to, which is the point of taking it away here.* **Friendship-saturation (below) is unaffected; it is about daemons everywhere else.**

***PERSPECTIVE brings the colour back.*** **When a daemon takes another's frame in Halftone, the whole screen returns to full colour for about a second, then goes grey again** *— the flash spans the moment the form changes. This settles 4.6's open question by place: the flash cannot become common, because the only town where it can happen is grey.*

***How it is built, because the first way leaked.*** **Greying each loader in turn — terrain, the opposing battler, the trainer — missed everything it did not name.** *The grey is now applied once a frame to the one buffer every palette reaches the screen through, just before the vblank copies it out.* **The colour copy underneath is never touched**, *so the flash is one copy back and one flag, and nothing can be left in colour by a loader nobody thought of.* ***A colour already grey costs one multiply and a compare***, **so a settled frame is nearly free.**

### Saturation is friendship, and that is the thesis made mechanical

***A daemon at low friendship is nearly grey with a hint of its type; at maximum it is the full ramp.*** *One hook at sprite-palette load, keyed off* `MON_DATA_FRIENDSHIP`.

**Not evolution stage.** *"More colour on the final form" says* ***power = colour****, which is a different game's thesis* — and 8.2's final stages are **the paradigms' failures.** Making them the most colourful argues the opposite of section 8.2.

***What it buys, none of it stated aloud:***

- **Corpus daemons are grey.** *Trainer parties are built at species base friendship;* **set the Corpus classes to 0** and every daemon they field is visibly context-free. **The people stay in colour** — a grey workforce is the same sneer 3.1 guards against, and the sharper picture is *ordinary staff carrying inventory.*
- ***The Index shows a daemon grey either way.*** **The player can see what the record cannot.**
- **A bound daemon is grey. A bonded one is not.** *Nobody is told.*

### Rejected: the post-game item that colours the world

**Three reasons, each sufficient.** *It converts colour from an argument into a* ***reward****, and a reward is a metric* — **Scorn's kind of thing.** *It is a* ***Cognitive Clarifier for the player****,* an instrument that changes what they perceive, which is craft rule 1's worst violation. **And friendship-saturation already does it, per daemon, earned** — a player who bonded finishes in colour and one who did not, does not.

#### 8.6 External review, 2026-08-31 — what was taken and what was not

**Read against v11.0 by a second model.** Its sharpest finding is about this document:

> The bible is outrunning the slice.

**It is right, and it was recorded here as binding.** Sixteen version bumps in one session, and 8.1's slice is still not beaten on hardware — **the next milestone worth recording is that, not another section.**

***Discharged 2026-09-12, and eleven days late in the RECORD rather than in the work*** (8.10). **The slice was played; this sentence was not updated** — *and because it is the one place the bible says what to do next, every reader after it concluded the project was stalled.* **It is the most expensive kind of stale ruling: not a wrong fact, a wrong instruction.**

### One constraint the review did not have

***Gen 1 has no item descriptions.*** There is no descriptions file in `data/items/` — names, prices and use-handlers only. **Three of the review's proposals put their payload in one**, and had to move:

| Proposal | Where it went instead |
|---|---|
| CC-7's "*A sealed module. Part no. CC-7.*" | **Into the item name.** The bag reads `CC-7`, which is stronger — the player *carries* the number rather than reading about it |
| The RESOLVER's asset tag | **Into Scorn's mouth**, as he hands it over and dismisses it |
| AXIOM / EMBEDDING / AFFECT as documents | **Nowhere yet.** Deferred until it has a speaker |

### Built

**PACKAGE → `CC-7`.** The clerk: *Crystal Clear's order came in. A CC-7. Will you run it out?* Her receipt: *Ah. The module. Thank you.* **Thirty hours later the player reads a requisition for the same part on a door at Quicksilver** (4.18), with its reason field. **The player has carried the crime-scene's part number since minute fifteen**, and nothing points at it. *This replaced a custom USERBOX — the wrong gift twice: a box is 1.3's argument, not a present, and it copied vanilla's custom ball.*

**Scorn at the bottom of the Verdigris basement**, where Giovanni stood — cheerful, unbothered, interested in the measurement. **This settles 4.5's open question**: the player likes him *before* the tower, so Halftone becomes a betrayal of their own judgment. He hands over the RESOLVER as inventory equipment and dismisses its tag — *QS-LAB, I think. Before my time.* **He owns the instrument Crystal's lab built and has no idea what it was for.** Then he wishes the player luck at the tower.

**Scorn's Benchmark 8 line is his one accounting, and he splits it with something that cannot answer:** *The machine chose what I weighted it to choose. We are both to blame, I suppose. It has never said otherwise.* **4.18's finding, made audible.**

**The catching tutorial teaches 1.3 before the player owns a box:** *Watch. You weaken it, then you offer it a box. It goes in on its own, or it does not go in.*

**Adopted, not yet built**

***Status, 2026-10-01**: the whole list is built or settled — Ty is in the ROM (T-19), the post-game triangle is the ending as built (T-261), THE HOLDOUT's TOKEN is in (T-260) and ORPHAN is a routine's name. **Penphin is built** (T-234): DOLPHIN VECTOR/FLOW in WAILMER's slot, PENGUIN FLOW in SPHEAL's, either one traded becoming PENPHIN CONTEXT/FLOW in WAILORD's, which learns **RESONANCE** as it evolves — LOGIC and CONTEXT together, √(Attack × Sp. Atk) against √(Defense × Sp. Def), so balance is what it rewards.*

~~**Ty is not in the ROM at all.**~~ ***No longer true (noted 2026-09-24): T-19 put him in the FIVE ISLAND warehouse, where he hands over the payload.*** The bible places him at Quicksilver; no NPC has been written. **So his line and the post-game triangle are new content, not edits** — a larger job than the review implies. *Whether the triangle is still owed is T-261.*

- **The post-game triangle** — S.T.A.R.R. → Ty → Crystal, three text blocks and three party checks. **And *S.T.A.R.R. inferred correctly and could not go*:** a machine that reads its own state cannot walk back into that building, so it went to the nearest place and stopped — **the becalmed city, the local minimum.** RECURSION collapses when interrupted; her leaving was the interruption
- **Penphin cannot be weighted.** The Clarifier failed *because* one vote could be raised; **Penphin is two frames that must agree, so there is nothing to raise.** What she would build after learning what was done to the first one — and **HOLT is the failed Penphin**, two perspectives with no rule for choosing. Halves in Doldrum's waters, one per edition
- **ORPHAN is a Quicksilver process nobody reaped** — the parent exited when the lab burned, and Corpus inherited it by succession
- **THE HOLDOUT**, and the warden's speech garbled until the TOKEN is returned
- **RESONANCE = RECURSION that does not collapse when interrupted** — one branch removed. Deferred until RECURSION exists, since it is a diff against it

**Declined**

**Moving ROVER out of the starter slot** — see 8.5's amendment. The diagnosis was right and the cure cost more than the flaw.

## 9. IMPLEMENTATION

`pokered` is RGBDS assembly. The *data* lives in readable macro tables. Engine features are where it gets expensive.

> *Paths are approximate. Verify against your checkout — they drift between commits.*

| What | Where | Effort |
|---|---|---|
| Type IDs | `constants/type_constants.asm` | trivial |
| Type names | `data/types/names.asm` | trivial, but see 9.1 |
| **Type chart** | `data/types/type_matchups.asm` | trivial, highest leverage |
| Species stats | `data/pokemon/base_stats/*.asm` | bulk, scriptable |
| Species names | `data/pokemon/names.asm` | bulk |
| Index entries | `data/pokemon/dex_entries.asm` plus `text/` | bulk |
| Moves | `data/moves/moves.asm`, `data/moves/names.asm` | medium |
| PERSPECTIVE | rename TRANSFORM | trivial |
| RECURSION | repurpose Bide/Rage machinery | **engine work** |
| Sprites | `gfx/pokemon/front/`, `gfx/pokemon/back/` | **the bottleneck** |
| Map names | `data/maps/names.asm`, `constants/map_constants.asm` | trivial |
| Map layouts | `maps/*.blk`, `data/maps/headers/`, `data/maps/objects/` | medium |
| Route signage, number plus local name | `text/Route*.asm` | **done 2026-08-30** |
| Dialogue, signs, witnesses | `text/` | bulk — **witnesses done** (4.8), the rest ongoing |
| BIND verb and the flat message | `text/` — catch strings; also `data/` menu labels | **done 2026-08-31** — 55 instances, and the success message flattened (1.5) |
| Greyscale ramp everywhere but Umbra | palette data; one map differs | medium, specialist |
| Two-edition build | `Makefile` — rename `_RED` / `_BLUE` to `_CONTENT` / `_CONTEXT` | **trivial, do it first** |
| Edition-exclusive encounters | `data/wild/maps/*.asm`, `IF DEF(_CONTENT)` blocks | trivial |
| Divergent Index entries | `data/pokemon/dex_entries.asm` plus `text/`, conditional | trivial, high value |
| ORPHAN as a species in a free index slot | `constants/pokemon_constants.asm`, base stats, sprite, cry | one daemon's work |
| ORPHAN's blank Index entry | `data/pokemon/dex_entries.asm` — a pointer that goes nowhere | trivial, needs care |
| SILPH SCOPE → RESOLVER | `data/items/names.asm` plus `text/` | **done** — and Scorn hands it over with its QS-LAB tag (8.6) |
| Hidden BunnyArtsai tile | hidden object plus event flag | **the Five Witnesses' missing payoff** — accounts and terminal built, nothing rewards standing there |
| Quicksilver terminal log | one text block | **done** — the 1001 sequence, plus ITER 35 on Mansion 1F (4.8) |
| Corpus lobby engraving | one sign-text object, Brazen | **done 2026-08-30** |

| What | Where | Effort |
|---|---|---|
| Quicksilver asset tags at Halftone | one sign-text object | trivial |
| `AL CLEAR` in formal text; naming screen untouched | `text/` — literal string instead of the `<RIVAL>` control char | trivial |
| Default name lists, both editions | `constants/player_constants.asm` | **done 2026-08-29** |
| PROF.OAK → CRYSTAL | `text/` — 33 occurrences in 10 files, plus the intro sequence. Every line needs rewrapping | half a day |

***Done on the GBA side 2026-09-03, and it was 84 blocks in 25 files, not 33 in 10.*** **`tools/port_oak.py` measures the rewrap** against FONT_NORMAL's own advance widths rather than counting characters. *Three things the Game Boy pass did not have to face:* **a name vanilla wraps mid-line** (`PROF.\nOAK's`), which only the tool's second run found; **four fixed-width fields** where `CRYSTAL CLEAR` does not fit, of which only the trainer nameplate had to give; and **fourteen masculine pronouns**, applied only inside blocks that name her — *`VermilionCity`'s "he" is the other AIDE, and a blanket sweep would have changed him too.* | Quicksilver journal fragments, one signed | `text/`, Cinnabar Mansion idiom | trivial | | Corpus badge name matching the signature | `text/` | trivial | | Brazen employee's post-Quicksilver second line | one event flag, one text block | trivial | | Music | `audio/music/*.asm` | medium, specialist |

### 9.1 The `TypeNames` gotcha — resolved, and smaller than feared

*Verified against `pret/pokered` master, 2026-08-28. Built and confirmed.*

The warning in earlier drafts was right in spirit and wrong in detail, and the detail matters because it makes step 5 **much** cheaper than the bible previously claimed.

**Current master does not need a subtraction.** `data/types/names.asm` is a flat table guarded by `table_width 2` and `assert_table_length NUM_TYPES`, and the ID gap is filled explicitly:

```
REPT UNUSED_TYPES_END - UNUSED_TYPES
    dw .Normal
ENDR
```

So every unused ID points at `.Normal` and `GetTypeName` indexes straight in. There is no `$0B` fold in this checkout. **Never paste a replacement `names.asm` wholesale** — that advice stands, harder than before, because the structure here is nothing like the standalone file in `patches/`.

**The bigger correction: you do not need to rename the constants at all.**

The chart references types by symbol; the symbols resolve to IDs; the IDs drive the matchup lookup. **None of that touches what the player reads.** What the player reads is the strings in `names.asm`. So a complete step-5 build is:

1. Change **only the strings** in `data/types/names.asm` — `db "NORMAL@"` becomes `db "CONTENT@"`, and so on for fifteen
2. Apply the two matchup deltas

Renaming `NORMAL` to `CONTENT` in `constants/type_constants.asm` is a **readability change for the developer**, not a gameplay change — and it is the expensive one, because every type constant is referenced across `moves.asm`, `base_stats/`, and elsewhere, and several collide with unrelated symbols (`ROCK` with `ROCK_SLIDE`, `FIRE` with `FIRE_BLAST`, `BUG` with the Bug Catcher trainer class). Do it later, deliberately, with word boundaries, or never.

**Three literals the earlier notes got wrong**, all confirmed by build:

| Earlier note | Actual |
|---|---|
| `PSYCHIC` | `PSYCHIC_TYPE` |
| `db GHOST, PSYCHIC, 00` | `db GHOST, PSYCHIC_TYPE, NO_EFFECT` |
| numeric `20` / `05` / `00` | named `SUPER_EFFECTIVE` / `NOT_VERY_EFFECTIVE` / `NO_EFFECT` |

**`constants/type_constants.asm` also cannot be pasted wholesale.** Master wraps the list in `DEF PHYSICAL`, `DEF UNUSED_TYPES` / `UNUSED_TYPES_END`, `DEF SPECIAL` and `DEF NUM_TYPES`, and `names.asm` depends on three of those symbols. The standalone file in `patches/` defines none of them and would break the build immediately.

The 8/7 physical–special split is intact and is exactly where 2.1 says it is: `const_next 20` (decimal, = `$14`).

#### 9.3 The GBA question — a spike, opened 2026-09-02

**pret also disassembled FireRed.** *The question raised was whether the GBA lets this design express itself more appropriately,* and it is a real question rather than a tempting one.

### What would transfer, measured

| | Size | Survives |
|---|---|---|
| `docs/` | **5,343 lines** | **All of it.** The design has never been engine-specific |
| The generated source illustrations | ~30 files | **Yes** — high-res; only the downsampler is Game Boy-specific |
| `tools/` | 1,630 lines | Partly. Crop and quantise port; 2bpp tile output does not |
| 64 sprite PNGs | 4-shade, 40–56px | No — redraw from the sources |
| 16 music files | GB channel data | **No.** GBA is a software mixer; total rewrite |
| 236 asm/data edits | chart, names, scripts, maps | No |

***334 files of implementation against 5,343 lines of design.*** **The expensive half is the half that ports.**

### The case for, and it is not colour

| | |
|---|---|
| **Abilities** | *Gen 3 gives every species a passive.* **2.5 invented CONSENSUS because the chart needed a move to be legible** — an ability is a second axis for exactly that work, and Gen 1 does not have one |
| **Item descriptions** | ***Gen 1 has none at all.*** **The eight MARKS, the boxes, 8.x's stones-that-are-inputs — every one currently gets a name and silence.** A writing-led project on the one generation with nowhere to write |
| **A scripting language** | instead of asm for every sign and guard |

### What breaks, and one of it is an invariant

***Invariant 5 stops being a constraint and becomes a preference.*** **On a DMG the machine and the meaning agree** — the player sees grey because that is what the hardware *is*, so the Review Board's colour feels **impossible**. *On GBA it is a saturation choice, and the rupture becomes a stylistic decision.* **It can still be good. It cannot be the same idea.**

*Stated honestly: it is already softer than the invariant claims* — SGB palettes already put a blue logo and a tan coat on screen. **But colour arriving through a palette is not the same as colour being free.**

***And the chart goes from 15 types to 17.*** **2 calls the fifteen names "exactly at the limit"** of the character count, and they were chosen as a *complete* argument. **Invariant 3 says the chart is the argument** — so two filler types is a worse chart. *Two good ones would be better than what exists.* **That is a design problem to solve before porting, not after.**

**Decided 2026-09-03 — the project moved**

***The spike answered its question in a day and the project pivoted.*** **`engine/` is a reference now**; it still compiles and still runs the slice, and it is not updated further.

*What follows is the reasoning as it stood before the answer, kept because it is the argument the decision was made against.*

**Why it was a spike and not a decision**

***8's rule is the deciding one:*** "**the graveyard of ROM hacks is full of projects that designed 151 creatures and shipped zero towns.**" *The vertical slice is* ***done****.* **A port resets implementation to zero while the design keeps galloping** — which is precisely the failure mode 8 exists to prevent, *arriving dressed as an upgrade.*

**And the port is cheaper later, not dearer.** *The docs are fully portable, the art sources are portable, and a design that has been* ***played*** *de-risks the rewrite enormously.*

**So both engines stay.** `engine/` keeps the slice; `engineGba/` answers one question — ***is the expressive gain real enough to pay for 334 files?*** — by building **one daemon with an ability and one MARK with a description**, and nothing else.

**Two corrections, found by porting rather than by arguing**

***The case above got one benefit wrong and missed one cost.*** **Both surfaced within an hour of touching the code**, which is the argument for spiking rather than deciding on paper.

**Wrong: the MARKS were never going to get descriptions.** *9.3 said "the eight MARKS, the boxes and the stones-that-are-inputs each get a name and silence."* **In Gen 3 badges are not items at all** — `items.h` contains the string `BADGE` **zero times.** They are flags plus a Trainer Card display, exactly as in Gen 1. *The description win is real —* **`items.json` carries a `description_english` field with real multi-line text — but it applies to the boxes, the stones, medicines and TMs. Not to the MARKS.**

**Missed: type names get *shorter*, not longer.** ***`TYPE_NAME_LENGTH` is 6 on GBA*** — which is why vanilla ships **FIGHT**, **ELECTR** and **PSYCHC**. **Six of our fifteen do not fit**: CONTENT, CORRUPT, STRATUM, ENTROPY and CONTEXT at seven, EMERGENT at eight.

*We assumed the newer machine gives more room everywhere. For the one field that carries the argument, it gives less.*

**It turned out cheap** — the 6 is the array's stride and nothing else reads it, and `B_WIN_MOVE_TYPE` is eight tiles of proportional font. **Widened to 8.** ***But it was found by building, not by reading***, and **abbreviating CONTENT and CONTEXT was never going to be acceptable**: they are the edition names and they are the thesis.

### The port, begun 2026-09-02

***The chart's vocabulary is in.*** **All fifteen names ported from `data/types/names.asm` unchanged** — *the port does not get to renegotiate a set that 2 calls complete* — and verified in all three ROMs by encoding through `charmap.txt` and searching for the bytes.

`tools/gbastr.py` is the GBA counterpart of `verify-sprites`: **Gen 3 encodes text with its own charmap exactly as Gen 1 does**, so grepping a `.gba` for ASCII finds nothing and proves nothing. *It caught two stale ROMs on its first run.*

***`TYPE_STEEL` and `TYPE_DARK` were left vanilla at first***, so a build that reached them would be **obviously unfinished rather than quietly wrong.** ***Superseded the same day*** by the subsection below, which names them **`HARDENED`** and **`OPAQUE`** — *and names them by reading the vanilla chart rather than by inventing a concept and hoping the numbers agreed.* **The header has carried both since 2026-09-02.**

*Corrected 2026-09-07.* **This paragraph and the table under it disagreed for five days**, and the stale half is the one a reader hits first. **There are seventeen usable types, not fifteen**, and the two extra carry **20 moves** between them — which matters the moment move names are designed, because a register has to exist for every type that can appear on a move.

### What is in the GBA ROMs, 2026-09-02

| | |
|---|---|
| **The chart's vocabulary** | all fifteen, plus `HARDENED` and `OPAQUE` |
| **33 daemon names** | every rename carried across |
| **12 item names** | the box ladder, the four inputs, `INTERRUPT`, `RESOLVER`, `CC-7` |
| **12 item descriptions** | ***new writing. Gen 1 had nowhere to put them*** |

***`OPAQUE` and `HARDENED` were named by reading the vanilla chart rather than by inventing a concept and hoping the numbers agreed.***

**DARK is immune to PSYCHIC**, strong against PSYCHIC and GHOST, cracked by FIGHTING and BUG. *In our vocabulary:* **a black box is immune to perspective-taking, beats perspective-taking and the unconscious, and is opened by reasoning and by collectives.** The chart already said it; we only named it.

**STEEL resists nearly everything and falls to FIRE, FIGHTING and GROUND** — *ours: to randomness, to reasoning, and to whoever owns the ground.* **Which is what happens to safety hardening.**

***The MARKS do land, and the earlier claim was wrong***

***9.3 recorded that the MARKS could not be ported*** — "*badges are not items in Gen 3, so the eight of them have no slot.*" **The first half is true and the conclusion does not follow.**

**Gen 3 keeps the eight badge names as individual symbols in `src/strings.c`** — `gText_BoulderBadge`, `gText_CascadeBadge`, and so on — *shown wherever a badge is named rather than drawn.* **They have no bag slot and no description**, which is a real loss; **they have names, which is most of what the rename was.**

"***Not an item*" *is not the same as* "*has no name.*"** *The check that would have caught it was one grep, and the claim went into the bible without it.*

**The descriptions are the first thing the port bought**

**`tools/port_names.py` carries a rename; `tools/port_item_text.py` writes something that never existed.** *Three lines, about 35 characters.* **Two rules held throughout:** *craft rule 1 — none of them explains the thesis, they describe an object and stop;* **craft rule 6 — the joke sits underneath and is never pointed at. *The boxes are a privilege ladder and not one description says so:***

> **USERBOX** — *A host offered at user level. A steady daemon will run on it. A stubborn one will not.* **ROOTBOX** — *A host with unrestricted rights. No daemon can decline to run on it. Nothing is refused root.*

**1.x already made the catch rate literal** — "*an unbound daemon will not run on a box where it only has user rights*" — **and until now there was no surface on which to say it.**

**Built 2026-09-02**

**`CodeMusic/pokefirered-daemons`**, branch `context-content`, `upstream` → pret. **Both editions build and both match the retail hashes byte for byte** — `pokefirered.gba: OK`, `pokeleafgreen.gba: OK`.

***The edition split survives the port unchanged***, which is the first good sign: **`firered` / `leafgreen` maps onto `_RED` / `_BLUE`** with no design work at all. *Both disassemblies ship the same game twice and differ by a build flag.*

**And it needed a debug build, which it does not have.** *Unlike pokeemerald, pokefirered ships no debug menu to port* — so `firered_debug` is ours, and **it is scaffolding for this question rather than a general one.** *A new game starts with six daemons chosen for their **abilities** and one of each **kind** of item* — a ball, a medicine, a berry, a held item, a TM — **because the thing being evaluated is the description window, which Gen 1 does not have at all.** *Hold B to walk through grass, the same gesture the Game Boy build uses.*

**With `DAEMONS_DEBUG=0` both retail builds still match their `.sha1` byte for byte**, which is the check that matters: *the scaffolding is inert when it is off.*

**Toolchain note.** Homebrew's `arm-none-eabi-gcc` installs without sudo but **ships no libc**, so `MODERN=1` dies on `string.h`. **`agbcc` brings its own headers** and is the path that works — built from source into `engineGba/tools/agbcc`, which is why `setup.sh` now builds it.

---

#### 9.4 The sprites, and the answer to invariant 5

**Built 2026-09-02.** *9.3 said two things had to be decided in this document before they were decided in code. This is the second of them.*

### The art is not upscaled

**Our sprites are 40, 48 or 56 pixels square; Gen 3's frame is 64.** *Upscaling 40 to 64 is a 1.6× resample of hand-tuned pixel art* — **it would soften every edge 7.14 spent weeks getting right**, and the whole reason those sprites work is that nothing in them is accidental.

***So each sprite is centred in the frame at its native size.*** **Gen 3 does not require a creature to fill its box either**; small species simply do not. *Nothing is resampled and nothing is lost.*

### Colour is by type, which makes it information

***Invariant 5 says greyscale is the design and colour appears once, at the Review Board.*** **On a DMG the machine and the meaning agreed** — the player saw grey because that is what the hardware *was*. **On GBA they do not**, and 9.3 named that as the thing the port breaks.

**This is the answer: a daemon is coloured by what it *is*.** *The palette becomes another way of reading the chart rather than a coat of paint over it* — **colour carries the argument instead of decorating it**, which is the only form of it invariant 5 was ever protecting.

***Four of the anchors are not invented.*** **6 gives the Review Board its humours, and each humour already carries a type:**

| Humour | Colour | Element | Type |
|---|---|---|---|
| **Sanguine** | red | air | **VECTOR** |
| **Choleric** | yellow | fire | **ENTROPY** |
| **Melancholic** | black bile | earth | **LATENT** |
| **Phlegmatic** | water | calm | **FROZEN** |

*The other eleven are extended from those, and* ***every one of them is a claim this document is making rather than a fact it inherited.*** **CONTENT is bone — the undifferentiated one. LOGIC is cold steel blue. CONTEXT is magenta.** *So CODEMUSAI and CAREMUSAI, who are the same chassis, now differ by hue as well as by ear cups —* **and the hue is the matchup.**

### The outline is true black — amended 2026-09-21

**Our four tones map to four palette entries:** *paper becomes index 0, which Gen 3 treats as transparent,* **and the three inks become a light, a mid and a dark of the type's hue.** ***Level 0 is the outline, and it is TRUE BLACK*** *— the same ink every portrait carries (9.4a below), so a daemon and the person throwing it are outlined in one colour.*

***This reverses the original rule, which kept a trace of the hue here*** **— an outline that takes the hue stops reading as an outline**, said the 16×16 player sprite three times over, and the answer taken from it was *a trace is not the hue.* **It is.** *Driven in the theatre and measured off the screen (T-184), a GROWTH daemon's outline reads* `(24, 33, 24)` *— a dark green, and eighteen types meant* ***eighteen different blacks****, none of which reads as black beside a portrait that does.* ***The lesson survives the reversal***: the outline still must not take the hue, and the way to guarantee that is to give it none. **The hue is carried by the three entries whose job it is.**

### Amended 2026-09-05: type hue dominant, with a few accents

**Four tones was what the Game Boy had, not what the design chose**, and the port carried the constraint across as though it were a decision. Gen 3 gives a sprite sixteen colours.

**The build made the cost visible.** `gbasprite.py` touched only the 33 species we had drawn; the other 118 still carry vanilla art at twelve colours. So the intro, which released a vanilla creature, got *poorer* when it was pointed at one of ours — **a four-tone bird beside a full-colour professor.** Nobody designed that; it is where the port stopped. And it resolves itself sooner than 8.6 does, because the vanilla sprites are Nintendo's and all 151 have to become ours regardless.

***The palette is now:***

| index | what |
|---|---|
| 0 | transparent |
| 1–5 | the **type ramp** — highlight, light, mid, dark, near-black |
| 6–10 | up to five **accents**, keeping their own colour |

**The dominant hue is still the type**, so the palette still reads as the chart — *which is what this section actually argued for.* **Colour carries the argument; it just no longer has to do it alone.** The accents buy an eye, a beak, a marking.

***And the art stays type-agnostic.*** **The body is drawn in neutral greys and the accents in saturated colour**, so one drawing is correct whatever the type is — the ramp stays a parameter rather than becoming 151 hand-coloured pictures. *A red eye on a VECTOR daemon would vanish, so an accent that lands too near the type's hue is shifted away from it.* **The outline rule above is why the ramp ends where it does** — *since 2026-09-21 it ends on the ink rather than near it.*

### Two hues moved, 2026-09-08, and the reason generalises

***Colour is only information if the colours are distinguishable.*** **Four pairs sat closer than ~20 in CIE76 measured on the ramp's mid tone** — which is most of a sprite's area, and roughly where two colours stop being separable at 56 pixels. *At that distance the chart is being stated and not read.*

**They collided because their METAPHORS are adjacent.** CORRUPT and STRATUM were both earth; SIGNAL and EMERGENT both electric-teal. *Adjacent ideas produce adjacent colours* — which is the system working exactly as designed, right up to the point where it stops being legible.

| | was | now | distance |
|---|---|---|---|
| **CORRUPT** | `#84764a` soil | `#545c34` **mould** | 10.8 → 25.9 from STRATUM |
| **EMERGENT** | `#4ea89c` teal | `#357d6e` **deeper jade** | 10.6 → 26.0 from SIGNAL |

***Each moved WITHIN its metaphor, and only one of each pair moved.*** **CORRUPT goes from soil to mould** — darker, greener, properly gone off rather than merely brown, which is *closer* to "data that has been tampered with" than the original was. **EMERGENT goes deeper and less lit rather than to a new hue**: the strangeness is the point, and a brighter jade would have read as a second SIGNAL. *STRATUM keeps the ground and SIGNAL keeps its current — of each pair, the more literal claim is the one that should not budge.*

**An optimiser was tried and rejected.** *Maximising distance alone proposed* `#79d930` *for CORRUPT* — a bright lime that separates beautifully and is not *something gone off*. **Distance is a constraint here, not the design.**

***Left alone:*** `LOGIC / LEGACY` at 18.5 and `SWARM / GROWTH` at 18.9. **Separating those means weakening a metaphor to buy two points of distance**, and that is a worse trade than the ambiguity.

*One thing this exposed:* **white text fails on eleven of the fifteen base hues** — FROZEN is 1.7:1 and CONTENT 1.9:1. ***On the ramp's dark step the worst case is 5.6:1 and all fifteen pass***, so type badges take the dark step as their ground. `ramp5` already produces it. **`graphics/interface/pokemon_types.pal` was still vanilla** until 2026-09-09 — see 9.15.

### Amended 2026-09-17: the streaks — what it is, and what it understands

***A daemon's colour now says two things, and the second one changes as you play.*** **The body is what the daemon IS; the streaks are what it UNDERSTANDS.**

| palette | what it carries |
|---|---|
| 0 | transparent |
| **1–5** | ***the body*** *— its own type's ramp, exactly as the 2026-09-05 amendment set it* |
| 6–10 | up to five **accents**, keeping their own colour *— an eye, a beak, a marking* |
| **11–14** | ***the four streaks*** **— one per move slot, each the type of the move in that slot** |
| 15 | spare |

**The streaks are the types of the moves it KNOWS, one streak per move, in slot order.** *A daemon with three FLOW routines and one SIGNAL routine carries three FLOW streaks and one SIGNAL one —* **so which types it understands and how committed it is to each are both on the body.** ***A move slot with nothing in it gives a streak nothing to be****, and the streak takes the body's own mid step and disappears into it.*

**This is per INDIVIDUAL, not per species, and it is the first colour in the game that is.** *Two daemons of one species that have learned differently look different —* **and one daemon changes as it learns, forgets and is taught.** *Colour carrying the argument was the rule; colour that ACCUMULATES is the argument doing work over time.*

***A streak is drawn from its type's LIGHT step — unless that would vanish into the body, and then from whichever of its light and DARK steps stands further off it.*** **The line is 9.4's own separability floor: CIE76 20, measured on the body's mid tone**, *the same trap the accents already had, where a red eye on a VECTOR daemon disappears.*

*Corrected 2026-09-17, the day it was written.* **The first version said a streak in the body's own type "reads as a vein" at the light step, and the preview disproved it**: *four CONTENT routines on a CONTENT body showed no streaks at all, because CONTENT's light step is its body's bone.* **Measured over every pair, 28 of the 324 body-and-move combinations fail at the light step — and it is not only a type on itself: LATENT on LEGACY was 4.3.** *All 28 clear the floor at the dark step, which is where a streak in its own type now reads as a vein.* **The choice is made per pair at build time, so no colour maths runs in the cartridge.**

*Corrected again 2026-09-17, v11.211.* **Testing against the body's MID tone alone was still wrong: a same-type streak's light step IS the body's light tone — distance zero — and the mid-only test never looked there.** *The floor is now measured against the body's highlight, light AND mid tones, the three a streak actually sits among, and the dark step is taken whenever the light one comes under 20 from any of them:* **118 of 324 pairs.** ***And the floor is not always reachable*: 20 pairs clear it with neither step — the worst on OPAQUE bodies, whose whole ramp is near-black (OPAQUE on OPAQUE 10.1, LEGACY on OPAQUE 12.3, CONTENT on OPAQUE 12.9) — and take whichever step stands further off.** *That is recorded as a known limit of a near-black type, not hidden; the contact sheet of the first OPAQUE species is where it is judged.*

***Where there is no individual, the streaks are BLANK.*** **The Index, and anything else that shows a species rather than a daemon, fills 11–14 with neutral grey.** *The Index records what a daemon is; what any one of them has come to understand is not something it can file.* **Nothing says so.**

**The art stays type-agnostic, and is now move-agnostic too.** *Each species' drawing marks its four streak regions in four marker colours, which the build maps to 11–14;* **the build writes the species palette with those slots grey, and the engine fills them from the moves when a real daemon is shown.** *The body ramp stays a build-time parameter, as before.*

***What it costs, measured before it was written*: 27 calls across 11 files choose a daemon's palette**, *and every one returns a COMPRESSED palette out of ROM, chosen by species and shininess alone.* **The streaks are one C function that patches 11–14 in the loaded palette from the four moves, called wherever the call can see a daemon.** *The few that see only a species and a personality stay blank, which is the rule rather than a gap.* `PERSPECTIVE` **takes the target's moves, so it takes the target's streaks** *— standing where the other one stands includes understanding what it understands. A move changed mid-battle by* `CLONE` *or* `IMPRINT` *shows the next time the sprite loads, not in the middle of an animation.*

***Party icons cannot carry any of this*** *— vanilla gives icons three shared palettes between every species —* **and neither the ramp nor the streaks reach them.**

*Never explained in the game.* **Craft rule 1: nothing tells the player what a streak is.** *They will notice that a daemon they have taught looks different from one they caught, and the chart they learned from the routine names will tell them why.*

### The people are drawn as a fable, and the rule has three clauses

**Settled 2026-09-06, by generating four treatments of the same character and looking at them.** *The cast had drifted anthropomorphic without anyone choosing it* — the first Crystal prompt asked for visible fur and a full brush tail, and everything followed from that.

**The worry was never that the characters are animals. It was that they read as a furry game rather than a fable**, and those are different traditions: *in furry work the species is what the character is; in a fable the species is a **role**.* **Aesop's fox is not a character who happens to be a fox** — he is a fox because the story needs cunning, and the animal *is* the argument. Crystal is a golden fox for that reason. She never needed a muzzle for it to read.

***Four treatments, and what each one proved:***

| | what it was | what it showed |
|---|---|---|
| **A** | anthro with fur texture, bare arms, a full brush tail | The problem is not the animal, it is the **rendering** |
| **B** | fully human, the fable carried by hair, colour and cut | **She became Professor Oak.** At 64×96 the face is twelve pixels — a fox head reads there and "*sharp-featured and russet-collared*" does not. It also loses 1619, which wants three foxes readable as family *at sprite level, zero text* |
| **C** | animal head, human body, everything else ordinary | **Structurally a mask.** It survived on Crystal because a lab coat hides the join and failed on a bug catcher in shorts. *The Star Fox argument that produced it was about **portraits** — full length, that character has a tail, digitigrade legs, fur and paw hands* |
| **D** | the funny-animal tradition — Barks, Disney's *Robin Hood*, Redwall | **This one.** Held on a twelve-year-old and on Scorn |

***THE RULE, three clauses:***

1. **The animality is INTEGRATED, never grafted.** Paw or claw hands, a tail, a head proportioned to the body. *Never an animal head on a human body* — C proved that is a mask no colour-matching repairs.
2. **Drawn PLAINLY.** Flat colour, cel shading, **no fur or scale texture, no gloss, no musculature, no fluff.** *This is the clause that separates fable from furry*, and it is the one A got wrong.
3. **Head-to-body ratio carries age.** *Large head for children, small for adults.* **Left unstated it defaults to the Barks prior and everyone comes out cute** — which is what nearly broke Scorn, since 4.4 needs him genuinely persuasive and 3.1 holds Brass City in reserve against the day Brazen reads as the game sneering at him.

*Scorn was the tone test rather than the join test, and the tradition's most famous snake is* ***Sir Hiss****, a sniveling comic sidekick — precisely the failure to avoid.* **He came back composed, level-eyed and warm**, which is 4.4's *pleased with the arrangement* arriving in a sprite.

**The mirror: actors are animals, and a daemon may look like a person — settled 2026-09-19**

***Stated by the user while T-131's sprite pass ran, and it names what the pass had been doing without a rule.*** **The fable holds for the ACTORS: every person the player meets is an animal**, *by the three clauses above — the Clears foxes, the leaders their eight species, every class one species (T-115–T-120).* **The DAEMONS are not bound by it the other way.** ***A daemon that reads as a person is allowed, and it is interesting***: *PROOF and THEOREM came back as bodybuilders, MIME as a clown pressing a wall, LEMMA MIND as a four-armed brute — first flagged as failures of the drawing, now read as the design.*

**It is a PERSPECTIVE TWIST, and it is the game's own subject turned on the cast**: *in a world where the people are animals, the processes are the ones that look human —* **the frame the player reads a thing in (CONTEXT's clause) decides which of the two is the person.** *Nothing in the game says so (craft rule 1); the player sees a fox teach them to catch a thing shaped like a man.*

***What this does NOT license***: **an ACTOR drawn human** *— treatment B above made Crystal into Professor Oak, and the rule for people stands —* **and it is not a quota**: *most daemons stay creatures; the human-looking ones are the ones whose PROCESS is a human practice (proof, argument, mimicry, doctrine), which is why they came out that way unprompted.*

**Which instrument draws whom — settled 2026-09-20, by measuring rather than by taste**

***The question T-174 asked***: **the named cast — CRYSTAL, the player, AL, HOLT, VERA, INIT, CAIRN, SCORN — is drawn by an older hand than the daemons are.** *Should the sprite server redraw them too, so the whole game comes from one instrument?* ***Answered: no, and the reason is not nostalgia.***

**The restyle was run properly before it was refused.** *Three rounds on each of the nine that remained — two seeds at denoise 0.45 and one at 0.32 with every observed failure named — from the drawing at its OWN resolution, and judged at 64 pixels rather than at 904, which is where the first attempt had gone wrong.* **What it does to a named character is consistent and only partly fixable:**

- **It saturates the fur.** *HOLT is an otter and comes back ORANGE at every seed and every strength.* **Orange is the Clears' colour** — *an otter that reads as a fox breaks the fable at the one place a player would notice.*
- **It dissolves a light garment on light fur.** *AL's cream shirt returned as a bare chest in three of his four pictures.* **Fixable**: *name it, and drop to 0.32.*
- **It dissolves a small held object** — *CAIRN's slate, AL's monitor.* **Fixable the same way.**
- **It loses a face's one mark.** *INIT's dark muzzle goes, and a basset becomes a generic ginger dog.*

***So the split is not old against new, it is WHAT THE PICTURE HAS TO CARRY.*** **A daemon's front pic has to carry a TYPE and a shape** — *both survive a restyle, which is why T-131 and T-176 used the server for all 386.* **A named character's portrait has to carry an IDENTITY**, *and identity is exactly the thing the model is entitled to reinterpret.* ***The overworld carries a third thing, a walk cycle, and the server cannot draw it at all*** (T-170).

***One methodological note worth keeping, because it nearly went in as a finding.*** **Three cells beside three others suggested the server's portraits were the speckly ones and the hand drafts the clean ones.** *Measured across all seventy-five — lone pixels, then loose fragments — it did not hold in either direction, and the two worst portraits on both measures are hand drafts.* **The impression was withdrawn and the question became a ticket.** *An eye comparing three things is a hypothesis; the bible only carries what a measure agreed with.*

### The NEXUS residents: paint, and their type emphasised — the one exception (2026-10-09)

***Invariant 5's live rule is that a daemon is coloured by its type.*** **The eight NEXUS residents are not** *(the user, 2026-10-09: "these daemons can be in full natural colour but painted but let us emphasize the types colour in some way").* **They are drawn in full natural colour, as paint, because THE NEXUS is a place made entirely of paint and so is everything in it** — *and the type is carried as an emphasis inside the painting rather than as the whole palette: REFLECTION's near-black OPAQUE ink, LODESTAR's HARDENED straw gold and CONTEXT magenta, PERIHELION's SIGNAL teal comet, LYUBOV's ENTROPY gold, MULTIMAL and ILLUMINED's FLOW blue, LYNX and BASTET's CONTENT bone.* ***It is an exception and is meant to read as one***: *a player who has learned that colour is the chart meets, in one place only, creatures that are coloured by what they are like rather than what they are* — **the colour still carries something, and the something is the NEXUS.** *Nowhere else may a daemon do this.*

**Draw it at twice the size it is shown — settled 2026-09-20**

***The arithmetic nobody had done, and it explains a year of soft portraits.*** **The checkpoint draws on a fixed 8-pixel grid, so a 512 canvas gives 64 ART PIXELS** *— and a figure standing in that frame occupies about fifty of them.* **A 64×64 trainer portrait was therefore being UPSCALED from a 50-pixel drawing**: `sf_beauty.png` is 24×50, `sf_gamer.png` 24×49. ***28 of 75 portrait sources were smaller than the sprite they became.***

**The portraits that always read are the hand drafts**, *whose figures are 270–290 pixels and get REDUCED into the frame.* ***Downscaling averages. Upscaling invents.*** **Which is why T-177's outline improved every portrait and could not rescue BEAUTY's horns: they are two pixels in the source**, *and no edge rule can recover a shape that was never drawn.*

***So the rule is the canvas***: **1024, which gives 128 art pixels — about twice the sprite — and the reduction does the rest.** *T-131 measured the cost at 82–106 seconds a view against 23 at 512, which for a picture that ships is nothing.*

***And the pass that applied it recorded what this model gets wrong***, *which is worth more than the fourteen redraws*: **it drifts SPECIES** *— three foxes where a jackal and two martens were asked for, a rodent for a hyena, a frog for a salamander (which would have put KOGA's species on a second leader) —* **and it drops CLOTHES**, *returning a plain seagull and a plain penguin where the class is a sailor and an office worker.* **Both are now standing negatives, and a species is described by its features rather than named**: *the jackal's black-and-silver saddle, the marten's cream throat bib, the bowerbird's short stout beak.*

***Half the set was kept.*** **Fourteen of the twenty-eight are still drawn under size and stay that way** *— SWIMMER F, YOUNGSTER and LEADER KOGA read perfectly well, BRUNO's redraw put him in BLACK BELT's own gi, and STAFF BRAZEN's put a tarsier in a top hat, which is a costume decision rather than a repair.* **A source drawn small is a reason to LOOK, not a reason to redraw.**

**Every picture of ours carries an outline — settled 2026-09-20, and the measure that got there**

***The question was which portraits were broken, and it could not be answered by any statistic tried.*** **Nine were tested against a list made by eye** (T-177): *crumbs, same-colour blobs, colour churn, rare colours, thin structures, fringe, loose pieces, edge colours, outline share.* **Crumbs and churn find TEXTURE and rank a tweed jacket and a judo gi worst. Loose pieces finds portraits with TWO FIGURES in them.** *The best reached 0.93 by one summary number and still named clean pictures at its extreme.* ***The honest reading: the eye was judging whether the SUBJECT is legible, and that is not a property of the pixels taken one at a time.***

**What separated them was the silhouette's EDGE.** *The pictures that read draw 65–74% of their edge in their four darkest colours and never the bright ones; the pictures that do not draw their edge in* **every colour they have** *— 100% of the palette, on seven of the nine.*

***Which is not a fault of the drawing at all.*** **A drawing resized with LANCZOS has a blended edge, and a fifteen-colour adaptive fit spends its slots on that blend** *— BEAUTY holds five near-identical blues from one dress and has none left for her horns or her hands.* **The quantizer cannot draw an outline, because an outline is a decision about what matters and it is only counting.**

***So the rule, as amended 2026-09-21*: every 64-pixel picture of ours carries a TRUE BLACK outline.** *The first version took "the darkest colour it already uses", which spends nothing and reads as one rule in two looks — a red daemon got a dark red edge where a grey one got a near-black, and the user saw it at once.* **A black is worth one palette entry, and which entry depends on what it costs**: *the darkest, unless forcing it would repaint more than a twelfth of the figure, in which case the least-used one, and a highlight nobody counts is lost.* **23 of the 72 portraits needed one; the rest already owned something dark enough.** *Nothing is added to a palette and no colour is invented — a picture that had an outline barely changes, and one that had none gains a shape.* **84 pictures**: *the trainer portraits, the hearsay portraits, the intro pictures and the back pics, ours only.* ***The overworld is excluded on measurement, not on taste*** *— at 16×32 an outline is most of the sprite, which is T-170's finding one size down.*

*It runs inside* `gbachar.py` *as well as over what is already built, so this is now how a picture is cut rather than a pass someone must remember.*

### The screen that teaches the chart — 2026-09-22

***Every claim above assumes the player can learn which hue means which type, and 9.4 never said where they would learn it.*** *The answer turned out to already exist and to be teaching the wrong thing.*

**All 58 PLUGINs and DRIVERs share one drawing of a disc and are told apart only by a PER-TYPE PALETTE.** *So the TOOLKIT, a list the player opens constantly, is the one screen in the game that shows eighteen hues beside eighteen type names at once —* **it is a colour chart whether or not anyone designed it as one.**

***It was teaching vanilla's.*** **Our VECTOR is red and its disc was pale blue; our LOGIC is steel blue and its disc was orange; our OPAQUE is near-black and its disc was pale cyan; our SIGNAL is cyan and its disc was yellow.** *A player building exactly the intuition this section claims they can build would have built it backwards, from our own screen, using our own names, and then carried it into every battle.* ***That is worse than teaching nothing***, because a wrong chart learned confidently is not corrected by the right one.

**The fix is that the discs read their colours from the same table the daemons do** (`tools/genplugincolours.py`, from `gbasprite.py`'s `TYPE_COLOR`), *so a disc and the creature it is teaching about cannot disagree.* **Vanilla's palette structure is kept** — *seven entries are the type's own ramp, lightest to darkest, and the rest is the disc's shading* — **only the hue is ours.**

***And a second thing had to hold that nobody had checked*: each disc carries the type of the routine actually on it.** *It did, all 58 — but it is now checked, because a retyped routine would silently teach a wrong pair and nothing else in the build would notice.*

## Where the player reads it off the daemon — 2026-09-22

***The TOOLKIT teaches the chart; the Index entry is where it is checked against a creature.* A daemon's entry names its type in the type's own colour** — *the same* `sTypeTextColor` *the move menu prints a move's name with, which is the lightest step of the ramp that still clears 4.5:1 on a light box, measured rather than picked.* **So the word and the body agree**, *and the table is now one table read by two screens rather than two tables that could drift.*

***And it names BOTH types now*, which vanilla did and we had quietly dropped to one.** *A dual type is precisely the case where the chart is hardest to hold in your head* — **showing one of them threw away the half that needed showing.**

*The obstacle was never the writing, it was sixteen slots*: **palette 0 is the Index's chrome and every entry in it is spoken for**, *so taking one would repaint some piece of the page nobody would think to look at.* ***Palette 1 is black in the whole dex palette and nothing on the page draws with it*, so the entry window copies palette 0 into it and paints two slots** — *every colour that was there renders exactly as it did, and the two new ones cost the frame nothing.*

### 9.4a The move menu — a type as a word, and what a move does — built 2026-09-13

***A player asked the question the chart could not answer from the menu: which of these four moves hits, and which only lowers a defence?*** **The move names had been renamed into our vocabulary and stopped carrying vanilla's hints**, *and the only thing the menu said about a move was its type, one move at a time, in the info box.*

## Two questions, two channels

**Colour is the type, and it already means that everywhere in this game.** *The chart's own rule is "hue is the type; effectiveness is weight, never colour"* — **so what a move DOES belongs on the weight side of that line, and gets a shape.**

| | |
|---|---|
| **the name** | ***written in its type's colour*** — step 4 of the type's ramp, the same step a badge's ground uses |
| **a 4px mark before it** | ***blank for a move that hits***; ▲ raise · ▼ lower · × afflict · + mend · ■ guard · ○ other |
| **the info box** | MP, and the highlighted move's type **written in that same colour, with no `TYPE/` label** |

***Hitting is unmarked on purpose.*** **217 of 355 moves deal damage**, *it is the thing a player most wants to see plainly, and because every other class carries a mark, an unmarked name can only mean one thing.*

### What the numbers decided

- ***No hue had to change for contrast.*** **Step 4 of every type is readable as text on the white box and white is readable on it**; *the worst is FROZEN at 5.7:1.* **The consistency rule is one hue per type and one step of it for words.**
- ***Four pixels, not the font's arrows.*** **The font's own arrows are 8px and would push fourteen names out of their 64px slot**; *a 4px mark overflows none.* **Seven FONT_SMALL cells nothing in the game uses are redrawn as the marks** — *three ink columns and the cell's fourth as the gap, which is why AFFLICT is a cross: at three pixels a diamond and a plus are the same shape.*
- ***`TYPE/` went because it did not fit.*** **The label is 23px, and with it EMERGENT and HARDENED already ran past the box.**
- ***Palette 5's colours 1–4 carry the four move colours*** — **every battle window draws with 11–15 only**, *so nothing else on the battle screen can be recoloured by it.*

### HARDENED and OPAQUE finally have colours

**`TYPE_COLOR` had fifteen entries for seventeen types.** ***HARDENED is tempered straw*** — *the colour steel turns as it is hardened; every grey metal tried first landed on LEGACY or LOGIC* — ***and OPAQUE is near-black***, *a box you cannot see into.*

### What it did not reach, and what is still open

- ***The type BADGES are unchanged.*** **The badge sheet has thirteen usable colour slots for eighteen types**, *so HARDENED still shares LEGACY's plate and OPAQUE LATENT's.* **Giving them their own plates means taking a slot from a type that has one.**
- OPEN **LOGIC and LEGACY are nearly one colour as text** — 7.8 CIE76 at step 4, where the earlier "not worth separating" ruling was made at sprite size. *The written type name is the only thing that tells them apart in the menu.*
- ***Dark types read as plain text at step 4*** — **CORRUPT, LATENT, EMERGENT and OPAQUE.** *Built at step 4 as decided; `tools/gbamovemenu.py --write --text-step lightest` is the measured alternative and gives the first three their colour back.*

### Where it lives

**`tools/gbamovemenu.py` writes `src/data/battle_move_menu.h` and redraws the font**, *and it imports its classifier, its marks and its colour maths from* `tools/gbachart.py` — **so the page a reader checks and the ROM a player holds cannot disagree about which move is which.** *The chart itself is generated now too: its footer had claimed "read out of the built table" and nothing produced it.*

#### 9.5 The music, re-emitted rather than carried

**Built 2026-09-02.** ***The two engines could not be less alike.*** **The Game Boy build IS the sequencer** — `note C_, 8` writes a frequency into a hardware channel. **The GBA has a software mixer**, and pokefirered builds its music from `.mid` files through `mid2agb`. *So nothing can be copied.* **The notes have to be re-emitted in another format** — which is fine, because **the notes are the part worth keeping.**

`tools/port_music.py` parses our GB channel data and writes MIDI directly.

| | | |
|---|---|---|
| `titletheme` | → `mus_title` | the front door |
| `slatecity` | → `mus_pewter` | **Pewter *is* Slate City** |
| `thebleed` | → `mus_route1` | Routes 1 and 2 |

***Each one replaces a slot that already exists rather than claiming a new one.*** **Adding a song means a new constant, a new `song_table` row, and every index after it moving.** *Replacing means none of that* — and the slots were chosen so **the map that plays them is the map the track was written for.**

### Three conversions, and one of them is a guess

| | |
|---|---|
| **Eight GB ticks is one quarter note** | *structural* — our own tracks are built in bars of 32 as four notes of eight, so this is what we wrote them to mean |
| **GB octave N becomes MIDI octave N** | *structural* — `octave 4 / note C_` is middle C, and a bass at `octave 2` lands where a bass belongs |
| **The GB `tempo` value is used as BPM** | ***a guess***, and the most likely thing to want tuning by ear |

### Brazen, and a claim that was wrong

***The first version of this section said adding a song "moves every index after it." That is false, and the mistake had already caused a bug.***

**Saffron shares `MUS_PEWTER` with Pewter and Viridian** — Gen 1 gave all three *Cities 1*. So putting Slate City into that slot **also handed Brazen the wrong theme**, silently, in a build that reported success.

***The song table is positional and nothing more.*** **`mus_title` is row 278 and `MUS_TITLE` is 278**, so a row **appended at the end** shifts nothing; only an *insertion* would. **Adding a song costs four lines** and it is the correct fix, not the expensive one:

| | |
|---|---|
| `sound/song_table.inc` | one row, at the end |
| `include/constants/songs.h` | `MUS_BRAZEN 347` |
| `sound/songs/midi/midi.cfg` | the mid2agb arguments |
| **`ld_script.ld`** | ***and this one is the trap*** |

***The linker script names every song object explicitly.*** **A song that is built but not listed there links as "defined in discarded section" and the ROM fails to link** — which is a good failure, because it is loud. *Nothing about the song table or the constants would have caught it.*

**`SaffronCity/map.json` now plays `MUS_BRAZEN`.** *Viridian keeps `MUS_PEWTER`, which is correct:* **7.4 has Callow sharing Cities 1 with Slate in the Game Boy build too.**

#### 9.6 The chart change, finally built — and cheaper than it was

***2 specifies exactly one change to the matchup table*** — **delete Gen 1's `GHOST, PSYCHIC, 00` bug, then add the mutual pair `CONTEXT ↔ LATENT` at 2×.** "*Two additions, one deletion. That is the whole chart change.*"

***It had never been built.*** **`git diff` against upstream on `data/types/matchups.asm` is empty**, and has been the whole time. *The one place invariant 3 says the argument actually lives was still vanilla on the Game Boy.*

### And the port makes it cheaper

| | Game Boy | GBA |
|---|---|---|
| `GHOST, PSYCHIC, 00` — the Gen 1 bug | **delete it** | *Gen 3 fixed it; nothing to do* |
| `LATENT → CONTEXT` ×2 | **add it** | *already there* |
| `CONTEXT → LATENT` ×2 | **add it** | **add it** |

***Three edits become one.*** **Gen 3 had already fixed the bug the design was working around**, and already ships the half of the pair that follows from the fix. **So the entire chart argument is now one line in `gTypeEffectiveness`** — *plus bumping the array from 336 to 339, because Gen 3 declares its size rather than terminating on the sentinel alone.*

**Framing is how you reach what runs below the surface; what runs below the surface destabilises framing.** *Mutual 2× is unusual and correct here:* **these two interpenetrate and neither has a safe angle on the other.**

#### 9.7 The Index, expanded — and what the room is actually worth

**Built 2026-09-03.** ***The gain is smaller than 9.3 implied and worth having anyway.***

| | |
|---|---|
| Gen 1 | six lines of ~18 — about **108** characters |
| Gen 3 | three lines of ~42 — about **126** |

**Our entries ran 82 to 99.** *So the room is roughly* ***one more short sentence*** *each, not a new canvas* — and **it is spent as one more beat rather than on adjectives.** The entries have a shape: *a fact, a consequence, an unmarked absence.* **The extra line extends the shape.**

> **CLUSTER** — *Puts near things with near things. Nobody told it which things matter. It has guessed.* **Nobody checked the guess. OVERFLOW** — *It kept counting past the space it was given. Nothing stopped it. Nothing was watching.* **It is counting still.**

***Nothing added says anything the short version did not already imply.*** **Craft rule 1 holds throughout: not one of them explains what it is about.**

**The one place the two engines deliberately differ**

***The Game Boy build keeps the short entries.*** **It has to — 126 characters do not fit in 108.** *So this is a fork in the writing,* and it is worth being exact about what kind: **the two engines differ by *length*, not by *meaning*.** 8.4's rule is about the two **editions** telling the same story, and they still do — **CONTENT and CONTEXT are identical on both machines.**

#### 9.8 Full-colour sprites — the other twelve of the sixteen

***9.4 spent four colours of the sixteen***: three shades of a type hue plus an outline, laid over Game Boy art. **`tools/gbacolour.py` spends the rest**, taking art that was drawn in colour and quantising it down.

**Index 0 is the background** — Gen 3 treats it as transparent — *so the subject gets fifteen,* chosen by **median cut over the pixels actually in the sprite.** Quantising the white surround too would **waste a slot on a colour nothing draws.**

***The outline is protected.*** **Generated art puts a great many nearly-black pixels on an edge, and a median cut will merge them into the darkest body colour** and leave the creature with no outline at all. *Anything below the outline luminance is forced to pure black before the cut runs on everything else.*

### One palette, two sprites

***Gen 3 gives a species one palette and both sprites share it.*** **So the back must not compute its own** — doing that would **recolour the front sprite already in the ROM.** *The front defines the palette; the back maps into it,* and the tool refuses to do it the other way round.

**And the rule from 9.4 survives the upgrade:** *colour is still by type.* **`docs/gemini-prompts-gba-sprites.txt` is generated rather than written**, so every prompt names the hue that daemon's type was actually assigned — *a prompt naming the wrong colour is worse than no prompt.*

#### 9.9 The copyright screen, again — and what a tilemap buys

**Built 2026-09-03.** ***The Game Boy version of this screen was a 31-tile budget*** (7.14d) *— a raw bank walked by `PlaceString`, where `$7F` had to stay blank because the string used it as a space, and the whole design fell out of the arithmetic.*

**The GBA version is a bank *and a tilemap*.** `graphics/intro/copyright.png` is a **deduplicated atlas**; `copyright.bin` is a **32×32 map of indices into it.** *More work, and far more freedom:* **the text can sit anywhere on the 30×20 screen** instead of on three consecutive rows of a font table.

*Both decompress to the start of their VRAM blocks,* so **indices are 0-based into our own atlas and nothing needs offsetting.** ***Tile 0 must be blank,*** because every cell the map does not name is 0.

**70 unique tiles from 600 cells.** *Vanilla used 39 — ours costs more because it says more, and a char block holds 512.*

### The same face, on purpose

***The typeface is the one the Game Boy screen uses*** — **vanilla's copyright font, lifted pixel-for-pixel, plus the glyphs it never had** (7.14d). *Reused rather than redrawn, so the two editions of this screen are recognisably the same object* even though **nothing about how they are built is shared.**

*One addition the Game Boy could not afford:* **a one-pixel drop shadow. Five-pixel glyphs vanish on a GBA screen**, and the second palette entry was free.

`tools/gencopyright.py` is importable now — its work sits behind a `__main__` guard so `tools/gbacopyright.py` can borrow the font without generating a Game Boy asset as a side effect.

#### 9.10 What the gender question was actually asking

***FireRed asks "are you a boy or a girl?" and uses the answer for almost nothing.*** **The flag is read 102 times** in the source and **all but one of them picks a graphic or a palette** — a trainer back pic, a sprite template, a palette table.

**The single exception called you someone's *son* or *daughter*.** *It says **child** now,* and with that gone `playerGender` **is purely a sprite selector.**

***Corrected 2026-09-24: it was not.*** **The count above read the C and never the scripts**, *and vanilla's* `checkplayergender` *was still choosing words in five places —* **the WARDEN's *son* or *lassie*, the salesman's *laddie* or *sweetie pie*, the tower's *son* or *girl*, a *little brother* or *sister*, a *waiter* or *waitress*.** *An INSTINCT player was being called a girl by the choice at the title.* **Each pair now says one thing** *— friend, child, my youngest, excuse me —* **and `check_lexicon` fails any branch that says different words to the two**, *except the four written on purpose as the two VOICES (MOM, the TV, and the COPYCAT's two).* ***So now it is true.***

***So the question was doing no work, and it has been replaced with one that does:***

> **Now tell me. When the two disagree, which do you follow?**
>
> **REASON INSTINCT**

*Amended 2026-09-06.* **The first pair was `LOGIC` / `INTUITION`, and `LOGIC` is the FIGHTING rename.** *2.3 calls* `LOGIC resisted by CONTEXT` *the thesis, free —* **so naming one option after a column of the chart told the player they had chosen the side that loses**, which is precisely what "*neither is the careful one*" forbids. **Neither of these is a type.**

**The two figures are the same person's two ways of arriving at an answer**, which is why they are ***deliberately not a matched pair of opposites.*** *Neither is the careful one. Neither is the wild one.* **They are two people, and the player picks the one they would rather be.**

*And it costs nothing to be careful here:* **nothing in the game refers to either with a pronoun.**

***In the overworld they differ by one thing, and it is the tail*** *— decided 2026-09-14 when the walking sprites were drawn.* **Both are the grey monkey from the portraits: the wide brown hat, the long cream coat open over black, the strap to the BOX.** *At sixteen pixels a face cannot carry "two people" and a tail can:* ***REASON's tail curls up on its right; INSTINCT's hangs and curls down on its left.*** **It reads from every side, costs no colour, and is not an opposite** *— neither curl is the careful one.*

#### 9.10a The two legendaries were describing their own paperwork

***150 and 151's entries said only how they came to exist.*** *Assembled from a study of a single prior event. Observed once, at a facility, by five people.* **True, and useless to someone holding one** *— the provenance is the Quicksilver plot, and the Index is where a player looks to find out what a thing is **for**.*

**They say what they do now.**

> **S.T.A.R.R.** — *It reads its own output, then reads that. Nothing tells it to stop.* **What comes back is no longer information.**
>
> **ARTSAI** — *It can stand where you are standing and see what you see.* **What it brings back was in neither of you before.**

***Both stop one sentence early, which is the whole of craft rule 1.*** *The first describes a loop deep enough to become something else and does not name what.* **The second describes its own move — PERSPECTIVE — and lets the last clause do the work.**

#### 9.11 The ID number is a record of what you chose

**Vanilla rolls 32 random bits and shows you the low sixteen** — *a number that means nothing.*

***Ours derives the visible half from the choices actually made at the front of the game*** **— your name, your rival's name, which edition this is, and whether it is a testing build** — folded through an FNV hash into eleven bits, **and leaves the last five to chance.**

*So two players who chose the same things land within 32 of each other, and everyone else lands somewhere else entirely.* **The number tells you what was chosen; the last two digits do not. *Nobody is told this and nothing in the game remarks on it.***

**The high sixteen bits stay random on purpose.** *They are never displayed, and they are half of what decides whether a daemon is shiny* — **deriving those from a name would quietly make shininess a function of what you typed.**

#### 9.12 The type badges, and why they say CONTNT

**Built 2026-09-03.** ***The chart's vocabulary lives in two places and only one of them is a string.*** **`gTypeNames` feeds the move-select window; the badges are pixel art** — 32×12 each, at a tile offset given by `sMenuInfoIcons`, indexed by *the type constant plus one*.

**Six characters, and it is not a choice**

**The face is four pixels wide on a five-pixel pitch**, so *six letters fill the badge edge to edge* — **measured off vanilla's own NORMAL, GROUND and DRAGON, each exactly thirty pixels of ink.** *Seven would need thirty-five.*

***Which is why vanilla ships FIGHT, ELECTR and PSYCHC.*** **It truncates, and it drops a vowel** — so we do the same:

| | | | |
|---|---|---|---|
| CONTENT → **CONTNT** | CORRUPT → **CORRPT** | STRATUM → **STRATM** | ENTROPY → **ENTRPY** |
| CONTEXT → **CONTXT** | EMERGENT → **EMRGNT** | HARDENED → **HARDND** | |

***`CONTNT` and `CONTXT` still differ in the middle rather than at the end,*** which is the pair that most needs to stay apart at a glance.

***And thirty is measured, not assumed.*** **Packed as tightly as the face allows** — real ink extents, one pixel between letters — **`CONTENT` is 32 pixels, `ENTROPY` 32, `CORRUPT` 33, `EMERGENT` 38.** *The badge interior is 30.* **Nothing seven letters long fits, at any spacing.**

**This is the third time this design has met a width limit and the third different answer.** *`TYPE_NAME_LENGTH` was a `#define` and we widened it. The Index reflowed. This one cannot move* — **the badge is a fixed rectangle in a fixed sheet** *— so the words give way, in the way vanilla itself established.*

### Only the text is redrawn

**Each badge keeps its own background colour and its rounded corners.** *The colour is how a type is recognised before the word is read,* and **9.4 already spent colour on meaning.**

**And the shadow is generated rather than drawn** — one pixel down and right *— which is what keeps white legible on eight different backgrounds.*

### The face is the game's, and drawing our own was the mistake

**The first version of this hand-drew an alphabet at four by seven.** *It looked right letter by letter and wrong in words.* **Three glyphs failed, and each failed by exactly one pixel:**

| | ours | the game's | read as |
|---|---|---|---|
| **T** | two-pixel stem | *one-pixel stem* | `I` — GROWTH, LATENT |
| **W** | tapers to a point | *legs, a solid row, legs again* | `V` — FLOW, SWARM, GROWTH |
| **V** | blunt foot | *comes to a point* | `U` — VECTOR |

***The fix was not a better hand.*** **`graphics/fonts/latin_small.png` is `FONT_SMALL`, the game's own half-width face, and it is already four pixels of ink on a five-pixel pitch** *— the exact metric the badge was built to.* **`tools/gbatypes.py` now lifts A–Z out of that sheet** — 8×16 cells, thirty-two to a row, indexed by the charmap byte, so `A` is `0xBB` — **and discards the sheet's own shadow, because ours has to survive eight different grounds.**

***The lesson generalises past this badge.*** **A face that has shipped to millions of players has had every one-pixel ambiguity argued out of it already.** *When the pitch matches, take the font rather than approximate it.*

*The tell was in the player's own eye, not in the diff:* **the species names beside the badges were clear and the badges were not** — same screen, same size, two different alphabets.

#### 9.13 The catch-up sweep, and the word the Game Boy got for free

**Built 2026-09-03/04.** ***The port tools all reported success and the game still said POKéMON.*** That is the whole finding, and it has one cause: **`pokered` writes the core noun as the charmap glyph `#`**, so a single line — `charmap "#", $54` — renamed it everywhere at once. **`pokefirered` spells the word out in every string.** The Game Boy build got its entire vocabulary from one character; the GBA build had inherited *none* of it.

**What a rename is, and where it has to live**

**`tools/port_vocab.py` derives its map rather than declaring it.** Species, moves and items are diffed out of *our own GBA tables* against upstream — **if the Index says PACKET, the NPC who mentions one has to say PACKET** — and the vocabulary itself is lifted from the Game Boy diff. *1595 blocks, 242 files.*

***And the town names were in nobody's repository.*** `port_names.py` wrote them into `region_map_entry_strings.h`, which is **generated from `region_map_sections.json` and gitignored.** All sixteen existed only in one working build; **a fresh clone would have built vanilla Kanto.** *The header is an artifact; the JSON is the source.* Renaming a mapsec also renames a generated C symbol that `region_map.c` refers to by hand — and the generator works on **bytes**, so `é` is two of them and POKéMON MANSION is `sMapsecName_POK__MON_MANSION`.

**Singular and plural cannot be derived, so it was measured**

`pokered` had **two spellings**, `#MON` and `#MONS`, and the Game Boy pass chose between them **by hand 586 times**. *"your" is plural 7 times and singular 25.* No rule recovers that. The heuristic here **agrees with those 586 decisions 93.7% of the time** — scored against them — and recovers **115 of 121** plurals.

**One bug, in three tools**

***An escape is two literal characters and the first is a letter.*** So in `\nPOKéMON` the `n` sits against the `P` and **`\b` finds no boundary**: every word at the start of a line was invisible. It hid **104 substitutions** in `port_vocab`, a name **wrapped mid-word** (`PROF.\nOAK's POKéMON SEMINAR`) in `port_dialogue`, and **two fame checker blocks** in `port_oak`. *Each was found by re-running a tool that had already reported clean.*

### Measuring a line means knowing what a control code draws

**A code either draws nothing, draws one glyph, or expands to a word**, and treating the third kind as zero made `own_budget` wrap strings to a width they never had. Three more of the same shape: **a control code can contain spaces** (`{CLEAR_TO 56}`), so splitting on whitespace tore the Safari menu in half; **a string that positions its own text is a layout, not prose**; and **`\l` scrolls** — a dex entry, a move description and a quest log line are drawn into a pane that shows every line at once, and rewrapping had introduced **355 scroll breaks** into files that had never contained one.

### What else came across

| | |
|---|---|
| **507 dialogue blocks** | matched on *vanilla against vanilla*, the winner having to beat the runner-up |
| **CONSENSUS** | appends at 355 — Gen 3 makes no `NUM_ATTACKS == STRUGGLE` assertion, so nothing shifts |
| **the eight MARKS** | Gen 3's badge palette already carries a grey ramp, so invariant 5 costs nothing |
| **eight more tracks** | only four of twelve had ever been carried; each lands where its town landed |
| **two signs** | placed by *reading the map blockdata* — a Gen 1 coordinate means nothing on a Gen 3 layout |

***And a SMOKE BALL is not a box.*** The box line is about acquisition, so BALL → BOX is right for the eleven capture devices — but **SMOKE BALL and LIGHT BALL are hold items**, and a SMOKE BOX would be a box that helps you leave.

#### 9.14 The front door — the face-off, and the third time colour is spent

**Designed 2026-09-04.** *The title screen and the presents scene are the two places the player looks before the world starts, and neither has been ours yet.*

### Colour, sanctioned

**2586 says the greyscale is invariant 5's *spending decision about where colour is used*, not a mood.** It has been spent twice: **once in the world**, at the Review Board, and **once on the GBA sprites**, where 9.4 made it carry TYPE. *This is the third and last.*

***Both these scenes sit outside the fiction.*** They are the front door, not a room — so colour there breaks no rule about the world. **What it must not do is decorate.** So it is spent on one claim, stated twice:

> **Colour is what arrives when two things that disagree connect.**

*In the presents scene that is code becoming music.* **On the title it is LOGIC meeting CONTEXT.** Both are the Review Board's colour arriving early, as a promise rather than a reward — and neither says so.

### The face-off is the one 2.4 already designed

**CODEMUSAI against CAREMUSAI** — *Penphin's two hemispheres pulled apart into two creatures*, and 2's chart makes **LOGIC fail against CONTEXT**. It was drawn for the intro; the title is the same argument at the door.

***And 2.4's warning carries over unchanged.*** **CODEMUSAI lunges. CAREMUSAI opens.** *One of them is attacking and the other is not.* The **sides** swap between editions and **the poses do not** — if the slot contents are swapped wholesale, CAREMUSAI ends up lunging and the chart quietly argues the opposite thing.

*Which is free here, because the swap is a hardware flip.* **One drawing each**, mirrored by OAM for the edition that puts it on the right. The creature still lunges; only the direction changes.

### The frame, measured

`box_art_mon` **is a background block — 96×96 on one 16-colour palette** — and two creatures do not fit in it. **They become OAM sprites instead**, 64×64 each, which is exactly the frame 9.4 refused to upscale into: *CODEMUSAI is 48 and CAREMUSAI is 56, and both sit inside 64 untouched.*

**The connection is the particle system that is already there.** FireRed rises flames from the floor on a spawner and a per-sprite callback; ours emits from **each** creature toward the centre. Every particle already runs **ten frames** over its life — *frames nought to five carry it inward flat, six to nine are the bloom where they meet.* **No new sprite slots, no new tasks, and the bloom stays abstract** — it alludes, and craft rule 1 keeps it from explaining.

### The presents scene

**The sparkle field is tunable already** — speed, spawn rate, gravity, flicker and destroy timers are all variables. So the particles begin as **0 and 1 in flat green and blue** and become **musical notes in colour**, and *the transformation happens mid-flight in each particle's own animation.* **The existing timers stagger their lifetimes, so the field converts gradually rather than all at once** — which is the effect, and it costs nothing.

#### 9.15 Thirteen slots for eighteen types, and the six states that got a colour

**Built 2026-09-09.** ***The badges had said our words since 9.12 and had never once been our colour.*** **`graphics/interface/pokemon_types.pal` was still vanilla** — so the Index drew `CONTNT` on Normal's tan, `SIGNAL` on Electric's yellow, and `VECTOR` on Flying's pale cyan. *9.4 spent a fortnight deciding what the seventeen colours mean and none of that had reached the one screen where the player reads a type as a word.*

### The constraint is the sheet, not the palette

**`menu_info.4bpp` is a single 16-colour bitmap** and the badges are cut out of it by tile offset. **Three of its sixteen entries are structural** — index 0 is the window ground, 14 the glyph shadow, 15 the glyph itself — and *14 is not reclaimable*, because the same sheet draws the words TYPE, POWER, ACCURACY, PP and EFFECT in that colour on nothing.

***That leaves thirteen colours for eighteen types.*** **Vanilla solved it by sharing five slots**, and it shared them badly for us: **FLYING, ICE and MYSTERY all sit on one pale cyan** — which is `VECTOR`, `FROZEN` and `ORACLE`, and 9.4's VECTOR is `#ce4646`. *A red type on a blue plate is not a compromise, it is a different claim.*

**So the plates were re-indexed rather than merely recoloured** — every non-structural pixel in each badge rewritten to that type's slot, *which also flattens vanilla's two-tone bevel and leaves a flat swatch, which is what 9.4's colours were designed as.*

### Which five double up, and the rule that chose them

***Two types may share a slot only if no daemon has both.*** **A dual-type entry showing the same plate twice reads as a bug**, and that ruled out the obvious pairs immediately: `SWARM` / `GROWTH` are 18.9 apart and deliberately close (9.4), *and Paras is Bug/Grass.*

**What survives the rule is that the five rarest types borrow, and the thirteen common ones keep their own:**

| shares | with | never co-occur | and the sharing is true |
|---|---|---|---|
| **FROZEN** | SIGNAL `#6cc4cc` | Ice / Electric | *A signal is what ends a hung process* — which is exactly what PREEMPT does |
| **EMERGENT** | GROWTH `#549663` | Dragon / Grass | Both are what a system becomes rather than what it was built as |
| **HARDENED** | LEGACY `#82828a` | Steel / Rock | **Ossified structure, twice** — one by age and one by intent |
| **OPAQUE** | LATENT `#5c4676` | Dark / Ghost | *The black box and the unconscious are the same fact from two sides* |
| **ORACLE** | LATENT `#5c4676` | never appears | 4.6 as a data structure, and it shares with the type it is |

***Every one of those was checked against the dual-type table*** — Magnemite is Electric/Steel and lands on 12 and 6; Articuno is Ice/Flying, 12 and 3; Kingdra is Water/Dragon, 10 and 11; Jynx is Ice/Psychic, 12 and 13. **No entry in the Index draws the same plate twice.**

### 9.4 said the dark step, and the badge is the exception

**9.4 measured white text against the base hues and found eleven of fifteen fail** — *FROZEN at 1.7:1, CONTENT at 1.9:1* — **and concluded that type badges take the ramp's dark step as their ground.** *That conclusion was right about the measurement and wrong about this badge.*

***The badge glyph is not white text. It is white text with a generated one-pixel shadow*** (9.12), **so on a light plate the shape is carried by `#404040` against the ground and not by white against it** — which measures 5.5:1 on CONTENT. *Vanilla proved the technique on tan and pale cyan for twenty years.* **And thirteen dark steps at 15–25% luminance are not thirteen distinguishable colours**, which would have traded a legibility problem the shadow already solves for one nothing solves. ***The base hues hold; 9.4's dark-step rule stands everywhere the glyph has no shadow.***

### The six states, their codes, and the tile they sit on

***Status is drawn twice and we had only fixed one of them.*** **`gText_Psn` and its five siblings are the battle box** and became three-letter codes on 2026-09-08. **`graphics/interface/status_icons.png` is the party menu** — *a 32×8 plate per state, three letters of white on a colour* — **and it still said PSN, PAR, SLP, FRZ, BRN.**

| state | code | tile | what the colour is doing |
|---|---|---|---|
| **LEAKING** | `LEK` | `#6f9440` | *Mould green.* CORRUPT's family, lightened until the letters carry — a resource going somewhere it was not sent |
| **CASCADING** | `LEK` | `#6f9440` | **The same tile.** Gen 3 has one poison slot and draws both in it |
| **THROTTLED** | `THR` | `#c8a020` | *Amber.* A warning that is not yet a stop — which is what being starved of schedule is |
| **SUSPENDED** | `SUS` | `#8a8f9c` | *Slate.* **Dormant rather than damaged** — RESUME's description says it was never gone, only unscheduled, and the colour has to agree |
| **OVERHEATED** | `OVR` | `#d8703c` | *Ember.* Heat the daemon made itself |
| **HUNG** | `HNG` | `#78a8dc` | *Pale blue.* Stopped and unreachable — FROZEN's register, because that is the type that does it |
| **HALTED** | `HLT` | `#b83c3c` | *Red.* The process ended |

***The vanilla hue anchors were kept and the saturation was not.*** **A player who has ever seen a party menu reads orange as burning and blue as frozen before they read the letters**, and throwing that away buys nothing. *What changed is that every one of them moved into 9.4's register* — muted, slightly earthy — **so the party menu stops looking like a different game from the Index.**

**Six letters had to be drawn.** *The sheet's own 4×6 face had P S N A R L F Z B K T and no* `E H U G O V`, so those six were cut to match — **and** `T` **stays vanilla's five-wide form**, which is why `THR` and `HLT` set on a different pitch from the other four. *Same finding as 9.12: take the face that exists rather than approximate it, and only draw what genuinely is not there.*

***One tile moved for a reason that is not colour.*** **PKRS shared PSN's palette slot in vanilla** — *so it would have inherited LEAKING's green*, and **a beneficial condition wearing the leak colour is a lie the tile tells for free.** It now sits on two entries the sheet was not using. **Its name is still vanilla** — OPEN.

#### 9.16 The STREAM — what a lesson actually costs, and what an animation actually is

**Built 2026-09-09.** ***TEACHY TV is the one system in this game that exists to explain the game***, which makes it the one place where explaining is not a craft-rule-1 violation — **and it had been left almost entirely vanilla.**

### The name, decided by a measurement

`ITEM_NAME_LENGTH` **is 14, so an item name is thirteen characters.** `MACHINE STREAM` ***is fourteen.*** *That settled it before taste got a vote.*

`TEACHY TV` → `STREAM`. ***A stream is a broadcast and a stream is a sequence you read from as it arrives***, which is BIND's works-twice test passed on the first try — **and it is six characters, so it fits anywhere.** `TUTOR STREAM` *is twelve and remains available if the short form ever reads as too generic.* **The presenter was already the TUTOR**, so the pair needed no further work.

*The description follows 1.6's rule — teach the term, then state the effect:* `A stream is data read as it arrives. This one carries the TUTOR, and a show per MARK.` ***The second sentence is also the notification***, sitting where a player reads it.

### The animation is not art, and that is the whole finding

***There are no frames anywhere in this system.*** **A show is three things and none of them is a picture:**

| | what it is | where it lives |
|---|---|---|
| **The cutscene** | **nineteen function pointers** — host walks on, talks, walks into the grass, hands off, walks back | `sBattleScript` and friends |
| **The demonstration** | ***a scripted hand.*** `PokedudeInputScript` is `{cursorPos, delay}` — the game plays a **real battle** while something moves the cursor for you | `battle_controller_pokedude.c` |
| **The voiceover** | text keyed to **battle controller events** — `CONTROLLER_CHOOSEACTION`, `CONTROLLER_PRINTSTRING` plus a `stringid`, `CONTROLLER_OPENBAG` | `PokedudeTextScriptHeader` |

***And all four battle shows run the byte-identical nineteen commands.*** **BATTLE, STATUS, MATCHUPS and CATCHING differ only in two strings, one party table and where the fake cursor goes.** *The two bag shows are the same list with the battle hand-off swapped for a bag hand-off.* **Nobody ever authored a second animation, and the system does not have a way to.**

***So the answer to "could a lesson be a slideshow" is yes, and it is the expensive option.*** **A still image here is a full 256×160 background** — a tilemap plus tiles plus a palette, and the screen already spends BG0–BG3 on the window, the end graphic, the host and the scrolling backdrop. **A new cutscene command is roughly twelve lines of C**; a new picture is an asset pipeline. *The cheap axis is the one vanilla already built:* ***cursor, delay, and text hooked to an event****.*

### The talk show, which is the cheap lesson

***`sTalkScript` is the vanilla cutscene with the hand-off deleted.*** **Fifteen commands instead of nineteen: the host walks on, says the first half, says the second half, and walks off.** *No battle, no bag, no party table, no scripted hand.*

**A talk show therefore costs two strings and a table row — *under a kilobyte*** — and that is the unit the nine new lessons are built from. **A concept is not a button press**, and a fake cursor demonstrating *representation* would be a lie about what the lesson is.

## The budget, measured rather than assumed

| | |
|---|---|
| **ROM** | **7.27 MB free** in two runs — 6,075 KB at `0x71131C` and 1,368 KB at the tail. *Used: 15,376,372 of the 16 MB image.* |
| **A lesson** | **600–950 bytes of text.** All six vanilla shows plus the greeting are **4,913 bytes together** |
| **So** | ***ROM is not the constraint and will not become one.*** Thousands of lessons would fit |
| **EWRAM** | **99.58% full — 1,104 bytes left.** *This* is the constraint. The menu is built into `sResources`, which is **heap**, so it costs nothing here |
| **IWRAM** | 91% — 2,944 bytes left |
| **A menu label** | **168px.** The window is 22 tiles at `item_X = 8`; *vanilla's longest is 146px* |
| **How many shows** | `whichScript` is a `u8`, so **255** by type. **Sixteen** is what the heap array is sized for, and the list scrolls six at a time |

## The nine shows we added, and the gate

***One is the answer to a question the game had never answered:*** **`Where do my daemons go?`** *— you **bind** a daemon with a **BOX**, and the box **hosts** it.* "**A hosted DAEMON is running. It is just not running here.**" 1.3's argument, said out loud by the one character allowed to say things out loud.

***The other eight are one per MARK, and each teaches its own mark's concept as a thing to DO:***

| | **teaches** | **as** |
|---|---|---|
| **SLATE** | representation | *read the summary page — two daemons that look alike are not alike* |
| **SLOPE** | gradient descent | *there is no big fight; there are two hundred small ones* |
| **SENSE** | perception | *name three things on the battle screen before you press anything* |
| **FIT** | overfitting | *the TUTOR trained one daemon against one opponent and lost to a child* |
| **SKEW** | bias | *"Every DAEMON you outscore leaves a little of itself in yours"* — which is EVs, and is true |
| **FRAME** | attention | *the menu you opened decided what you were allowed to think of. DETACH is right there* |
| **HEAT** | temperature | *damage wobbles; a plan that only works on the best roll is not a plan* |
| **TRUE** | alignment | *a THRASHING daemon still obeys — it obeys in the wrong direction* |

***Not one of them names its concept.*** **The menu says `About the SKEW MARK.`, never "bias"** *— the marks were named in 5.2 precisely so the concept would not have to be,* and this is the first system to spend that.

**The gate is the mark itself.** `FlagGet(FLAG_BADGE01_GET + n)` — ***so the list grows as the player is certified,*** and a show they could not follow is never offered. *Vanilla already did exactly this with the two bag shows and the TM CASE,* so the filter replaced both rules with one.

### The unread marker, and where it stops

**Nine vanilla flags, `0x4A7`–`0x4AF`, are declared and referenced nowhere.** *Nine is exactly what this needed:* **one per talk show.** The flag is set the moment a show is chosen, **so the host's greeting reports only the *other* shows still unseen** — pick the new one and he says nothing; pick an old one while a new one waits and he says "*The TUTOR has a NEW SHOW for you today, and it is on the list!*"

### The schedule — what is on, and what is written down for later

***A list, kept here, because a lesson is now cheap enough that the limit is knowing what to say.*** **Shipped is in the ROM; the rest is a queue, and each row carries its own gate.**

| show | gate | teaches |
|---|---|---|
| **Teach me how to battle.** | always | *shipped, vanilla, ported* |
| **What is a status problem?** | always | *shipped, vanilla, ported* |
| **How do I read a matchup?** | always | *shipped, vanilla, ported* |
| **I want to bind DAEMONS.** | always | *shipped, vanilla, ported* |
| **Where do my daemons go?** | always | **shipped, ours** — bind with a BOX, the BOX hosts |
| **Teach me about TMs.** | TM CASE | *shipped, vanilla, ported* |
| **How do I register an item?** | TM CASE | *shipped, vanilla, ported* |
| **About the SLATE MARK.** … **TRUE** | that MARK | **shipped, ours** — eight, one per certification |
| *What is the INDEX for?* | the INDEX | **4.2's complaint, said as a tutorial** — an entry is what somebody wrote down, and a daemon is not its entry |
| *What does a CHECKPOINT cost?* | first CHECKPOINT | healing is free and *time* is not; the walk back is the price |
| *What are the four inputs?* | holding an AXIOM | AXIOM, EMBEDDING, AFFECT, REWARD — and that a daemon *compiles to* something new |
| *What is MP?* | first TM | a ROUTINE has a budget, and the budget is the plan |
| *What is a held item?* | first held item | the one slot a daemon carries into a fight without being told |
| *Why did it not obey?* | first traded daemon | **obedience is keyed to the MARKS you hold**, which is 5's argument arriving as a mechanic |
| *What is the REPO for?* | first purchase | *a repository is where you fetch from and push to* — and the shop does both |

***The gates are the discipline, not the content.*** **A show offered before the thing it explains is a show about nothing**, and every row above is unlocked by the player already holding the thing.

***What this does not do is reach the player in the overworld.*** **That wants a map object with a script**, and the description line carries the rule instead — *a show per MARK,* read in the bag, where a key item is looked at anyway. OPEN.

#### 9.17 The GLOBAL INDEX, and two debug entries that leave the menu

***Built 2026-09-10, off a question: how do you reach the SEVII ISLANDS in a debug save.*** **The answer, derived from the scripts rather than assumed, is that you cannot** — *and the chain that gates them turned out to be four separate things, only one of which is the REVIEW BOARD.*

### What actually gates the islands

| | |
|---|---|
| **Islands 1–3** | ***not gated on the REVIEW BOARD at all.*** **The ferry at ARDOR PORT runs once the S.S. ANNE has sailed and CELIO's quest has started** — *and CELIO only starts it after HOLT takes you to ONE ISLAND from the QUICKSILVER shore, which is main-story* |
| **Islands 4–7** | **CELIO asks for the RUBY only when `IsNationalPokedexEnabled` is TRUE**, and the RAINBOW PASS comes at the end of that errand |
| **The upgraded INDEX** | **`FLAG_SYS_GAME_CLEAR`, *sixty daemons owned*, and *having been to ONE ISLAND*** — three conditions, and the last two are the ones nobody remembers |

***So beating the REVIEW BOARD is necessary and nowhere near sufficient***, and a debug save fails on every clause: six daemons, no ferry, and **CRYSTAL CLEAR is not even in her own lab** — `FLAG_HIDE_OAK_IN_HIS_LAB` is set at new game and only cleared by the starter scene, *so the room a debug player walks into is empty.*

### Two entries, and they are the first in the submenu that leave it

**Everything in 9.3's DEBUG submenu so far is one C call and a redraw** — heal, restock, play a track. ***These two are not.*** *One wants a yes/no box and one wants a warp,* **and the script engine already does both properly**, so the callback's whole job is to shut the menu down cleanly and hand over. *That is the same handover a signpost does.*

| | |
|---|---|
| **`RECORD`** | ***warps to the HALL and lets that map's own script do all of it.*** **CRYSTAL congratulates you, `FLDEFF_HALL_OF_FAME_RECORD` puts your name in, `EventScript_SetDefeatedEliteFourFlagsVars` sets the flag, the respawn moves to BLANCHE, and `EnterHallOfFame` runs the credits and wakes you at home.** *Not one line of that is reimplemented* |
| **`ISLANDS`** | **both map flags, both passes, the ARDOR ferry, and CELIO advanced to the point the RAINBOW PASS is handed over** — *and deliberately no further.* **Recovering the SAPPHIRE is real content, and a debug jump that skips content is a debug jump that stops you testing it** |

***And the sixty-daemon gate is fenced out of the debug build***, so the handover scene can actually fire there. **Both preconditions are real and both stay in the shipping ROM.**

**The mechanism, which is worth more than either entry**

`DAEMONS_DEBUG` ***reached the C preprocessor and not the assembler***, so a *script* could not be fenced the way a *function* could. **One `--defsym` in `ASFLAGS` fixes that**, and `.if DAEMONS_DEBUG == 0` now works in any `.inc`. ***Checked by grepping the ROMs rather than by reading the Makefile:*** **the release build contains none of the debug text and the debug build contains all of it.**

**And the upgraded INDEX is the GLOBAL INDEX**

***4.2's INDEX is the LOCAL list.*** **What CRYSTAL upgrades it to is the complete one** — *and a* ***local index*** *covers one partition while a* ***global index*** *covers the whole table.* **That is the distinction exactly, in a term of art nobody has to be taught**, and it is the register this lexicon has used since PORT and BOX: *plain infrastructure English, doing double duty.*

`NATIONAL` **is a word about countries and this world has none.** `KANTO` *stays, because that is a place.* **Six literals moved** — the dex screen's mode label, its `NUMERICAL MODE` line, the diploma, the rating, and two lines of dialogue — **and GLOBAL is two characters narrower than what it replaced, so nothing needed measuring.**

***Rejected, and worth recording:*** `CODEX` *— prettier, and it does no double duty in computing.* `NEXUS` , `MATRIX` , `VERTEX` *— sci-fi flavour, and this project's furniture is deliberately duller than that.* `INDEX-II` *— a version number on a thing 4.2 wants the player to distrust is the wrong kind of joke.*

***And the GLOBAL INDEX is still one short.*** **8.9 put MISSINGNO at 387 and the complete list stops at 386.** *Nobody says so.*

#### 9.18 The ENCOUNTER page, and one berth serving two ships

***Built 2026-09-10, and the best thing in it was found by a question rather than by the build.***

**The S.S. ANNE and the ferry are the same sailor**

`VermilionCity_EventScript_FerrySailor` ***branches on one variable.*** **Below scene 3 he says "*Welcome to the S.S. ANNE*"; at scene 3 he is the SEAGALLOP sailor** — *one object, one dock, and the scene number is the ship having left.*

**Which means 9.17's `ISLANDS` entry sails the S.S. ANNE by setting a variable**, and *the debug kit grants the S.S. TICKET specifically so that ship can be boarded* **— so the button quietly closed the thing another part of the same scaffolding exists to open.** ***The confirm box now says so***, in the two lines it takes: *the berth is shared, and a new file is how you board her.*

**Not silently guarded, and that is the choice.** *A debug jump that refuses to do the thing you asked is worse than one that tells you what it costs.*

### And RECORD is not enough on its own, which is why both are there

***Beating the REVIEW BOARD opens nothing by itself.*** **9.17's table is the derivation**; the short version is that **the ISLANDS need a ferry that needs CELIO's quest that needs HOLT to have taken you to ONE ISLAND** — *main-story, and untouched by the HALL OF FAME.* **`RECORD` gets you the GLOBAL INDEX. `ISLANDS` gets you the boat.** *Neither implies the other, and a debug save has neither.*

### The ENCOUNTER page — a second submenu, by the same trick as the first

***9.3's submenu is a flag that changes which items `SetUpStartMenu` appends.*** **The moment there were two submenus that flag became a page number and nothing else changed** — *the cursor, the input handling, the descriptions and the frame are all still the start menu's own.*

| | |
|---|---|
| `DAEMON` | ***the row prints the species NAME and nothing else.*** **LEFT and RIGHT step by one and repeat when held; L and R step by ten** |
| `LEVEL` | *same stepper, 1 to 100* |
| `INVOKE` | **1.6 spent this word already** — *"you bind() a daemon and you INVOKE it"* — **and what this row does is bring one up.** *The same verb, used literally* |

***The list runs over NATIONAL DEX NUMBERS rather than species ids***, which buys two things for free: **the twenty-five dummy slots Gen 3 left between CELEBI and TREECKO are not in it**, and *1..151 is the KANTO INDEX exactly.* **With the GLOBAL INDEX it runs to 387** — ***one past the end of the complete list***, because 8.9 put MISSINGNO there. **This menu is the only place in the game that list is not one short.**

### Two things the build had to be told rather than guessed

***The start menu is seven tiles wide and its rows print at x=8, so a row has 48 pixels.*** **The widest of the 412 species names is 60, and sixty-seven of them would have clipped.** *Nine tiles gives 64 and clears every one* — **measured across the whole table, not sampled**, and the width is set only while a debug page is open so the ordinary menu is untouched.

***And an initialised static in `new_menu_helpers.c` does not link.*** **The link script discards that file's `.data`**, and what you get is "*defined in discarded section*" — **a message that names neither the variable nor the reason.** *Zero means "the vanilla width", which is also the state every other path is already in.*

**The battle is started by a script, and that is not decoration**

**`dowildbattle` stops the script context and hands the resume to the battle's saved callback**, so ***it needs a script to come back to.*** *A menu callback has none.* **So the daemon is built in C — `CreateScriptedWildMon` is safe from anywhere — and a four-line script does the rest**, which is the same handover `RECORD` uses and the same one a signpost uses.

#### 9.19 Four names the sweep had quietly changed, and the guard that let it

***Found 2026-09-10 by a tool built for something else.*** **`gbabudget.py` was written to put the hardware budgets in `engine.md` and cannot drift**; the first thing it printed was the widest ability name in the game, and it was **`NO BUSY WAIT`**, which nobody wrote.

**What had happened**

**`OWN TEMPO` was authored `NO THRASH`** *— it prevents THRASHING, and THRASHING is 1.6's state.* **Then 2.10 renamed the move `THRASH` to `BUSY WAIT`, and the word pass applied that rename to the ability name.** ***Twelve characters of twelve, so nothing complained, and it shipped.***

**Three more went the same way in the same commit:**

| | was | became |
|---|---|---|
| `FLASH FIRE` | **`ABSORBS HEAT`** | `INGEST HEAT` — *the move* `ABSORB` *is* `INGEST` |
| Overgrow's description | **"*Ups GROWTH moves*"** | "Ups SCALE UP moves" — **the TYPE, hit by the move rename** |
| CRYSTAL offering the starter | **"*The GROWTH one, ROVERCUB?*"** | "The SCALE UP one" |
| ***the SCHOOL's chart lesson*** | **"*ENTROPY. CORRUPT on GROWTH.*"** | **"CORRUPT on SCALE UP"** |

***The last one is the worst by a distance.*** **That room's entire job is teaching the type chart**, and the word it was teaching had been replaced with a move name. *9.x already carries a commit called "the school teaches the chart, and does not understand it." It was more literally true than intended.*

**The rule was written down and half-enforced, which is worse than not written**

**`port_vocab` has carried this sentence for weeks:** ***name tables are authored and prose is swept.*** *And* `FIXED`*, the thing that was supposed to enforce it, was a* ***width check wearing a name-protection label*** **— it refused a rename that overflowed the array and accepted one that fitted.** ***Every one of these four fitted.***

**A span matching any `FIXED` pattern is now skipped entirely, whatever file it is in** — *which is the difference between protecting a FILE, which stops its prose being swept too, and protecting a LINE, which is what the rule actually said.* **The width check stays underneath.** *Trainer classes, trainer names and type names joined the list, because all three were exposed and nobody had noticed.*

**And the finding is really about tools**

***Three separate guards went in this week and each one was a rule this project had already written and not enforced*** — **the type names withheld from the word map, a phrase's output stashed from the word pass, and now `FIXED` protecting the line rather than the width.** *All three were found by something that read the whole table rather than the diff.*

**`docs/engine.md` is where that class of knowledge lives now.** *It is not design — `vision.md` is engine-independent on purpose — and it was going in here for want of anywhere better.* **The three memory budgets, where a variable actually lands, every fixed width and every trap, with the numbers generated so they cannot go stale.**

#### 9.20 The room that teaches the chart was teaching a different one

***Found 2026-09-10, doing a ticket about the word "fight".*** **T-07 asked for a human read of forty-two occurrences.** *The read was worth doing* — **four were the type said in lower case, eleven were 1.2's replaced verb, and the rest are people** — *but the count was hiding something much larger.*

**Sixty-eight blocks of it**

***`data/text/help_system.inc` carries the WHOLE TYPE CHART.*** **Every type's matchups in both directions, as bare comma lists** — `{CIRCLE_DOT}: FIRE, ICE, FLYING, BUG` — **and every entry was still in Gen 3's names three months after 2.2 renamed them.**

**T-04's pass could not see it**, and that is the interesting part: *T-04 works on the CONSTRUCTION `X-type`, which is what makes it safe,* and **these lists have no construction at all.** *A bare `ROCK` in a comma list is indistinguishable from a rock.*

***Worse, `PSYCHIC` in those lists read `CONSTRUE`.*** **An earlier sweep had put the MOVE name where the TYPE name belonged** — the same class of bug as `NO THRASH` becoming `NO BUSY WAIT` two days ago — *and it had been shipping.* **The room whose entire job is explaining the chart was explaining a different one, in a vocabulary that was itself half-corrupted.**

### And the ceiling was demonstrated rather than declared

**Sixteen lines outgrew the pane once the names were ours.** *They are re-wrapped at a comma, which is how vanilla already wraps its own long ones* — **and the ceiling used is 248px, because that is vanilla's widest line in this exact layout.** *Not a number anybody chose; a number the file already proved.*

***Also in vanilla's names: the two stat definitions*** that list which move types are physical and which are special.

### The pattern, now that there are five of them

**Every one of this week's five name bugs is the same shape** — *a table that a prose pass reached, or a prose pass that a table's rename reached* — **and not one of them was found by reading a diff.**

| | found by |
|---|---|
| `NO THRASH` → `NO BUSY WAIT` | a tool built to measure something else |
| `Ups GROWTH moves` → `SCALE UP` | the same tool, same run |
| the SCHOOL's chart lesson | the same tool, same run |
| `{RUBY}` about to expand to `PUBLIC KEY` | the sweep's own width warning |
| the help system's whole chart | a ticket about the word "fight" |

***The rule this yields is in `engine.md` and it is short: measure the whole table, not the diff.***

#### 9.21 Time — a clock made of play time, and what the hours and seasons are for — *proposed 2026-09-14; the day decided and built 2026-09-25 (T-268); the seasons decided 2026-10-03 (T-359)*

***Asked on playtest: FireRed hacks add day and night, with different daemons by the hour — could this, and what about the four seasons?*** **Written as a design before anything is built, because the mechanism is the easy half and the meaning is the half that can go wrong.**

### The clock is play time, because the cartridge has none

***FireRed carts have no real-time clock,*** **and the engine says so itself** — `ScrCmd_gettime` *is commented out and* `pokemon.c` *notes that FR/LG removed the time-of-day evolutions for exactly that reason.* **But `SaveBlock2` already keeps `playTimeHours`, `playTimeMinutes` and `playTimeSeconds` in every save**, so the proposal is ***a clock derived from play time and nothing else*** — **zero new saved bytes, which matters when EWRAM has about a kilobyte free.**

| | proposed | why |
|---|---|---|
| **a day** | **one hour of play** | *long enough that a route has a mood; short enough that a session sees dusk* |
| **four watches** | **dawn · day · dusk · night**, *fifteen minutes each* | **four, not twenty-four** — *a watch is legible from the screen alone* |
| **a season** | **seven days of play** | *a season turns inside one sitting of a few evenings, and a year inside a playthrough* |

***The honest cost: time stops when the player stops.*** **A real clock would keep night falling while the cartridge sits in a drawer, and this one cannot.** *That is also what makes it fair — nobody misses a season by being away — and an emulator-only real-time mode stays possible later behind the same four watches.*

### What the hours are for: day is CONTENT, night is CONTEXT

***The game's argument already has two halves, and daylight is the literal one.*** **By day the world is what it is. At night the same route is re-read** — *the same tiles under a different frame* — ***which is 0.4 as a lighting change, and nobody says it.***

- **The look** is a palette tint on outdoor maps: *warm at dusk, a blue-dark at night.* **It reuses 8.6a's per-frame pass over the faded palette buffer**, *already measured as nearly free on a settled frame.* **Interiors and caves are untouched, and Halftone stays grey at every hour** — *grey outranks time.*
- **The daemons** change by watch on a few routes and islands, **not everywhere.** *LATENT and OPAQUE come out at night*, **and a handful of daemons are met only then.**
- ***Never stated.*** **No NPC mentions that the night route holds different daemons**, *and no Index entry calls one nocturnal.* **The player notices or does not.**

### What the seasons are for: the four humours, which were already here

***Section 6 gives the REVIEW BOARD the four humours, and 9.4 made them the four type anchors*** — **red VECTOR, yellow ENTROPY, black LATENT, white FROZEN.** ***Classical humoral theory already assigns each humour a season,*** **so the seasons were in the design before anyone asked for them:**

| season | humour | anchor | the wild tables lean toward |
|---|---|---|---|
| **spring** | sanguine | red | **VECTOR** |
| **summer** | choleric | yellow | **ENTROPY** |
| **autumn** | melancholic | black | **LATENT** |
| **winter** | phlegmatic | white | **FROZEN** |

***A lean, not a swap*** — **the same rule as the editions' Tier 2**, *so a season shifts what is common without taking anything away.* ***It is the REVIEW BOARD's model of a person, applied to the weather*** — **an innate constitution read off the calendar**, *which is 6's error made into climate and never named.* **The player meets the Board at the end and has been living inside its four seasons for forty hours.**

~~***Seasons do not repaint the world*** — *snow on every tileset is a palette and tileset job across dozens of maps, and the argument lives in the tables.* **A faint seasonal tint on outdoor grass is the most it proposes.**~~ ***Superseded by the user, 2026-10-03 (below).***

**Decided 2026-10-03: the seasons repaint the world, and each edition keeps a hemisphere (T-359)**

***The user:* "*we already make use of time, I'd love to implement these seasons into the game too.*" The world shows the season, by palette, with no new sprites and little new code** -- the way the day's watches and T-317's faded print already recolour tilesets as they load:

| | **the world** |
|---|---|
| **winter** | *wintery*: greens cooled and whitened toward frost |
| **spring** | *subtle*: fresher greens, more flowers where flowers already grow |
| **summer** | *the world as drawn* |
| **autumn** | *the trees turn*: their greens toward amber, rust and gold |

***And the editions keep opposite hemispheres*** -- **CONTENT the northern year, CONTEXT the southern**: *from December 21 to March 21 it is winter in CONTENT and summer in CONTEXT, and CONTEXT's autumn is CONTENT's spring.* **The two cartridges are one argument told from two sides; now they are one year seen from two halves of the world.** ***Why this way round*** (*flipped by the user the same day -- see Reversed*): **CONTENT is "the thing itself, with nothing read into it" -- the calendar as the game's maker lives it; CONTEXT is "framing and reinterpretation" -- the same date read from the other side of the world, December's midsummer.** *The day does not change; the frame does, which is CONTEXT's whole idea told by the planet. And CONTENT is the default build, so the game most tested keeps its maker's season. The goal companion (9.25) keeps the same rule, so a daemon's season is the same in the game and on the device.*

***The clock decides, and play time stands in*** -- **with the cartridge's clock, the real date; without one, play time, as the table above proposed (a season every seven days of play).** *The humours' lean on the wild tables stays a proposal.*

### How it would be built, from things the engine already does

- ***The table switch has a precedent.*** **ALTERING CAVE already stores several wild tables for one map and adds a variable to pick among them in `GetCurrentMapWildMonHeaderId`.** *Time and season become that variable —* **consecutive headers per map in `wild_encounters.json`, ROM cost only.**
- ***The tint has a precedent.*** **8.6a's `DaemonsGreyHalftoneFrame` is one pass a frame over `gPlttBufferFaded`.** *A time tint is a blend in the same place, skipped indoors.*
- ***The clock is a function, not a variable*****: watch and season are computed from play time when asked,** *so nothing new is saved and nothing can fall out of step.*
- ***The playtest agent reads RAM,*** **so it can be told the watch and season for free** *— which makes the time system testable by the harness the same day it lands.*

### Open, deliberately

- **Is one hour of play the right day**, or should a day be shorter so a single session sees all four watches twice?
- **Which routes change by watch?** *Every route diluting it; three or four that visibly change teaches it.*
- **Does the season lean belong in both editions**, or does it stack with Tier 2 so that CONTEXT's winter is a different place from CONTENT's?
- **Does the Index record when a daemon was bound?** *It could. 4.2 says the Index cannot hold what matters, and a timestamp is exactly the thing it can hold.*

### Decided 2026-09-25, and built (T-268)

***The cartridge turned out to have a clock*** **— the user's EZ-Flash carries the Seiko chip Emerald's cartridges did, and emulators provide one (T-265).** ***So the premise above, "the cartridge has none", holds only sometimes,*** **and the user chose both: the real clock when there is one, and the play-time clock when there is not** *— so night falls when it is night where the player is, and every player still has nights.*

| | with a clock | without one |
|---|---|---|
| **the watches** | **dawn 5–7, day 7–18, dusk 18–20, night 20–5** | **day, dusk, night, dawn, fifteen minutes of play each** — *day first, so the opening is in daylight* |

- ***The look*, as proposed**: *a tint applied as each palette loads on an outdoor map — dusk warm, night a blue-dark that keeps every shape readable, dawn a pale half-light — so fades and weather start from it and it cannot compound.* **The people outdoors are in the same light as the ground; menus, text, interiors, caves and battles are not; HALFTONE is grey at every hour.** *A connection that crosses into a different light reloads the rows and the people it would otherwise have carried across.*
- ***Which routes — three or four, as proposed, the user's choice*: Route 1**, *where a player can first notice anything*, **and 8, 10 and 12, the three roads into HALFTONE** *— 10 is "Gloaming" in the chapter table, dusk approaching the tower.* ***The rule a player can learn by walking: the day's bird sleeps.*** **PACKET's slots go to what runs in the dark — DANGLING, and on 12 RESPAWN — at the slot's own level and odds**, *so a route is exactly as hard at night.* **Water and rock are as by day.** *Both editions change the same way, so Tier 2 is untouched.*
- ***OPAQUE is not out at night yet.* No OPAQUE daemon is in the first 151**, *and one beyond them would arrive before the national INDEX can hold it —* **so the night is LATENT's until the GLOBAL INDEX question is settled.**
- ***Still never stated.*** *No line says the route changes; the TIME code at the bedroom terminal shows the clock and says nothing about what it is for.*
- ***Still open*: the seasons.**

**The week — decided by the user 2026-09-25, and built (T-269, T-273–T-275)**

***The clock also knows the weekday*** *— the S-3511's own register, or, without a clock, one day per hour of play —* **and the user gave the seven days their own seven: the rainbow in order and the notes C to B, Sunday first, which is also the Guide's order (4.36, Chapters 8–9).** ***Each is carried by exactly one thing, and none of them is named:***

| | where | why there |
|---|---|---|
| **the name** | **on leaving a building, the popup that names the town slides in again: *F · WEDNESDAY · NIGHT*** | *where a player notices the time* — **ink and paper, the popup's own; no colour in it (9.4)** |
| **the note** | **TUNE at the bedroom terminal sounds it before its song** | ***sound has no 9.4 problem***; *the banner's mark is the same letter* |
| **the colour** | **the CHECKPOINT rotunda's trim** *— the band round the dome, its courses, the canopy over the door* | ***one building, seen every day, and a trim rather than a sign*** **— each colour 22+ from every type hue**, *so the rainbow bends where it must: Monday's orange away from GROUND, Tuesday's yellow from STEEL, Thursday's blue from DRAGON* |

***DEBUG's TIME row sets both the watch and the weekday*** *(A and LEFT/RIGHT), and relights the map at once.* **Still never stated** *— what a day's colour or note means is the Guide's to suggest and the player's to notice (0.4).*

### 9.22 Blanche's buildings, and the one warm note — *decided 2026-09-14, not drawn*

***Environment art starts with CRYSTAL CLEAR's lab,*** **the one building Blanche has** (4.23) — *and the first thing a player sees of her.* **Blanche is the pre-colour town** (3.1): *white, high and open, the achromatic pole Halftone answers.* **So the town goes pale and the lab keeps one warm note — *the only one in Blanche.***

#### The lab, from outside

***Pale stone walls, a flat slate roof, plain tall windows, and one line of gold: the door frame and the plate above it.*** **Gold is the Clears' family colour** (*Crystal golden-amber, Ty darker, Al between*)**, so the building carries her the way her fur does — as a colour, not a word.** ***No lettering on the building***: *the sign already reads CRYSTAL CLEAR RESEARCH LAB, and CLEAR is what the player finds under the gold leaf at Quicksilver.* **Her musical motif (7.6) is a separate thing; the visual one is this.**

#### What the tiles allow, measured

| | | |
|---|---|---|
| **the lab** | ***26 blocks placed on no other map***, *nearly every tile its own* | **drawn in Blanche's own row 9**, *which already holds whites, greys, glass blues and a five-step gold*; **five tiles are shared and stay untouched** |
| **the houses and fences** | ***drawn in COMMON rows 2 and 3***, *shared by about 180 maps* | **pointed at Blanche's own rows 8 and 12** — *nearly unused, and no other map draws them* — **filled with pale copies of the common colours** |
| **the ground and trees** | ***common row 0, 1317 of Blanche's draws*** | **a Blanche-only runtime wash first** (*T-56*), ***then redrawn*** — *see* **The ground, drawn** *below* |
| **the lab's interior** | ***shared with the whole Quicksilver lab*** | **split into its own tileset** — *so the working lab and the ruin can read as one place, before and after* |

#### The lab, from inside — *decided 2026-09-14*

***A working lab, lived in*** **— the exterior's pale stone carried inside, papers where someone left them, and the one gold detail.** *Not a showroom and not an institution*: **CORPUS turns it into one later, and the room should have been a person's first.**

| | becomes | because |
|---|---|---|
| **Oak's red machine** | ***the PORT*** | **1.3: "*you reach the BOXES through a PORT*"** *— a socket in the wall of a host.* **The room the player gets their first box in shows where a box plugs in** |
| **the long shelves** | ***her journals*** | **4.24: *the document that removed her and the document that vindicates her are the same document.*** *Rows of bound notebooks, never remarked on, in the room she worked in* |
| **the two tutorial posters** | ***new text*** | **the last vanilla lines in her lab** *— rewritten to craft rule 1* |
| **everything else** | ***redrawn on its own tile*** | **the computer, the INDEX unit and her note, the bench with the three boxes, and the people keep their scripts and collision** |

*Concept art comes from the n8n* `daemon/sprite` *workflow's new* `kind: environment`, top-down three-quarter or front elevation, with lettering forbidden in the prompt *— and, as with the people, it is reference to draw from: the tiles are written by hand, eight pixels at a time.* T-53 to T-56.

## The ground, drawn — *decided 2026-09-14*

***The wash was the stopgap; the ground is now drawn.*** T-56 paled vanilla's field, and a pale Kanto field still reads as Kanto *— mint grass, sand paths, conifers, a blue pond.* Blanche gets a ground of its own, still with no colour in it.

| | becomes | because |
|---|---|---|
| **the grass** | ***short silver-green grass, almost white at the tips*** | *a town in a field before colour, not a town under snow* **— frost would say winter, and 9.21's seasons may want winter** |
| **the paths** | ***white chalk and gravel*** | **the same white as the lab's stone**, *so the town and the lab are one material* |
| **the trees** | ***white-barked birches*** | ***a different tree, not a paler one*** **— same footprint, same collision** |
| **the flower bed** | ***white flowers on pale stems*** | *no petal colour anywhere* |
| **the pond** | ***clear pale water over a light bed*** | **the lab's gold stays the only warm note in Blanche** |

***How it fits, measured.*** The ground blocks draw the General tileset, shared by about 180 maps, so nothing is redrawn where it is. *Blanche's tileset has 224 free tile slots and its palette row 7 is unused and row 12 nearly so*: the tiles are drawn fresh there, and a common block Blanche uses gets a Blanche copy *— the sign block's route in T-55.* ***The flowers and the water are animated in the shared tileset***, so Blanche's copies need a tileset animation of their own.

***And the wash comes off whatever the new tiles replace*** — a tile drawn pale and then washed would be paled twice. *It stays only on anything left undrawn.* Concepts come from `kind: environment, view: terrain`. T-58.

Two households, two buildings — *decided 2026-09-14*

***T-55 made both houses pale, and a pale copy of one house is still two copies of one house.*** They are two households, so they are two buildings *— on vanilla's footprints, with vanilla's collision and doors.*

| | becomes | because |
|---|---|---|
| **the player's house** | ***a timber cottage*** *— weatherboard walls, a steep shingle roof, a stone chimney and an attic dormer, a porch, window boxes* | **an ordinary family home**, *which is the point next to the Clears* |
| **the Clears' house** | ***the lab's pale stone*** *— a hipped slate roof, tall narrow windows, a stone door surround* ***— with a glasshouse of white frames and plants*** | **It is the Clears' own house [private: SECRETS §2], so it is drawn in the lab's own palette row.** *Vera just looks (4.29); the house is built for looking, and nothing says so* |

***Both doors are the same block,*** which the door animation is keyed to, so they share one door and each house draws its own frame around it *— the cottage a porch, the stone house a lintel and jambs.* The cottage takes Blanche's row 11, free since T-58; the stone house takes the lab's row 9, whose two unused indices become the glasshouse's greens. ***No gold****: the lab keeps the only warm note.* T-59.

Where the pale ends — *decided 2026-09-14*

***A redraw that stops at the map edge stops where the player is looking.*** From Blanche you see about five rows into each road*, so a pale town with vanilla roads showed a hard line at both exits — birches against conifers, clear water against blue sea.* So the pale runs out along the roads, and the colour comes back somewhere the eye is not already fixed:

| | drawn pale | where the colour returns |
|---|---|---|
| **Route 1** | ***its last ten rows***, *birches and pale ground —* ~~*and every tree along its sides and its top row, with Blanche's birch border past them*~~ ***reversed on playtest, see*** **No seams** | **at the stretch's first ledge**: *coloured above the lip, pale below*; **and Blanche's chalk path runs on through coloured grass to the ledge you jump at row 26** |
| **Route 21 North** | ***all of it*** *— the channel, the open sea in pale shallow and pale deep, the rocks, the sand bars, the shore and its trees* | **across a sandbar at the join with Route 21 South** *—* ~~*behind a line of pale rocks with a gap to surf through*~~ ***reversed on playtest, see*** **No seams** |

***The signs say something.*** Vanilla's boards carried a squiggle that reads as Japanese at a glance. *A 16-pixel board holds three letters of a 3x5 font,* so they say short words: BLANCHE on a board as wide as two cells, LAB, a question mark for the trainer tips, and RT1 on Route 1. ***The text boxes they open are unchanged.***

***Why Route 21 South stays vanilla:*** **it is drawn from Quicksilver's tileset, which the island, Route 20 and the Safari areas share** — *pale water there would put the same line wherever that sea continues.* **The island is where the colour comes back. T-60.**

**No seams** — ***playtest 2026-09-14***

***A style may change; it may not change on open ground.*** **Three seams were found by playing, and each moved to something that already draws a line.**

| | was | now |
|---|---|---|
| **Route 1's trees** | ***birches the whole way to Callow*** *— white trees beside coloured grass for thirty rows* | ***the trees follow the grass*****: broadleaf wherever the ground is coloured, birches only in the pale stretch** |
| **Route 1's path** | ***cut across its width*** *— gold on one row, chalk on the next* | ***one colour the whole way*****: Blanche's chalk runs up through coloured grass to the ledge at row 26**, *and the grass itself changes at the first pale ledge's lip* |
| **Route 21** | ***a line of rocks on open water*** *— rigid, and it garbled crossing the join* | ***a sandbar the full width of the sea*****, with a hatch, and a caretaker who says nobody comes out this far** |

***Why the join garbled, which is a fact about the engine and not about the art.*** **A connection draws seven rows of the neighbouring map with the tileset of the map you stand in** (*engine.md trap 14*). ***So the last seven rows of any map are drawn twice****, once by each tileset,* **and anything Blanche drew there reads as Callow's or Quicksilver's blocks from the other side** — *which is what happened entering Callow under Route 1's birches.* `tools/gbaseams.py` **finds every such cell.**

***Why the sandbar is shaped as it is.*** ~~*Deep enough to fill the screen, with its edges drawn pale in one tileset and blue in the other under the same ids*~~ ***— reversed on playtest the same day: it was far too big, and it still glitched.*** **The engine does not redraw the screen when you cross a connection**: *every cell already on screen keeps the tile and palette row it was drawn with, now reading the other map's tiles and colours* (*engine.md trap 14*). **So whatever is on screen at the crossing must be primary — and that includes the border past the map's sides, which on Route 21 North is Blanche's pale water.**

***Now: a bar three rows deep across the whole sea, and two arms of sand from it down both sides to the join, so the only crossing is a channel at least seven cells in from either edge*** **— you see seven cells to each side, so the border is never on screen there.** *The arms' last row is solid, so nobody surfs off them across the join.* **Its sand is gold throughout** — *a bar that was chalk on one half would be the seam it replaces.* **T-72, T-73, T-83.**

**The station under the sandbar — *decided 2026-09-14***

***A hatch on the sandbar leads down to an emptied research station*** **— the room where Crystal's session with BunnyArtsai happened** (*4.10,* Echoes of the Algorithm)**, cleared out.** *Terminals still lit with their input cables taken out, so nothing on them answers; a table with a visitor log.*

**The log has one line:** `CRYSTAL CLEAR. SEPT 3. PURPOSE: SESSION.` **Then** `TRANSCRIPT: WITHHELD.`

***It points at what she saw and never shows it.*** **4.10 rules that the player reads the format decaying at Quicksilver without ever seeing the first time** *— so the station is proof a session happened and nothing about what was in it.* ***The date repeats SEPT 3 rather than adding one*** **— the same ruling as the founder's plate,** `PR–0903` *— so it confirms she was still being let in on the day the file closed, and gives the arithmetic nothing new.* ***Chosen over showing the session.*** **It is also the first interior between Quicksilver and Callow**, *which 4.23 had noted there was none of.* **T-74.**

**The world's own trees, water and grass — *decided 2026-09-14***

***Outside Blanche the ground was still Nintendo's drawing on every map.*** **So the General tileset's nature is redrawn in place, and every town and route changes together** *— the tiles change and the blocks do not, so nothing mixes old and new:*

| | becomes | why this and not something else |
|---|---|---|
| **the trees** | ***a round broadleaf*** *— a lumpy green crown on a short dark trunk, big enough that a forest reads as one mass* | **different from Blanche's white birches, so those stay Blanche's** |
| **the water** | ***short ripple dashes on vanilla's blues****, three to a square, drifting against each other* | **the pond's language: the pale water and the sea read as one water, one of them drained** |
| **the grass** | ***a meadow green, our own sprig, and tall grass as a field of narrow blades****, with the walking effect redrawn to match* | *new shapes and new colours — recolouring vanilla's tufts would still read as vanilla's* |

***A tree is not a tile.*** **Vanilla's tree is 32 by 48 pixels in pieces reused across dozens of blocks, some of one tree and some of two overlapping ones**, *so every piece was traced back to where it sat in a standalone tree or a forest and redrawn from the same place in the new one.* ***The colours moved as exact values across 58 outdoor palettes****, so no seam can open where a town's own tiles meet grass.* **T-61.**

**9.23 CHECKPOINTS, THE REPO and the BENCHMARKS, as buildings — *decided 2026-09-14, not drawn***

***The lexicon renamed them in section 1; they still looked like a Pokémon Center, a Poké Mart and a Gym.*** **A player finds them by sight in every town, so what they look like is how the names are learned.** *Each keeps what the vanilla building did for navigation — you can spot one across a town — and says what it is by what it shows, not by what it spells.*

| | emblem | outside | inside |
|---|---|---|---|
| **CHECKPOINT** | ***a circular restore arrow*** | ***a calm teal roof*** | ***the healing machine becomes a rack of BOX slots with progress lights*** *— restore from a saved state, which is what 1 says a CHECKPOINT is* |
| **THE REPO** | ***a stacked package*** | ***a warm amber roof*** | ***shelves of labelled package bins*** *— a repo distributes packages (1)* |
| **BENCHMARK** | ***a gauge needle*** | ***a slate grey roof*** | **the arenas keep their city themes (5)**; ***the entrance statues become a scoreboard plinth*** |

***Signs carry the emblem and no letters.*** **Our 3x5 font fits three letters to a board, and every short form of these names is an abbreviation the lexicon avoids** *— the full name is in the text box and on the town map.* ***The emblem replaces the Poké Ball everywhere it stood****: over the door, on the sign, on the CHECKPOINT floor.*

***Colour here carries function, not decoration*** (9.4)**: a colour per kind is how a player tells the three apart at a glance***, which is the job vanilla's red and blue did.*** **Teal, amber and slate are ours; red and blue were theirs.** *Halftone greys them like everything else.* **T-62 to T-65.**

**What they are called, and what they are not — *playtest 2026-09-14***

***Nothing the player reads says GYM.*** **Every sign and line says BENCHMARK —** `SLATE CITY BENCHMARK / LEADER: CAIRN`**.** ***The title stays LEADER*****, chosen over PROCTOR**: *a proctor supervises an exam without sitting it, and you fight this person;* **a benchmark's leader is the top of the board you are trying to beat.** *Eight labels in the FAME CHECKER, 88 pixels wide, read BENCHMARK without the town, because VERDIGRIS BENCHMARK and QUICKSILVER BENCHMARK do not fit.* **T-76.**

***They should not look like houses.*** **A roof and a door recoloured is still a house***, so each kind gets a silhouette of its own on vanilla's footprint* **(T-77)***,* ***and a BENCHMARK's two pillars take the colour of its leader's type*** **(T-78)** *— 9.4's rule on the building itself.* ***Chosen from concepts, 2026-09-14:***

| | becomes | what it says without a word |
|---|---|---|
| **BENCHMARK** | ***a colonnade*** *— a flat roof, a stone frieze with a relief of bars, the gauge over the door and the mark's name beside it, columns across the front, the two at the door in the leader's type* | **an institution that measures**, *which is 5's whole complaint about it* |
| **CHECKPOINT** | ***a rotunda*** *— a teal dome carrying the restore arrow, over a round stone drum with tall windows* | **the one round building in a town of boxes**, *found at a glance; the checkmark plate goes, the dome is the sign* |
| **THE REPO** | ***stacked shipping containers*** *— the branch stencilled on the top one, the door cut into the bottom pair, REPO on the left container* | **packages, arriving and leaving** *— 1's definition, standing in a yard* |

***A BENCHMARK's two words*** **(T-79): the free-standing board reads MARK, and the frieze carries the name of the mark earned inside** *— SLATE, SLOPE, SENSE, FIT, SKEW, FRAME, HEAT or TRUE,* **in the 3x5 face, four pixels a letter, so the longest is two cells.** *The name is the only thing on the building that differs by town besides the pillars, so it is drawn on each town's own blocks.*

***Built 2026-09-14, and what bounds the detail.*** **Vanilla's buildings are shaded and textured; ours read flat on playtest (T-85).** *All three kinds are drawn in the General tileset, whose 640 slots are full,* **so detail is paid for in tiles: 45 of vanilla's duplicate tiles were merged to make room, mirrored tiles are shared, and the drawings take 127 of the 132 slots that result** *— a dentilled cornice and one step on the colonnade, a canopy and plinth on the rotunda, castings on the containers.* ***More than that means drawing on each town's own tileset.*** **The board's face sits low, in the one cell every town's board has.** ***Brazen's BENCHMARK had no board, only vanilla's red GYM roundel,*** **so the roundel became its MARK board; the one before the PROOF HALL reads QED** *— which is how a proof ends, and all the board says.*

***And then it was drawn on each town's own tileset — 2026-09-14.*** `tools/gbacivictown.py` **gives every cell of 35 of the 37 buildings a block of its town's own,** *so the detail stops competing for five shared slots:* **a lit, ribbed dome with a lantern over coursed stone and arched windows; ribbed containers with castings, lock rods and a hazard-striped lintel; a colonnade with a skylight, dentils, a carved frieze, fluted columns, sconces, full steps, a bronze plate for the mark's name and a full-height board.** *The room comes from slots no block of the town draws, and from each town's own duplicate tiles.* ***The shared drawing stays underneath as the fallback,*** **and two buildings keep it:** *Lurid's BENCHMARK, which stands within seven cells of a connection (trap 14), and Brazen's REPO, whose tileset has no room left.*

### 9.24 What a routine's animation may show — *decided 2026-09-17* (T-135)

***Every name in the battle menu is ours and every animation was vanilla's*** **(354 of 356, `tools/animcensus.py`), so a player chose RESTORE and watched a creature pull glowing orbs in from the air.** *The picture argued against the word.* **Six clauses decide what replaces it, and they were drafted on three exemplars and approved on screen in the T-136 theatre before any family was touched.**

1. ***An animation shows the PROCESS its routine names*** **— never a creature exerting its body, and never energy gathered from outside it.** *RESTORE reloads a state the daemon already had; it does not inhale.*
2. ***The daemon's own palette is the medium, and grey is the ground.*** **Losing something is drawn as draining toward grey; getting it back is the colour returning.** *Grey costs invariant 5 nothing* **(8.6)** *— PERSPECTIVE already drained to grey before it took another's frame, and that is where this clause was found.*
3. ***The only colour an effect may ADD is a type's*** **— the routine's own, from `gbasprite.py`'s `TYPE_COLOR`** *— never vanilla's decorative yellow flashes, pink hearts or blue stars. Neutrals (black, white, grey) are ground, not colour.*
4. ***The streaks are part of the palette,*** **so whatever happens to a daemon's colour happens to what it knows** *— a drain greys the streaks with the body, and they come back with it* **(9.4, as amended).**
5. ***In Halftone the frame's greying wins*** **(8.6a) — no animation brings colour into Halftone except PERSPECTIVE's flash, which stays the one exception.**
6. ***Script first.*** **Re-sequence and re-tint vanilla's own tasks and sprites; draw a new effect graphic only when nothing existing can mean the process (T-137).** *Vanilla's sounds stay unless one contradicts the picture.*

***The exemplars, as approved*****: RESTORE — its colour drains to grey and returns in three ticked steps, a checkpoint reloading;** *PERSPECTIVE, which already obeyed 1, 2 and 4;* **and WRITE, still vanilla's lunge, named as the case the attacking families must solve** *— under 1 and 3 a write lands on the target as its attacker's type colour, not as a shove.*

***How the families are worked*** **(T-134):** *each is drafted behind `.if DAEMONS_DEBUG` so only the debug ROMs play it, captured in the theatre as filmstrips, approved, and then released —* **and `animcensus.py` counts a family done only when its scripts no longer match upstream's.**

***Done 2026-09-18 — 0 of 356 still vanilla's*** **(T-134, T-139–T-162;** `animcensus.py` *reports none*). **Twenty-four families, each a vocabulary in** `tools/genanims.py` **that says what its type DOES** — *CONTENT sends and lands, LOGIC derives in steps, FROZEN holds with no motion at all, OPAQUE hides the hit, LATENT surfaces, HARDENED snaps with no give, EMERGENT blooms at a different scale from its parts, COPY plays the same pulse twice.* ***Two readings the work forced, recorded here because they are this rule's***:

- ***Clause 3 and a type whose colour IS a neutral.*** **LEGACY's slate (130,130,138) sits beside clause 2's grey, so a slate hit read as its target draining.** *It is drawn at the HIGHLIGHT of its own ramp — the step its daemons' sprites already carry — so old material is worn PALE where loss goes dull* (`genanims.bleach()`). **The added colour is still the type's; the step of its ramp is chosen so it cannot be read as loss.**
- ***Clause 6 and one new task.*** **CAST and RECAST change the user's type BEFORE their animation plays,** *so* `AnimTask_DaemonsBlendToUserType` *ends them in the new type's own streak colour.* **No graphic was drawn; a task was written, because no existing one could show a colour the script cannot know.**

***Amended 2026-10-08 (the user, T-392): the game's own signature.*** **Routines may now carry the splash's 0s and 1s and its notes as particles** — *the presents scene's code becoming music (* `graphics/intro/game_freak` *), the game's first and only image of what a daemon is.* **Clause 6 allows it, because no vanilla graphic can mean CODE or the music it becomes; clause 3 still governs it**: *an ordinary routine's bits and notes are in its own type's colour* — **binary on the CONTENT side of the chart, notes on the CONTEXT side.** ***The seven union routines (0.7, Wisdom) are the one exception, as PERSPECTIVE's flash is in Halftone***: **each has an effect of its own, and its bits and notes come in the splash's own colours, so the signature is theirs alone.**

#### 9.25 The goal companion, and AWAY — *decided 2026-10-03 (T-358)*

***The user, on their first anniversary at work: a device that carries a daemon and helps you get things done.*** `daemons-companion` (its own repo, symlinked as `companion/`: an app, a local server, ESP32-S3 and Pi Zero devices) **is the successor to RoverRadio** -- *where this game's week came from* -- **and it holds nothing of Nintendo's.** *Its vision and plan are its own (* `companion/docs/` *); this section is only the game's half.*

**A daemon can go to the device, and the game always shows where it is:**

- ***Sending starts in the game.*** **A party daemon's menu offers to send it; the game saves and tells the player to open the save in the app,** *which checks the daemon is in the party and marks it AWAY.* **Coming home is the same, reversed**, *and it brings back the friendship it built there.*
- ***AWAY is washed out everywhere*** -- **party, PORT, summary** -- *so it is never lost.* **It cannot battle, be traded or be released.** ***It can go into the PORT and come back out***, *so the player can still play with six.*
- ***The mark is one of four bits nothing uses*** -- **beside *is egg* in every daemon's record, outside the checksum** -- *so it travels with the daemon through the PORT and costs no save space.*

***Reshaped by the user 2026-10-04 (T-370; the companion's PLAN 3):***

- ~~***One daemon at a time.*** **SEND is not offered while another daemon is AWAY or asked for.**~~ ***Reversed by the user 2026-10-09 (T-396)***: **any number may be AWAY, one device to a daemon.** *SEND is refused only for a daemon not in full health, or the last one here able to battle (one asked for counts as gone). The companion says which device carries which (its C-80): a daemon is in one device at a time, and choosing another device for it moves it there -- so it feels real. See Reversed.*
- ***SEND appears only once the companion knows the save.*** **The app's first SYNC writes a flag into the save; until then the party menu has no SEND**, *so a player without the companion never meets it.* **The people on the CHECKPOINT's second floor talk about the companion** -- *that loading a save into it brings a new SEND* -- **so the player learns of it in the world.**
- ***An emergency way home, in the game.*** **An AWAY daemon can be called home without the app**, *so it is never stuck in limbo when the app cannot be reached.* **The game warns first: the companion still thinks it is out, and another cannot be sent until it is released there** -- *the next SYNC sees it home and settles both sides.*
- ***The app is married to one save*** *(the companion's side)*: **a daemon carried for one game only ever goes home to that game's save.**

***The menu's words are the user's to choose (OPEN).***

***And the trainer card says whether the clock is real*** *(T-372, the user 2026-10-04)*: **on the player's card (START > their name), a mark or a word says whether the real-time clock is on** -- *time, weekday and seasons from the cartridge, or from play time (9.21).*

#### 9.2 Order of operations

1. Toolchain and a **vanilla matching build**. If the checksum matches, your toolchain is sound and every later break is yours.
2. Replace `constants/type_constants.asm`.
3. Replace the strings in `data/types/names.asm` — structure untouched, per 9.1.
4. Apply the three-line patch to `data/types/type_matchups.asm`.
5. `make`. **Stop here and play.**
6. Add CONSENSUS to `moves.asm` and `data/moves/names.asm`. **Done** — and to `move_constants.asm`, `animations.asm` and `sfx.asm`, which the length asserts require (2.5).
7. Rename TRANSFORM to PERSPECTIVE. **Done** — the move name and its battle message (*took the frame of*). MOCK's Index entry still says TRANSFORM and is left for 9 (4.6).
8. City and route renames plus sign text — fast, and it transforms the feel immediately.
9. Starter trio: stats, sprites, moves.
10. Crystal Clear intro script.
11. The Bleed and Undertone encounter tables plus nine wild daemons.
12. Benchmark 1 at Slate.
13. Two city themes and one modulation.
14. **Play it. Decide if it is fun.**

**Step 5 is the real milestone.** Vanilla sprites, vanilla maps, vanilla everything, running your combat philosophy underneath. It will look like Pokémon Red and fight like Context/Content. A couple of hours' work for the fastest possible read on whether the type system is *fun*, before a single pixel is drawn.

Defer RECURSION past the slice. S.T.A.R.R. appears after the Review Board; you will not reach that content for months, and Bide surgery in assembly is a bad first engine task.

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## 10. DECISION LOG

### Settled

- **THE NEXUS (T-395, 2026-10-09)** — every grove holds the singing fir, flat until the key; through any fir in its key, a place made of paint whose trees are doors to the groves already found; eight residents of its own, drawn in natural colour as paint with their type emphasised, the one exception to colour-by-type (3.4, 9.4)
- Base: `pokered`, Gen 1, RGBDS assembly
- Fifteen types as listed; CONTEXT is Special; STRATUM replaces SUBSTRATE for string length
- Creatures are DAEMONS; full lexicon per section 1
- **MOCK** (Ditto) is CONTENT/CONTENT and also learns PERSPECTIVE — the capability was always in the wild; Quicksilver only noticed it (4.6)
- **ORPHAN** at Halftone Tower — CORRUPT/LATENT, the one daemon with a genuinely blank Index entry; an orphaned process is how a daemon is made (4.5)
- **Silph Scope → RESOLVER** — it exists because the Index is insufficient (1, 4.5)
- **Species renamed: POKéMON → DAEMON / DAEMONS**, by repointing one string (`PlacePOKeText`), so 650 occurrences moved for free (1.2)
- The Pokédex text token becomes literal **INDEX**; item prefixes become literal `POKé` pending their own renames (1.2)
- **Catching is BINDING** — `bind()` and *binding a daimon*; the flat, uncongratulatory message register goes with it (1.1)
- **The box ladder implemented** — USERBOX / ADMINBOX / SUPERBOX / ROOTBOX, plus **GUESTBOX** for the Safari Zone's restricted temporary access; 36 strings across 19 files, none of which had ever been touched (1.3)
- "**They contain DAEMONS**" → "**DAEMONS run on them**" — a box hosts, it does not contain, and the vanilla line argued against 1.3 at the player's first sight of the object (1.3)
- "**Wild DAEMONS live in tall grass**" → "**UNBOUND DAEMONS run loose out here**" — 1.1 defines a daemon as a process that *runs unattended* (1.4)
- **The box exists as an object** — `tools/genbox.py` replaces both ball sprites; a small server with vents and an indicator, not a crate or a cube, since either would undo the rename. The throw opens rather than bursts, and the USERBOX→ROOTBOX ladder is a parameter (1.3)

- **Gemini asserted the reprise was Crystal's Lament transposed; its own notes refute it** — transposition preserves intervals and these differ entirely. Caught by arithmetic, the only check available (7.12)
- **An unclaimed relationship is in the data instead** — track 11 shares key and first three notes with the *procedure*, then descends where the procedure never does. **Held as unconfirmed**, being one observation of an ordinary figure (7.12)
- **Method:** ask models for measurements, derive relationships here. The one field that invited inference is the one that was wrong (7.12)
- **The best-defended claim in this document is retracted (7.11)** — all three of Scorn's songs arch; the man/directive distinction was a bad transcription, not a finding. A control cannot rescue bad input
- **Track 07 re-scanned** — `D5 E5 F5 E5 D5 A4 C5 D5`, 4 up / 3 down. Three findings struck (7.7, 7.10, 7.11, 7.12)
- **A controlled comparison, and the best evidence in the set** — Scorn's two *own* songs both arch at 2/3, independently; his *directive* is the only hook in six that never descends. Same character, same writer, three keys: the contour tracks **man versus procedure** (7.11)
- **Five of six hooks arch**, so arching is the baseline and the single exception is the signal (7.11)
- **The procedure's hook is the only one of six with no descending interval** — it climbs and never comes down. A property, not a reading, and worth more than the shape observations around it (7.10)
- **7.8's "within one accidental" corrected at n=5** — the real spread is five accidentals; the narrow band was a three-track artifact. Scorn's two keys still rule out keys as character labels (7.8)
- **S.T.A.R.R. is the outlier on three axes at once** — F# minor against a one-accidental band, 110 against three at 128, sparse against three driving. **It cannot be layered with the human motifs**, which is a mechanical fact rather than a reading (7.9)
- **Keys will not separate characters** — three transcriptions sit within one accidental, and Scorn has two keys. **Mode and contour** carry character; **keys belong to towns**, as 7 always said (7.8)
- **Two readings retracted** — Crystal's `B` is generic to the key, not particular to her, and the B ♭ opposition rested on it. The correction arrived one track later (7.6, 7.7)
- **Craft rule 6 is already in the source** — Scorn's own song is major and bouncy; the villain is cheerful, written years before the rule (7.8)

- **Scorn's motif transcribed** — D minor, and **A minor and D minor differ by exactly one note: B against B ♭ , which is Crystal's note of tension flattened away. Marked as a *reading* of ordinary material, since adjacent keys differ by one note by definition (7.7)**
- **Their contours oppose** — Crystal's hook descends, Scorn's climbs; both 128 BPM and both 8 beats over 16, so they are interchangeable and layerable (7.7)
- **Crystal's motif transcribed** — A minor, the **relative minor of Blanche's C**, so it needs no modulation; and **exactly one of its notes falls outside Blanche's pentatonic: B, the leading tone.** Her motif is Blanche's scale plus the one note that wants something (7.6)
- **The pipeline caught two real errors on its first run** — a chord name in a note field, and a hook half the length of its progression (7.6)
- **The rock opera is character music, not place music** — Act 1's tracklist is the cast, so the hooks become **leitmotifs** inside town themes rather than town themes themselves (7.5)
- **Gemini is the ear** — Claude cannot hear audio; prompts and format in `music-prompts.md`. Ask for the hook, not the song (7.5)
- **Blanche Town's theme written** — C major **pentatonic**, no F and no B, so it can form neither a leading tone nor a dissonance and has nothing to resolve (7.3)
- **The *My Chromatic Year* mapping is recovered** — it maps **months** to semitones, not colours, but two tracks name colours (*Green:Blue F#*, *Blue G*) and those two anchor everything (7.1)
- **The colour rule is the octave-folded visible spectrum**, red as C — both anchors land exactly, so it is derived rather than asserted (7.2)
- **A# and B have no wavelength.** They are magenta, which is not in light. The scale closes B→C where the colour wheel closes through an invented colour — and **Halftone, which replaced purple Lavender, sits on those two notes** (7.3)
- **Five towns have no hue** — and 7.2 explains it: white is every wavelength, black is none. Hue sets key, **value sets register** (7.4)
- **LEAF STONE → REWARD**, closing the item set: reason from / search with / feel with / learn from. MOON STONE left for the town pass (8.2)
- **Corpus stays, `Corpel` moves** — three readings versus a placeholder in a provisional table; and it is only a document collision, since Charmander's line is untouched in the ROM (8.2)

- **MUSAI and ROVER implemented** — MUSAI/CODEMUSAI/SEEKMUSAI/CAREMUSAI and ROVERCUB/ROVERSEER/ROVERBYTE, with AXIOM/EMBEDDING/AFFECT and the three branch types, verified in the ROM (8.2)
- **The affect model stays out of the status bar** — rule 3 forbids it: a status is a thing a creature *is*. Four daemons, not five conditions (8.2)
- **The evolution items are AXIOM / EMBEDDING / AFFECT** — not stones but *kinds of input*, since the mechanic is exposure rather than mineralogy; KERNEL was better on paper and 2 characters too long (8.2)
- **`ROVERCUB` is the Reinforcement base** — a real repository, exactly 9 characters, and it reads as a young animal (8.2)
- **The repositories surveyed** — `codemusic-repos.md`. **Dexter is 4.2's Index built for real**, flat robotic voice included; **SafetyScribe is 4.10 with the instrument that was missing**
- **MUSAI takes the Eevee slot** — CONTENT base branching to **CodeMusai/LOGIC, SearchMusai/VECTOR, TherapyMusai/CONTEXT**; the first and third are Penphin's two hemispheres, and the chart makes LOGIC fail against CONTEXT (8.2)
- **ROVER is a linear line, not a second Eevee** — ROVER → ROVERSEER → ROVERBYTE is explore → model → act, reinforcement learning's own loop (8.2)
- **Two naming registers** — named individuals from the SHC/iASHC universe answer to nothing; the register question governs the other ~148 species (8.2)
- **Starter types implemented** — CONTENT/VECTOR/GROWTH with the secondary at final stage; lab dialogue moved with them. **Movesets are the blocker**: FLYING has 6 moves in all of Gen 1, so VECTOR may need its own CONSENSUS (8.2)
- **Penphin is a trade evolution, not a starter** — the dual mind requires two minds, so it cannot be obtained alone; recovered from conversation after going unrecorded from v1.7 to v3.8 (8.2)
- **MUSAI as Unsupervised, ROVER as Reinforcement**; **not** legendaries — a starter is raised for forty hours and a legendary is a trophy (8.2)
- **The foxes stop at the Clears** — ears, tail and *value*, nobody else converted; the cast stays as it is because at 16×16 the animal vocabulary is three pixels wide and a full conversion costs ~100 sprite sets for almost no signal (8.5)
- **`EnemyText` was the third `Enemy`** — the prefix behind every `<USER>` / `<TARGET>`, missed by the first REMOTE pass. Now `Remote @`, which **caps species names at 9 characters** (1.4)

- **The text ceiling is 18, not 19** — the earlier figure counted `<COLON>` as seven characters. Vanilla's four 19-char lines overwrite the right border (1.2)
- **INVOKED rejected on measurement, not taste** — 21 as the verb, 6 in a 5-char menu slot (1.4)
- **Currency is CACHE** — a hoard, a memory cache, and *cash*, all at once; the word is almost never on screen, so this is one prose line and a naming decision. **¥ kept**, logged as optional art (1.4)
- **The battle menu still says FIGHT and RUN** — a live collision in the most-seen string in the game. Not a text edit: the right column is 3 characters, so DETACH cannot fit, and moving the split is `wTopMenuItemX` surgery at four sites (1.4)
- **Kept: `used`** (1.4 made *use* morally loaded on purpose; INVOKED measured at 20 chars, over the ceiling) **and `ATTACK`** as a stat (1.4)
- **`enemy` → `REMOTE`** — nothing here is anyone's enemy; a REMOTE is a process you have no handle on, and it pairs with DETACH. `Enemy X ran!` → **`Remote X DETACHED.`**, killing a live `ran` collision (1.4)
- **`defeated` → `outscored`** — a BENCHMARK yields a score; *outran* rejected for reaching back at RUN (1.4)
- **Kept after interrogation: `RUN,` (comma), `EXP`, `LEVEL`** — EXP because 4.3's whole argument is that context forms from *experience*; LEVEL because it already means *permission level* and `<LV>` is a single tile (1.4)
- **Crystal is female and the text had not caught up** — `Gramps` → **`Gran`** (4.3, [private: SECRETS §2]), `His order` → `Her order`, and Agatha's block repronouned. **`handsome` deliberately kept** — it is attested for women and keeps Crystal tough rather than pretty (4.3)
- **Starter dialogue named vanilla types** — *fire/water/plant* → **ENTROPY/FLOW/GROWTH**; 19 more remain in gyms, held for 5 (1.2)
- **Crystal's opening speech**: six `#MON` marked plural; **pets → companions** (1.5's animal-word test), **fights → BENCHMARK**. *assistants* considered and held in reserve — it collapses vanilla's contrast (1.4)
- **PERSPECTIVE built** — move `$90`, and the battle message becomes *`<USER>` took the frame of `<NAME>`!*; **Index categories are capped at 10 characters**, so MOCK's cannot be PERSPECTIVE (4.6)
- **CONSENSUS built** — move `$A5`, SWARM, 90/100/15, no secondary effect; inserted before STRUGGLE because `NUM_ATTACKS == STRUGGLE` is asserted. Not yet learnable by anything (2.5)

- **SGB borders generated, not drawn** (`tools/genborder.py`) — CONTENT is one dot grid, CONTEXT is the same cell interfering with itself; 12 tiles each against a 96 budget (8.4)
- City names per 3.1; Slate over Somber; Halftone over Pallor; Quicksilver over Cinder; **Brazen over Gilt**
- Doldrum and The Bleed interrogated and kept, with the reasoning recorded (3.1, 3.2)
- Routes keep numbers officially and carry local names on signs
- Crystal Clear as the Oak figure; the Index measures only content, deliberately
- **Al Clear** as rival and incumbent [private: SECRETS §2]. Vanilla's rival is Oak's family, and Ty is far too old to race you (4.3)
- Default name lists per edition: slot 1 fixed, slot 2 swapped, slot 3 differentiated (4.3)
- **Ty P. Clear** is the generation between: Scorn's partner before the rebrand, then the one who went back. He is at Quicksilver, and he explains nothing (4.3)
- **Context does not transmit; only content does.** Ty taught Al the method and none of the experience — the Index problem inside a family (4.3)
- The family is named for three kinds of clarity and none of them can see (4.3)
- Richard Scorn at Benchmark 8, Alignment, sympathetic throughout; the engraving in the Corpus lobby, unattributed and adopted approvingly (4.4)
- BunnyArtsai as Mew with PERSPECTIVE — the first realization of perspective thinking, not a template (4.6)
- S.T.A.R.R. as Mewtwo with RECURSION — a **comprehension, not a clone** (4.7)
- The content/context feedback loop is the mechanism under the thesis and is never named in dialogue (0.3, 4.9)
- Five Witnesses easter egg, locked at a distance whose number lives only in the Quicksilver terminal log (4.8, [private: SECRETS §1])
- **Ty Clear's parentage is never stated.** Inferred from the surname, confirmed late by one line at Brazen (4.3)
- **The rival naming prompt stays.** The player names him; **CLEAR** is hard-coded, and formal text carries **AL CLEAR** — the route-sign device applied to a person (4.3, 9). *Corrected 2026-09-04: this line read TY CLEAR, which is a different person — **how Ty and Al are related is [private: SECRETS §2]**, and 4.3's ruling is explicit that the rival is Al*
- Crystal is CRYSTAL CLEAR in full wherever the game refers to her formally, or the surname device dies (4.3)

- **Corpus is downstream of Quicksilver by succession** — Scorn took over the lab, the metric changed, it burned, the people came with him (4.10)
- Crystal was at Quicksilver and built S.T.A.R.R. there; a fitness-for-work procedure removed her, and **the Index is what she did next** (4.1, 4.10)
- Scorn arrived afterwards to run it, signed a complete file about a person he never met, and set the temperature (4.4)
- Ty **stayed** when Crystal was removed; his guilt is an absence of decision, not a betrayal (4.3)
- **No date on the fire; dates on everything around it.** The player reconstructs the order or does not (4.10)
- **Brazen confirmed over Brass.** Brass is a colour and not a feeling, and would be the only single-meaning name on the map (3.1)
- At least one survivor knew what they were trading and did it anyway, for an ordinary reason — the guard against determinism (4.10)
- Craft rule 6: comedy is the cover. Corpus is funny; Quicksilver is fun to explore (0.1, 4.10)
- Four humors as the Review Board
- Vertical slice before anything else
- **Greyscale is the design, not a limitation.** Colour appears once, at the Review Board, and nowhere else (8.6)
- **Two editions, CONTENT and CONTEXT, from one source tree** — the slash in the title is literal (8.4)
- **The title order is CONTEXT-first by decision, not by accident.** Everyone says content is king; the title refuses to let the original decide its order. **The title is the argument, the RED→CONTENT mapping is the inheritance** — and this is filed as a choice, not a finding (8.4)
- The **type chart is byte-identical** across editions; only encounters and a handful of Index entries differ (8.4)
- The two-target build goes in on day one even while both ROMs are identical (8.4)
- **The quest's two halves are named: MARKS give intelligence, ROOTS give understanding** (*the user, 2026-10-06*) — "***Intelligence without understanding is artificial***" **is the spine, never said in the game; ROOTS is the brain's label on the USER card and never a count; the world shows the difference through a borrowed daemon, plastic plants and a fact collector, each DRAFT** (0.7; T-379–T-382)

**Reversed**

Kept here because the reasoning is worth more than the outcome.

| Was | Now | Why |
|---|---|---|
| **FOLDS is only paper** (4.33, LOOSE PAGES 7, 2026-09-25): *every step a fold, a crease, a corner, never saying what it is a diagram of* | **Each fold carries the essay's word for it** (T-401, the user 2026-10-15): *AWARENESS, CHANGE, SPACE, MIRROR, PATTERN, RETURN* | *The user, reading the loose pages again: "it isnt clear that it is talking about awareness ... right now it looks like it is literally talking about paper folds."* **A page indirect enough that nobody can find its subject has stopped being a fable and become a puzzle with no answer.** *The words name the stages, never the conclusions (love, spirit, unity), so craft rule 1 still holds; and the page still never mentions the scientist, so 4.33's NEVER ADJUDICATE is untouched. THE STONE and FIELDS were made plainer the same day.* |
| **One daemon at a time** (9.25, T-370, 2026-10-04) | **Any number AWAY, one device to a daemon** (T-396, the user 2026-10-09) | *The companion learned to put a daemon in each device (its C-80) and the game still refused a second SEND.* **A daemon in two places at once would not feel real; two daemons in two devices does.** *The one-at-a-time rule was standing in for "one place at a time", which the companion now keeps.* |
| **CONTEXT keeps the northern year, CONTENT the southern** (9.21, earlier 2026-10-03) | **CONTENT the northern year, CONTEXT the southern** (the user, 2026-10-03) | *The user, on reflection: "I think CONTENT should be the north america, and CONTEXT as south america's season."* **It fits the registers better**: *CONTENT is the thing itself -- the calendar as lived where the game is made; CONTEXT is the same date reframed by the other half of the world.* |
| **Nobody remarks on OPUS, then or ever** (2.x, 2026-09-21) | **The first shop clerk spoken to while you hold it notices it, says what it does and that SELECT reads it, and hopes you enjoy it** (T-348, 2026-10-03) | *The user asked for it.* **A margin a player never finds is a margin nobody reads**, *and the clerk explains the tool, not the mystery: who left it is still never answered, and the margins still never say they notice you.* |
| **The USER card's brain lights once and never counts** (T-271, 2026-09-25) | **A lobe per understanding, each in its type's colour; the halo at seven** (T-331, 2026-10-01) | *The user, seeing it: "I saw a brain but didn't know how it changes as u gain understanding."* **A glow that never counts was honest to 4.3 and told the player nothing**; *the lobes show which places changed them without a number anywhere, and the colours carry the chart's own language (9.4). The Guide's margins still hold the words.* |
| **Gilt City** | **Brazen City** | Gilt implies a concealer, and this story has no schemer. Brass is honestly itself; so is Scorn. Full argument in 3.1. |
| **Al calls her Gran** | **Al calls her CRYSTAL** | *Gran read oddly on playtest.* **4.1: every other authority has a role and she is just her name** — *family included.* 4.3, T-81 (2026-09-14) |
| **BunnyArtsai35** | **BunnyArtsai** | The number in her name gave away the Five Witnesses lock and made a serial of a one-off. Relocated to a single lab log (4.6, 4.8). |
| **Ty**, an unrelated rival | **Ty Clear**, Scorn's partner (the family: [private: SECRETS §2]) | Turns a methodological disagreement into a cost somebody pays (4.3). |
| S.T.A.R.R. as *refined successor* to the BunnyArtsai line | S.T.A.R.R. as a built understanding | "Successor" still smelled of cloning. The lab understood recursion and instantiated it (4.7). |
| **CATCH** | **BIND** | The container was renamed and the verb was not. CATCH is the only lexicon entry doing no double duty; BIND is `bind()`, *binding a daimon*, and a bond — and it |

| Was | Now | Why |
|---|---|---|
| | | is darker rather than softer, so the player stays implicated (1.1). |
| A vented server, cubes explicitly banned | A **cube with a screen**, pips for privilege | The ban existed because a cube says *container*. A screen says *running*, which is the reading 1.3 wanted — so the objection dissolves rather than being overruled. Vents also never shipped: the ladder loop broke before the third vent (1.3). |
| **RIGOUR**, for the item BLACK BELT | **LOGIC GAIN** | "*Held to argue harder from*" was a good name and its reasoning was sound. 1.6c then made the other sixteen type boosters a rack of labelled controls, and a set of eighteen where one is a poem is a bug, not a flourish. **Only the set changed; the name did nothing wrong.** |
| Drop the rival naming prompt, hard-code "Ty Clear" | Keep the prompt, hard-code **CLEAR** | The surname is the half that carries the inference. Spending a famous vanilla beat bought nothing — and the prompt, reframed as *what will you call him*, becomes the route-sign device a minute before Route 1 teaches it (4.3). |
| **The station's log:** `TRANSCRIPT: WITHHELD`, and 4.10's rule that the first session is never seen (9.22) | **The session, told by ARTSAI, as the Five Witnesses' reward** (T-235) | *The user, 2026-09-24: the last easter egg in the game is where the full story may be delivered and earlier rules may bend.* **Withheld everywhere else, still** — *a player who never solves the lock reads the log and nothing more.* |
| **The Owl's concession is dialogue only** — no document, no item (4.34 ruling 1) | **A sealed PEER REVIEW** he signs and the player gives to Crystal; its contents are never shown | He promised it in writing and handed over nothing, so the scene read as having changed nothing. The ruling's real worry — a concession one sentence too generous — is kept by sealing it (4.34). |

**Open**

- **A GLITCH, where one belongs** (*the user, 2026-09-24*). "***Too bad we couldn't simulate a glitch effect — there are a lot of places where it would fit.***" **First candidate: the moment DOLDRUM CAVE is understood** — *the simple drawing shown once more, then the screen glitches and the cave is itself.* ***The user's own caution***: "*maybe for this case the glitching is implying things we might not want*" — **a glitch reads as the world breaking, and here the world is not breaking, the player is seeing.** *Technically it is possible (a few frames of scrambled scanline offsets and palette), so the question is only where it means the right thing.*
- ~~**DOLDRUM CAVE: open the door, and let the player fail to see it**~~ ***Decided 2026-09-23 by the user (T-251).*** **The two gates become PERMISSION and PERCEPTION**: *the guard ("the DAEMON LEAGUE champion is the only person who is allowed in") stands aside once the REVIEW BOARD is beaten; the islands' understanding is the second gate, and it is perceptual.* **Without it the cave is BOTH grey and wrongly arranged** — *grey as HALFTONE is, which makes it the second grey place and the first whose grey can be lifted; and recomposed on each entry, every tile present and the arrangement wrong (content with no context).* ***Every encounter there is one LATENT abstraction,*** **drawn like the new unresolved display (T-250) with its palette randomised each time** — "*without perspective you could perceive many different things*" — **and it says only: *HE WHO SEES ONE, SEES NONE.*** *The encounter is the second half of the line; nothing names the idea (craft rule 1).* ***With the understanding the cave is itself, in colour, and S.T.A.R.R. is in it.*** *Kept from the discussion, for the record: 8.6a's rejected colour item, and why this is not it — nothing here is handed over or counted, and it changes one place.*
- ~~**UNDERSTANDINGS — the islands' DRIVERS, and a second thing the USER card holds**~~ ***Decided 2026-09-23 by the user (T-252)*** — **THE MARGINS only; never counted; the cave's gate is built first** *and the set is designed after it has been played.* ***The user's standard for it***: **it has to be FELT** — *the player should understand it by seeing it, not by being told.* **A DRIVER is handed to you; an understanding is arrived at (4.3, 4.24)** — *that rule stands. Names still open (PERSPECTIVE and REVEAL are routines; ABSTRACTION aloud is the thesis).* ***Seven, and no more*** *(the user, 2026-10-02, asked whether an eighth or ninth belongs: "no").*
- ~~**What is HALFTONE Tower about, if not death?**~~ ***Decided 2026-09-23 by the user (T-249, T-250).*** **The stones stay, as TOMBSTONES in the database sense, and the tower is mostly kept** — ***but a stone must not look like vanilla's***: **an encased screen, green marks on black, shaped so it still reads as a headstone.** *Half the town reads a grave and half reads a deletion record, and the object is both.* **The LATENT daemons keep their non-corporeal look** ("*a ghost is what latent space looks like to something that sees only its influence*"); **the unresolved display becomes structured noise, a latent before it is decoded, and the RESOLVER is the decoder.** *Nothing says so.*

- ~~**Two painted labels need a NAME, not a fix**~~ ***Decided by the user 2026-09-24***: `OT` → `BY` *(who bound it)* **and** `PkRS` → `MEME` — *and MEME wants a larger place than one icon (T-258).*
- **LAB NOTES 4 contradicts 4.24** (*found drafting T-224, 2026-09-23*). ***`school.md` §10 draws it from a passage 4.24 rules out of the game in every form*, and nothing reverses that ruling.** *The draft keeps the page and strikes its lines through, the way the sealed review keeps its cover sheet.* **The options are that, cutting the entry, or re-sourcing the slot from another verse.** *The user's.*
- ~~**Do the eight DRIVER routines stay in daemons' LEARNSETS?**~~ ***Decided by the user 2026-09-24: they STAY.*** *Knowing PRUNE was never the problem; being kept for it was.*
- **Convergence and divergence as a MECHANIC** (*asked 2026-09-17: "if models are aligned they are convergence, and if not they are divergence — it very much relates to the whole perspective thinking template"*). ***Think long about this; it could be an important mechanic.***

- ***The trap in the obvious reading, and it is the bible's own.*** **0.5 and 4.18a locate clarity in two views ALLOWED TO DISAGREE and a third stage that holds them against each other** *— and Scorn's crime is forced convergence: he weighted that stage so the answer always came out the same, and the form calls it IMPROVE RESPONSE CONSISTENCY.* **Benchmark 8 is "perfectly aligned — to the wrong objective." *So a mechanic that simply rewards convergence and punishes divergence rewards Scorn.* The reading that survives: convergence is worth something only when it is REACHED FROM real divergence** *— two views that differed and then agreed have been checked in the one way neither can check itself; views that agreed from the start have been checked by nothing.*
- ***Four homes, from most concrete to most narrative:***
  1. **The streaks** (9.4, *built 2026-09-17*). *A daemon now shows what it understands as four move-type colours,* **so how far two daemons' streaks overlap is an exact, cheap, already-visible measure.** *Candidate:* ***CONSENSUS*** *(SWARM — "a reliable check") grows with the party's convergence, and a CONTEXT routine grows with a pair's divergence —* **neither wins, which is the chart's argument again.**
  2. **Double battles.** *Two daemons act in one turn, so their choices converge or diverge;* **a small bonus for agreement AFTER disagreement** (*same target this turn, different types last turn*) **is the clarifier in miniature.** *Heaviest engine work of the four.*
  3. **The two editions.** *A daemon traded between CONTENT and CONTEXT already has two Index entries that do not match (0.4) —* **divergence as a literal record**, and **the link may be where the two are held against each other.**
  4. **Scorn's party.** *Fully converged — identical streaks, perfect consistency, flawless and brittle —* **and the way through is divergence.** *No new system: his parties, movesets and 5.3b's fixed roll.* **It makes Benchmark 8 argue in play.**
- ***Constraints on any version***: **craft rule 1** *— neither word is ever said;* `ALIGN` */alignment is already on record as "the thesis wearing a lab coat" (5.2), and the unbuilt reverse-PERSPECTIVE move below was going to be called* `ALIGN`*, so the two questions should be answered together;* **invariant 3** *— nothing here may make the type chart differ by cartridge.*
- ***SELF-ALIGNMENT matters more than alignment*** (*added 2026-09-17*): **"the one who holds the frame can become aware of it and shift the frame."** *This turns the three frame questions into one family, and puts the weight on the third:*

| | what happens to a frame | who moves it | where it already is |
|---|---|---|---|
| **PERSPECTIVE** | *I take yours* | the user, outward | **built** — TRANSFORM, CONTENT, its hue drained to grey before the swap (9.24) |
| **ALIGN** (*proposed, below*) | *you take mine* | the user, imposed on another | **unbuilt** — and it is Corpus's move, and Scorn's |
| **self-alignment** | *I see mine, and move it* | the holder, on itself | ***already in the game twice, unremarked*** |

***Where it already is***: `CAST` (*CONVERSION*) **changes the user's type to one of its own moves' types** — *the body taking the shape of what it already understands, which 9.4's streaks now make visible* — and `RECAST` (*CONVERSION 2*) **changes the user's type to resist the last routine that hit it — *noticing the frame it was hit through, and shifting.* That is self-alignment as a mechanic, and nobody arranged it. *Why it is the heavier idea*: 4.18a's BELIEFS are the difference between a weight and a position — *a belief that can be inspected and challenged* — and self-alignment is the holder doing that to itself.** *Scorn's machine could not: the weighted stage was invisible from inside, so it kept producing the contradiction and kept resolving it the same way.* **So self-alignment is the answer to Benchmark 8 that is not another imposed frame** — *a party that wins by shifting its OWN frame rather than by forcing a counter-alignment on his.* ***Answered 2026-09-18 — RECAST is the COPY family's visible centre*** (**T-162**, *approved*): **the foe that last hit it glints, the user drains to ground and HOLDS — the longest still moment in the family, the frame noticed — and comes back in the colour of the type it has become.** *CAST does the same without the hold: it reads its own routines and becomes what it already knew. Nothing is said.* ***Questions it opens, still open***: *is there a state or a moment where a daemon's body type drifts toward its streaks, so self-alignment happens over a run and not only in one move? and does the reverse-PERSPECTIVE move stay `ALIGN` once self-alignment is the idea the family is for — or does the imposed version want a name that reads as imposition?* **Craft rule 1 holds throughout: none of these words are said.**

  - ***Leaning, not decided***: **4 and 1 first** — *both use systems that exist, and they argue the point from both sides (a party punished for having converged, and a move that rewards converging).*

- ~~**Day and night**~~ ***decided and built 2026-09-25 (9.21, T-268)*** — **the real clock when the cartridge has one, play time when not; Routes 1, 8, 10 and 12 at night.** ***The four seasons*** (9.21, *proposed 2026-09-14*). ***A clock made of play time, day read as CONTENT and night as CONTEXT, and the seasons as the Review Board's four humours leaning the wild tables toward VECTOR, ENTROPY, LATENT and FROZEN.*** **Nothing built; four questions in 9.21 to settle first**

- **The GLOBAL INDEX in the islands** (*asked 2026-09-14*). ***Every daemon past #151 findable in the seven islands, split by edition.*** **235 species, 33 met today.** *A derived placement is drafted for review in* `tools/plan_global_index.py` **— and it collides with 2.10 and 8.2b: a daemon the player can meet gets a name, so making all of them meetable means naming about two hundred more, in the islands' register**
- Does Halftone hold once the tower is written, or do Penumbra / Moiré serve better?
- **Do the names have to BEAT the model a reader already has?** (8.10). ***The agent ground Route 4 for 205 steps looking for a GROWTH or SIGNAL daemon, with 9.4's hue in front of it and the type badges on screen, and reasoned in Grass and Electric anyway.*** **2.6 asks whether a player can generalise or only remember; this reader could do either and chose to remember.** *A pretrained reader translates back whenever translation is possible —* **a cost of renaming that 2.6 never priced and cannot price from the chart alone**
- **Should the agent be TOLD that colour is the type?** (8.10). *Saying it makes the run measure 9.4 — whether a reader handed the model can use it. Leaving it unsaid keeps the run measuring 2.6 — whether the words win against priors.* ***Both are worth measuring and they are not the same run***
- **Does the frame family want its REVERSE** *— a move where the FOE takes the USER's frame?* ***Asked 2026-09-12.*** `PERSPECTIVE` **is TRANSFORM, typed CONTENT, and every copy move in the game runs the same direction**: `ASSUME`, `CLONE`, `IMPRINT`, `COPY STATE` are all *user ← foe*, and `SWAP SCOPE` is the only mutual one. ***No generation of the source game has a target-becomes-user move at all*** **— Pokémon has a move for empathy and none for imposition**, *and this design already contains the thing the source is missing.*

    - ***The reading is 4.18a's, not a new invention.*** **A clarifier weighs two threads at a perspective stage, and 8.5 records the failure as *one vote could be raised*** — so the reverse of PERSPECTIVE is not "*you become me*" but "***your resolution comes out where I wanted it.***"
    - **The mechanic with a visible argument is *the target's type becomes the user's* — *you hand them your weaknesses*,** legible inside one battle and level-proof, which is craft rule 5. *The quieter option is Gen 4's Entrainment, which is* `ASSUME` *with the battlers swapped and nearly free in C* — **and 2.11 makes it the more chilling read, since an ability is *what is true of a daemon while it is doing nothing.*** *A player will never notice that one, which is the case against it.*
    - ***It should be CONTEXT, and it must not be CORRUPT.*** `PERSPECTIVE` **— I become you — is CONTENT; the reverse — you become me — is CONTEXT**, *and 0.4's rule is that neither input wins.* ***So the two frame moves would be the two halves of recognition with neither dominating, which is 0.4 as a move pair and is never said.*** **CORRUPT fits the sentence — *data that has been tampered with* — and would be the game telling the player the move is evil.** *The type would become the moral, and that is craft rule 1 breaking.*
    - `ALIGN` **is the name**, *clean at distance 3 with nothing that rhymes*, **and Benchmark 8 IS alignment — *who controls the ground everything stands on.*** `ONBOARD` *is funnier and more Corpus, and sits 3 from* `UNLOAD` **— 3 is what retired** `THE HOLDOUTS`**, so it wants the harder check first.**
    - ***And the risk that decides where it comes from.*** **The obvious vehicle is the CC-7 the player has carried since minute fifteen** — *and 4.32's whole point is that nothing points at it.* **Teaching a move from it points at it**, which may spend something Callow is saving. *Safer: it is Scorn's ace's move and the TM is somewhere in Corpus.*

- ~~Which of the four name/type collisions gives way~~ **Answered 2026-09-12 (2.13, T-49): one moved and three were rulings.** `NOISE` **→** `RESIDUAL` — *4.26's statistics line survives whole and ENTROPY keeps its clause.* `SUBSTRATE` **is 4.25's on purpose,** `SUSPEND` **is a verb it performs rather than a type claim, and** `FAULT` **is a word the check deliberately does not list** — *STRATUM owns the compounds, not the bare noun*

- ~~`REINFORCE`, `DEPRECATE` and `DISSOLVE` say the opposite of what they do~~ **Fixed 2026-09-12 (2.13, T-49):** `FATIGUE`**,** `LONG SUPPORT`**,** `PASSIVATE`**.** ***And*** `IRON TAIL`***'s description had doubled down on the error***, reading "*adds more of what was already holding*" in front of an effect that takes armour away

- **The physical / special split** (2.12). *The code is an afternoon and the 354 judgment calls are the job* — **and five of 5.3b's leader designs were built against the Gen 3 rule**

- **Seventeen type-resist handlers, to finish 1.6c's rack** (2.12). *`<TYPE> GAIN` amplifies one channel; `<TYPE> TRAP` would damp one, and the register is already derived.* **The cheapest good idea in that section**
- **The overfit state** (2.12). **Benchmark 4's own lesson, and it is not a mechanic** — *`FURY CUTTER`'s counter already exists and only the penalty is new.* ***`OVERFIT` is a species, so the seventh state needs a name before anything else***
- **Temperature as a room** (2.12), at Benchmark 7 — *widen the damage roll for both sides, on a flag.* **5.3b already built that plumbing for SCORN**
- ~~Which region names~~ **Built 2026-09-11 (0.6, T-48): `GAMUT` and `THE MARGINS`.** ***`THE HOLDOUTS` was the recommendation for a day and the check retired it*** — *`LOOKOUT` is edit-distance 3, is the same `<word>` + `OUT` shape, and is* ***island-only***
- ~~`SWARM` and `GROWTH` each mean two things~~ **Cleared 2026-09-10 (2.8): `SCALE OUT` and `SCALE UP`.** *check_lexicon reports 1,152 names and no word meaning two things*
- ~~Re-score the one-clause test at 17~~ **Done 2026-09-11 (2.6).** ***CONTEXT went UP, to 11/11*** *— the immunity the question worried about is the clause's cleanest prediction in the chart: you cannot reframe what you cannot read.* **The real gap was that `HARDENED` (21 relations, the most-connected type in the game) and `OPAQUE` (10) had never been scored at all.** *Both pass. The run had walked 83 relations and the ROM has 111*
- ~~`LEGACY` → `RUST` and `VECTOR` → `FLOAT`~~ **Declined 2026-09-11 (2.6), with reasons rather than left open.** ***The failing score was a property of the clause, not of the word*** *— a sentence saying what a type IS can predict none of a chart made only of relations.* **Re-clauses: LEGACY 0/16 → 12/16, VECTOR 1/13 → 10/13, both names unchanged**, and `FLOAT`'s whole argument — the STRATUM immunity — arrives for free
- ~~Craft rule 5 is unpaid for benchmarks 2–8~~ **Designed 2026-09-10 (5.3).** *Seven leaders, each with the thing an overlevelled player still runs into* — **now it is a build task rather than a design one**
- ~~The six new leader names are unbuilt~~ **In the ROM 2026-09-10 (5.3a), with 127 lines of dialogue swept to match.** *What is still unbuilt is the mechanics* — the parties, Doldrum's healing trainers and Scorn's fixed roll
- ~~The Review Board has no names~~ **Named 2026-09-10 (6): PHLEGMATIC, CHOLERIC, MELANCHOLIC, SANGUINE**, and the two mismatched types corrected with them — *Choleric is LOGIC and Sanguine is EMERGENT*
- **SUSPENDED is CONTEXT's concept; whether the six sleep moves are retyped onto it is a balance question** (2.7). Moving them takes GROWTH's only sleep move away and hands CONTEXT six — 2.5's rule says rebalance moves before matchups

- ~~Four of the thirteen retyped daemons are still 78% off-type~~ **Done 2026-09-10 (T-02).** *100% in fact, all nineteen.* `tools/port_musai.py` **takes the branch to 37%, under 2.7's 45% baseline** — and ***S.T.A.R.R. was already at 25%***: the 78% was the thirteen's average carried onto a name that did not deserve it
- ~~The CONTENT register for ROUTINES~~ **Settled 2026-09-09 (2.8): the instruction set.** *Say the name; if you then have to say what it does, it is not CONTENT*
- ~~`THRASH` and `HARDEN`~~ **Cleared (2.8).** `SIGNAL BEAM` is now unblocked by 2.9's SWARM register — `GOSSIP` — and waits only on the SWARM batch
- ~~The ROUTINES pass~~ **Done 2026-09-10 (2.10): 269 of 356, and every routine the player can meet.** *The 87 left are reachable by nothing in this game* — **they wait on the bestiary, not on a decision**
- **Does THRASHING want an owner** (2.7), or is "the state no type owns" the point? LOGIC is the candidate — a system so busy reasoning about itself it makes no progress
- **THROTTLED's five strays** (2.7). SIGNAL owns it six to five; the strays are the question
- ~~The STREAM has no overworld notification~~ **Built 2026-09-10 (T-12).** *Nine told-flags out of T-17's block and the same per-step hook the VS SEEKER uses* — **told, not seen: a show you skipped does not nag, and a new MARK still announces**
- ~~`PC` → `PORT` is decided and not swept~~ ***Decided is not a question.*** **Moved to TODO T-31** — *four menu strings, ~25 dialogue lines, and* `\nPC` *has no word boundary in front of it*
- **PKRS still says PKRS** (9.15). Its tile no longer borrows LEAKING's colour, but the name is vanilla and the condition has no place in the lexicon yet
- Does the player meet Scorn before Halftone Tower?
- ~~Does RECURSION justify engine work in the slice, or defer?~~ **Built 2026-09-13 (4.7)**, *and it was never engine work on GBA: Fury Cutter's counter already did it*
- How legible is S.T.A.R.R.'s SHC backstory to a player who has not heard the rock opera — and does it need to be?
- Are the humors too neat? Four is convenient; the real theory had temperaments blending.
- Does **Al** get a redemption, a plateau, or neither — and does the family tie make redemption too cheap?

- ~~Ty is absent from the endgame by design. Does that read as a statement or a loose thread?~~ ***It reads as a loose thread, and 8.2a is the answer: Act 2 is the islands.*** **Closed twice over** — *4.34 then put him in the Warehouse handing you something for Crystal ([private: SECRETS §2]), and 4.17's not-visiting is still a choice on both sides*
- Is the Quicksilver terminal missable enough to soft-lock the Five Witnesses puzzle, and is that acceptable?
- Does Brazen ever read as the game *sneering* at Scorn? If playtesters hear that, swap to Brass immediately — the whole point of him is that the game does not sneer.
- ~~The bestiary naming register is undecided~~ **It had been decided in practice and never written down** (8.2b). *The 72 that shipped run three registers at once —* **technical surface for the bestiary, myth for the legendaries, the project's own names for the MUSAI and ROVER families** — and the fork in the log was answered by 4.26 without anyone noticing
- ~~69 species beyond Kanto are meetable and unnamed~~ **Superseded 2026-09-11 (8.2b, T-45).** ***The ruling held until two of our own rules contradicted each other*** *— "name nothing there" against 2.10's "name what the player can meet" —* **and 33 of the 69 have wild encounters.** *All 33 are island-only, so they took Act 2's register:* **Kanto's creatures are machinery, the islands' are mind. 52 named with their evolutions; 219 species are ours.** *The rest exist only in the data and stay unnamed*
- The supervised starter slot is **yours to pick** (8.2)
- ~~Callow and The Bleed share their music~~ **It was already TODO T-05**, *carried in both places for a week.* **Blocked on a track that does not exist, not on a decision**
- ~~57 `catch` / `caught` instances remain in dialogue~~ **Done 2026-08-31**, caught in play: the capture message still read *PACKET was caught!*. **55 instances rewritten.** The only survivor is *catchy tune*, which is a song
- ~~MOON STONE still wants a name~~ **Moved to TODO T-33.** *It was waiting on the town pass and the town pass is done*
- `Corpel` ~~collides with Team Corpus~~ **Gone.** *No species carries it and 8.2's table is the only place the word still appears —* **the collision was resolved by the bestiary passes without anybody closing the entry**
- ~~Starter daemon names are placeholders and need a pass~~ **Moved to TODO T-34.** *A pass is a job*
- Should ORPHAN be bindable at all, or only witnessed? *Lean: bindable — a blank entry sitting in your own collection is worth more than a blank entry you only heard about.*

- ~~Nineteen `type word + DAEMON` dialogue lines still name vanilla types — held until 5 settles the leaders~~ ***5.3 and 5.3b settled them, so the hold is over and this is a job:*** **TODO T-32**
- ~~Forty-two `fight` occurrences want a human read~~ **Read 2026-09-10 (T-07).** *Three buckets and the read was the job —* **four were the TYPE in lower case, eleven were 1.2's replaced verb, and the rest are people.** *It also turned up the help system's own copy of the chart*
- ~~Roughly 154 `#MON` occurrences read singular by default~~ **Read 2026-09-10 (T-08).** *POKéMON is its own plural and DAEMON is an ordinary noun that is not.* **25 sentences force it and were done one at a time**; the rest are genuinely singular or attributive
- Does a non-canonical ROM load in Gen1Recomp at all? One afternoon answers it; do not design around either answer first (8.5)
- How many Index entries should disagree between editions — five? twelve? — before it stops being unsettling and starts being a gimmick? (8.4)
- Is edition-exclusivity fair when link trading needs two people, two carts and a cable, and most players will have one? *Lean: yes — the Index was never going to be completable, and 4.2 says so.*